File 17G

Scott AFB
Illinois

ADMINISTRATIVE RECORD
COVER SHEET

AR File Number________1777_____

Final
Operation & Maintenance Manual
Site LF-01, Former Base Landfill
Scott Air Force Base, Illinois

Contract Number W91238-06-D-0021-DK02

Prepared for

375th Airlift Wing, Air Mobility Command
Scott Air Force Base
and
U.S. Army Corps of Engineers
Omaha District

Prepared by

HydroGeoLogic, Inc.
6340 Glenwood, Suite 200
Building #7
Overland Park, KS 66202

June 2014

FINAL

OPERATION & MAINTENANCE MANUAL
SITE LF-01, FORMER BASE LANDFILL

SCOTT AIR FORCE BASE, ILLINOIS

Prepared for

375th Airlift Wing, Air Mobility Command
Scott Air Force Base
and
U.S. Army Corps of Engineers
Omaha District

Contract No. W91238-06-D-0021-DK02
CLIN 0001

Prepared by

HydroGeoLogic, Inc.
6340 Glenwood, Suite 200
Building #7
Overland Park, KS 66202

June 2014

This page was intentionally left blank.

TABLE OF CONTENTS

Section **Page**

1.0 INTRODUCTION 1-1
1.1 SITE DESCRIPTION 1-1
1.1.1 Physiography and Climate 1-2
1.1.2 Surface Water Hydrology 1-3
1.1.3 Hydrogeology 1-3
1.1.4 Current and Anticipated Future Land Use 1-4
1.2 REMEDIAL ACTION OBJECTIVES 1-5
1.3 REMEDIAL ACTION SUMMARY 1-6
1.4 REPORT ORGANIZATION 1-8

2.0 GENERAL INFORMATION 2-1
2.1 LANDFILL COVER SYSTEM 2-1
2.1.1 Protective Layer 2-1
2.1.2 Geocomposite Drainage Layer 2-1
2.1.3 Geomembrane 2-2
2.1.4 Landfill Gas Collection System 2-2
2.1.5 Foundation Layer 2-2
2.2 EROSION CONTROL MEASURES AND SLOPE STABILIZATION 2-3
2.3 ENVIRONMENTAL MONITORING SYSTEM 2-3
2.3.1 Groundwater Monitoring 2-4
2.3.1.1 Potentiometric Conditions 2-4
2.3.1.2 Landfill Network Monitoring Well Network 2-5
2.3.1.3 Groundwater Remedy Monitoring Well Network 2-6
2.3.1.4 Wells Not Included In Long Term Monitoring Network 2-7
2.3.1.5 Monitoring Well Installation 2-8
2.3.2 Surface Water Monitoring 2-9
2.3.3 Landfill Gas Management System 2-9
2.3.4 Ambient Air Monitoring 2-9
2.4 EOD FACILITY 2-10
2.5 CONSTRUCTION 2-10
2.6 FINAL INSPECTION 2-10

3.0 POSTCLOSURE INSPECTIONS, MAINTENANCE, AND REPAIRS 3-1
3.1 SITE INSPECTIONS 3-1
3.1.1 Landfill Cover 3-2
3.1.2 Landfill Gas System 3-2
3.1.3 Drainage System 3-2
3.1.4 Groundwater Monitoring Wells 3-3
3.1.5 Landfill Cover System Settlement 3-3
3.1.6 Access Road 3-4
3.1.7 Surveys 3-4
3.2 MAINTENANCE AND REPAIRS 3-4
3.2.1 Landfill Cover 3-5
3.2.2 Vegetative Cover 3-5
3.2.3 Landfill Gas System 3-6
3.2.4 Groundwater Monitoring Wells 3-6

TABLE OF CONTENTS (continued)

Section **Page**

3.2.5 Access Road 3-7
3.3 LAND USE CONTROL 3-7
3.3.1 Land Use Control Implementation 3-8
3.3.2 Land Use Control Inspection 3-9
3.3.3 Future Use Restrictions 3-9

4.0 POSTCLOSURE MONITORING 4-1
4.1 GROUNDWATER MONITORING 4-1
4.1.1 Groundwater Sampling 4-1
4.1.2 Sampling Frequency 4-3
4.1.2.1 Landfill Network Monitoring Wells 4-3
4.1.2.2 Groundwater Remedy Monitoring Wells 4-4
4.1.3 Sample Analysis 4-4
4.1.3.1 Landfill Performance Monitoring Well Network 4-5
4.1.3.2 Groundwater Remedy Monitoring Well Network 4-5
4.1.4 Groundwater Field Parameters 4-5
4.1.5 Field Records 4-6
4.1.6 Potentiometric Surface Maps 4-6
4.1.7 Data Analysis 4-6
4.1.7.1 Applicable Groundwater Quality Standards 4-7
4.1.7.2 Evaluation of Groundwater Quality Data 4-7
4.1.7.3 Groundwater Quality Assessment Program 4-9
4.1.7.4 Quality Control Program 4-10
4.2 SURFACE WATER MONITORING 4-11
4.2.1 Surface Water Sampling 4-11
4.2.2 Sampling Frequency 4-11
4.2.3 Sample Analysis 4-12
4.2.4 Surface Water Field Parameters 4-12
4.2.5 Field Records 4-12
4.3 LANDFILL GAS MONITORING 4-12
4.3.1 Landfill Gas Field Parameters 4-13
4.3.2 Field Sampling Procedures 4-13
4.3.3 Field Records 4-13
4.4 AMBIENT AIR MONITORING 4-14
4.4.1 Ambient Air Monitoring Locations 4-14
4.4.2 Field Sampling Procedures 4-14
4.4.3 Field Records 4-14
4.5 IDW MANAGEMENT 4-15

5.0 RECORD-KEEPING AND REPORTING REQUIREMENTS 5-1
5.1 ANNUAL REPORT 5-1
5.2 FIVE YEAR REVIEWS 5-3

6.0 REFERENCES 6-1

LIST OF APPENDICES

Appendix A	As-Built Drawings for Landfill Cover
Appendix B	Construction Details for Monitoring Wells, Gas Wells, and Cover Settlement Monuments
Appendix C	Standard Operating Procedures
Appendix D	Supporting Documents

LIST OF TABLES

Table 2.1 Landfill Cover Components
Table 2.2 Groundwater Long Term Monitoring Network

Table 3.1 Summary of Monitoring, Inspection, and Maintenance Activities
Table 3.2 Postclosure Site Inspection Checklist

Table 4.1 Analytical Summary of Groundwater and Surface Water Monitoring

LIST OF FIGURES

Figure 1.1 Site Location Map
Figure 1.2 Landfill LF-01 Site Overview Map

Figure 2.1 Groundwater and Surface Water Monitoring Network
Figure 2.2 Postclosure Landfill Gas and Ambient Air Monitoring Network

Figure 3.1 LUC Boundary

LIST OF ACRONYMS AND ABBREVIATIONS

AFB	air force base
ANOVA	analysis of variance
bgs	below ground surface
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
COC	contaminant of concern
DCE	dichloroethene
DoD	U.S. Department of Defense
EOD	explosive ordnance disposal
ERM	Environmental Resources Management
ERP	Environmental Restoration Program
ft/ft	feet per foot
FSP	Field Sampling Plan
HGL	HydroGeoLogic, Inc.
IAC	Illinois Administrative Code
IDW	investigation-derived waste
IEPA	Illinois Environmental Protection Agency
LEL	lower explosive limit
LF-01	Landfill 01
LLDPE	linear low-density polyethylene
LUC	land use control
LUCIP	Land Use Control Implementation Plan
MCL	maximum contaminant level
MNA	monitored natural attenuation
mph	miles per hour
MWH	Montgomery Watson Harza Americas, Inc.
NCP	National Contingency Plan
O&M	operation and maintenance
PAH	polynuclear aromatic hydrocarbon
PVC	polyvinyl chloride
QA	quality assurance
QAPP	quality assurance project plan
QC	quality control

RA	remedial action
RAO	remedial action objective
RG	remediation goal
ROD	Record of Decision
RPM	Remedial Project Manager
SOP	standard operating procedure
TAL	target analyte list
TCE	Trichloroethene
USACE	U.S. Army Corps of Engineers
USAF	U.S. Air Force
USEPA	U.S. Environmental Protection Agency
VOC	volatile organic compound
WWTP	wastewater treatment plant

FINAL
OPERATION & MAINTENANCE MANUAL
SITE LF-01, FORMER BASE LANDFILL
SCOTT AIR FORCE BASE, ILLINOIS

1.0 INTRODUCTION

This Operation and Maintenance (O&M) Manual was developed to provide guidance for the postclosure O&M activities required for the Environmental Restoration Program (ERP) Site Landfill 01 (LF-01), Former Base Landfill located at Scott Air Force Base (AFB) in St. Clair County, Illinois. This O&M Manual has been prepared by HydroGeoLogic, Inc. (HGL) on behalf of Scott AFB under contract to the U.S. Army Corps of Engineers (USACE) Omaha District, Contract No. W91238-06-D-0021, Delivery Order DK02.

This O&M Manual details the postclosure requirements for maintaining the landfill and monitoring groundwater contamination as part of the groundwater remedy at LF-01. The manual complies with the postclosure care requirements specified in 35 Illinois Administrative Code (IAC) 811 as well as other pertinent Illinois applicable or relevant and appropriate requirements specified in the Record of Decision (ROD) addressing surface soil, waste, and sediment (Scott AFB, 2013a) and the ROD addressing groundwater at LF-01 (Scott AFB, 2013b). Postclosure O&M activities at LF-01 include performing site inspections, completing maintenance and repair actions, implementing land use controls (LUCs), and conducting environmental monitoring. The site inspections consist of general site condition inspections that include the condition of the protective cover, site monitoring wells, and verification of compliance with LUCs. At each five year review, the frequency of future O&M inspections and monitoring will be re-evaluated. Adherence to the monitoring, inspection, maintenance, and repair procedures contained in this O&M Manual will facilitate compliance with federal and state regulations.

1.1 SITE DESCRIPTION

Scott AFB is located in St. Clair County, Illinois, approximately 25 miles southeast of St. Louis, Missouri (Figure 1.1). The city of Belleville, Illinois, is located approximately 6 miles southwest of Scott AFB and the city of O'Fallon, Illinois, is located approximately 5 miles northwest of the base. Scott AFB property encompasses 3,545 acres of U.S. government-owned and easement land. Airbase facilities and runways compose the largest portion of the installation, which also includes living quarters and recreational facilities. Agricultural land, retail development, and residential properties border the base (Environmental Resources Management [ERM], 1992). The MidAmerica Airport is located east of and adjacent to Scott AFB.

Scott AFB was established in 1917 as an aircraft pilot training facility known as Scott Field. Between 1920 and 1937, dirigible airships were assigned to Scott Field. A basewide construction program began in 1938 and, between 1940 and 1942, four concrete runways were constructed. During and after World War II, Scott AFB was the headquarters for various military commands (ERM, 1992). Today, the headquarters for U.S. Transportation Command and Air Mobility Command, a reserve aeromedical airlift wing, and an Illinois Air National Guard aviation support facility are stationed at the base. The primary mission of Scott AFB is global mobility support (Scott AFB, 2010).

Site LF-01 consists of two landfill cells that total 35.9 acres (areal extent); the North Cell comprises 20.4 acres, and the South Cell comprises 15.5 acres (Figure 1.2). Landfill operations at LF-01 began in the early 1940s and ended in 1976 (Montgomery Watson Harza Americas, Inc. [MWH], 2003). Since then, base refuse has been taken off site for disposal. Hardfill and earthen fill materials continued to be placed on the landfill until 2012. Sewage treatment plant sludge was placed on the landfill until 1989. Additionally, nonhazardous soil from previous Scott AFB RA activities was transported and staged at the landfill before the landfill cover was constructed. This material was used to construct the foundation layer of the landfill cover. Records searches revealed that the historical materials disposed of at LF-01 were wastes generated at Scott AFB, including domestic refuse, hardfill and construction rubble, coal ash from the Scott AFB power plant, wastewater treatment plant (WWTP) sludge, and industrial wastes. Industrial wastes reportedly disposed of at the landfill included a quantity of paint (exceeding 1,000 gallons) in cans, pesticides, oils, transformers, and drums of unknown contents.

The explosive ordnance disposal (EOD) proficiency range (0.1 acre) that was previously located on the South Cell was decommissioned before the landfill cap was constructed. Scott AFB plans to construct a new range on the landfill cap at the South Cell in an area of the cover engineered to accommodate the EOD proficiency range.

1.1.1 Physiography and Climate

Scott AFB is within the Springfield Plain subdivision of the Till Plains section of the Central Lowlands Physiographic Province (MWH, 2003). The upland till plain is relatively flat; however, it is interrupted by a belt of low ridges and mound-shaped hills west and northwest of the base and by the floodplain of Silver Creek to the east. The Silver Creek floodplain is located on the east side of the base and includes wetlands, meanders, and terraces.

Southwestern Illinois, including Scott AFB, has a continental climate. Average temperatures range from the mid-teens on winter nights to the low 90s on summer days. The wettest months are May and June, with each averaging 4 inches of rainfall. The average annual precipitation is 39 inches. Winds at Scott AFB are predominantly from the north at speeds of approximately 4.6 to 11.5 miles per hour (mph). Winds seldom occur from the southwest or at speeds in excess of 24 mph. Calm conditions (less than 1 mph) prevail nearly 15 percent of the time.

1.1.2 Surface Water Hydrology

The areas located to the north and east of both landfill cells are predominantly wooded wetlands. The two landfill cells are separated by Mosquito Creek, which flows from the northwest and discharges into Silver Creek approximately 1,500 feet to the east of the cells. South Ditch flows northeastward before turning right 90 degrees and discharging into Mosquito Creek just north of the entrance bridge to the North Cell. The Scott AFB WWTP outfall discharges into a small drainage ditch that flows 300 feet eastward and discharges into Mosquito Creek. An Unnamed Tributary is located adjacent to the east side of the South Cell and connects the wetlands area to the east and south of the South Cell to Mosquito Creek. The Unnamed Tributary connects to Mosquito Creek about 1,000 feet west of Mosquito Creek's confluence with Silver Creek. Flow direction in the Unnamed Tributary is dependent on intensity and duration of local precipitation events and can vary from north to south.

The wetlands area east of the landfill cells is flat, with the exception of Mosquito Creek and the Unnamed Tributary, which incise the wetlands approximately 7 to 9 feet. Several small seasonal ponds have been identified in the wetlands area north of the North Cell. The wetlands area is heavily saturated and flooding occurs during the spring thaw and periods of heavy and prolonged rainfall. Approximately 2 to 3 feet of standing water has been observed in the wetlands north and east of the North Cell and approximately 1 to 2 feet of standing water have been observed in the wetlands east of the South Cell. Floodwaters have been observed in contact with the slopes of the two landfill cells. As a result, accessibility to these areas can be difficult during these conditions.

1.1.3 Hydrogeology

No significant regional aquifers exist in the area of southwestern Illinois where Scott AFB is located; however, groundwater within unconsolidated units and bedrock can be used locally as a source of potable water. Scott AFB and nearby communities purchase water from a municipal water distribution system that obtains water from the Mississippi River.

The hydrogeologic zones examined as a part of previous investigations have included waste materials within the landfill cells and the unconsolidated deposits of the Cahokia Alluvium (the upper and lower clay units with underlying and interbedded silt and sand units). Bedrock was not considered as a hydrogeologic unit because it does not appear to influence shallow groundwater.

To distinguish among the different hydrogeologic zones encountered across Scott AFB, the water-bearing units for other Scott AFB sites have been divided into the following zones:

- Zone 1 (shallow clay/silt zone);
- Zone 2 (shallow silt/sand zone);
- Zone 3 (lower clay/silt zone); and
- Zone 4 (lower silt/sand zone).

Similar to the other Scott AFB sites, LF-01 has a series of nontransmissive and transmissive units, although the depths at which they occur may vary from those observed for other Scott

AFB sites. The first hydrogeologic zone at LF-01 occurs in the uppermost clay/silt layer, also called Zone 1. Within the landfill boundary, Zone 1 is beneath the fill and waste. Beneath the nontransmissive saturated clay/silt of Zone 1 is the shallow sand unit, or Zone 2, which is the uppermost transmissive zone. Based on geologic cross sections at LF-01, Zone 2 is present at an average depth of 18 feet below ground surface (bgs). A second nontransmissive clay/silt layer, or Zone 3, is located directly under Zone 2. Beneath the nontransmissive layer of Zone 3, another deeper sandy/transmissive zone, or Zone 4, is present at approximately 50 to 60 feet bgs.

Zone 2 is recharged from infiltration of precipitation, from upgradient contributions, and from leakage from landfill leachate. Discharge from this unit is through evapotranspiration, lateral outflow to downgradient areas, and leakage through the lower clay unit to Zone 4 (MWH, 2003).

1.1.4 Current and Anticipated Future Land Use

LF-01 was used as a landfill from the 1940s to 1976. Currently, the site is used as open space (hunters occasionally cross the cells to reach hunting areas). The EOD structure was recently demolished as part of the landfill cap construction activities. The EOD unit is planning construction of a new facility within the boundary of the South Cell and on top of the newly constructed landfill cover.

Current and anticipated future land use for LF-01 is nonresidential. Scott AFB has no plan to change the existing land or resource use in the future. LUCs will be applied that will (1) prevent intrusive activities, thereby protecting the integrity of the landfill cover and preventing contact with contaminants, and (2) ensure that the site is not developed for residential or commercial purposes. Because the landfill cover will limit infiltration and precipitation, the amount of contaminants migrating from the landfill waste and soil to the underlying groundwater will decline over time.

Landfill-related activities have contaminated the groundwater at the site. Illinois classifies groundwater in this area as Class I, which means that it is potentially usable as a potable water source. Scott AFB and other nearby communities purchase potable water from municipal water distribution systems. The sources of the municipal water supplies are the Mississippi River and the Kaskaskia River. No drinking water wells exist on the base. In addition, the St. Clair County code contains an ordinance for entities under county jurisdiction requiring the use of public water where it is available. The ordinance reads as follows:

> ***19-6-3*** ***PUBLIC WATER SUPPLY USE.*** *In those locations where a public supply is reasonably available, that supply will be the sole source of water for drinking, culinary and sanitary purposes. A public water supply will be deemed reasonably available when the subject property is located within* ***three hundred (300) feet*** *of the public water supply to which connection is practical and is permitted by the controlling authority for said water supply.*

1.2 REMEDIAL ACTION OBJECTIVES

This section summarizes the remedial action objectives (RAO) for ERP Site LF-01; the RAOs are formalized in the decision documents for the site. The remedies for this site were selected to address human health and ecological risks associated with the exposure of potential receptors to the contaminants of concern (COCs) in landfill waste, soil, sediment, and groundwater at LF-01. Site COCs are as follows:

Soil

- Benzo(a)pyrene
- Benzo(a)anthracene
- Benzo(b)fluoranthene
- Dibenzo(a,h)anthracene
- Benzyl butyl phthalate

Sediment

- Benzo(a)pyrene
- Aroclor 1260

Groundwater

- Trichloroethene (TCE)
- cis-1,2-Dichlorethene (DCE)
- Vinyl chloride
- Arsenic
- Iron

The remedial actions for landfill waste, soil, and sediment have been completed. COCs in soil and sediment were remediated at the site and no longer pose a risk to human health or the environment. Soil and sediment remediation included excavation, placement on the landfill, and covering with an engineered landfill cover. To ensure that protection of human health and the environment is maintained and to limit contaminant migration, Scott AFB will conduct O&M activities to maintain the integrity of the landfill cover and ensure compliance with LUCs. Scott AFB is responsible for implementation of O&M activities, and will operate, maintain, monitor, review, and enforce the selected remedy in accordance with Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) and the National Oil and Hazardous Substances Pollution Contingency Plan, known as the National Contingency Plan (NCP) to ensure protection of human health and the environment.

Because of the presence of the landfill cover, LUCs will be maintained to ensure that the site remains nonresidential and prevent disturbance of the engineered cover to protect human health and the environment. LUCs also will be required to ensure that groundwater within the landfill footprint and the groundwater attenuation zone (within 100 feet of the outer landfill cover boundaries) is not used as potable water or for other purposes. Administrative controls also will be established to prevent drinking water wells from being installed within the affected groundwater zones beyond the attenuation zone. Signs will be posted and other notifications will be made to raise awareness about the potential exposure to groundwater contamination at the site.

For contamination present in landfill waste, surface soil, sediment, and groundwater at LF-01, the following RAOs were developed:

RAOs for Human Health

- Prevent exposure to buried waste and to surface soil contaminated with benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, and dibenzo(a,h)anthracene at concentrations that pose an unacceptable cancer risk.
- Prevent exposure to sediment contaminated with benzo(a)pyrene at concentrations that pose an unacceptable cancer risk.
- Prevent future residential land use and intrusive activities.
- Prevent exposure to contaminated groundwater within the landfill footprint and attenuation zone.
- Prevent exposure to groundwater beyond the attenuation zone contaminated with TCE, cis-1,2-DCE, vinyl chloride, arsenic, and iron at concentrations that pose an unacceptable cancer risk and/or noncancer hazard.
- Install a groundwater monitoring well network capable of detecting new releases of contaminants from waste materials in both the North and South Cells of LF-01.

RAOs for Environmental Protection

- Prevent exposure of ecological receptors to buried waste.
- Prevent exposure of terrestrial receptors to benzyl butyl phthalate in surface soil at concentrations that pose a threat via direct contact.
- Prevent exposure of insectivorous birds to high-molecular-weight polynuclear aromatic hydrocarbons (PAHs) present in surface soil at concentrations that pose a threat via the food web.
- Prevent exposure of flying insectivores (bats) to Aroclor 1260 in sediment at concentrations that pose a threat via the food web.

RAOs for Limiting Contaminant Migration

- Minimize infiltration of precipitation through the landfill waste to minimize leaching of waste constituents to the underlying groundwater.
- Prevent migration of surface soil contaminated with Aroclor 1260 and benzo(a)pyrene to Mosquito Creek via erosion and overland flow.

1.3 REMEDIAL ACTION SUMMARY

The remedial action (RA) construction activities for the landfill cover at LF-01 were conducted from November 2011 through October 2012, and final inspection of the RA construction was completed on July 17, 2013. The remedial approach developed for LF-01 involved (1) excavating contaminated soil and sediment, (2) on-site disposal of the contaminated media within the footprint of the landfill cover, (3) constructing a landfill cover over the waste disposed on site, (4) installing supplemental monitoring wells, and (5) restoring the site by revegetating the disturbed areas. Specific elements of the RA for LF-01 included the following:

- Installing erosion control devices and employing stormwater best management practices within the disturbed areas;
- Removing trees from LF-01 before April 1, 2012, to minimize the construction impact on the roosting habits of the Indiana bat, a federally endangered species;
- Clearing, grubbing, and disposing of woody and nonwoody plants and grasses in support of landfill cover construction;
- Demolishing structures and clearing and grubbing to the limits of disturbance in the LF-01 North and South Cells (work included demolishing the existing EOD bunker located in the South Cell, removing selected sections of existing fencing, clearing and stripping of surface vegetation, and abandoning existing groundwater monitoring wells and landfill gas monitoring wells within the footprint of the work area);
- Excavating contaminated sediment from Mosquito Creek, disposing of the material at the landfill before the landfill cover was placed, and restoring the Mosquito Creek streambed;
- Excavating PAH-contaminated soil from areas immediately adjacent to LF-01 South Cell and disposing of the soil at the landfill before the landfill cover was placed;
- Grading nonhazardous soil generated from past remedial activities that had been temporarily stockpiled at the North and South Cells;
- Preparing the subgrade in the soil cover area by excavating and consolidating materials from outside the limits of waste, and spreading, grading, and compacting the subgrade;
- Installing a passive landfill gas collection system at both landfill cells;
- Installing a geosynthetic layer (geomembrane) and drainage system over the waste cells;
- Grading and placing the protective cover soil and vegetative layers within landfill cell boundaries;
- Grading and installing the slope stabilization materials;
- Armoring the side slopes of the finished cover with stone to control erosion and allow for surface runoff;
- Restoring and revegetating all disturbed areas including the landfill cover, borrow area, and temporary soil stockpile area; and
- Installing a groundwater monitoring well network capable of monitoring known contaminants and detecting new releases of volatile organic compounds (VOCs) and metals from waste materials within both landfill cells of LF-01.

For the implementation of the groundwater remedy, a monitoring well network was installed and completed by (*date to be included upon completion*). The well network is capable of monitoring the migration and attenuation of known contaminants.

LUCs will be implemented at the site. Scott AFB will post signs near the landfill as notification of the location of the landfill, its restricted use, and restrictions on groundwater use. The Base General Plan will also be revised by Scott AFB to include the LUCs outlined in the ROD addressing landfill waste, surface soil, and sediment (Scott AFB, 2013a) and the ROD addressing groundwater (Scott AFB, 2013b). Institutional controls can be placed on U.S. Air Force (USAF) property through a Land Use Control Implementation Plan (LUCIP), which will be prepared separately following the approval of the Final ROD addressing groundwater. For groundwater contamination that has migrated off base and has affected property that is not under the control of Scott AFB, the USAF will work with the property owner(s) to impose activity and groundwater use restrictions on the affected property in order for the remedy to be protective of human health and the environment.

1.4 REPORT ORGANIZATION

The remainder of this O&M Manual is organized as follows:

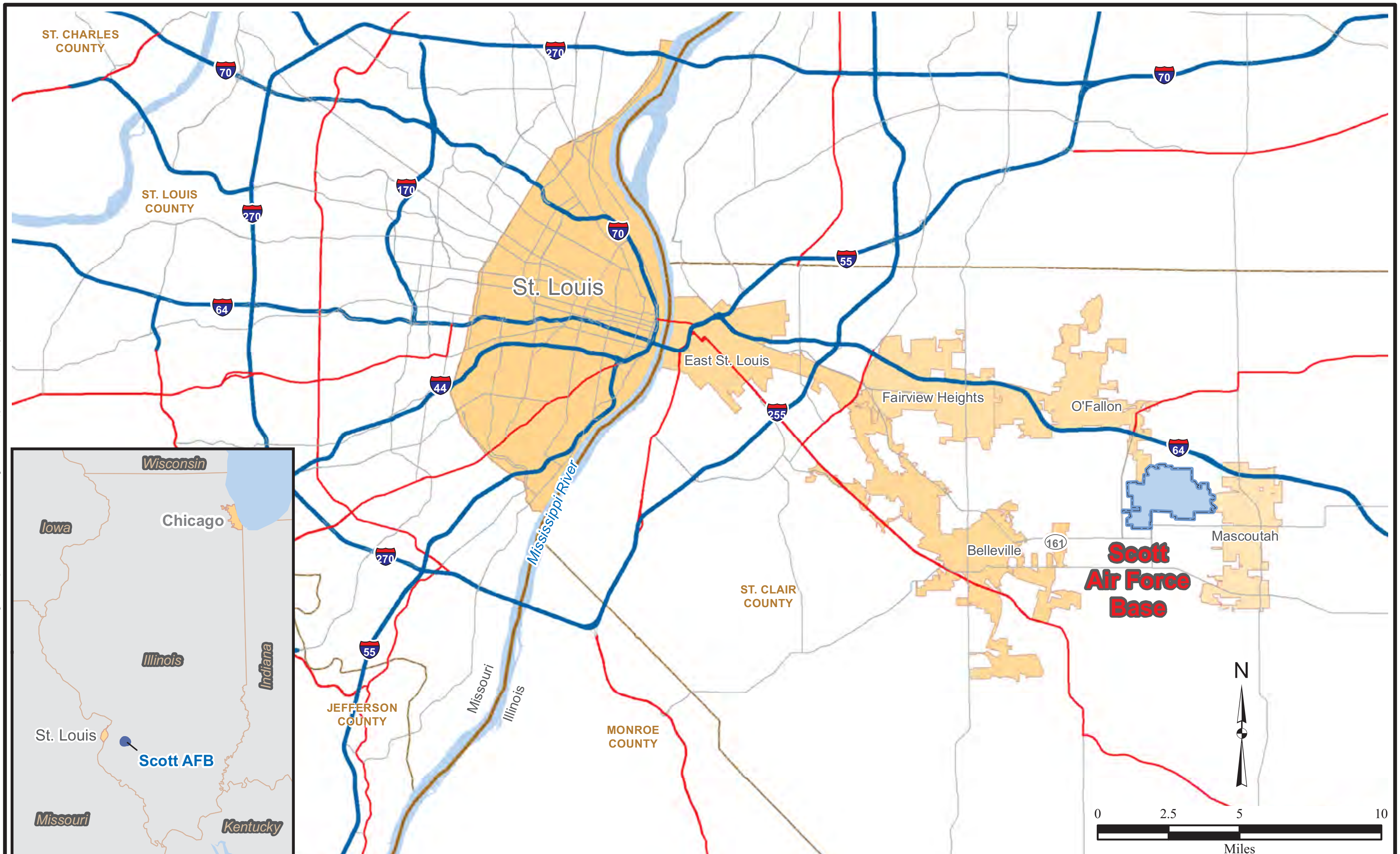
- **Section 2.0** – describes the general information of the site, landfill cover system, landfill performance monitoring system, and the groundwater remedy monitoring system.
- **Section 3.0** – describes the inspections and maintenance operations to be performed and addresses repair activities.
- **Section 4.0** – describes the environmental monitoring activities to be performed.
- **Section 5.0** – addresses reporting requirements.
- **Section 6.0** – provides references.
- **Appendix A – As-Built Drawings for Landfill Cover**
- **Appendix B – Construction Details for Monitoring Wells, Gas Wells, and Cover Settlement Monuments**
- **Appendix C – Standard Operating Procedures**
    - Sampling Groundwater with the HydraSleeve
    - Surface Water Sampling
    - Field Logbook Use and Maintenance
    - Landfill Gas Measurements
    - Landfill Gas Log Sheet
    - Ambient Air Log Sheet
- **Appendix D – Supporting Documents**
    - LTM Well Installation at LF-01 Technical Memorandum (HGL, 2013b)
    - IEPA approval letter dated April 8, 2013 for the LTM Well Installation at LF-01 Technical Memorandum
    - Proposed Long Term Monitoring Well Locations at LF-01 Technical Memorandum (HGL, 2013a)

- IEPA approval letter dated January 28, 2013 for the proposed LTM groundwater monitoring network
- Scott AFB Basewide Groundwater Dataset Summary

This page was intentionally left blank.

FIGURE(S)

This page was intentionally left blank.



U.S. Army Corps of Engineers, Omaha District

HGL—LF-01 O&M Manual — Scott AFB, Illinois

Filename: X:/SA13/LF01/FS Addendum/
Site_Location_Map.mxd
Project: SA1302.01.03
Revised: 03/23/2012 ST
Source: HGL Database

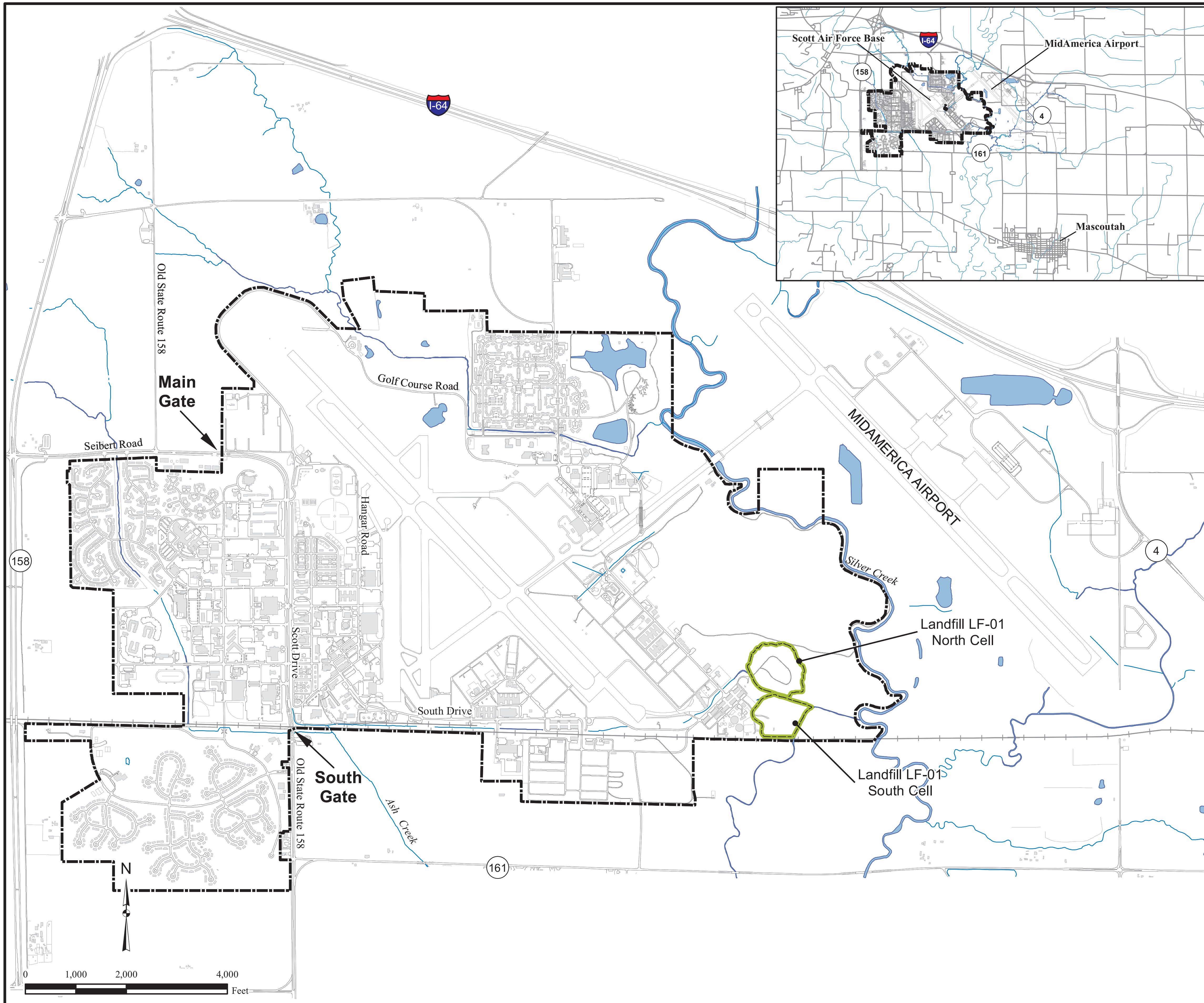
HGL HydroGeoLogic, Inc

Scott AFB

USACE
Omaha District

Figure 1.1
Site Location Map
Scott Air Force Base, Illinois

Figure 1.2
Landfill LF-01
Site Overview Map



Legend

- Scott AFB Boundary
- Existing Lake/Pond
- Existing Structure
- Railroad Tracks
- Stream/Drainage Channel
- Landfill LF-01

Filename: X:/SA13/LF01/FS_Addendum/LF01_Site_Overview.mxd
Project: SA1302.01.03
Revised: 03/23/2012 ST
Source: HGL GIS Database, 2001

HELP FROM ABOVE

HGL HydroGeoLogic, Inc.

2.0 GENERAL INFORMATION

This section provides general information on the components of the landfill cover and the postclosure environmental monitoring system. Construction drawings from the final Remedial Action Work Plan (HGL, 2012c) for the landfill are provided in Appendix B. As-built drawings for the landfill cover system are included in Appendix A.

2.1 LANDFILL COVER SYSTEM

The landfill cover system has a minimal thickness of 24 inches. The landfill cover system was designed and constructed to meet the applicable substantive requirements specified in 35 IAC 811, *Standards for New Solid Waste Landfills*. Detailed discussions of the landfill cover design, surface water drainage, and landfill gas management system are presented in the *Design Analysis Report for LF-01 Closure, Scott Air Force Base* (HGL, 2012b). The primary components are discussed in the following subsections in order from shallowest to deepest and are listed in Table 2.1.

2.1.1 Protective Layer

A final protective layer is installed over approximately 20 acres in the North Cell and 15 acres in the South Cell. It consists of a 24-inch thick vegetated earthen infiltration layer, which exceeds the estimated frost depth of 21 inches in this region. The top deck of the soil cover is graded to a minimum 2 percent slope to promote proper runoff of precipitation and to limit ponding of water on the cover system.

The primary means of protection against erosion is proper grading and establishing a surface layer of vegetation. This step was accomplished through seeding and mulching all disturbed areas in accordance with Specification 31 32 11 of the RA Work Plan (HGL, 2012c). A blend of annual and perennial plant seed was applied on the landfill protective layer and side slopes to provide a permanent vegetative cover. Separate seed mixes were used for the landfill cover and the side slopes and both seed mixes were compatible with local vegetation, topography, weather, sunlight, and soil type. After seeding, straw mulch was uniformly broadcast over the landfill cover at a rate of 2 tons per acre. Seed mixes and application rates are listed in Appendix B, Table B.5.

2.1.2 Geocomposite Drainage Layer

A drainage layer is installed beneath the final protective layer and above the geomembrane. The purpose of this system is to remove surface water infiltration from the final protective layer to keep that water from ponding on the geomembrane. The drainage system is made up of the following:

- A drainage geocomposite, composed of geonetting wrapped in nonwoven geotextile fabric underlying the final protective layer;
- Plastic flat pipe drainage headers that were installed at 200-foot intervals over the geocomposite on the top deck;

- A gravel-wrapped drainage header located along the perimeter edge of the cover; and
- Outfall pipes that convey the collected water from the headers off the landfill cover system.

2.1.3 Geomembrane

The geomembrane system is a 40-mil (0.040 inch) linear low-density polyethylene (LLDPE) textured, geosynthetic membrane (geomembrane) overlying the gas collection layer. This material acts as a low-permeability (less than $1x10^{-7}$ centimeters per second) vertical barrier to surface water infiltration, yet is flexible and elastic to accommodate differential settlement of the landfill waste.

2.1.4 Landfill Gas Collection System

The landfill gas management system vents gas from underneath the landfill cover to prevent damage to the cap and minimize potential lateral subsurface migration of landfill gas. Separate gas management systems are installed at the two landfill cells. Based on historically low concentrations of methane and the small volume of gas produced by the landfill waste, an active gas collection system was not necessary to comply with 35 IAC Section 811.311. The landfill gas collection system at LF-01 is a passive system with vertical vents to the atmosphere. The passive collection system was designed to be readily retrofitted to become an active system with treatment to meet the requirement of 35 IAC 811.311(d)(12), if necessary.

The landfill gas collection system is embedded within the foundation layer, below the low permeability geomembrane. The landfill gas collection system includes a 12-ounce nonwoven geotextile blanket layer, lateral trenches with perforated high density polyethylene pipes encased in gravel and geotextile, and vertical vents to the atmosphere. The permeable geotextile layer enables landfill gas to travel laterally to the landfill gas collection trenches, to avoid areas of landfill gas buildup underneath the LLDPE geosynthetic membrane of the landfill cover. This layer also provides a cushion for the low-permeability LLDPE geosynthetic membrane. As gas is generated in the landfill, the lateral collection pipes provide preferred pathways for gas migration. Collection pipe trenches are spaced 100 feet apart within the north landfill cell and 150 feet apart within the south landfill cell. Gas vents are located along the high points on both landfill cells, as well as near the perimeter. The landfill gas, which is not under vacuum, travels through the path of least resistance, migrating vertically and laterally. Gas vents located near high points allow gas that has migrated to the highest points of the cell to be vented to the atmosphere. Gas vent locations are depicted on the as-built drawings included in Appendix A.

2.1.5 Foundation Layer

The foundation layer reduces the potential for differential settlement, provides the bearing capacity for overlying layers, and provides grade control and a smooth surface on which to establish the remaining cover system layers. This layer is composed of a minimum 12-inch-thick layer of compacted soil constructed on the North and South Cells to separate the waste and debris from the landfill cover system. Material used to construct the foundation layer consisted of the following:

- Nonhazardous soil stockpiles generated from previous Scott AFB RA activities that were temporarily staged on the North and South Cells;
- Soil generated from the RAs conducted at Sites FT-03 and OT-09 (Scott AFB, 2013a and Scott AFB, 2013b); and
- Soil imported from the off-site borrow area.

2.2 EROSION CONTROL MEASURES AND SLOPE STABILIZATION

Before the landfill foundation layer was constructed, the existing slopes around the perimeter of the North and South Cells were pulled back to provide a uniform toe and base for the installation of the landfill slopes per the design. Soil excavated during perimeter grading and slope pullback operations was spread on the North and South Cells and incorporated into the landfill subgrade. Refuse encountered during slope excavation was transported and placed in low areas of the North and South Cells in accordance with the site reconsolidation plan. The refuse was spread in 12-inch lifts, compacted with the dozer, and covered with 12-inches of subgrade material.

Structural erosion controls were installed around the perimeter of the cover system, to protect the integrity of the landfill slopes during flooding events. Riprap was installed immediately following slope pullback, thus minimizing the potential for erosion of the landfill bank during construction of the cover. In addition, the cap construction was moved out of the flood zone making it less problematic to install the subgrade, geomembrane, and final cover during high water conditions in Mosquito Creek. Approximately 5,700 linear feet of new riprap slope was installed around the perimeter of the North and South Cells. Before the riprap was installed, a 12-ounce nonwoven geotextile (EroTex 275EX) was installed along the base and vertical wall of the side-slope excavation to strengthen the subbase. The slopes were then constructed at a 2:1 grade using Illinois Department of Transportation Class RR3 stone.

Before seeding of the landfill cover, turf reinforcement mat was installed along the edges of the landfill slopes to facilitate vegetative growth. The turf reinforcement mat extended onto the landfill top deck and was anchored 10 feet from the top edge of the slope. From the edge of the turf reinforcement mat, a vegetative cover was established over the entire top deck surface to provide erosion protection from surface water runoff. Additionally, an erosion control blanket was installed in the drainage swale constructed on the South Cell as part of the final grading of the protective layer.

2.3 ENVIRONMENTAL MONITORING SYSTEM

Environmental monitoring will occur to determine the long-term effectiveness of (1) the landfill cover and (2) the groundwater remedy. The following media will be included in postclosure environmental monitoring activities:

- Groundwater,
- Surface water,

- Landfill gas, and
- Ambient air.

The following subsections discuss the monitoring networks for postclosure care of the landfill and of the groundwater remedy.

2.3.1 Groundwater Monitoring

This section describes the long term groundwater monitoring that will occur at LF-01 to demonstrate that the landfill cover system is properly functioning to protect human health and the environment as designed. Groundwater monitoring also will monitor the attenuation of groundwater COCs to ensure remedial goals and RAOs have been attained. To accomplish these objectives, three well networks are established: (1) the landfill performance monitoring network, (2) the groundwater remedy monitoring network, and (3) baseline monitoring network.

The landfill performance monitoring well network focuses on evaluating the long term performance of the landfill cover system in accordance with the postclosure care requirements specified in 35 IAC 811.318(b). The cover system is required to reduce leachate production from the waste to the underlying groundwater and which will, over time, lead to reduced contaminant concentrations in groundwater leaving the landfill cells. The groundwater remedy monitoring network will monitor the attenuation of groundwater COCs beyond the attenuation zone. This well network includes monitoring wells located downgradient and within areas of high contaminant concentrations. All monitoring wells that function as landfill performance wells also function as groundwater remedy monitoring wells to monitor the migration of groundwater COCs from the landfill cells. The baseline monitoring network contains wells installed upgradient with respect to groundwater flow of the landfills to establish baseline values for comparing to downgradient concentrations.

The combined long term monitoring well networks consist of 48 wells (28 existing wells and 20 proposed wells) and are discussed in detail in the following subsections. Groundwater will be routinely sampled and analyzed from the groundwater monitoring network as described in Section 4.1. The postclosure groundwater monitoring network is shown on Figure 2.1 and presented on Table 2.2.

2.3.1.1 Potentiometric Conditions

To distinguish among the different hydrogeologic zones encountered at Scott AFB, the water-bearing units at LF-01 and Scott AFB have been divided into the following zones:

- Zone 1 (shallow clay/silt zone);
- Zone 2 (shallow silt/sand zone);
- Zone 3 (lower clay/silt zone); and
- Zone 4 (lower silt/sand zone).

The first water-bearing unit at LF-01 occurs in the uppermost clay/silt zone, also called Zone 1. Within the landfill boundary, Zone 1 is beneath the fill and waste. Leachate is present in the fill/waste under normal water table conditions. Beneath the saturated clay/silt zone is the shallow sand unit, or Zone 2, which is the uppermost transmissive zone. Based on geologic cross sections at LF-01, Zone 2 is present at an average depth of 18 feet bgs. A clay/silt layer, or Zone 3, is located directly under Zone 2. Beneath the nontransmissive layer of Zone 3, another deeper sandy/transmissive zone, or Zone 4, is present at approximately 50 to 60 feet bgs.

The uppermost water-bearing unit, or Zone 2, is recharged from infiltration of precipitation and from upgradient contributions. Discharge from this unit is through evapotranspiration, lateral outflow to downgradient areas, and leakage through the lower clay unit to the deep water-bearing unit, or Zone 4.

Before the landfill cover was constructed, potentiometric conditions across the site in November 2011 are shown on Figure 2.1 and described as follows:

- At the north landfill cell, shallow groundwater in Zone 2 flows radially toward Mosquito Creek and toward Silver Creek. Wells located east and beyond the attenuation zone of the north landfill cell indicated a relatively low hydraulic gradient of 0.0012 feet per foot (ft/ft), which is typical of wetland areas.
- For the south landfill cell, preconstruction potentiometric conditions indicated that shallow groundwater in Zone 2 flowed radially toward the north, east, and south. Based on November 2011 groundwater elevations, the horizontal gradient in Zone 2 flows eastward across the south landfill cell at 0.005 ft/ft.
- For shallow groundwater east of the unnamed tributary, the flow demonstrated a southerly component, with an estimated average horizontal gradient of 0.00098 ft/ft. East of the unnamed tributary, the gauging data suggested a predominantly eastward flow direction toward Silver Creek.

Postclosure groundwater levels will be monitored in all wells prior to purging during each sampling event.

2.3.1.2 Landfill Network Monitoring Well Network

At the request of IEPA, the Monitoring Analysis Package (Version 1.1) modeling program was used to facilitate the design of the landfill performance monitoring well network. Results from the model simulations were presented in the *Proposed Long Term Monitoring Well Locations at LF-01* technical memorandum (HGL, 2013a). According to the IEPA's guidance document LPC-PA2, a monitoring efficiency of 99 percent is required for well networks designed for new landfills. In a letter dated January 28, 2013, IEPA approved the Monitoring Analysis Package modeling configuration from Run #6 (Appendix D) as the landfill performance well network. Run #6 yielded a modeled monitoring efficiency of 99.5 percent, which exceeds IEPA's efficiency criterion of 99 percent. The 30 wells associated with Run #6 compose the groundwater monitoring well network to monitor the effectiveness of the landfill cover. The *Proposed Long Term Monitoring Well Locations at LF-01* technical memorandum outlining the model results and the IEPA approval letter are available in Appendix D.

In accordance with 35 IAC Section 811.318(b)(2), the groundwater monitoring wells will be screened in both of the water-bearing units that could serve as contaminant migration pathways from the landfill. The landfill performance monitoring well network will consist of 30 wells (or 15 nested well pairs), with the shallow (S) wells screened in Zone 2 groundwater and the deep (D) wells screened in Zone 4 groundwater. Monitoring wells will be constructed in accordance with the Basewide Field Sampling Plan (FSP) criteria (HGL, 2009a) and all federal, state, and local requirements (35 IAC 811[d] and 77 IAC 920.170). Well construction details (for existing and proposed wells), including well screen intervals and total well completion depths, are provided in Appendix B.

The landfill performance monitoring network will consist of 30 wells: 26 wells (13 well pairs) surrounding the landfills cells and located downgradient and crossgradient of groundwater flow and 4 wells (2 well pairs) located upgradient to groundwater flow. The well network is summarized in Table 2.2 and includes the following:

- Ten existing monitoring wells (five well pairs)
  - LF01-MW01S/D
  - LF01-MW03S/D
  - LF01-MW12SR/D
  - LF01-MW21S/D
  - LF01-MW25S/D
- Sixteen new monitoring wells (eight nested well pairs)
  - LF01-MW30S/D
  - LF01-MW31S/D
  - LF01-MW32S/D
  - LF01-MW33S/D
  - LF01-MW36S/D
  - LF01-MW37S/D
  - LF01-MW38S/D
  - LF01-MW39S/D

The anticipated locations of the new monitoring wells will be located as close as possible to the attenuation zone, taking into account accessibility, feasibility, and implementability.

To meet the intent of 35 IAC 811.320(d)(4), four wells (two nested well pairs) will be installed upgradient to groundwater flow of the landfill cells. These wells will include:

- MW34S/D, and
- MW35S/D.

The upgradient wells will be sampled concurrent with the other monitoring networks to provide baseline concentration data for the duration of the monitoring period. The baseline data generated from the upgradient wells will be used to determine when statistically significant increases in downgradient chemical concentrations occur during long term monitoring per 35 IAC 811.320(e)(1).

2.3.1.3 Groundwater Remedy Monitoring Well Network

A long term monitoring network has been established to monitor the attenuation of groundwater COCs (TCE, cis-1,2-DCE, vinyl chloride, iron, and arsenic). The groundwater remedy network consists of 44 wells and is designed to monitor in-plume characteristics of groundwater COCs and identify whether the contaminant plumes are expanding beyond their current boundaries. Twenty six of the landfill performance wells also function as part of the groundwater remedy monitoring network. In addition to the 26 landfill performance wells mentioned above, 18 wells serve as groundwater COC compliance monitoring wells. To monitor expansion and attainment of cleanup levels, the groundwater remedy network wells are located within the contaminant plumes or downgradient and outside of groundwater contamination. The purpose and location of the long term monitoring wells are presented in Table 2.2. In addition to the 26 landfill performance wells, the groundwater remedy network wells are as follows:

- LF01-MW11
- LF01-MW17SR/D
- LF01-MW18S/D
- LF01-MW19S/D
- LF01-MW20S/D
- LF01-MW23S/D
- LF01-MW24S/D
- LF01-MW26S/D
- LF01- MW27
- LF01-MW28
- LF01-MW29

The highest VOC concentrations were observed at monitoring well LF01-MW11 (located within the South Cell landfill cover boundary). This well will be monitored as part of the groundwater remedy to provide source area contaminant concentrations. As presented in Table 2.2, seven wells located downgradient and outside of the COC plumes with concentrations below cleanup levels will serve as sentinel wells. These sentinel wells are intended to confirm that COC concentrations meet regulatory cleanup levels. If at any point groundwater COC concentrations increase and/or exceed cleanup levels in the sentinel wells, this will trigger an action to manage potential expansion of the plume.

2.3.1.4 Wells Not Included In Long Term Monitoring Network

Several existing monitoring wells at LF-01 are not included in the long term groundwater monitoring network based on at least one of the following reasons:

- Another nearby well provides similar or more effective data on contaminant migration and concentrations, or
- The well is screened in an inappropriate, nontransmissive layer.

Based on the above rationale, the following existing monitoring wells are not part of the long term monitoring well network:

- MW02,
- MW12S,
- MW17S,
- MW22S, and
- MW22D.

By eliminating wells from the prescribed monitoring program that will provide redundant data, the data collection process will be streamlined while maintaining overall data quality. These wells will not be abandoned at this time so that, if in the future the distribution of groundwater contamination or flow direction changes, these existing wells could provide valuable data without the need to install new wells. These five wells will be inspected and maintained in accordance with this O&M Manual to ensure their integrity in case future groundwater sampling is necessary.

2.3.1.5 Monitoring Well Installation

The proposed monitoring wells will be installed, developed, and surveyed in accordance with the Basewide Field Sampling Plan (HGL, 2009a) and all federal, state, and local requirements (35 IAC Section 811.318 and 77 IAC Section 920.170). Each well will be installed using hollow-stem auger drilling methods. Each well pair will consist of a shallow well screened in Zone 2 and a deep well screened in Zone 4. At each well pair location, continuous sampling (from ground surface to the total depth of the borehole) will be conducted at the Zone 4 (deep well) location using split spoon or 5-foot continuous sampling methodology. Soil cores will be provided to the field geologist for field logging. The Zone 2 borehole will be blind augered and will be set based on the lithology observed in the Zone 4 well. The drill log of the Zone 4 well that was used to set the Zone 2 well will be referenced on the Zone 2 drill log. Monitoring well construction details are summarized below:

- The total depths of the Zone 2 wells will range from 30 to 35 feet bgs and the total depths of the Zone 4 wells will range from 60 to 70 feet bgs. However, totals depths may vary based on observations made by the field geologist.
- Each monitoring well will be 2 inches in diameter and will be screened to yield representative groundwater samples from the water-bearing zone being monitored (that is, Zone 2 or Zone 4). Each screen (0.010-inch slot) and blank riser pipe will be composed of schedule 40 polyvinyl chloride (PVC). Screen lengths will be approximately 10 feet and will be sealed with a schedule 40 PVC end cap. The blank riser will extend from the top of the screen to approximately 3.5 feet above the ground surface.
- The sand pack will consist of No. 20-40 silica sand and will extend approximately 2 feet above the top of the screen.
- A buffer pack consisting of a 1-foot-thick seal of #100 fine silica sand will be placed above the sand pack.
- A seal consisting of 1/4-inch coated bentonite pellets will be placed on top of the buffer pack and will extend approximately 2 feet above the buffer pack.
- High-solids (30 percent) bentonite grout will then be installed from the top of the bentonite seal to ground surface using a tremie method to emplace the grout from the bottom up.
- To alleviate the concern of standing water surrounding the wells, well protectors that are 7 feet in length will be used. The bottom 3 feet of the protector will be set below ground surface into the grout seal with the remaining 4 feet of the protector above the ground surface. Each well also will be sealed with a locking, compression plug and surrounded by a 2-foot by 2-foot concrete pad. Additionally, three bollards will be set around the concrete pad at each individual well location.

Each monitoring well will be developed within 10 days of completing construction activities. The development will remove any sediment that may accumulate in the bottom of the well during construction activities, as well as settle and remove sediment from the filter pack for clear inflow of groundwater for sample collection. Anticipated well construction details, including well screened intervals and total well completion depths, are available in Appendix B.

All new and existing monitoring wells will be surveyed by a State of Illinois certified land surveyor to maintain accuracy in elevation. The coordinates of the new monitoring wells will be surveyed to an accuracy of 0.01 foot. The top of casing elevation and rim elevation of each well will be included in the survey.

2.3.2 Surface Water Monitoring

The Unnamed Tributary (adjacent to the South Cell) is a gaining stream that potentially receives groundwater that passes through and under the LF-01 South Cell. To ensure the protectiveness of human health and the environment and to monitor the effectiveness of the cover to limit contaminant migration, postclosure surface water monitoring will be conducted in the Unnamed Tributary and Mosquito Creek.

To obtain a representative sample of upstream and downstream characteristics of Mosquito Creek and the Unnamed Tributary, six surface water samples (SW01 through SW06) will be collected from the following locations (Figure 2.1):

- Mosquito Creek upstream of the North Cell,
- Mosquito Creek upstream of the confluence of South Ditch outfall,
- Mosquito Creek downstream of the confluence of South Ditch outfall,
- Unnamed Tributary upstream of South Cell,
- Unnamed Tributary upstream of the confluence with Mosquito Creek, and
- Mosquito Creek downstream of the confluence with the Unnamed Tributary.

2.3.3 Landfill Gas Management System

The landfill gas generation analysis presented in the Design Analysis Report indicated that a low volume of landfill gas is being generated at LF-01 as a result of the low waste volume and age of waste (HGL, 2012b). As a part of construction activities, the landfill gas wells located within the landfill boundary were properly abandoned. Five existing gas wells (LF01-GW20, -GW27, -GW32, -GW38, and -GW41) located outside the landfill boundary to the west were retained for postclosure monitoring. Within the landfill boundary, a passive landfill gas management system was installed with 29 passive gas vents. To monitor for the migration of landfill gas and meet the intent of 35 IAC Section 811.310, postclosure landfill gas monitoring will be conducted at the 5 existing gas wells and the 29 passive gas vents. The locations of the landfill gas wells and passive gas vents are illustrated on Figure 2.2. Appendix B summarizes the landfill gas well construction details.

2.3.4 Ambient Air Monitoring

In accordance with 35 IAC Section 811.310(b)(8), methane concentrations will be monitored in ambient air at a minimum of three locations, 100 feet downwind from the perimeter of the landfill boundary. Real-time samples will be taken no higher than 1 inch above the ground surface. At least one upwind sample will be collected to obtain background conditions. A location is considered "downwind" or "upwind" based on the wind direction. Eight potential ambient air monitoring locations are depicted on Figure 2.2. Depending on the wind direction at the time of sampling, three ambient air sampling locations that are downwind of the landfill may be chosen from the eight potential locations shown on Figure 2.2. The wind direction and the three sampling locations that were used will be recorded on the field sheet.

2.4 EOD FACILITY

The remedial action for LF-01 included subsurface preparation for the construction of a new EOD facility. The EOD facility will be built and maintained separately from the remedial O&M of the landfill cover.

2.5 CONSTRUCTION

The remedial action for LF-01, including construction of the landfill cover, began in November 2011 and was completed by HGL in October 2012. As-built drawings for the cover system are included in Appendix A.

2.6 FINAL INSPECTION

Final stabilization of site restoration occurred at the site on July 17, 2013 when perennial vegetation was established at a density of 70 percent over the landfill cover area. Final acceptance of the completed project was based on an inspection of the North and South Cells conducted in the presence of USACE and Scott AFB representatives. During the inspection conducted on November 14, 2012, the HGL Construction Manager maintained a punch list of items that needed to be completed. After all items on the punch list were addressed, the site was reinspected on July 17, 2013 in the presence of USACE and Scott AFB representatives. Based on this reinspection, all the tasks identified on the punch list and all tasks detailed in the contract were successfully completed resulting in final acceptance of the RA phase of the project.

TABLE(S)

This page was intentionally left blank.

Table 2.1
Landfill Cover Components
Landfill LF-01
Scott AFB, Illinois

Layer Name	Component Material	Layer Thickness (inches)	Design Purpose
Protective Soil Layer	Imported Clean Fill	24	Provide a matrix capable of sustaining plant growth; protect the infiltration layer from wind, erosion, and frost; provide an infiltration barrier between the ground surface and the drainage layer.
Drainage Layer	Drainage Geocomposite	0.2	Convey infiltrated surface water off the geomembrane barrier layer.
Barrier Layer	40-mil Geomembrane	0.04	Ultra-low permeability layer to prevent infiltration of surface water into the landfill waste layer.
Gas Collection Layer	Geotextile/ Landfill Gas Collection Piping	0.2	Capture and convey landfill gas to prevent structural damage to the cover and prevent lateral (off site) migration of landfill gas.
Foundation (Subgrade) Layer	Fill	variable	The bottom layer of the cover system designed to separate the waste from the cover and to establish adequate surface slope to promote drainage from the finished cover surface. Gas collection system embedded within foundation layer.

Notes:
mil = 0.01 inch

This page was intentionally left blank.

Table 2.2
Groundwater Long Term Monitoring Network
Landfill LF-01
Scott AFB, Illinois

Monitoring Well Identification	Easting (feet)	Northing (feet)	Monitoring Purpose	Landfill Performance Monitoring Network	Groundwater Remedy Monitoring Network	Located Within Zone of Attenuation	Located Within VOC Plume	Located Within Iron Plume	Located Within Arsenic Plume	Upgradient of Plumes - COCs Below Cleanup Levels	Downgradient of Plumes - COCs Below Cleanup Levels
LF01-MW01D [1]	2392295.48	681804.56	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	X	--
LF01-MW01S	2392284.96	681807.85	Landfill Performance / Groundwater COC Compliance	X	X	--	--	X	--	--	--
LF01-MW02	2393031.48	681563.43	Not Included	--	--	X	--	X	--	--	--
LF01-MW03D	2393339.45	681186.67	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF01-MW03S	2393324.78	681195.52	Landfill Performance / Groundwater COC Compliance	X	X	X	--	X	--	--	--
LF01-MW11	2393146.53	679861.01	Groundwater COC Compliance	--	X	X	X	--	--	--	--
LF01-MW12D	2393292.10	679810.42	Landfill Performance / Groundwater COC Compliance	X	X	X	X	--	X	--	--
LF01-MW12SR	2393140.74	679764.76	Landfill Performance / Groundwater COC Compliance	X	X	X	X	X	--	--	--
LF01-MW12S	2393301.55	679805.86	Not Included	--	--	X	X	X	--	--	--
LF01-MW17D	2393658.96	679893.73	Not Included	--	--	--	--	--	X	--	--
LF01-MW17SR	2393499.60	679851.24	Groundwater COC Compliance	--	X	X	X	X	--	--	--
LF01-MW17S	2393669.48	679902.57	Groundwater COC Compliance	--	X	--	--	X	--	--	--
LF01-MW18D	2394364.02	680281.31	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW18S	2394359.46	680278.95	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW19D	2394661.50	681004.44	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW19S	2394655.16	681009.96	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW20D	2394358.31	681376.34	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW20S	2394353.36	681373.23	Groundwater COC Compliance	--	X	--	--	X	--	--	--
LF01-MW21D	2393499.48	681573.16	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF01-MW21S	2393501.95	681565.20	Landfill Performance / Groundwater COC Compliance	X	X	--	--	X	--	--	--
LF01-MW22D	2392280.24	679717.13	Not Included	--	--	X	--	--	--	--	X
LF01-MW22S	2392272.83	679714.43	Not Included	--	--	X	--	--	--	--	X
LF01-MW23D	2393317.48	679724.81	Groundwater COC Compliance	--	X	--	--	--	X	--	--
LF01-MW23S	2393323.27	679727.53	Groundwater COC Compliance	--	X	--	X	X		--	--
LF01-MW24D	2393038.17	679529.18	Groundwater COC Compliance	--	X	--	--	--	X	--	--
LF01-MW24S	2393040.54	679522.00	Groundwater COC Compliance	--	X	--	X	X	--	--	--
LF01-MW25D	2392925.74	679555.28	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF01-MW25S	2392927.24	679561.82	Landfill Performance / Groundwater COC Compliance	X	X	--	X	--	--	--	--
LF01-MW26D	2392903.63	679341.88	Groundwater COC Compliance	--	X	--	--	--	X	--	--
LF01-MW26S	2392891.17	679345.30	Groundwater COC Compliance	--	X	--	X	X	--	--	--

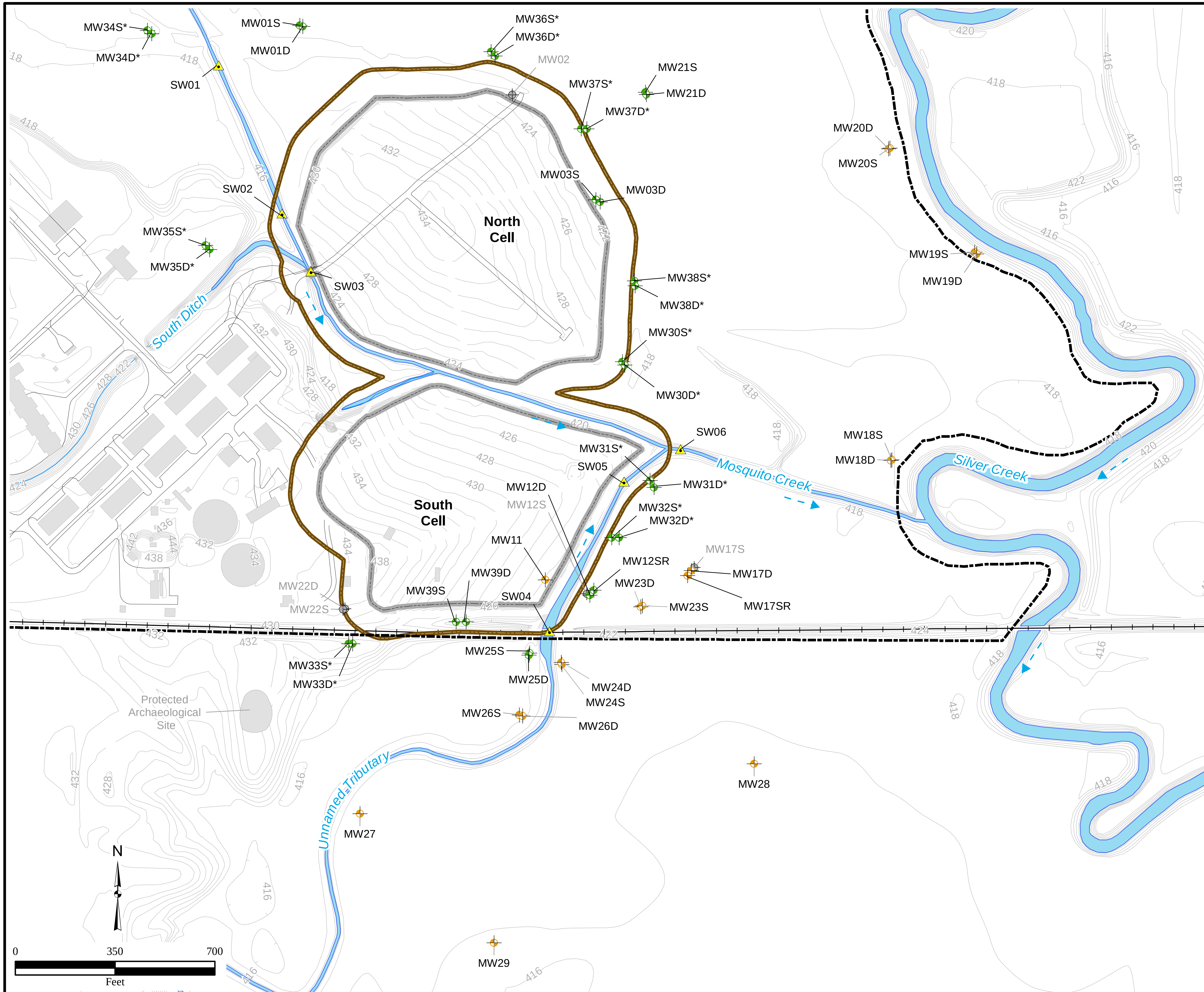
Monitoring Well Identification	Easting (feet)	Northing (feet)	Monitoring Purpose	Landfill Performance Monitoring Network	Groundwater Remedy Monitoring Network	Located Within Zone of Attenuation	Located Within VOC Plume	Located Within Iron Plume	Located Within Arsenic Plume	Upgradient of Plumes - COCs Below Cleanup Levels	Downgradient of Plumes - COCs Below Cleanup Levels
LF01-MW27	2392332.67	678998.13	Sentinel Well / Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW28	2393714.84	679173.42	Groundwater COC Compliance	--	X	--	--	X	--	--	--
LF01-MW29	2392803.11	678545.41	Sentinel Well / Groundwater COC Plume Expansion	--	X	--	--	--	--	--	X
LF01-MW30D*	2393261.69	680572.30	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF01-MW30S*	2393253.24	680584.50	Landfill Performance / Groundwater COC Compliance	X	X	X	--	X	--	--	--
LF01-MW31D*	2393364.15	680142.84	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF01-MW31S*	2393350.44	680167.96	Landfill Performance / Groundwater COC Compliance	X	X	X	X	X	--	--	--
LF01-MW32D*	2393241.64	679966.75	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF01-MW32S*	2393216.37	679966.75	Landfill Performance / Groundwater COC Compliance	X	X	X	X	X	--	--	--
LF01-MW33D*	2392306.60	679594.65	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF01-MW33S*	2392292.90	679594.65	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF-01-MW34D*	2391602.23	681733.89	Upgradient Well	X	--	--	--	--	--	X	--
LF-01-MW34S*	2391587.05	681746.10	Upgradient Well	X	--	--	--	--	--	X	--
LF-01-MW35D*	2391791.27	680992.12	Upgradient Well	X	--	--	--	--	--	X	--
LF-01-MW35S*	2391805.11	680977.85	Upgradient Well	X	--	--	--	--	--	X	--
LF-01-MW36D*	2392806.09	681656.86	Landfill Performance / Groundwater COC Compliance	X	X	--	--	--	--	--	X
LF-01-MW36S*	2392792.26	681671.13	Landfill Performance / Groundwater COC Compliance	X	X	--	--	X	--	--	--
LF-01-MW37D*	2393127.97	681400.29	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF-01-MW37S*	2393108.29	681400.89	Landfill Performance / Groundwater COC Compliance	X	X	X	--	X	--	--	--
LF-01-MW38D*	2393297.99	680849.73	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF-01-MW38S*	2393295.69	680867.35	Landfill Performance / Groundwater COC Compliance	X	X	X	--	X	--	--	--
LF01-MW39D*	2392705.27	679672.27	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
LF01-MW39S*	2392668.55	679675.44	Landfill Performance / Groundwater COC Compliance	X	X	X	--	--	--	--	X
		Totals	**48 Monitoring Wells**	**30 LF Wells**	**44 GW Remedy Wells**	**21**	**11**	**19**	**5**	**5**	**22**

Notes:
[1] LF01-MW01D is upgradient with respect to groundwater of the North Cell and contains elevated levels of arsenic attributed to background conditions. However, because this well is close to the landfill and the shallow nested well LF01-MW01S contains iron concentrations above cleanup levels, this well pair is monitored as part of the groundwater remedy and landfill performance network.
-- = not applicable
COC = contaminant of concern
S = shallow well
D = deep well
SR = shallow replacement well
* Proposed monitoring wells to be installed.

FIGURE(S)

This page was intentionally left blank.

Figure 2.1
Groundwater and Surface Water Monitoring Network



Legend

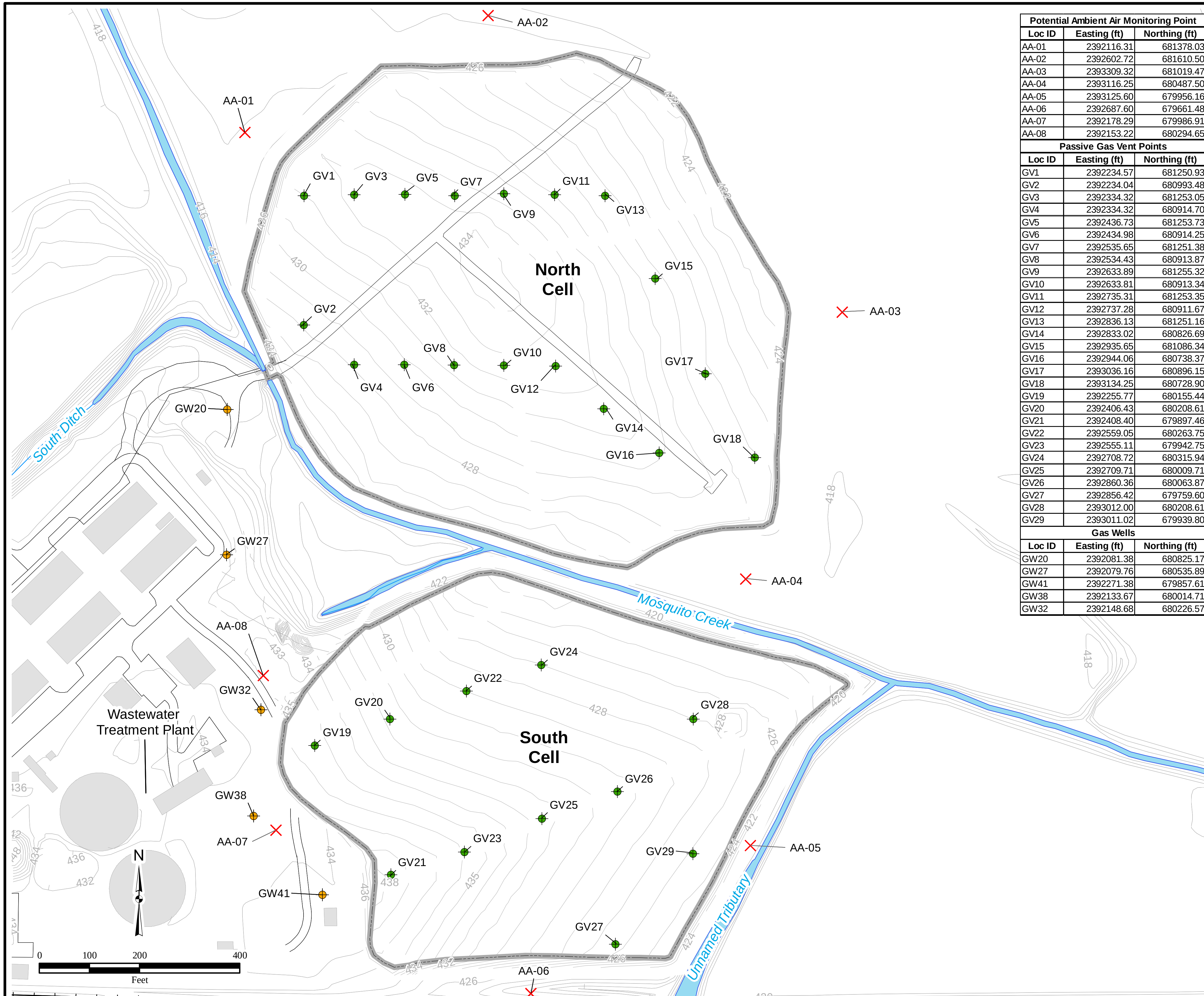
- Landfill Performance Monitoring Well
- Groundwater Remedy Monitoring Well
- Not Included in Groundwater Monitoring Network
- Surface Water Monitoring Point
- Surface Water Flow Direction
- 422 Topographic Contour Line (ft amsl)
- Rail Line
- Road
- Scott AFB Boundary
- Landfill Cover Boundary
- Existing Structure
- Attenuation Zone
- Surface Water

Notes:
*=proposed long term monitoring well
ft amsl=feet above mean sea level
D=deep well
S=shallow well
SR=shallow replacement well
Well pairs consist of a shallow well screened in Zone 2 groundwater and a deep well screened in Zone 4 groundwater.

\\Gst-srv-01\hglgis\Scott_AFB\LF-01\O_M_Manual\(2-1)Monitoring_Landfill_Performance_GW_COCs.mxd
6/11/2013 ST, QC: 9/26/2012 PD
Source: HGL

HGL HydroGeoLogic, Inc.

Figure 2.2 Postclosure Landfill Gas and Ambient Air Monitoring Network



Potential Ambient Air Monitoring Point		
Loc ID	Easting (ft)	Northing (ft)
AA-01	2392116.31	681378.03
AA-02	2392602.72	681610.50
AA-03	2393309.32	681019.47
AA-04	2393116.25	680487.50
AA-05	2393125.60	679956.16
AA-06	2392687.60	679661.48
AA-07	2392178.29	679986.91
AA-08	2392153.22	680294.65
Passive Gas Vent Points		
Loc ID	Easting (ft)	Northing (ft)
GV1	2392234.57	681250.93
GV2	2392234.04	680993.48
GV3	2392334.32	681253.05
GV4	2392334.32	680914.70
GV5	2392436.73	681253.73
GV6	2392434.98	680914.25
GV7	2392535.65	681251.38
GV8	2392534.43	680913.87
GV9	2392633.89	681255.32
GV10	2392633.81	680913.34
GV11	2392735.31	681253.35
GV12	2392737.28	680911.67
GV13	2392836.13	681251.16
GV14	2392833.02	680826.69
GV15	2392935.65	681086.34
GV16	2392944.06	680738.37
GV17	2393036.16	680896.15
GV18	2393134.25	680728.90
GV19	2392255.77	680155.44
GV20	2392406.43	680208.61
GV21	2392408.40	679897.46
GV22	2392559.05	680263.75
GV23	2392555.11	679942.75
GV24	2392708.72	680315.94
GV25	2392709.71	680009.71
GV26	2392860.36	680063.87
GV27	2392856.42	679759.60
GV28	2393012.00	680208.61
GV29	2393011.02	679939.80
Gas Wells		
Loc ID	Easting (ft)	Northing (ft)
GW20	2392081.38	680825.17
GW27	2392079.76	680535.89
GW41	2392271.38	679857.61
GW38	2392133.67	680014.71
GW32	2392148.68	680226.57

Legend

- Landfill Gas Monitoring Well
- Landfill Cover Passive Gas Vent Point
- Potential Ambient Air Monitoring Point
- Topographic Contour Line (ft amsl)
- Landfill Cover Boundary
- Scott AFB Boundary
- Rail Line
- Road
- Surface Water
- Existing Structure

Notes:
Monitor ambient air at three locations 100 feet downwind of landfill boundary.
Coordinates listed in North American Datum 1983, StatePlane Illinois West, US Feet coordinate system.

ft amsl=feet above mean sea level

\\Gst-srv-01\hglgis\Scott_AFB\LF-01\O_M_Manual\(2-2)Post_Closure_Landfill_Gas_Monitoring_Network.mxd
5/9/2013 ST
Source: HGL

3.0 POSTCLOSURE INSPECTIONS, MAINTENANCE, AND REPAIRS

This section provides the details of the O&M program including:

- Site inspections,
- Maintenance actions,
- Corrective measures,
- Environmental monitoring, and
- LUCs for LF-01.

This section addresses foreseeable, near-future situations that are commonly encountered in landfill cover maintenance. The limits of the new cover, the LUC boundary, and LUC sign locations are shown on Figure 3.1.

3.1 SITE INSPECTIONS

The stability of the landfill cover system relies on the successful propagation of a permanent stand of robust vegetative cover, identifying and quickly repairing protective layer erosion, and restricting unapproved ground-breaking or intrusive activities. The overall goal of the inspection program is to identify situations that would compromise the structural and vegetative stability of the protective layer in the cover system, as this will extend the service life of the system.

Unless otherwise noted, inspection of the landfill cover and all vegetated surfaces will be conducted on a quarterly basis for the first 5 years and annually thereafter. Table 3.1 provides the schedule of monitoring, inspection, and maintenance events for LF-01. The inspections also will evaluate the status of the LUCs and how any LUC deficiencies or inconsistent uses have been addressed. The inspections will be used to verify whether the use restrictions and controls were documented and maintained in Scott AFB's Base General Plan, and whether use of the property has conformed to such restrictions and controls. The general Postclosure Site Inspection Checklist (Table 3.2) details the items at the site to be inspected and will be used with the inspection details described in this section.

Inspections will be conducted by personnel with experience in landfill cover or cap inspections and maintenance programs. Unscheduled inspections may be performed at the discretion of the Scott AFB remedial program manager (RPM), or designee/equivalent, following a severe weather event, such as a major rainstorm event (specifically, more than 1.5 inches during a 24-hour period or 2.5 inches during a 48-hour period).

Any activity that is inconsistent with the LUC objectives, or any other action that may interfere with the effectiveness of the LUCs, will be documented with the cover inspection results and provided to the IEPA for informational use. In addition, the inconsistent activities will be addressed by Scott AFB as soon as practicable, but in no case will the process be initiated later than 30 calendar days after Scott AFB confirms the activity has occurred.

The site inspection reports will be included in the Annual Report that will be prepared and submitted as specified in Chapter 5.0.

3.1.1 Landfill Cover

The top surface and side slopes of the vegetated protective cover will be carefully inspected for evidence of the following:

- Ponded water;
- Erosion, rills, and gullies;
- Inadequate vegetative cover – An area larger than 100 square feet with no vegetative cover will be considered a problem;
- Displacement of erosion control materials;
- Daylighting of cover components that should otherwise not be exposed, such as normally concealed drainage piping, geomembrane, drainage mat, or similar elements;
- Slope stability failures or symptoms of impending failure (applicable for portions of the landfill cover with slopes steeper than 3:1 and for portions of the landfill cover adjacent to a waterway);
- Vegetative disease considerations;
- Animal burrow holes in the soil of the protective cover;
- Any intrusive human activity; and
- Invasion of soil cover by undesirable plants (such as trees, woody brush, wetland species).

3.1.2 Landfill Gas System

The landfill gas system vent pipes at LF-01 will be inspected for integrity and to detect any evidence of the following:

- Clogging, which could result from pests building nests in vent pipes and obstructing flow;
- Damage to vent pipes;
- Vandalism and/or tampering; and
- Damage to the high visibility posts.

3.1.3 Drainage System

The drainage system will be inspected for the following:

- Evidence of erosion or localized depressions;
- Any other conditions that may prohibit or impede flow of run-off or otherwise affect operational efficiency such as clogged drainage valves;
- Large debris on slopes (such as logs) that could create sediment buildup and/or impact drainage features; and

- Damage to the high visibility posts.

3.1.4 Groundwater Monitoring Wells

The groundwater monitoring wells at LF-01 will be inspected for the following:

- Locks (should be in closed or locked position);
- Locking well caps;
- Locking metal covers;
- Protective casing, bumper, pads, and surface seal; and
- Note any use of lubricants.

In addition to inspections for the external physical condition the monitoring well, the integrity of the monitoring well can also be checked by the following criteria:

- Changes in well depths,
- Trends in water levels,
- Changes to yield, and
- Increase in turbidity.

3.1.5 Landfill Cover System Settlement

Inspection and monitoring for landfill cover settlement will be conducted semiannually for the first 2 years of postclosure care (2013 and 2014) and annually thereafter. Applicable regulations do not specify an inspection schedule. The landfill cover and settlement monuments also should be inspected after extreme weather events. Surveying the settlement monuments will be conducted every 2 years and the data included in the monitoring report. The settlement monuments will be inspected for the following:

- Settlement monument damage;
- Evidence of damage from a severe weather event; and
- Damage to the high visibility posts.

The locations of the 24 settlement monuments have been established on the landfill cover system. The locations of the 14 North Cell settlement monuments (LF01-NC-01 through LF01-NC-14) and the 10 South Cell settlement monuments (LF01-SC-01 through LF01-SC-10) are presented on the as-built drawings found in Appendix A. Each settlement monument consists of a 3.5-inch brass cap set in a base, surrounded by four reflective markers. Upon completion of cover construction, each settlement monument was surveyed. Each marker is stamped with point identification and baseline elevation. These baseline elevations will be used to compare to future surveys to determine whether settlement is occurring on the cover system. The coordinates and baseline elevations of each of the 24 settlement monuments are presented in Appendix B, Table B.4, for future comparison. The accuracy of the elevation measurements is rounded to the nearest 0.01 foot.

3.1.6 Access Road

A mulch access road has been constructed within the fenced area of the North Cell. The access road will be inspected for the following:

- Excessive rutting, ponded water, and/or potholes; and
- Erosion and/or loss of material that may impair or restrict access.

Inspection and repair of the mulch access road is not critical to the integrity of the cover system; however, maintaining the road will increase the ease with which the other monitoring, inspection, and maintenance activities are performed.

3.1.7 Surveys

A postclosure topographic map of the site showing the location of key installed features and the boundary of the disturbed areas after the construction of the cover was prepared by a professional surveyor licensed to practice in the state of Illinois. The existing features surveys for the landfill cells are available as Tables B.6 and B.7 in Appendix B. The as-built survey is provided in Appendix A.

3.2 MAINTENANCE AND REPAIRS

Maintenance activities will be performed after the inspection has been conducted and the findings documented. This approach allows for timely identification, repair, and/or redesign of features with recurring problems. The majority of postclosure maintenance and repair activities will be conducted on an "as-needed" basis. If any significant problems are identified requiring non-routine repairs or maintenance, the USAF will notify IEPA in writing within seven days. In the unlikely event a catastrophic failure of the LF-01 remedy is observed, the USAF will notify IEPA as soon as possible (electronically or via telephone) and follow-up with a written notification within seven days.

The features that require repair will be identified and documented through inspections, as previously described. For all areas requiring repairs, the cause of the damage will be documented, if known. Upon approval by the Scott AFB RPM, or designee/equivalent, repairs will be performed in accordance with the specifications for the original design, or an approved alternative. At the completion of repairs, the Scott AFB RPM, or designee/equivalent, is responsible for ensuring work performed is satisfactory and is properly documented.

The materials and equipment needed for corrective actions will depend on the nature and extent of the required repair. The repairs will be consistent with construction industry standard practices.

This section serves as a guide for most of the repairs that may be needed. For more serious damage (such as significant erosion gullies or slides), a qualified civil engineer may be consulted at the discretion of the Scott AFB RPM or designee/equivalent to inspect the damage and recommend corrective measures.

3.2.1 Landfill Cover

The landfill protective cover is stabilized by vegetation consisting of a mix of grasses and plants. The cover and other vegetated projective areas will be maintained to minimize soil washout areas and erosion gullies.

Damages to the soil cover and their corrective action could include the following:

- Erosion rills, slides, settlement, or other depressions (greater than 6 inches deep) – If these situations occur, clean fill and/or topsoil will be placed, seeded, and covered with approved mulch. The Scott AFB RPM or designee/equivalent and a qualified professional civil engineer will be contacted immediately if the erosion rills or slides expose waste materials or if they are severe or excessive in extent.
- Burrowing animals – If evidence of burrowing animals is present, it may be necessary to eradicate or relocate the animals to avoid potential damage to the cover system.
- Intrusive activities – LUCs restrict intrusive activities (such as, digging or trenching) at LF-01 without prior approval. Section 3.3 discusses LUC inspections and the corrective actions to be taken should intrusive activities be observed and documented.

3.2.2 Vegetative Cover

When vegetation has been established, vegetated areas at the site, including the landfill cover, will be mowed and maintained four times per year, preferably spring, early and late summer, and fall. The blade will be set at 4 inches. Grass and vegetation along fences and areas not accessible to mowers will be edged or trimmed as needed. Mowing will be avoided after periods of precipitation where equipment traffic may cause ruts. Any woody plants encountered during inspections will be dug out and removed, the roots destroyed, and any depressions resulting from plant removal will be filled with suitable soil and the ground surface reseeded. Various control methods can be used to remove invasive species on the landfill cover surface including the use of natural herbicides (per manufacturer's specification) or hand pulling (U.S. Environmental Protection Agency [USEPA], 2006). No burning is allowed.

An area larger than 100 square feet with no vegetative cover will be considered deficient. In such instances, the following corrective actions will be performed:

- Rake the area to remove vegetation and debris (material larger than 3 inches in diameter) and to scarify soil.
- As needed, apply additional topsoil to eliminate low spots and match the surrounding grade.
- Apply the appropriate seed blend and fertilizer and rake into topsoil (see Table B.5 in Appendix B for seed blends).
- Cover the area with approved mulch per manufacturer recommendation.
- Water the area and keep moist until vegetation sprouts and no drought conditions are forecast.

Reseeding will be performed until grass is established and as needed if deficient vegetative cover is observed. Seeding times for cool season grass mixes are as follows:

Seasonal Planting Dates:

Autumn	August 15 - October 15
Spring	March 15 - May 15
Temporary Seeding	October 16 - March 14
	May 16 - August 14

Seed applied outside the above time frames will consist of temporary or annual seed blends and will be used as a means to provide temporary erosion control. Areas in which temporary or annual seeding have been applied will be reseeded with the prescribed cool season grass blend during the next planting timeframe listed above. All seeding rates are per acre of pure live seed. The pure live seed should be specified when ordering seed blend. Appropriate seed blends and application rates are listed in Appendix B, Table B.5.

At the discretion of the Scott AFB RPM or designee/equivalent, and upon consultation with the inspector and/or other relevant professional, soil samples from these areas may be collected and analyzed to determine whether the soil needs to be amended to promote successful revegetation. Results from the analyses and any recommendations from the professionals consulted will be documented and incorporated into future maintenance activities.

3.2.3 Landfill Gas System

Damage to gas vents or landfill gas monitoring wells that require repairs and their respective corrective actions could include the following:

- Damaged or missing vent cap – Repair or replace as needed.
- Damaged vent riser pipe – Repair or replace as needed.
- Damaged vent riser pad – Repair or replace as needed.
- Damaged or missing gas sample valve – Repair or replace as needed.

3.2.4 Groundwater Monitoring Wells

The monitoring wells will be inspected for their structural integrity and will be maintained to provide representative groundwater samples. Monitoring wells can accumulate fine particles and sediments over time. Additionally, the casing and screen can become corroded or plugged and cause a loss of hydraulic connection. Wells that are deteriorating or are becoming plugged can affect groundwater data. Changes in hydraulic connection of the well (such as gradual plugging of the well screen) can be evaluated by the following issues:

- A significant drop in yield during purging,
- Increased turbidity,
- Changes in total well depth, and

- Long-term water level trends fluctuations (such as a long-term decline).

To provide a representative sample, wells that are becoming plugged or occluded may need to be redeveloped. At minimum, wells should be redeveloped when 10 percent of the well screen is occluded by sediments, or records indicate a change in yield, well depth, and turbidity.

The locks and hinges on the protective casing cover of the groundwater monitoring wells may be cautiously lubricated to facilitate their use, taking special care to isolate the lubricant to the area of use to avoid well contamination. Before lubricating the hinges and locks, the interior well cap will be placed on the well casing and the top of the well cover. Spray lubricants, penetrants, or paints will not be used on any hardware associated directly with the monitoring wells. This approach is intended to prevent the lubricants from interfering with groundwater analytics. The lubricants used on any monitoring well housing will be documented.

Damage to the groundwater monitoring wells that require repairs and their respective corrective actions could include the following:

- Damaged or missing locks – Replace locks with durable key locks keyed to match the other well locks. Provide a duplicate key to the Scott AFB RPM.
- Damaged or missing locking well caps – Repair or replace as needed.
- Damaged or missing locking covers – Repair or replace as needed.
- Damaged protective casing, posts, or pads – Repair or replace as needed.

Maintenance of the monitoring well network also will include removing intrusive woody plants and weeds, removing sediment and debris from around the well, and refurbishing protective posts and protective casing. If damage to the monitoring well is severe or if groundwater cannot be sampled, the entire well or well casing may need to be replaced. The Scott AFB RPM will be contacted if this condition is found to exist.

3.2.5 Access Road

Maintain ingress/egress of access road. Fill the ruts in the mulch access road located within the limits of the North Cell with appropriate material (for example, wood mulch) and regrade the road as necessary to fill the void and promote drainage off the road.

3.3 LAND USE CONTROL

The LUC objectives for LF-01 include preventing contact with buried waste and contaminated soil by restricting land use to nonresidential, preventing unauthorized ground-disturbing or intrusive activities, and maintaining the integrity of the cover system. Groundwater use restrictions will be required to ensure that groundwater within the landfill footprint and attenuation zone is not used as potable water or for other purposes. For groundwater beyond the attenuation zone of LF-01, LUCs could be implemented to prevent potable use of the water for as long as the COC concentrations exceed the remediation goals (RGs). The LUCs for groundwater beyond the attenuation zone would be short-term and would only be required until RGs are met.

3.3.1 Land Use Control Implementation

The USAF is committed to effectively implementing, maintaining, monitoring, reporting on, and enforcing LUCs for ensuring protection of human health and the environment. Scott AFB's Base General Plan, in both text and graphical forms, provides pertinent information used in planning and decision-making regarding permissible current and future land uses and activities. Scott AFB will revise the Base General Plan to include the LUCs identified for LF-01 in the final ROD addressing landfill waste, soil, and sediment (Scott AFB, 2013a) and the final ROD addressing groundwater (Scott AFB, 2013b). The LUCIP for LF-01 will be prepared separately following the approval of the Final ROD addressing groundwater. The LUCIP is a component of the remedy for LF-01 and restricts the use of the property to nonresidential purposes and prohibits the use of groundwater, reducing the risks to human health and the environment. These restrictions meet the postclosure requirements found in 35 IAC Section 811.111(d). Overall implementation and inspection of LUCs will be the responsibility of the Scott AFB RPM or designee/equivalent. LUCs will be required for as long as conditions or contamination do not allow for unlimited use and unrestricted exposure (Scott AFB, 2013a).

For groundwater contamination that has migrated off base and has affected property(ies) that is(are) not under the control of Scott AFB, the USAF will work with the property owner(s) to impose activity and groundwater use restrictions on the affected property in order for the remedy to be protective of human health and the environment.

Scott AFB will post signs that notify the public of the location and restrictions of the areas subject to LUCs. These signs will be posted at the gates to the North and South Cells. Soil and buried waste are located beneath the LF-01 cover area; the cover area is the only portion of the site with soil contaminants at levels that pose unacceptable risks to human health and the environment. The area requiring institutional controls that restricts residential development and prevents intrusive activities for LF-01 corresponds to the attenuation zone limits, which extend 100 feet from the limits of the landfill waste. Within this boundary, the withdrawal or use of groundwater for potable, irrigation, industrial, or agricultural use is prohibited. LUC signs will contain, but are not limited to, the following information:

Warning
Closed Former Base Landfill LF-01
Environmental Liner in Place
No Digging, Excavation, or Intrusive Activities Allowed
Groundwater Use Prohibited
Authorized U.S. Government Employees Only
Contact Base Environmental Office (256-2092)

Signage locations and the LUC boundary are presented on Figure 3.1.

3.3.2 Land Use Control Inspection

The quarterly inspection will evaluate the status of the LUCs and how any LUC deficiencies or inconsistent uses have been addressed. An Annual LUC Report will be prepared to certify that all LUCs are in place and implemented in accordance with the LUC Memorandum of Agreement.

3.3.3 Future Use Restrictions

LUCs are required at the site to ensure the long-term integrity of the landfill cover system. The LUCs will (1) prevent intrusive activities, thereby preventing contact with contaminants, and (2) ensure that the site is never developed for residential or commercial purposes. The engineered landfill cover system components consist of the vegetated protective layer, the geocomposite drainage layer, the low permeability geomembrane, a landfill gas collection system, and the foundation layer. These individual components function as a whole, whereby the integrity of each component must be protected from any activity or element that would compromise the overall cover system. Any surface activity that would disrupt the vegetated cover or any surface features, such as the gas vents, gas probes, settlement markers, and similar elements, must be avoided. Intrusive activities also are unacceptable. The foundation for the future EOD Proficiency Range was constructed on the South Cell, in accordance with the design drawings in the Remedial Action Work Plan (HGL, 2012c), to allow open detonation training to continue in the future. The EOD facility will be maintained separately from the remedial O&M of the landfill cover but will abide by the requirements of this plan. Proposed changes in the future use of the landfill cover will be evaluated by a professional engineer where each function of the landfill components would be evaluated. Any changes in future use must be approved by the IEPA.

TABLE(S)

This page was intentionally left blank.

Table 3.1
Summary of Monitoring, Inspection, and Maintenance Activities
Landfill LF-01
Scott AFB, Illinois

Postclosure O&M Activity	Quantity	Required Frequency	Year 1	Year 2	Years 3 to 5	Years 6 to 30	Basis for Requirement
Environmental Monitoring							
Groundwater Sampling and Analysis - Landfill Performance (1) and Groundwater Remedy Parameters (2)	48 Monitoring Wells(1),(2)	Quarterly for 5 years; Semiannually thereafter	Quarterly	Quarterly	Quarterly	Semiannually	35 IAC 811.319(a)
Groundwater Analysis - Natural Attenuation Parameters (3)	37 Monitoring Wells (4)	Semiannually	Semiannually	Semiannually	Semiannually	Semiannually	35 IAC 811.319(a)
Groundwater Levels and Field Parameters	48 Monitoring Wells	Each groundwater sampling event	Quarterly	Quarterly	Quarterly	Semiannually	35 IAC 811.318(e)(6)
Surface Water Sampling and Analysis	6 Surface Water Locations	Quarterly for 5 years; Semiannually thereafter	Quarterly	Quarterly	Quarterly	Semiannually	35 IAC 302, Subparts A, B, and F
Landfill Gas Monitoring	5 Landfill Gas Wells; 29 Landfill Passive Gas Vents	Quarterly for 5 years; Semiannually thereafter	Quarterly	Quarterly	Quarterly	Semiannually	35 IAC 811.310(c)
Ambient Air Monitoring	3 Ambient Air Locations	Quarterly for 5 years; Semiannually thereafter	Quarterly	Quarterly	Quarterly	Semianually	35 IAC 811.310(b)(8)
Monitoring Well Depths	48 Monitoring Wells	Annually	Annually	Annually	Annually	Annually	35 IAC 811.318(e)(7)

Table 3.1
Summary of Monitoring, Inspection, and Maintenance Activities
Landfill LF-01
Scott AFB, Illinois

Postclosure O&M Activity	Quantity	Required Frequency	Year 1	Year 2	Years 3 to 5	Years 6 to 30	Basis for Requirement
Inspections							
Landfill Protective Cover and Vegetation	Where Applicable	Quarterly for 5 years; Semiannually thereafter	Quarterly	Quarterly	Quarterly	Semianually	35 IAC 811.111(c)
Drainage Areas	Where Applicable	Semiannually	Semiannually	Semiannually	Semiannually	Annually	35 IAC 811.111(c)
Landfill Gas Vents	29 Landfill Passive Gas Vents	Semiannually	Semiannually	Semiannually	Semiannually	Annually	35 IAC 811.310
Landfill Gas Monitoring Wells	5 Landfill Gas Wells	Semiannually	Semiannually	Semiannually	Semiannually	Annually	35 IAC 811.318
Groundwater Monitoring Wells	53 Monitoring Wells [5]	Semiannually	Semiannually	Semiannually	Semiannually	Annually	35 IAC 811.318
Stormwater Management Inspection	Where Applicable	After all extreme weather events	After all extreme weather events	After all extreme weather events	After all extreme weather events	After all extreme weather events	35 IAC 811.111(c)
Access Road	Where Applicable	Annually	Annually	Annually	Annually	Annually	35 IAC 811.111(c)
LUCs (Restrictions and Controls)	Where Applicable	Quarterly	Quarterly	Quarterly	Quarterly	Quarterly	LUC Memorandum of Agreement
Cover Settlement Monuments	24 Cover Settlement Monuments	Semiannually	Semiannually	Semiannually	Semiannually	Annually	35 IAC 811.205(c)
Survey Cover Settlement Monuments and Benchmarks	24 Cover Settlement Monuments; 2 Benchmarks	Every 2 years	Survey at Completion of Landfill Cap	Survey (year 2)	Survey (year 4)	Every 2 years	35 IAC 811.205(c)

Table 3.1
Summary of Monitoring, Inspection, and Maintenance Activities
Landfill LF-01
Scott AFB, Illinois

Postclosure O&M Activity	Quantity	Required Frequency	Year 1	Year 2	Years 3 to 5	Years 6 to 30	Basis for Requirement
Maintenance							
Groundwater Monitoring Wells	53 Monitoring Wells [5]	As Needed	As Needed	As Needed	As Needed	As Needed	35 IAC 811.318
Landfill Gas Vents	29 Landfill Passive Gas Vents	As Needed	As Needed	As Needed	As Needed	As Needed	35 IAC 811.310
Landfill Gas Monitoring Wells	5 Landfill Gas Wells	As Needed	As Needed	As Needed	As Needed	As Needed	35 IAC 811.318
Mow and Maintain Vegetation	Where Applicable	Four times per year (Spring, Summer, Fall), if needed	Four times per year (Spring, Summer, Fall)	Four times per year (Spring, Summer, Fall)	Four times per year (Spring, Summer, Fall)	Four times per year (Spring, Summer, Fall)	35 IAC 811.111(c)
Landfill Cover	Where Applicable	As Needed	As Needed	As Needed	As Needed	As Needed	35 IAC 811.503

Notes:
(1) The landfill performance well network (30 wells) will be analyzed for VOCs (method 8260C) and filtered and unfiltered metals (arsenic, boron, iron, lead, magnesium, and zinc).
(2) An additional 18 groundwater remedy wells that are not included in the 30-well landfill performance monitoring network will be sampled only for applicable groundwater contaminants of concern (trichloroethene, cis-1,2-dichloroethene, vinyl chloride, arsenic, and iron) as shown in Table 4.1.
(3) MNA parameters include sulfate, nitrate, chloride, ammonia, alkalinity, dissolved gases, ferrous iron, sulfides, total organic carbon, and total dissolved solids.
(4) Because 5 background wells and 7 sentinel wells will not require MNA parameter sampling, only 37 of 48 wells will require MNA analysis semiannually.
(5) Monitoring well inspections/maintenance will occur for all 53 permanent wells which includes five wells not sampled as part of the well network.

AFB = Air Force Base
IAC = Illinois Administrative Code
LUC = land use control
MNA = monitored natural attenuation
O&M = operation and maintenance

This page was intentionally left blank.

Table 3.2
Postclosure Site Inspection Checklist
Landfill LF-01
Scott AFB, Illinois

Name of Inspector:	**Inspection (Circle One): 1 2 3 4**
Temperature:	**Date and Time:**
Wind Direction:	**Date and Time of Last Inspection:**

A. LANDFILL COVER - NORTH AND SOUTH CELLS		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Is there ponded surface water or noticable depressions on the protective cover of the landfill cell?						
2	Are there significant erosion, rills, and gullies on the protective cover of the landfill cell?						
3	Is there inadequate vegetative cover? (An area larger than 100 square feet with no vegetative cover will be considered inadequate.)						
4	Is there any displacement of erosion control materials? (Erosion control blanket is installed at the swale on the South Cell. Turf reinforcement mats are installed under the riprap located along the slopes of the landfill cells. Check coverage.)						
5	Is there daylighting of system components (such as drainage piping, geomembrane, drainage mat, gas collection system, etc.) that should otherwise not be exposed?						
6	Are there indications of slope stability failures or symptoms of impending failure?						
7	features?						
8	protective cover of the landfill cell where vegetation has been established?						

Table 3.2
Postclosure Site Inspection Checklist
Landfill LF-01
Scott AFB, Illinois

A. LANDFILL COVER - NORTH AND SOUTH CELLS (CONTINUED)		NORTH CELL	SOUTH CELL	YES	NO	N/A	Comments/Corrective Actions
9	Is there any evidence of animal burrow holes in the soil on the protective cover of the landfill cell?						
10	Is there any evidence of intrusive human activity?						
11	Is there invasion of undesirable plants (such as trees, shrubs, wetlands species) onto the soil cover or side slopes?						
B. LANDFILL GAS SYSTEM		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Are the gas vents clogged?						
2	Is there damage to vent pipes?						
3	Is there any evidence of vandalism or tampering?						
4	Is there any damage to the high visibility protective posts? (Ensure they are in good condition and are standing upright.)						
C. DRAINAGE SYSTEM		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Is there evidence of erosion or localized depressions?						
2	Is there damage to high visibility protective posts? (Ensure they are in good condition and are standing upright.)						
3	Are there any other visible conditions that may prohibit or impede flow of runoff or otherwise affect operational efficiency? (Check drainage valve outlets for clogging or buildup of debris. Remove large debris such as logs.)						

Table 3.2
Postclosure Site Inspection Checklist
Landfill LF-01
Scott AFB, Illinois

D. LANDFILL NETWORK MONITORING WELLS		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
Are the following items damaged or in poor condition:							
1	Locks						
2	Locking well caps						
3	Locking metal covers						
4	Protective casing, bumper posts, and surface seal, and pad						
5	Has there been any use of lubricants?						
6	Peeling or chipping paint?						
7	Other (identify)						
E. LANDFILL COVER SETTLEMENT MARKERS		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Are cover settlement markers damaged or in poor condition? (Check to be flush against protective cover of landfill cell.)						
2	Is there damage to high visibility protective posts? (Ensure they are in good condition and are standing upright.)						
F. ACCESS ROAD		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Is there excessive rutting or large potholes on the access road?						
2	Is there excessive erosion or loss of material on the access road that may impair or restrict access?						
G. LAND USE CONTROLS		**NORTH CELL**	**SOUTH CELL**	**YES**	**NO**	**N/A**	**Comments/Corrective Actions**
1	Is there evidence of ground disturbance on the protective cover of the landfill cell?						
2	Are there deficiencies with LUC restrictions?						
3	Are LUC signs missing or is there damage to the sign?						

Table 3.2
Postclosure Site Inspection Checklist
Landfill LF-01
Scott AFB, Illinois

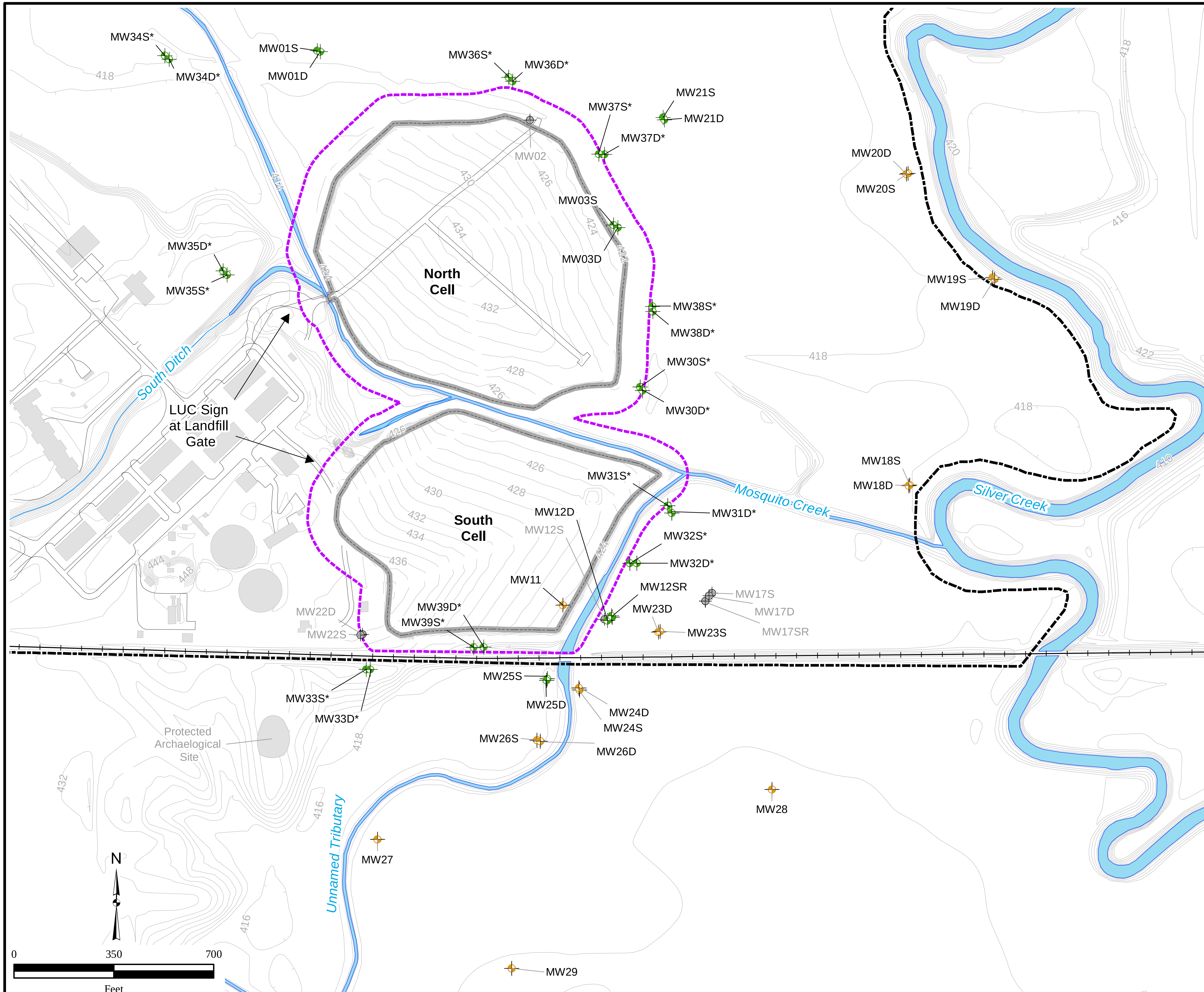
N/A = Not applicable LUC = Land Use Control

Additional Notes:

FIGURE(S)

This page was intentionally left blank.

Figure 3.1
LUC Boundary



Legend

- Landfill Performance Monitoring Well
- Groundwater Remedy Monitoring Well
- Not Included in Groundwater Monitoring Network
- 422 Topographic Contour Line (ft amsl)
- Landfill Cover Boundary
- LUC Boundary
- Rail Line
- Scott AFB Boundary
- Road
- Surface Water
- Existing Structure

Notes:
LUCs will be implemented on base to prevent residential development, building construction, intrusive activities, and groundwater from being used as drinking water.

*=proposed long term monitoring well
ft amsl=feet above mean sea level
D=deep well
LUC=Land Use Control
S=shallow well
SR=shallow replacement well

*\\Gst-srv-01\hglgis\Scott_AFB\LF-01\O_M_Manual\
(3-1)Monitoring_Landfill_Performance_GW_COCs.mxd
5/30/2013 ST
Source: HGL*

HGL HydroGeoLogic, Inc

This page was intentionally left blank.

4.0 POSTCLOSURE MONITORING

This section describes the postclosure monitoring that will occur at LF-01 to: (1) demonstrate that the landfill cover system is properly functioning to protect the human health and the environment as designed; (2) provide an early warning system in the unlikely event of leachate migration or landfill gas migration; and (3) ensure attainment of RAOs for the site.

Postclosure monitoring includes the following:

- Groundwater sampling and analysis for landfill performance parameters;
- Groundwater level measurements (potentiometric conditions);
- Surface water sampling;
- Landfill gas management system monitoring; and
- Ambient air monitoring.

All groundwater sampling, surface water sampling, landfill gas monitoring, ambient air monitoring, decontamination, surveying, and waste handling/investigation-derived waste (IDW) procedures will follow the protocols detailed in the Basewide FSP (HGL, 2009a).

The quality assurance project plan (QAPP) provides the overall quality assurance (QA)/quality control (QC) goals for all work conducted at Scott AFB. Sampling, sample analysis, data management, validation, and reporting will be documented and conducted in accordance with a Uniform Federal Policy QAPP (UFP-QAPP). The UFP-QAPP will be based on the 2009 Basewide QAPP Addendum (HGL, 2009b). All field QA/QC samples will be collected as outlined in the UFP-QAPP. Sample collection and management-related tasks will be documented on the appropriate field forms. Safe working conditions during investigation activities will be developed, maintained and implemented in accordance with the Final Site Safety and Health Plan (HGL, 2009c).

4.1 GROUNDWATER MONITORING

This subsection details the groundwater monitoring program that will occur to monitor the effectiveness of the landfill cover. The frequency and analytes monitored are discussed in detail in the following subsections.

4.1.1 Groundwater Sampling

At the first quarterly groundwater sampling event, all 48 wells will be sampled using the low-flow sampling technique outlined in the UFP-QAPP. Low-flow sampling will be used to conduct baseline evaluations and five-year reviews. However, while low-flow sampling is typically used to define the nature and extent of contamination and evaluate risk, passive sampling techniques can be used to evaluate trends. Therefore passive sampling will be used for quarterly and semiannual groundwater monitoring events using the HydraSleeve SuperSleeve, which has been accepted by IEPA. The HydraSleeve SuperSleeve standard operating procedure (SOP) is presented in Appendix C and discussed below. Table 4.1 summarizes the groundwater sampling and analysis plan to monitor landfill performance.

Beginning at the second quarterly groundwater sampling event, all wells will be sampled using passive sampling methods (Appendix C). Groundwater will be collected with a 2-liter HydraSleeve sampler (60 inches in length, 3-inch flat width, and 1.9-inch filled outside diameter). The suspension cord, sample sleeve, and the discharge tube are designed for one-time use and are disposable. A reusable stainless-steel weight (8 ounces and 4 inches long) with clip (2 inches long) is attached to the bottom of the sleeve to place the sampler at the bottom of the well. The spring clip, weight, and weight clip should be decontaminated after each use and before being reused.

Before deploying or retrieving a HydraSleeve sampler, the depth-to-water measurement from the well and the depth to the bottom of the well will be collected. These measurements will be used to determine the preferred position of the HydraSleeve in the well. Using a premeasured tether, the HydraSleeve sampler will be carefully deployed so that the weight rests at the bottom of the well. The top of the HydraSleeve should be 66 inches above the bottom of the well. This placement will ensure that the sample will be collected approximately 5 to 10 feet above the bottom of the well. Only groundwater from the screened interval will be collected as a sample.

The HydraSleeve sampler must be deployed for a minimum of 48 hours (preferably 96 hours) before sampling to allow conditions in the water column to stabilize. To eliminate separate mobilizations for deployment and recovery of the samplers for quarterly or semiannual sampling events, the HydraSleeve sampler for the next sampling event can be installed immediately after recovering the HydraSleeve sampler for the current event. This step ensures that a new HydraSleeve sampler remains in the well until the next sampling event, thus allowing for a long equilibration time and immediate recovery of the HydraSleeve sampler. HydraSleeves can be left in a well for an indefinite period of time without concern.

Per the SOP presented in Appendix C, collecting a sample from the HydraSleeve occurs as follows:

- Hold the tether while removing the well cap.
- Secure the tether at the top of the well while maintaining tension on the tether. Do not pull the tether upwards at this time.
- Measure the water level in the well.
- In one smooth motion, pull the tether up 3 to 4.5 feet at a rate of 1 foot per second (or faster). The motion will open the top check valve and will fill the HydraSleeve. When the HydraSleeve is full, the top check valve will close.
- Continue pulling the tether upward until the HydraSleeve is at the top of the well.
- Decant and discard the small volume of water trapped in the Hydrasleeve above the check valve by turning the sleeve over.
- Collect the sample immediately after the HydraSleeve has been brought to the surface to preserve sample integrity.
- Remove the discharge tube from its sleeve.

- Hold the HydraSleeve at the check valve.
- Puncture the HydraSleeve just below the check valve with the pointed end of the discharge tube.
- Discharge water from the HydraSleeve into the appropriate sample containers. The flow can be controlled by either raising the bottom of the sleeve or by squeezing it.
- Continue filling sample containers until all are full.

The HydraSleeve SuperSleeve will be used during sampling events to establish postclosure trends at LF-01. Low-flow sampling will be used at the initial groundwater sampling event (year 0) and during subsequent five year reviews to confirm HydraSleeve results and to make decisions concerning attainment of groundwater quality standards.

4.1.2 Sampling Frequency

To meet the intent of 35 IAC Section 811.319(a)(1), monitoring will begin within 1 year of the final inspection and approved installation of the landfill cover, which occurred on July 17, 2013.

The majority of the monitoring wells for both networks are located within or near wetland areas that frequently exhibit saturated conditions. These conditions present logistical challenges for equipment accessibility for sampling events. Large rain events occur during the spring, which create heavily saturated wetland areas that surround the North and South Cells. These events typically result in approximately 2 to 3 feet of standing water at the well locations on the north and east sides of the North Cell and approximately 1 to 2 feet of standing water at the well locations on the east side of the South Cell. The wetland areas to the north and east of the North Cell tend to remain saturated for longer periods of time as compared to the areas east of the South Cell. As a result, sampling activities and frequency may depend on the wetland conditions and accessibility. If wells are not accessible during a sampling event, the O&M contractor would notify the USAF and form a contingent plan for sampling when conditions allow. This plan would be developed in agreement with the IEPA and the USACE.

4.1.2.1 Landfill Network Monitoring Wells

For the initial 5 years (year 0 to year 5) following construction of the landfill cover, the landfill performance monitoring well network (30 wells) will be sampled on a quarterly basis and will be analyzed as specified in Section 4.1.3. Based on the trends observed from the actual groundwater data, the number of wells sampled, the frequency of sampling, and the chemical analyses performed may be adjusted at the five year review period. According to 35 IAC 811.319(a)(1)(A), after the initial 5-year groundwater sampling period, the sampling frequency can be reduced to a semiannual basis if sampling data demonstrates that monitoring effectiveness has not been compromised, that sufficient quarterly data has been collected to characterize groundwater, and that leachate from the monitored unit does not constitute a threat to groundwater. The source will be considered a threat to groundwater if contaminant concentrations within the zone of attenuation exceed the federal MCL or IEPA Class I GRO for that constituent.

According to 35 IAC 811.310(a)(1)(B), beginning 15 years after closure of the unit, or 5 years after potential discharge sources no longer constitute a threat to groundwater, the monitoring frequency may change on a well by well basis to an annual schedule if either of the following conditions exist:

- All constituents monitored within the zone of attenuation have returned to a concentration less than or equal to 10 percent of the maximum allowable predicted concentration; or
- All constituents monitored within the zone of attenuation are less than or equal to their maximum allowable predicted concentration for eight consecutive quarters.

However, in accordance with 35 IAC 811.320(e), if a statistically significant increase in the concentration of any constituent with respect to the previous sample is observed at a well, monitoring will return to a quarterly basis at that well. Applicable statistical methods include normal theory tests (where appropriate) or nonparametric statistical tests.

4.1.2.2 Groundwater Remedy Monitoring Wells

Monitoring will begin within 3 months of installing the groundwater remedy monitoring well network. To monitor the attenuation of groundwater COCs at the site, wells are located within the potential source area, the attenuation zone, and the respective contaminant plumes. Seven sentinel wells are included in the network and are located downgradient and outside the contaminant plumes to monitor potential plume expansion. The groundwater remedy monitoring network is presented in Table 4.1.

For the initial 5 years (years 0 to 5), monitoring will be performed on a quarterly basis. After the first 5 years of groundwater monitoring, it is assumed that the monitoring frequency will be reduced to semiannual sampling. Based on the trends observed from the actual groundwater data, the number of wells sampled and chemical suites analyzed may be adjusted. In addition, contaminant trends and site-specific decay rates will be used to evaluate the effectiveness of the remedy. In the unlikely event that it is determined that remediation goals will not be achieved within the 30-year remedial time frame, a contingency remedy will be implemented. The contingent remedy will consist of active treatment, such as the in situ bioremediation contemplated in Alternatives 3 and 4 that were presented in the Feasibility Study Addendum addressing Groundwater (HGL, 2012d), to expedite remediation of the remaining groundwater contamination. The specific type of active treatment will be determined based on site specific data at the time the contingent remedy is implemented.

4.1.3 Sample Analysis

A U.S. Department of Defense (DoD) Environmental Laboratory Accreditation Program accredited laboratory will analyze the collected samples and comply with the requirements of the DoD Quality Systems Manual for Environmental Laboratories Version 4.2. Samples for laboratory analysis will be shipped via overnight carrier to the laboratory.

4.1.3.1 Landfill Performance Monitoring Well Network

To meet the intent of 35 IAC 811.319 subsections (a)(2)(A) and (a)(3)(A), the landfill performance monitoring network will have groundwater samples collected and analyzed on a quarterly basis for the following parameters:

- VOCs (method 8260C)
- Field filtered and unfiltered metals (arsenic, boron, iron, lead, magnesium, and zinc) (method 6010C)

On a semiannual basis, groundwater samples also will be analyzed for natural attenuation parameters, which are as follows:

- Anions (sulfate, nitrate, and chloride) (method SM300);
- Ammonia (as nitrogen) (method SM350.2);
- Alkalinity (method SM310.1);
- Dissolved gases (method RSK-175);
- Ferrous iron (method 3500 DE-D);
- Sulfides (method 376.1);
- Total organic carbon (method SW9060A); and
- Total dissolved solids (method SM2540C).

Table 4.1 summarizes the groundwater sampling and analysis program.

4.1.3.2 Groundwater Remedy Monitoring Well Network

Monitoring wells within the groundwater remedy monitoring network will be used to monitor the attenuation of groundwater COCs (TCE, cis-1,2-DCE, vinyl chloride, iron, and arsenic). Wells included in both the landfill performance network and the groundwater remedy network will be analyzed for VOCs and field filtered and unfiltered metals (arsenic, boron, iron, lead, magnesium, and zinc). As outlined in Table 4.1, wells only included in the groundwater remedy network will be analyzed for only the applicable groundwater COCs (TCE, cis-1,2-DCE, vinyl chloride, arsenic, and iron). On a semiannual basis, groundwater samples will be analyzed for monitored natural attenuation (MNA) parameters. However, the seven sentinel wells will be only analyzed for applicable groundwater COCs and not MNA parameters. If COC concentrations show a statistical increase in a sentinel well, then MNA parameters will be collected from that well at the next sampling event. A summary of the groundwater sampling and analysis program is provided in Table 4.1.

4.1.4 Groundwater Field Parameters

During each groundwater sampling event, field parameters will be collected at all monitoring wells at the time of sample collection and before preserving the samples for shipment. Field parameters include the following:

- Depth to groundwater (groundwater elevations);
- pH;

- Temperature of the sample;
- Specific conductance;
- Oxidation reduction potential;
- Dissolved oxygen; and
- Turbidity.

Water level measurements will be taken from all permanent wells to determine the elevation of the water table surface at least once within a single 24-hour period. Static water levels will be measured each time a well is sampled, and before any equipment enters the well. When an airtight casing cap is removed, the pressure inside the well will be allowed to equilibrate before the measurement is taken. Measurements will be repeatedly taken until the water level has stabilized. Any conditions that may affect water levels will be recorded in the field log. Water level measurements will be taken with electronic sounders, air lines, pressure transducers, or water-level recorders (for example, Stevens recorder). Groundwater levels will be measured to the nearest 0.01 foot. During the water level measurement, the total well depth will be measured and recorded on the field sheet.

4.1.5 Field Records

All field activities will be recorded in a field logbook. Entries will be made in real time, in permanent ink, and include sufficient detail to reconstruct field activities in an accurate manner. All field forms to support this effort, including sample collection field sheets and IDW tracking forms, are provided in the Final Basewide FSP (HGL, 2009a). These forms will be considered extensions of the field logbook and will be kept in the project file. Section 9.0 of the Basewide QAPP discusses the documentation requirements for Scott AFB (HGL, 2009b). Copies of logbooks and field forms will be maintained in the project files and included in reports submitted to the USACE and the USAF, as requested.

4.1.6 Potentiometric Surface Maps

Groundwater elevations will be collected from all wells at the site to determine potentiometric conditions for Zone 2 and Zone 4. The depth below the ground surface of each well will be measured annually. Potentiometric surface maps will be generated from the elevation data and included in the annual report. The bottom depth of each well will be used to determine the percent occlusion of each well screen. Any well screen that is occluded will be documented and results will be provided to the USACE and the USAF. If the well contains an occlusion percentage of 10 percent or more of the screened interval, the well will be redeveloped. Redevelopment will be required before the next groundwater sampling event begins.

4.1.7 Data Analysis

The following subsections discuss groundwater quality standards, evaluation criteria, and the quality control program.

4.1.7.1 Applicable Groundwater Quality Standards

In accordance with 35 IAC Section 811.320(c), groundwater within the attenuation zone (defined as the area within 100 feet of the landfill waste) does not need to attain groundwater quality standards or the remedial goals (IEPA, 1997). At or beyond the attenuation zone, constituents must be maintained at applicable groundwater quality standards. In accordance with 35 IAC Sections 811.320(a) and 811.320(b), applicable groundwater quality standards qualify as the USEPA maximum contaminant levels (MCLs) for drinking water or Scott AFB basewide groundwater ranges (HGL, 2012a), whichever is appropriate. For groundwater COCs, the following remedial goals were established in the ROD addressing groundwater (Scott AFB, 2013b):

Contaminant of Concern	Remediation Goals (μg/L)	Source
cis-1,2-Dichloroethene	70	Federal MCL for Drinking Water
Vinyl chloride	2	Federal MCL for Drinking Water
Trichloroethene	5	Federal MCL for Drinking Water
Arsenic	19.1	Maximum value of Scott AFB basewide concentrations
Iron	26,000	Maximum value of Scott AFB basewide concentrations

Notes:
AFB = Air Force Base
μg/L = micrograms per liter
MCL = maximum contaminant level

Scott AFB basewide groundwater values are included in Appendix D.

4.1.7.2 Evaluation of Groundwater Quality Data

The groundwater quality data will be evaluated in accordance with 35 IAC Section 811.319(a)(4). The required evaluations include comparisons of the concentration of constituents as follows:

- Progressive increases in inorganic concentrations over eight consecutive monitoring events;
- Exceedances of applicable Scott AFB basewide groundwater values (Appendix D);
- Increases above the preceding measured concentration (for the organic constituents); and
- Exceedances of applicable USEPA MCLs.

Groundwater quality data will also be compared to site-specific baseline values. Baseline values will be defined by concurrent sampling of upgradient wells during the first year of O&M. To establish whether a statistically significant change occurred, the data will be analyzed using one or more of the normal theory statistical tests in accordance with 35 IAC

811.320(e)(1). Individual well comparisons are to be analyzed at a level of significance (Type I error level) of no less than 0.01. For multiple well comparisons, the level of significance will be no less than 0.05. Tests for normality distribution and equality of variance will be performed on the water quality data before performing other statistical tests. If normal theory tests are inappropriate, then nonparametric statistical tests will be used per 35 IAC 811.320(e)(4). Statistical analyses of data will include, but are not limited to, the following:

- Developing site-specific baseline concentrations;
- Determining whether statistically significant changes occurred for monitored constituents or to baseline concentrations over time; and
- Plotting time-series data (control chart) for water quality field parameters to provide a visual depiction for trends or abrupt changes in concentration levels.

To statistically analyze data that are below the limit of detection, the following will occur in accordance with 35 IAC 811.320(e)(3):

- If the percentage of nondetects is less than 15 percent, the nondetect values will be represented as one-half the method detection limit or practical quantitation limit (whichever is higher), and one or more of the normal theory statistical tests will be used (such as an analysis of variance [ANOVA]).
- If the percentage of nondetects is between 15 and 50 percent, and the data is normally distributed, the Cohen's or Aitchison's adjustment will be performed on the sample mean and standard deviation, followed by an applicable statistical procedure.
- If the percentage of nondetects is between 15 and 50 percent, and the dataset is not normally distributed, then the nonparametric ANOVA will be used and the nondetects will be treated as ties.
- If the percentage of nondetects is greater than 50 percent and the percentage of quantified observations is greater than 10 percent, then the test of proportions will be used. The test of proportions determines whether the proportion of compounds detected in the network monitoring wells differs significantly from the proportion of compounds detected in the baseline well dataset.
- If the percentage of nondetects is greater than 50 percent and the percentage of quantified observations is less than 10 percent, a Poisson distribution will be used for statistical evaluation of the dataset. The Poisson distribution arises when the occurrences of quantifiable constituents are randomly distributed in the monitoring well network.

The influence of seasonal and temporal effects to the statistical evaluation of groundwater quality data will be reduced by including all quarterly analytical results into the statistical evaluations and using the control chart method. The control chart method is an intra-well comparison method that monitors and displays constituents within a single well over time. The plotted time-series data will be visually inspected for trends or abrupt changes in concentration levels. This approach will primarily be used to evaluate changes in field parameters due to seasonal and temporal influences.

4.1.7.3 Groundwater Quality Assessment Program

If no statistically significant changes from baseline levels or previous measurements are identified, the O&M contractor will document this in the annual report. However, if statistically significant increases from previous measurements are observed, the USAF will notify the IEPA and USACE in writing within 14 days and will place a notice in the operating record indicating specific constituents have shown significant change (35 IAC 811.319(b)(5)(F)). The USAF will establish an assessment monitoring program in concurrence with the IEPA and USACE. The purpose of the program will be to determine the following:

- Whether additional landfill contaminants (leachate) have entered the groundwater;
- The rate and extent of migration of contaminants from the landfill in groundwater (plume expansion); and
- The concentrations of contaminants in groundwater.

To obtain the above objectives, the Groundwater Quality Assessment Plan will occur in a phased investigative approach. To meet the intent of 35 IAC 811.319(b), Phase I will consist of the investigation to confirm that the landfill is the source of the contamination and to provide information to perform a groundwater impact assessment. Phase I will include the following:

- Resample the groundwater monitoring network wells within 90 days of the initial sampling event;
- Present the analytical data in a report in accordance with 35 IAC 811.320(e);
- Review the site hydrogeologic conditions;
- Re-evaluate the construction and monitored intervals of the existing groundwater monitoring wells; and
- Evaluate the appropriateness of the statistical and graphical groundwater quality data evaluation procedures.

If the concentrations of the monitored constituents are at or below baseline values (using statistical procedures outlined in 35 IAC 811.320(e)) for two consecutive sampling events, the USAF will notify IEPA and USACE, stop the Groundwater Quality Assessment Plan, and return to detection monitoring. However, if concentrations continue to increase, then Phase II will be initiated as follows (35 IAC 811.319(c):

- Collecting and interpreting additional hydrogeologic data;
- Collecting and interpreting additional groundwater quality data;
- Installing additional assessment monitoring wells;
- Characterizing the rate and extent of contaminant migration using numerical models;
- Evaluating the current effectiveness of the remedial action (natural attenuation);
- Determining the current site-specific degradation rates and attenuation time frame for groundwater COCs; and

- Assessing the risk to human health and the environment posed by the statistically significant contaminant concentration increases.

Results of the Phase I and Phase II studies will lead to either (1) applying to reinstate the detection monitoring program or (2) retrofitting the current remedial action for groundwater. The objective of retrofitting the current groundwater remedial action (natural attenuation) or implementing a contingent remedy is to select a constructible, cost-effective remedial alternative that is technically and logistically capable of being implemented to achieve the cleanup goals. A contingency remedy may be implemented if the impact assessment determines that groundwater COCs will not achieve cleanup levels within the 30-year time frame.

4.1.7.4 Quality Control Program

Sampling will be performed in accordance with the procedures outlined in the UFP-QAPP. These plans specify the collection of field QC samples that can be used to assess whether the analytical data were affected by the sampling process. The field QC samples will include:

- One equipment blank will be collected each day sampling is conducted per sampling team.
- One trip blank will accompany each cooler of samples sent to the laboratory for analysis of VOCs.
- One duplicate sample will be collected for every 10 groundwater samples collected.

In addition to the field QC samples, other field procedures to promote QC will include the following:

- Preprinting sample labels to facilitate sample tracking from the field through the laboratory to the final report.
- Documenting sample collection in the field to ensure that sample labeling, chains of custody, and requests for analysis are in agreement and traceable back to the correct field sample.
- Placing custody seals on each cooler before shipment by an overnight carrier.

The laboratory QC program, including sample handling, laboratory QC elements, and data reporting, is fully documented in the 2006 Basewide QAPP (Booz Allen Hamilton, 2006) and the Final QAPP Addendum (HGL, 2009b). These documents will be updated and incorporated into the UFP-QAPP format. Sample handling includes documentation of sample receipt, placement in storage, controlled sample access, and disposal. Laboratory QC elements consist of instrument calibration and maintenance, laboratory control samples, method blanks, matrix spike/matrix spike duplicate samples, and method-specific QC checks. A matrix spike/matrix spike duplicate sample will be collected at a frequency of 1 per 20 samples (5 percent).

4.2 SURFACE WATER MONITORING

Postclosure surface water monitoring will be conducted for all surface water locations that may potentially serve as daylighting locations for groundwater migrating from LF-01. This sampling will monitor the long term effectiveness of the landfill cover and will be conducted in accordance with 35 IAC 302 Subparts A, B, and F. Sampling locations SW01 through SW06 are depicted on Figure 2.1. The frequency and analytes monitored are discussed in detail in the following subsections.

4.2.1 Surface Water Sampling

All surface water samples will be collected as grab samples using a clean disposable bucket, jar, bailer or other sample container placed in the middle of the depth of the water per the UFP-QAPP. The site-specific procedures developed for Scott AFB are consistent with USAF's Model FSP.

The UFP-QAPP provides the overall QA/QC goals for all work conducted at Scott AFB. Types of required analysis, sample volumes, appropriate sample containers, applicable holding times, sample preservatives, sample preparation methods, sample identification, custody procedures, and the correlating techniques are provided in the UFP-QAPP. All documentation of field activities and implementation of field procedures will be maintained to properly meet these data quality objectives. All field QA/QC samples will be collected as outlined in the UFP-QAPP. Sample collection and management-related tasks will be documented on the appropriate field forms. The field QC samples are as follows:

- One equipment blank will be collected each day sampling is conducted per sampling team.
- One trip blank will accompany each cooler of samples sent to the laboratory for analysis of VOCs.
- One duplicate sample will be collected for every 10 surface water samples collected.
- A matrix spike/matrix spike duplicate sample will be collected at a frequency of 1 per 20 samples (5 percent).

4.2.2 Sampling Frequency

Surface water monitoring will begin within 1 year of the effective date of the final inspection and approved installation of the landfill cover, which occurred on July 17, 2013. For the initial 5 years (year 0 to year 5) following construction of the landfill cover, surface water samples will be collected on a quarterly basis and analyzed as specified in Section 4.2.3. After the first 5 years of surface water monitoring, it is assumed that the long-term landfill monitoring frequency will be reduced to semiannual sampling. Based on the trends observed from the actual surface water data, the number of locations sampled, the frequency of sampling, and the chemical analyses performed may be adjusted at the five year review period.

4.2.3 Sample Analysis

To monitor for potential releases of groundwater COCs into the surface water, surface water samples will be collected and laboratory analyzed for the following:

- VOCs (method 8260C);
- TAL metals (field filtered) (method 6010C);
- Total dissolved solids (method SM2540C); and
- Total organic carbon (method SW9060A).

Results will be compared to the Illinois General Use Water Quality Standards (35 IAC 302.208 and 302.210). Table 4.1 summarizes the surface water sampling and analysis program. Samples for laboratory analysis will be shipped via overnight carrier to the approved laboratory.

4.2.4 Surface Water Field Parameters

During each surface water sampling event, field parameters will be collected at the time of sample collection and prior to preserving the samples for shipment. Field parameters include the following:

- pH,
- Temperature of the sample,
- Specific conductance,
- Oxidation reduction potential,
- Dissolved oxygen, and
- Turbidity.

4.2.5 Field Records

All field activities will be recorded in a field logbook. Entries will be made in real time, in permanent ink, and include sufficient detail to reconstruct field activities in an accurate manner. All field forms to support this effort are provided in the UFP-QAPP. These forms will be considered extensions of the field logbook and will be kept in the project file. The UFP-QAPP provides the documentation requirements for Scott AFB. Copies of logbooks and field forms will be maintained in the project files and included in reports submitted to the USACE and the USAF, as requested.

4.3 LANDFILL GAS MONITORING

To meet the intent of 35 IAC Section 811.310(c), landfill gas wells located outside the landfill boundary and passive gas vents within the landfill boundary will be monitored on a quarterly basis for a minimum of 5 years after closure. Five years after closure, the monitoring frequency may be reduced to a semiannual basis upon approval by the IEPA.

4.3.1 Landfill Gas Field Parameters

The following parameters will be measured at the landfill gas wells and passive gas vents:

- Methane,
- Hydrogen sulfide,
- Pressure,
- Oxygen, and
- Carbon dioxide.

4.3.2 Field Sampling Procedures

Landfill gas field measurements will be collected using a Landtec GEM2000 Plus landfill gas monitor, or equivalent, in accordance with the landfill gas monitoring SOP provided in Appendix C. To collect measurements, inert Tygon tubing will be connected to a sample connection port and the sampling device. Real-time measurements of percent methane, hydrogen sulfide, carbon dioxide, oxygen, static pressure, differential pressure, and the percent lower explosive limit (LEL) of methane will be recorded. The LELs for methane and hydrogen sulfide are 5 percent by volume and 4.4 percent by volume, respectively. If methane concentrations are detected outside the landfill boundary at concentrations above the 50 percent LEL (2.5 percent by volume) and are attributed to the landfill, the IEPA will be notified in writing, within 2 business days, of the observed exceedance.

4.3.3 Field Records

All field activities will be recorded in a field logbook. Entries will be made in real time, in permanent ink, and include sufficient detail to reconstruct field activities in an accurate manner. The landfill gas field measurement form is available in Appendix C to support this effort. These forms will be considered extensions of the field logbook and will be kept in the project file. Copies of logbooks and field forms will be maintained in the project files and included in reports submitted to the USACE and the USAF, as requested.

At a minimum, the following data will be recorded in the field logbook or on data log sheets for each measurement event:

- Name of person(s) taking the measurements;
- Date and time of measurements;
- Measurement location identification;
- Ambient temperature, barometric pressure, and weather conditions;
- Landfill gas meter make and model;
- Documentation of meter calibration;
- Methane, oxygen, carbon dioxide, and pressure measurements; and
- Duration of purging required to achieve stable readings.

4.4 AMBIENT AIR MONITORING

Per 35 IAC 811.310(b)(8), ambient air monitoring will be conducted around the perimeter of the landfill boundary, 100 feet downwind, to verify that the passive landfill gas collection system is functioning as designed. Real-time samples of percent methane in ambient air will be recorded at the same time landfill gas points are measured. Five years after closure, the monitoring frequency may be reduced from a quarterly basis to a semiannual basis, upon approval by the IEPA.

4.4.1 Ambient Air Monitoring Locations

Eight potential monitoring locations are depicted on Figure 2.2. During each sampling event, three downwind locations and at least one upwind location will be monitored. Whether a location is “downwind” or “upwind” will depend on wind direction. Wind direction and speed will be based on data obtained from the nearest climatic weather station. On the field sampling sheets, the locations will be documented as either “downwind” or “upwind” based on wind direction and speed. A GPS unit will be used to verify and document the location coordinates.

4.4.2 Field Sampling Procedures

Ambient air monitoring of methane concentrations near the ground surface will be conducted around the perimeter of the landfill boundary to verify that the landfill gas collection system is functioning as designed. The monitoring will be performed with a flame ionization detector, or equivalent, that is capable of measuring percent methane and percent LEL of methane. To meet the requirements of 35 IAC 811.310(b)(8), at least three monitoring locations will be chosen with ambient air samples collected no higher than 1 inch above the ground surface and 100 feet downwind from the edge of the landfill. The samples will be collected until measurements stabilize. Readings are considered stable when they vary by less than 0.5 percent by volume. The background concentration will be determined by analyzing the ambient conditions upwind and outside the landfill at a distance of at least 100 feet from the perimeter gas vents.

Per 35 IAC 811.310(d)(2), ambient air monitoring points will be sampled for methane only when the average wind velocity is less than 5 miles per hour at locations downwind of the landfill. If methane concentrations are detected above the 50 percent LEL (2.5 percent by volume) or 500 parts per million above background concentrations in the ambient air and are attributed to the landfill, the IEPA will be notified in writing, within 2 business days, of the observed exceedance.

4.4.3 Field Records

All field activities will be recorded in a field logbook. Entries will be made in real time, in permanent ink, and include sufficient detail to reconstruct field activities in an accurate manner. The ambient air measurement form is available in Appendix C to support this effort. These forms will be considered extensions of the field logbook and will be kept in the project file. Copies of logbooks and field forms will be maintained in the project files and included in reports submitted to the USACE and the USAF, as requested.

At a minimum, the following data will be recorded in the field logbook or on data log sheets for each measurement event:

- Name of person(s) taking the measurements;
- Date and time of measurements;
- Measurement location coordinates;
- Ambient temperature, barometric pressure, and weather conditions;
- Wind direction and speed;
- Multigas meter make and model;
- Documentation of meter calibration; and
- Methane concentration and percent LEL measurements.

4.5 IDW MANAGEMENT

IDW for this investigation will include the following:

- Soil generated during well installation activities;
- Purge waters generated during well development and groundwater sampling activities; and
- Used personal protective equipment (for example, nitrile/latex disposable gloves) and miscellaneous trash.

All containers of IDW will be properly labeled with the appropriate site name, work area designation (LF-01), contents, and date. IDW will be generated during well installation and sampling activities and will consist of soil and water. All IDW will be containerized, labeled, and staged at the designated Scott AFB 90-day IDW storage pad. A waste inventory tracking form will be completed to document IDW generated. Before disposal, a composite sample will be collected to characterize the contents of the IDW in accordance with the Basewide FSP (HGL, 2009a) and the QAPP (HGL, 2009b). A manifest will be completed for each load of material leaving the site. Scott AFB personnel will sign all manifests before the loads leave the site.

This page was intentionally left blank.

TABLE(S)

This page was intentionally left blank.

Location Identification	Monitoring Purpose	Quarterly Sampling Analyte List					Semiannual Sampling Analyte List - Natural Attenuation Parameters							
		VOCs (Method 8260C)	TCE, cis-1,2-DCE, and VC (Method 8260C)	Arsenic, Boron, Iron, Lead, Magnesium, and Zinc, Field Filtered and Unfiltered (Method 6010C)	Arsenic and Iron, Field Filtered and Unfiltered (Method 6010C)	Field Parameters [2]	Ferrous Iron (Method 3500 DE-D)	Total Dissolved Solids (SM2540C)	Sulfate, Nitrate, and Chloride (Method SM300)	Sulfides (Method 376.1)	Ammonia (Method SM350.2)	Total Organic Carbon (Method SW9060A)	Alkalinity (Method SM310.1)	Dissolved Gases (Methane, Ethene, Ethane) (Method RSK-175)
		QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY
Groundwater Monitoring Well Locations [1]														
LF01-MW01D	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW01S	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW03D	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW03S	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW11	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW12D	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW12SR	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW17D	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW17SR	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW18D	Sentinel Well	--	--	--	1	1	--	--	--	--	--	--	--	--
LF01-MW18S	Sentinel Well	--	--	--	1	1	--	--	--	--	--	--	--	--
LF01-MW19D	Sentinel Well	--	--	--	1	1	--	--	--	--	--	--	--	--
LF01-MW19S	Sentinel Well	--	--	--	1	1	--	--	--	--	--	--	--	--
LF01-MW20D	Sentinel Well	--	--	--	1	1	--	--	--	--	--	--	--	--
LF01-MW20S	Groundwater COC Compliance	--	--	--	1	1	1	1	1	1	1	1	1	1
LF01-MW21D	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW21S	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW23D	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW23S	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW24D	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW24S	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW25D	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW25S	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW26D	Landfill Performance / Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW26S	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1
LF01-MW27	Sentinel Well	--	1	--	1	1	--	--	--	--	--	--	--	--
LF01-MW28	Groundwater COC Compliance	--	1	--	1	1	1	1	1	1	1	1	1	1

Location Identification	Monitoring Purpose	Quarterly Sampling Analyte List					Semiannual Sampling Analyte List - Natural Attenuation Parameters							
		VOCs (Method 8260C)	TCE, cis-1,2-DCE, and VC (Method 8260C)	Arsenic, Boron, Iron, Lead, Magnesium, and Zinc, Field Filtered and Unfiltered (Method 6010C)	Arsenic and Iron, Field Filtered and Unfiltered (Method 6010C)	Field Parameters [2]	Ferrous Iron (Method 3500 DE-D)	Total Dissolved Solids (SM2540C)	Sulfate, Nitrate, and Chloride (Method SM300)	Sulfides (Method 376.1)	Ammonia (Method SM350.2)	Total Organic Carbon (Method SW9060A)	Alkalinity (Method SM310.1)	Dissolved Gases (Methane, Ethene, Ethane) (Method RSK-175)
		QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY
LF01-MW29	Sentinel Well	--	1	--	1	1	--	--	--	--	--	--	--	--
LF01-MW30D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW30S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW31D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW31S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW32D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW32S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW33D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW33S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW34D*	Upgradient Well	1	--	1	--	1	--	--	--	--	--	--	--	--
LF-01-MW34S*	Upgradient Well	1	--	1	--	1	--	--	--	--	--	--	--	--
LF-01-MW35D*	Upgradient Well	1	--	1	--	1	--	--	--	--	--	--	--	--
LF-01-MW35S*	Upgradient Well	1	--	1	--	1	--	--	--	--	--	--	--	--
LF-01-MW36D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW36S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW37D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW37S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW38D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF-01-MW38S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW39D*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
LF01-MW39S*	Landfill Performance / Groundwater COC Compliance	1	--	1	--	1	1	1	1	1	1	1	1	1
Sum of Groundwater Samples		30	12	30	18	48	37	37	37	37	37	37	37	37
Duplicates		3	1	3	2	0	4	4	4	4	4	4	4	4
Matrix Spike		2	1	2	1	0	2	2	2	2	2	2	2	2
Matrix Spike Duplicate		2	1	2	1	0	2	2	2	2	2	2	2	2
Trip Blank		1 per cooler of VOCs	1 per cooler of VOCs	0	0	0	0	0	0	0	0	0	0	0
Grand Total of Groundwater Samples [3]		**37**	**15**	**36**	**22**	**48**	**44**	**44**	**44**	**44**	**44**	**44**	**44**	**44**

Location Identification	Monitoring Purpose	Quarterly Sampling Analyte List					Semiannual Sampling Analyte List - Natural Attenuation Parameters							
		VOCs (Method 8260C)	TCE, cis-1,2-DCE, and VC (Method 8260C)	Arsenic, Boron, Iron, Lead, Magnesium, and Zinc, Field Filtered and Unfiltered (Method 6010C)	Arsenic and Iron, Field Filtered and Unfiltered (Method 6010C)	Field Parameters (2)	Ferrous Iron (Method 3500 DE-D)	Total Dissolved Solids (SM2540C)	Sulfate, Nitrate, and Chloride (Method SM300)	Sulfides (Method 376.1)	Ammonia (Method SM350.2)	Total Organic Carbon (Method SW9060A)	Alkalinity (Method SM310.1)	Dissolved Gases (Methane, Ethene, Ethane) (Method RSK-175)
		QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY	QTY
Surface Water Sampling Locations (1)														
LF01-SW01*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
LF01-SW02*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
LF01-SW03*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
LF01-SW04*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
LF01-SW05*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
LF01-SW06*	Surface Water Compliance	1	0	1	0	1	0	1	0	0	1	0	0	0
	Sum of Surface WaterSamples	6	0	6	0	6	0	6	0	0	6	0	0	0
	Duplicates	1	0	1	0	0	0	1	0	0	1	0	0	0
	Matrix Spike	1	0	1	0	0	0	1	0	0	1	0	0	0
	Matrix Spike Duplicate	1	0	1	0	0	0	1	0	0	1	0	0	0
	Trip Blank	1 per cooler of VOCs	0	0	0	0	0	0	0	0	0	0	0	0
	Grand Total of Surface Water Samples (3)	**10**	**0**	**9**	**0**	**6**	**0**	**9**	**0**	**0**	**9**	**0**	**0**	**0**

Notes:
*Proposed monitoring well or surface water location.
(1) Monitoring well and surface water sampling locations are shown on Figure 2.1.
(2) Field parameters include depth to water, pH, temperature, speicific conductance, oxidation reduction potential, dissolved oxygen, and turbidity.
(3) One equipment blank will be collected each day sampling is conducted per sampling team.
COC = contaminant of concern
S = shallow well
D = deep well
SR = shallow replacement well
DCE = Dichloroethene
TAL = target analyte list
TCE = Trichloroethene
QTY = quantity
VOC = volatile organic compound

5.0 RECORD-KEEPING AND REPORTING REQUIREMENTS

This chapter provides the reporting requirements associated with the postclosure O&M program activities described in Sections 3.0 and 4.0.

5.1 ANNUAL REPORT

Annual reports will be prepared presenting the results of site inspections, environmental monitoring (groundwater, surface water, landfill gas, and ambient air), and LUC inspections. Table 3.1 provides the schedule of events for LF-01. The schedule may be adjusted to incorporate other Scott AFB sites into a comprehensive O&M schedule, including inspection, monitoring, and reporting. If a professional is consulted or soil samples are analyzed as a result of soil or cover vegetation concerns, the recommendations reports, along with a map of each site area showing sample locations and sample identification, will be included as appendices to the next annual report.

Copies of the annual report will be provided to the IEPA. The annual report will include the following items:

- Completed Postclosure Site Inspection Checklists regarding site conditions (see Section 3.0 and Table 3.2 for details):
  - General site conditions;
  - Landfill protective cover;
  - Cover settlement inspection and survey results;
  - Vegetative cover;
  - Landfill network monitoring wells;
  - Landfill passive gas vents and gas wells; and
  - LUCs (landfill signs, restrictions, and ground disturbance).
- Narrative describing any maintenance and/or repairs performed during the associated reporting period;
- Narrative on items of concern, ongoing maintenance, recommended repairs, as appropriate, and a schedule for performing these repairs, if required;
- A map showing the locations of deficiencies, problems, and repair items;
- Cover settlement survey evaluation (performed every 2 years):
  - Date of surveying event;
  - Tabulated elevations of each control monument; and
  - Discussions of changes in elevation observed.
- Environmental monitoring results:
  - Date of sampling events;
  - Tabular presentation of groundwater monitoring results, complete with validated analytical data flags, where appropriate;

- Statistical data and trend analysis (using one or more of the normal theory statistical tests) following eight consecutive sampling events to establish whether a statistically significant change occurred;
- Tabular presentation of groundwater level measurements (collected every sampling event), percent occlusion, wells requiring redevelopment, and well boring depths (collected on a yearly basis);
- Updated potentiometric surface maps;
- Tabular presentation of surface water results, complete with validated analytical data flags, where appropriate;
- Tabular presentation of landfill gas monitoring results;
- Tabular presentation of ambient air monitoring results;
- Narrative on whether there has been any significant changes in concentrations, groundwater quality data, or flow direction of COCs from prior periods. A detailed trend analysis of the data will be presented in the annual reports (after sufficient data has been collected) and at the five year review;
- Discussion of problems associated with sampling and analysis, if applicable; and
- Field sheets, chain of custody forms, calibration documents, analytical results, and photographs.

Any activity that is inconsistent with the LUC objectives, or any other action that may interfere with the effectiveness of the LUCs, will be documented in the annual report and provided to the IEPA. In addition, the following will be observed:

- The inconsistent activities will be addressed by Scott AFB as soon as practicable, but in no case will the process be initiated later than 30 calendar days after Scott AFB confirms the inconsistency has occurred.
- Scott AFB will inform the IEPA regarding how Scott AFB has addressed, or will address, the LUC inconsistency within 30 calendar days of confirming a LUC inconsistency has occurred. The information will include a summary, description, and discussion of any problems or deficiencies in the LUCs or other controls and measures that may have caused or allowed the inconsistent activity, and any corrective measures taken or planned.
- Scott AFB will notify the IEPA 60 days in advance of any proposed land use changes that are inconsistent with LUC objectives and the selected remedy.
- Scott AFB will provide notice to the IEPA at least 6 months before any transfer or sale of LF-01 so that the IEPA can be involved in discussions to ensure that appropriate provisions are included in the transfer terms or conveyance documents to maintain effective LUCs. If it is not possible for the facility to notify the IEPA at least 6 months before any transfer or sale, then the facility will notify the IEPA as soon as possible, but no later than 60 days before the transfer or sale of any property subject to LUCs. In addition to the land transfer notice and discussion provisions above, Scott AFB further agrees to provide the IEPA with similar notice, within the same time frames, as

to federal-to-federal transfers of property. Scott AFB will provide a copy of executed deed or transfer assembly to the IEPA.

- Scott AFB will not modify or terminate LUCs, modify implementation actions identified in the ROD, or modify land use without first seeking concurrence from the IEPA on any anticipated action that may disrupt the effectiveness of the LUCs or any action that may alter or negate the need for LUCs. Within 60 days of approved changes, Scott AFB will update the Base General Plan and notify the IEPA.

5.2 FIVE YEAR REVIEWS

A five year review will be required at LF-01 because hazardous substances, pollutants, or contaminants remain beneath the landfill cover and within groundwater at levels requiring restrictive measures. In accordance with the Defense Environmental Restoration Program Management Manual (DoD, 2012) the first five year review will be completed no later than 5 years after the initiation of the remedial action (November 2011). Therefore, the first five year review will occur no later than November 2016.

The purpose of the five year review will be to determine whether the remedy continues to be protective of human health and the environment. The methods, findings, and conclusions of the review will be documented in a five year review report. In addition, the report will identify deficiencies found during the review, if any, and make recommendations to address them. The five year review will be consistent with the CERCLA and the NCP. CERCLA §121(c), as amended, states:

> *If the President selects a remedial action that results in any hazardous substances, pollutants, or contaminants remaining at the site, the President will review such remedial action no less often than each five years after the initiation of such remedial action to assure that human health and the environment are being protected by the remedial action being implemented.*

The NCP § 300.430(f)(4)(ii) in Title 40 of the Code of Federal Regulations states:

> *If a remedial action is selected that results in hazardous substances, pollutants, or contaminants remaining at the site above levels that allow for unlimited use and unrestricted exposure, the lead agency will review such action no less often than every five years after the initiation of the selected remedial action.*

The five year review will assess whether all RAOs contained in the ROD addressing landfill waste, soil, and sediment (Scott AFB, 2013a) and the ROD addressing groundwater (Scott AFB, 2013b) that are applicable to LF-01 continue to be met. This information will be compared to subsequent information collected since implementation of the RA. Site inspections and related interviews conducted after completion of the RA will also be presented to support the findings of the review process.

This page was intentionally left blank.

6.0 REFERENCES

Booz Allen Hamilton, 2006. Final Basewide Quality Assurance Project Plan, Revision 1, Scott Air Force Base (AFB). June.

Environmental Resource Management (ERM), 1992. Installation Program Stage I Remedial Investigation/Feasibility Study, Scott AFB, Illinois. April.

HydroGeoLogic, Inc. (HGL), 2009a. Final Basewide Field Sampling Plan, Scott AFB, Illinois. October.

HGL, 2009b. Final Quality Assurance Project Plan Addendum, Scott AFB, Illinois. June.

HGL, 2009c. Final Site Safety and Health Plan, Scott AFB, Illinois. March.

HGL, 2012a. Final Basewide Groundwater Dataset Evaluation Technical Memorandum, Scott AFB, Illinois. January.

HGL, 2012b. Final Design Analysis Report for Scott AFB LF-01, Scott Air Force Base, Illinois. June.

HGL, 2012c. Final Remedial Action Work Plan, Landfill LF-01, Scott Air Force Base, Illinois. June.

HGL, 2012d. Final Feasibility Study Report Addendum for Groundwater at Landfill LF-01, Scott Air Force Base, Illinois. December.

HGL, 2013a. Proposed Long Term Monitoring Well Locations at LF-01 Technical Memorandum, Scott AFB, Illinois. February.

HGL, 2013b. LTM Well Installation at LF-01 Technical Memorandum, Scott Air Force Base, Illinois. April.

Illinois Environmental Protection Agency (IEPA), 1997. Title 35 of the IAC, Subtitle G, Part 811, Standards for New Solid Waste Landfills.

Montgomery Watson Harza Americas, Inc. (MWH), 2003. Final Remedial Investigation/Feasibility Study for LF-01 and FT03, Scott Air Force Base, Illinois. April.

Scott AFB, 2010. Website of Scott Air Force Base at URL http://www.scott.af.mil/.

Scott AFB, 2013a. Scott Air Force Base, Final Record of Decision, Site LF-01, Landfill LF-01 (Addressing landfill waste, surface soil, and sediment), Scott AFB, Illinois. January.

Scott AFB, 2013b. Draft Record of Decision (addressing groundwater) at Site LF-01, Scott AFB, Illinois. January.

U.S. Department of Defense (DoD), 2012. Defense Environmental Restoration Program (DERP) Management Manual, Number 4715.20, March 9, 2012.

U.S. Environmental Protection Agency (USEPA), 2006. Revegetating Landfills and Waste Containment Areas. EPA 542-F-06-001. October.

APPENDIX A

AS-BUILT DRAWINGS FOR LANDFILL COVER

This page was intentionally left blank.

APPENDIX A

AS-BUILT DRAWINGS FOR LANDFILL COVER

This page was intentionally left blank.

WILLIAM CHARLES CONSTRUCTION COMPANY LLC

5290 NIMTZ ROAD
LOVES PARK, IL 61111
815.654.4700 FAX 815.654.4736

TO: HydroGeoLogic, Inc. 105 Executive Parkway, Suite 200 Hudson, OH 44236	**DATE: DECEMBER 6, 2012** | **JOB NO. 9110253** **ATTN: JEFF NEUMANN** **RE: SCOTT AIR FORCE BASE** **SUBMITTAL NO. 0230 R2**

COPIES	DATE	NO.	DESCRIPTION
1	12/06/12	230 R2	Final As-Built Survey – LF-01 South Cell

THESE ARE TRANSMITTED as checked below:

☒ For approval ☐ Approved as submitted ☒ Resubmit 1 copies for approval

☐ For your use ☐ Approved as noted ☐ Submit ______ copies for distribution

☐ As requested ☐ Returned for corrections ☐ Return ______ corrected prints

☐ For review and comment ☐ ______________________

☐ FOR BIDS DUE ______________________ ☐ PRINTS RETURNED AFTER LOAN TO US

REMARKS:

HGL HydroGeoLogic, Inc

Date Received: 12/6/12

X Approved

______ Approved with corrections as noted on the submittal data and/or attached sheet(s)

SIGNED: [signature]
Cody Stambaugh

______ Declined

Signature: [signature]

Title: Sr Construction Manager

Date: 12/6/2012

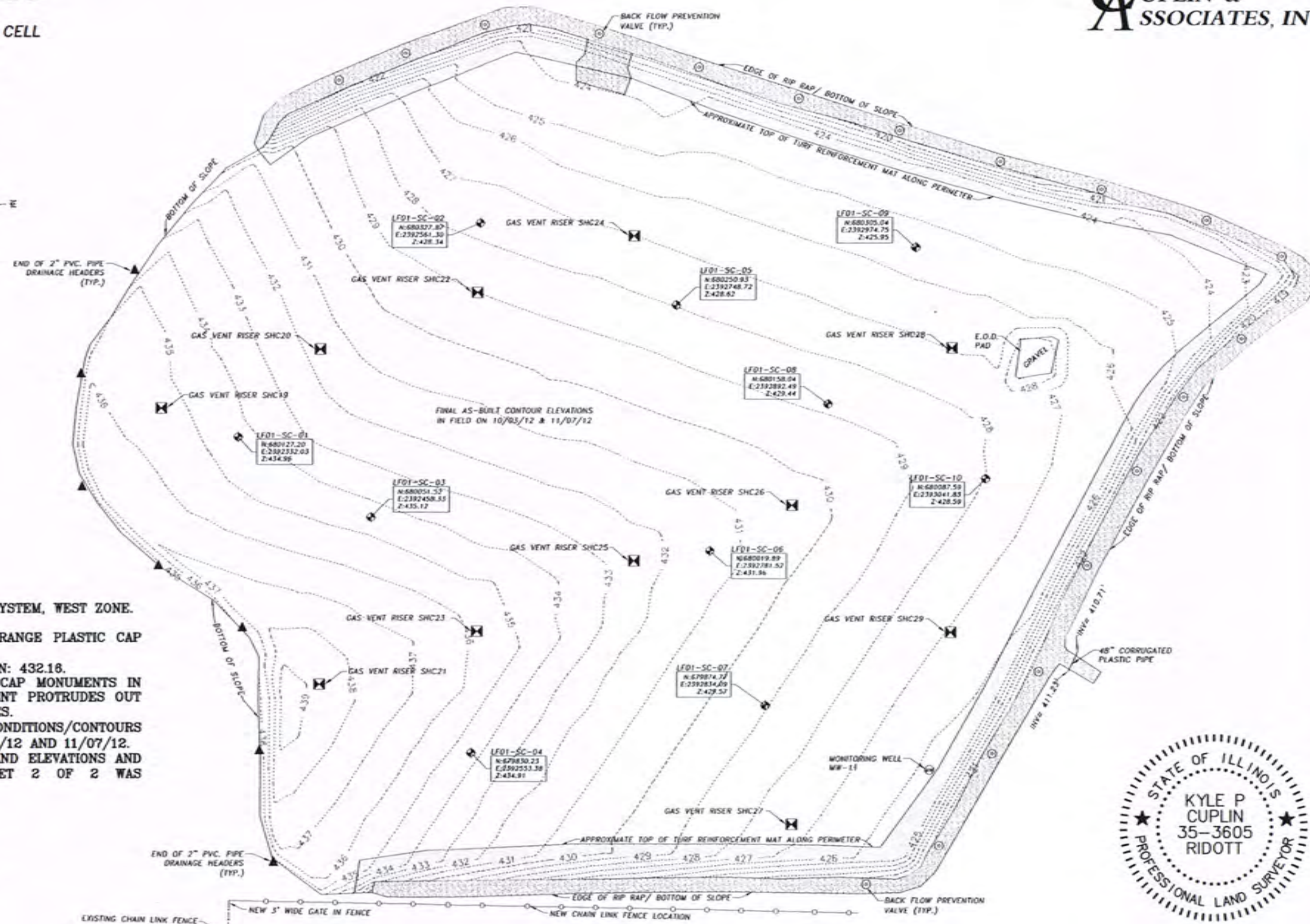
TOPOGRAPHIC SURVEY
OF
SCOTT AIR FORCE BASE LANDFILL SOUTH CELL
FINAL AS-BUILT CONDITIONS
& UNDERGROUND GAS PIPE
LOCATION AND ELEVATION

CUPLIN & ASSOCIATES, INC.

LOCATION MAP
NOT TO SCALE

LEGEND

- BACK FLOW PREVENTION VALVE
- PIPE SEAM
- END OF 2" PVC PIPE
- GAS VENT RISER
- BRASS CAP IN PVC PIPE
- MONITORING WELL
- END OF PIPE AND "TEES" OF PIPE
- CHAIN LINK FENCE
- × 433.87 TOP OF PIPE ELEVATION



EXISTING CONDITIONS IN FIELD
LOCATED ON OCTOBER 3, 2012
& NOVEMBER 7, 2012

SURVEYOR'S NOTES:
1. HORIZONTAL DATUM IS THE ILLINOIS COORDINATE SYSTEM, WEST ZONE.
2. VERTICAL DATUM IS THE NAVD 1988.
3. CONTROL UTILIZED IS A 5/8" IRON PIN WITH ORANGE PLASTIC CAP STAMPED "ABNA TRAV PNT"
NORTHING: 680392.21, EASTING: 2391833.82, ELEVATION: 432.16.
4. HORIZONTAL AND VERTICAL LOCATION OF BRASS CAP MONUMENTS IN PVC PIPE ARE TO THE TOP CENTER. EACH MONUMENT PROTRUDES OUT OF THE GROUND LEVEL APPROXIMATELY +/- 8 INCHES.
5. FIELDWORK FOR FINAL AS-BUILT EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 1 OF 2 WAS PERFORMED ON 10/03/12 AND 11/07/12.
6. FIELDWORK FOR UNDERGROUND PIPE LOCATIONS AND ELEVATIONS AND EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 2 OF 2 WAS PERFORMED ON 5/24/12.

I HEREBY CERTIFY THAT THIS SURVEY WAS PERFORMED ON THE GROUND AND WAS SURVEYED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT THIS PROFESSIONAL SERVICE CONFORMS TO THE CURRENT MINIMUM STANDARDS FOR A TOPOGRAPHIC SURVEY AS ESTABLISHED BY THE ILLINOIS BOARD OF PROFESSIONAL LAND SURVEYING.

Dated this 21ST day of NOVEMBER, 2012.

Kyle P. Cuplin

Kyle P. Cuplin
Illinois Professional Land Surveyor No. 35-3605
Cuplin & Associates, Inc.
742 North River Road
Ridott, IL 61067
(815) 990-8773

STATE OF ILLINOIS
KYLE P CUPLIN
35-3605
RIDOTT
PROFESSIONAL LAND SURVEYOR

EXPIRES 11/30/12

PROFESSIONAL DESIGN FIRM NO. 184.005197

SHEET 1 OF 2

PROJ NO. IL-11030
PREPARED FOR: WILLIAM CHARLES CONSTRUCTION CO.
TECH: C.CUPLIN
APPROVED: K.CUPLIN
FIELDWORK PERFORMED ON: 10/03/12 & 11/07/12

CUPLIN & ASSOCIATES, INC.
742 N. RIVER ROAD
RIDOTT, ILLINOIS 61067
PHONE:815-990-8773
FAX:815-232-4049

SCALE 1" = 100'
0 50 100

DATE	NO.	DESCRIPTION
	2	
	1	

REVISIONS

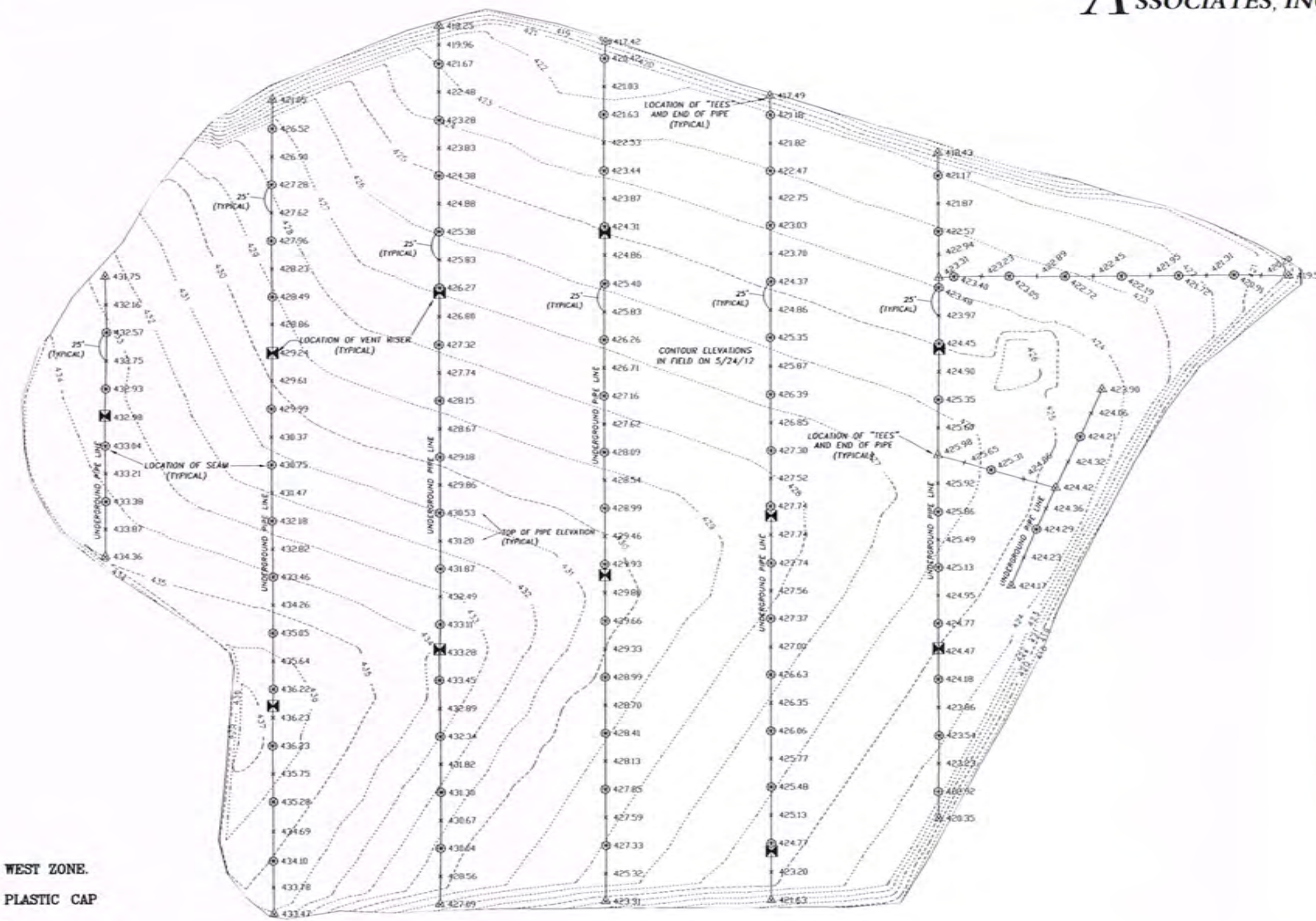
TOPOGRAPHIC SURVEY
OF
SCOTT AIR FORCE BASE LANDFILL SOUTH CELL
FINAL AS-BUILT CONDITIONS
& UNDERGROUND GAS PIPE
LOCATION AND ELEVATION

CUPLIN & ASSOCIATES, INC.

LOCATION MAP
NOT TO SCALE

LEGEND

- BACK FLOW PREVENTION VALVE
- PIPE SEAM
- END OF 2" PVC PIPE
- GAS VENT RISER
- BRASS CAP IN PVC PIPE
- MONITORING WELL
- END OF PIPE AND "TEES" OF PIPE
- CHAIN LINK FENCE
- x 433.87 TOP OF PIPE ELEVATION



EXISTING CONDITIONS IN FIELD
LOCATED ON MAY 24, 2012

SURVEYOR'S NOTES:
1. HORIZONTAL DATUM IS THE ILLINOIS COORDINATE SYSTEM, WEST ZONE.
2. VERTICAL DATUM IS THE NAVD 1988.
3. CONTROL UTILIZED IS A 5/8" IRON PIN WITH ORANGE PLASTIC CAP STAMPED "ABNA TRAV PNT"
NORTHING: 680392.21, EASTING: 2391833.82, ELEVATION: 432.16.
4. HORIZONTAL AND VERTICAL LOCATION OF BRASS CAP MONUMENTS IN PVC PIPE ARE TO THE TOP CENTER. EACH MONUMENT PROTRUDES OUT OF THE GROUND LEVEL APPROXIMATELY +/- 8 INCHES.
5. FIELDWORK FOR FINAL AS-BUILT EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 1 OF 2 WAS PERFORMED ON 10/03/12 AND 11/07/12.
6. FIELDWORK FOR UNDERGROUND PIPE LOCATIONS AND ELEVATIONS AND EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 2 OF 2 WAS PERFORMED ON 5/24/12.

PROFESSIONAL DESIGN FIRM NO. 184.005197

SHEET 2 OF 2

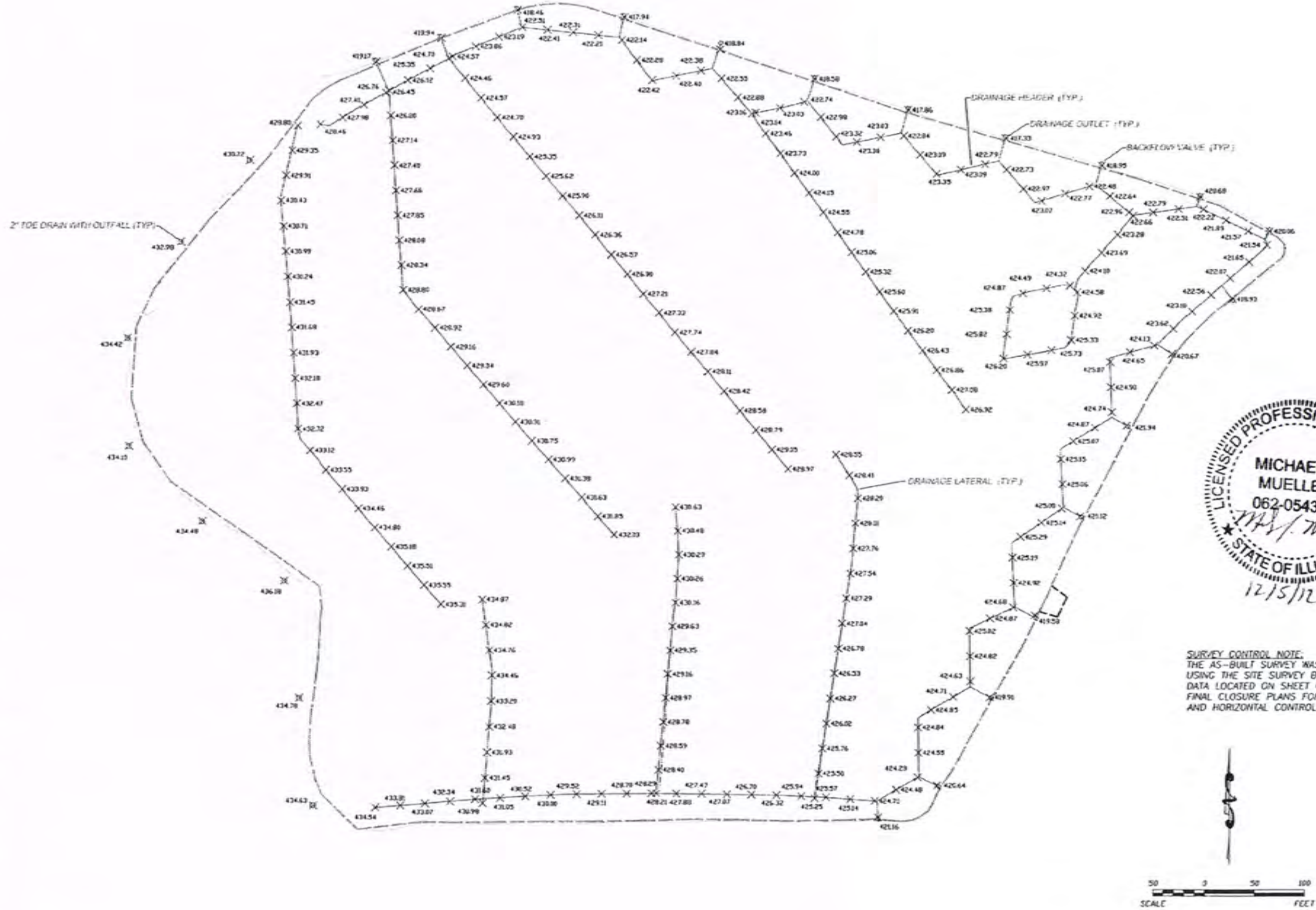
PROJ NO. IL-11030
PREPARED FOR: WILLIAM CHARLES CONSTRUCTION CO.
TECH: C.CUPLIN
APPROVED: K.CUPLIN
FIELDWORK PERFORMED ON: 5/24/12

CUPLIN & ASSOCIATES, INC.
742 N. RIVER ROAD
RIDOTT, ILLINOIS 61067
PHONE:815-990-8773
FAX:815-232-4049

SCALE 1" = 100'
0 50 100

DATE	NO.	DESCRIPTION
	2	
	1	

REVISIONS



SURVEY CONTROL NOTE:
THE AS-BUILT SURVEY WAS PERFORMED USING THE SITE SURVEY BENCHMARKS DATA LOCATED ON SHEET C-1 OF THE FINAL CLOSURE PLANS FOR THE VERTICAL AND HORIZONTAL CONTROL.

LICENSED PROFESSIONAL ENGINEER
MICHAEL J. MUELLER
062-054318
STATE OF ILLINOIS
12/5/12

AQUATERRA
ENVIRONMENTAL SOLUTIONS, INC.
13 Executive Drive, Ste. 1
Fairview Heights, Illinois 62208

WILLIAM CHARLES CONSTRUCTION COMPANY
SCOTT AIR FORCE BASE
BELLEVILLE, IL

SCOTT AIR FORCE BASE LANDFILL CLOSURE PROJECT
SOUTH CELL DRAINAGE
AS-BUILT

DATE 10/8/12
PROJECT NUMBER 5403.10
SHT. 1 OF 1

WILLIAM CHARLES CONSTRUCTION COMPANY LLC

5290 NIMTZ ROAD
LOVES PARK, IL 61111
815.654.4700 FAX 815.654.4736

TO: HydroGeoLogic, Inc.
105 Executive Parkway, Suite 200
Hudson, OH 44236

DATE:DECEMBER 6, 2012	JOB NO. 9110253
ATTN: JEFF NEUMANN	
RE: SCOTT AIR FORCE BASE	
SUBMITTAL NO. 0241	

COPIES	DATE	NO.	DESCRIPTION
1	12/06/12	0230	Final As-Built Survey – LF-01 North Cell

THESE ARE TRANSMITTED as checked below:

☒ For approval ☐ Approved as submitted ☐ Resubmit ______ copies for approval

☐ For your use ☐ Approved as noted ☐ Submit ______ copies for distribution

☐ As requested ☐ Returned for corrections ☐ Return ______ corrected prints

☐ For review and comment ☐ ________________________________

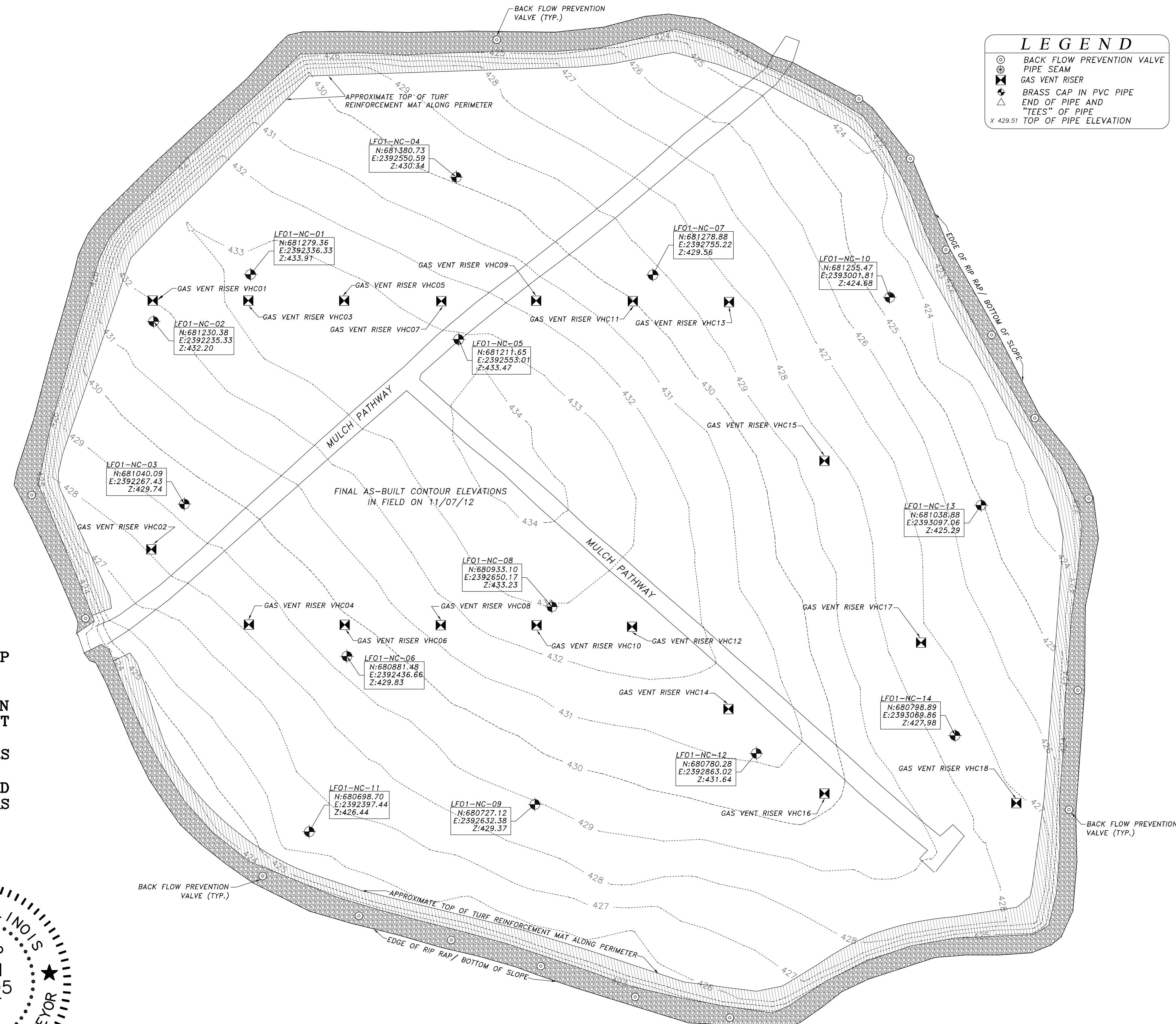
☐ FOR BIDS DUE________________________ ☐ PRINTS RETURNED AFTER LOAN TO US

REMARKS:

SIGNED: ______________________
Cody Stambaugh

TOPOGRAPHIC SURVEY
OF
SCOTT AIR FORCE BASE LANDFILL NORTH CELL
FINAL AS-BUILT CONDITIONS
& UNDERGROUND GAS PIPE
LOCATION AND ELEVATION

LOCATION MAP
NOT TO SCALE



EXISTING CONDITIONS IN FIELD
LOCATED ON JULY 12, 2012
& NOVEMBER 7, 2012

SURVEYOR'S NOTES:
1. HORIZONTAL DATUM IS THE ILLINOIS COORDINATE SYSTEM, WEST ZONE.
2. VERTICAL DATUM IS THE NAVD 1988.
3. CONTROL UTILIZED IS A 5/8" IRON PIN WITH ORANGE PLASTIC CAP STAMPED "ABNA TRAV PNT"
NORTHING: 680392.21, EASTING: 2391833.82, ELEVATION: 432.16.
4. HORIZONTAL AND VERTICAL LOCATION OF BRASS CAP MONUMENTS IN PVC PIPE ARE TO THE TOP CENTER. EACH MONUMENT PROTRUDES OUT OF THE GROUND LEVEL APPROXIMATELY +/- 8 INCHES.
5. FIELDWORK FOR FINAL AS-BUILT EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 1 OF 2 WAS PERFORMED ON 11/07/12.
6. FIELDWORK FOR UNDERGROUND PIPE LOCATIONS AND ELEVATIONS AND EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 2 OF 2 WAS PERFORMED ON 7/12/12.

I HEREBY CERTIFY THAT THIS SURVEY WAS PERFORMED ON THE GROUND AND WAS SURVEYED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT THIS PROFESSIONAL SERVICE CONFORMS TO THE CURRENT MINIMUM STANDARDS FOR A TOPOGRAPHIC SURVEY AS ESTABLISHED BY THE ILLINOIS BOARD OF PROFESSIONAL LAND SURVEYING.

Dated this 18TH day of JULY, 2012.

Kyle P. Cuplin

Kyle P. Cuplin
Illinois Professional Land Surveyor No. 35-3605
Cuplin & Associates, Inc.
742 North River Road
Ridott, IL 61067
(815) 990-8773

STATE OF ILLINOIS ★ KYLE P CUPLIN 35-3605 RIDOTT ★ PROFESSIONAL LAND SURVEYOR

EXPIRES 11/30/12

CUPLIN & ASSOCIATES, INC.

DATE	NO.	DESCRIPTION
	1	
	2	

REVISIONS

SCALE 1" = 120'
0 60 120

CUPLIN & ASSOCIATES, INC.
742 N. RIVER ROAD
RIDOTT, ILLINOIS 61067
PHONE:815-990-8773
FAX:815-232-4049

PROJ NO. IL-11030
PREPARED FOR: WILLIAM CHARLES CONSTRUCTION CO.
TECH: C.CUPLIN
APPROVED: K.CUPLIN
FIELDWORK PERFORMED ON: 7/12/12 & 11/07/12

SHEET 1 OF 2

PROFESSIONAL DESIGN FIRM NO. 184.005197

TOPOGRAPHIC SURVEY
OF
SCOTT AIR FORCE BASE LANDFILL NORTH CELL
FINAL AS-BUILT CONDITIONS
& UNDERGROUND GAS PIPE
LOCATION AND ELEVATION

LOCATION MAP
NOT TO SCALE

EXISTING CONDITIONS IN FIELD LOCATED ON JULY 12, 2012 & NOVEMBER 7, 2012

SURVEYOR'S NOTES:
1. HORIZONTAL DATUM IS THE ILLINOIS COORDINATE SYSTEM, WEST ZONE.
2. VERTICAL DATUM IS THE NAVD 1988.
3. CONTROL UTILIZED IS A 5/8" IRON PIN WITH ORANGE PLASTIC CAP STAMPED "ABNA TRAV PNT"
NORTHING: 680392.21, EASTING: 2391833.82, ELEVATION: 432.16.
4. HORIZONTAL AND VERTICAL LOCATION OF BRASS CAP MONUMENTS IN PVC PIPE ARE TO THE TOP CENTER. EACH MONUMENT PROTRUDES OUT OF THE GROUND LEVEL APPROXIMATELY +/- 8 INCHES.
5. FIELDWORK FOR FINAL AS-BUILT EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 1 OF 2 WAS PERFORMED ON 11/07/12.
6. FIELDWORK FOR UNDERGROUND PIPE LOCATIONS AND ELEVATIONS AND EXISTING CONDITIONS/CONTOURS SHOWN ON SHEET 2 OF 2 WAS PERFORMED ON 7/12/12.

PROFESSIONAL DESIGN FIRM NO. 184.005197

CUPLIN & ASSOCIATES, INC.

CUPLIN & ASSOCIATES, INC.
742 N. RIVER ROAD
RIDOTT, ILLINOIS 61067
PHONE:815-990-8773
FAX:815-232-4049

REVISIONS

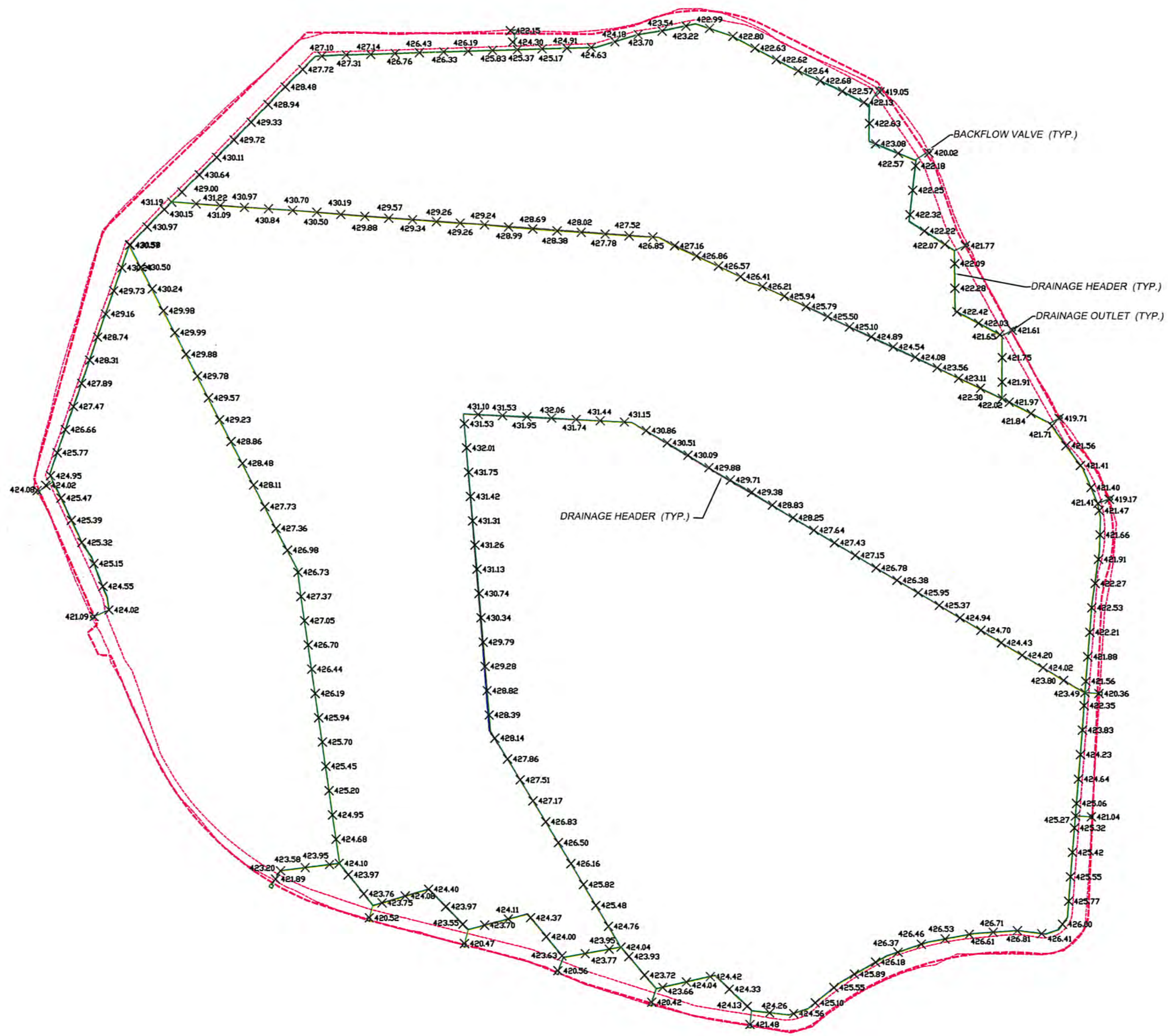
NO.	DATE	DESCRIPTION
1		
2		

SCALE 1" = 120'

0 60 120

PROJ NO. IL-11030
PREPARED FOR: WILLIAM CHARLES CONSTRUCTION CO.
TECH: C.CUPLIN
APPROVED: K.CUPLIN
FIELDWORK PERFORMED ON: 7/12/12 & 11/07/12

SHEET 2 OF 2



LICENSED PROFESSIONAL ENGINEER
MICHAEL J. MUELLER
062-054318
STATE OF ILLINOIS
12/5/12

SURVEY CONTROL NOTE:
THE AS-BUILT SURVEY WAS PERFORMED USING THE SITE SURVEY BENCHMARKS DATA LOCATED ON SHEET C-1 OF THE FINAL CLOSURE PLANS FOR THE VERTICAL AND HORIZONTAL CONTROL.

60 0 60 120
SCALE FEET

REV.	DATE	BY	DESCRIPTION
R6	-	6BY	-
R5	-	5BY	-
R4	-	4BY	-
R3	-	3BY	-
R2	-	2BY	-
R1	-	1BY	-

AQUATERRA
ENVIRONMENTAL SOLUTIONS, INC.
13 Executive Drive, Ste. 1
Fairview Heights, Illinois 62208

CLIENT: WILLIAM CHARLES CONSTRUCTION COMPANY
SCOTT AIR FORCE BASE
BELLEVILLE, IL

DRAWN BY: JRA DESIGNED BY: JRA PROJECT MGR: MJM
ELECTRONIC FILE NAME: SAFB AS-BUILT_AQ STANDARDS - NORTH 2F.DWG DATE: 10/8/12

SCOTT AIR FORCE BASE LANDFILL CLOSURE PROJECT
NORTH CELL DRAINAGE
AS-BUILT

REV. 0 DRAWING NUMBER: 1 PROJECT NUMBER: 5423.10 SHT. 1 OF 1

K:\WILLIAM CHARLES CONSTRUCTION\5423.10 SAFB LF-01 CQA\AS-BUILT DRAWINGS\NORTH CELL\SAFB AS-BUILT_AQ STANDARDS - NORTH 2F.DWG

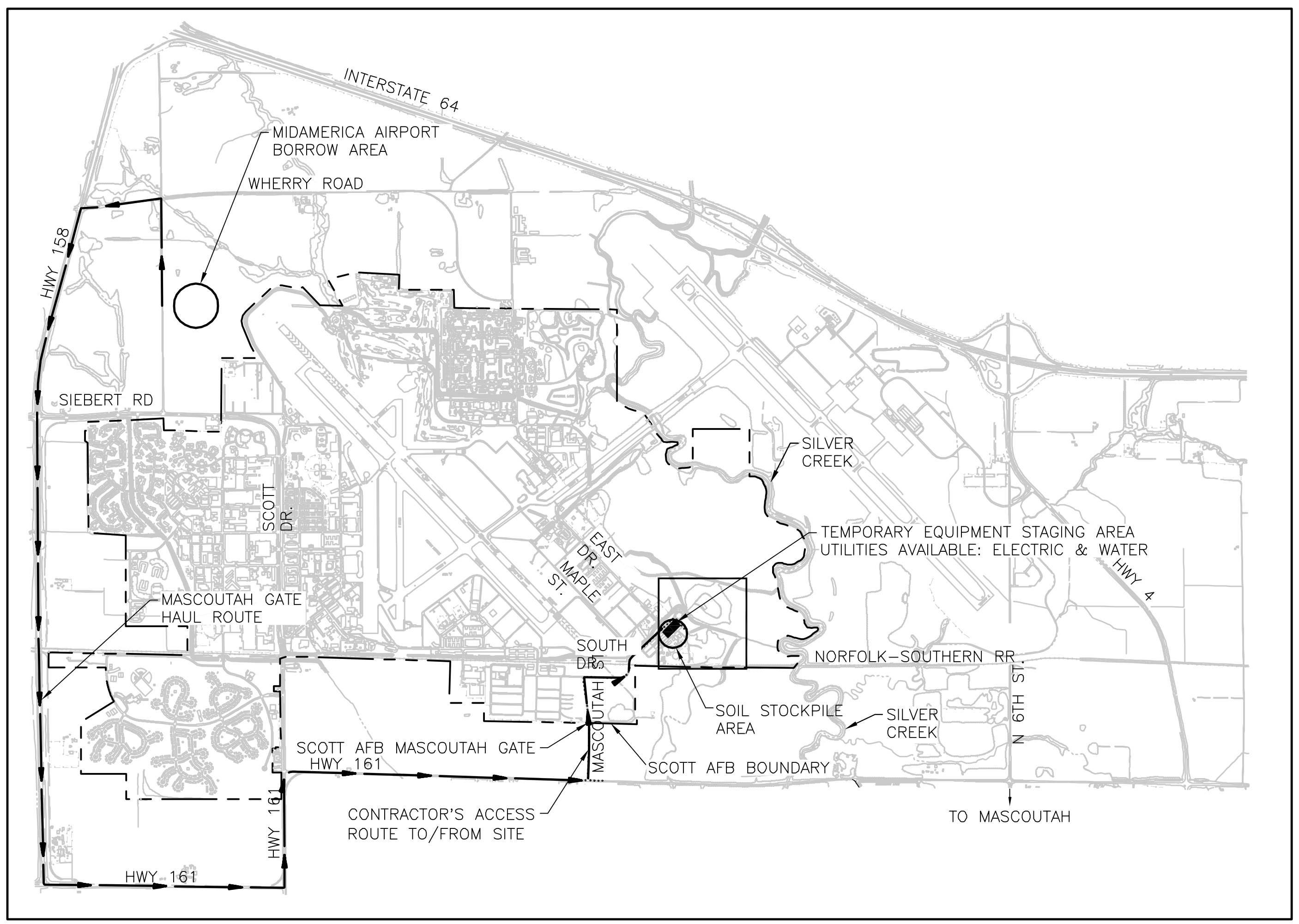
SCOTT AIR FORCE BASE LANDFILL
FINAL CLOSURE

SCOTT AFB, ST. CLAIR COUNTY, ILLINOIS

SITE- SCOTT AIR FORCE BASE

SCALE: NTS

VICINITY MAP
NORTH



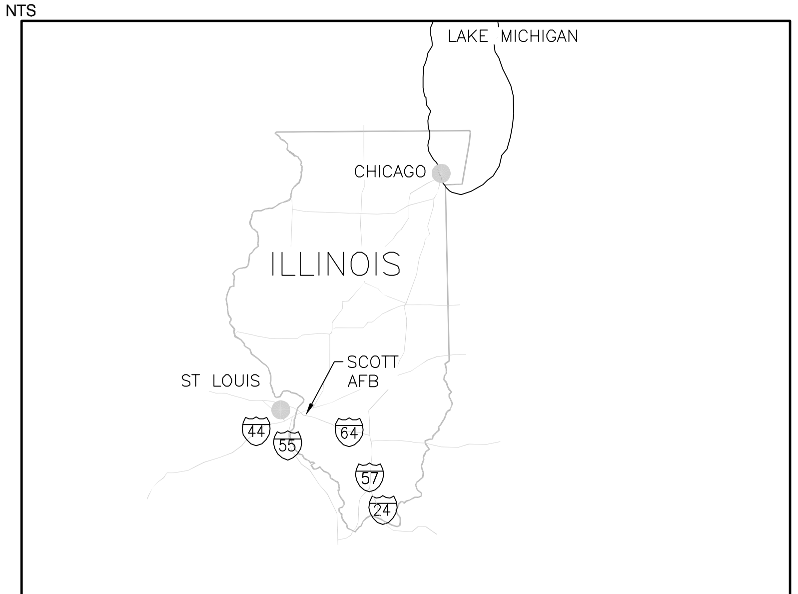
DRAWING INDEX

SHEET	DWG.	DESCRIPTION
1	T-1	TITLE SHEET
2	C-1	SITE PLAN
3	C-2	EROSION CONTROL / CLEAR AND GRUB PLAN
4	C-3	LANDFILL GAS SYSTEM PASSIVE VENTING PLAN
5	C-4	SUBGRADE PLAN - NORTH CELL
6	C-5	SUBGRADE PLAN - SOUTH CELL
7	C-6	FINAL COVER PLAN - NORTH CELL
8	C-7	FINAL COVER PLAN - SOUTH CELL
9	C-8	LANDFILL SECTIONS - NORTH CELL
10	C-9	LANDFILL SECTIONS - SOUTH CELL
11	C-10	EOD SECTIONS
12	D-1	DETAILS
13	D-2	DETAILS
14	D-3	DETAILS
15	D-4	DETAILS

LOCATION MAP

SCALE: NTS

LOCATION MAP
NORTH



DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

SYM	DESCRIPTION	DATE	APPR

DES C.H.M. DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

FINAL CLOSURE

TITLE SHEET

CODE ID. NO. 80091 SIZE D
SCALE: AS SHOWN
MAXIMO NO. TRACKING#
STA. PROJ. NO. #
WORK ORDER NO. #
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#
DRAWING NO. #
SHEET 1 OF 15

T-1

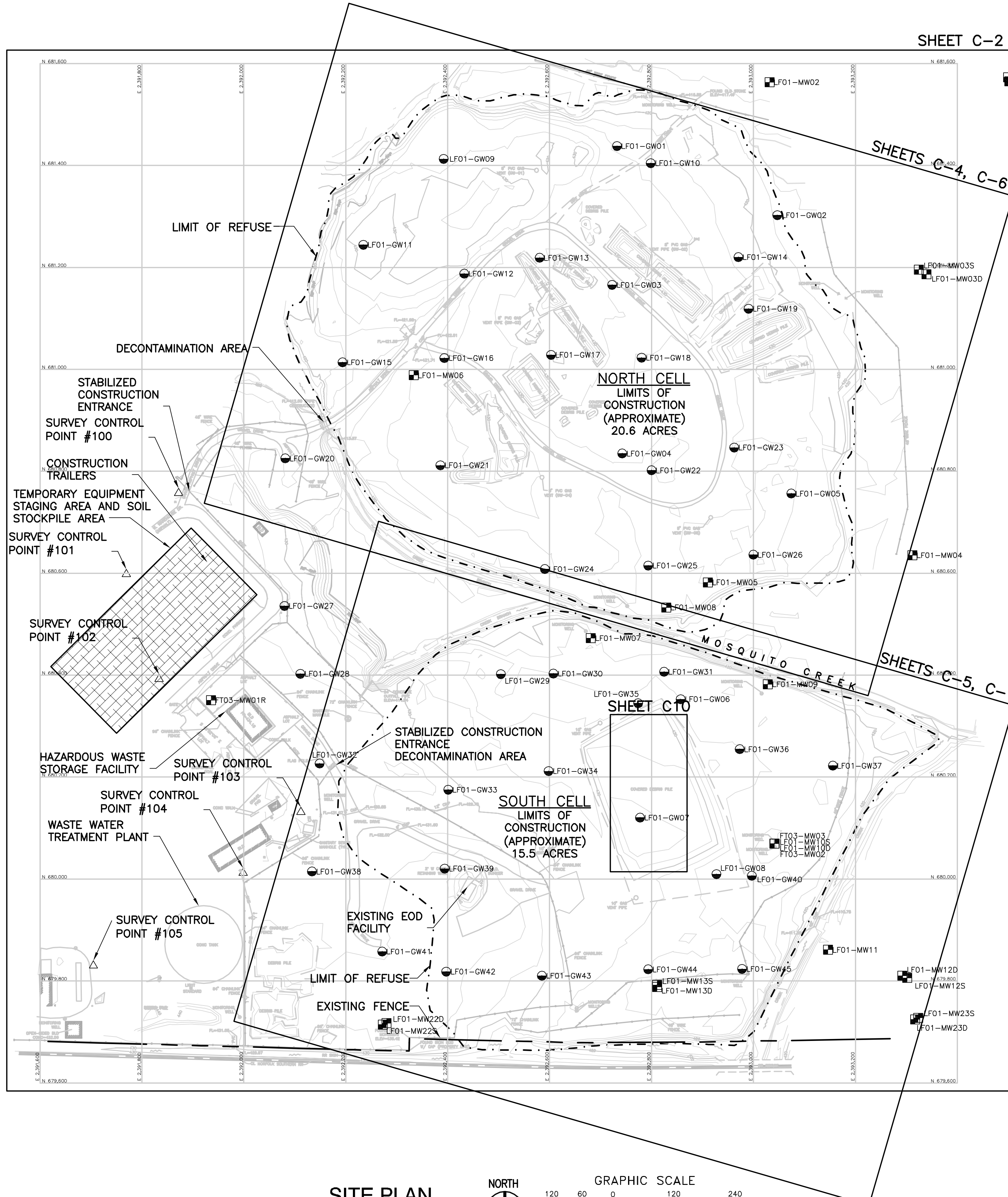
REVDATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\01-0001TTL.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 4:57pm

ABBREVIATIONS

BMPs	BEST MANAGEMENT PRACTICES
CF/LF	CUBIC FOOT PER LINEAL FOOT
CLR	CLEAR
CM/S	CENTIMETER PER SECOND
CONC	CONCRETE
DIA	DIAMETER
ELEV	ELEVATION
EOD	EXPLOSIVE ORDINANCE DISPOSAL
FL	FLOWLINE
HDPE	HIGH DENSITY POLYETHYLENE
HP	HIGH POINT
LLDPE	LINEAR LOW DENSITY POLYETHYLENE
LFG	LANDFILL GAS
LP	LOW POINT
MAX	MAXIMUM
MIL	ONE THOUSANDTH OF AN INCH
MIN	MINIMUM
NTS	NOT TO SCALE
OC	ON CENTER
OZ	OUNCE
OZ/SY	OUNCE PER SQUARE YARD
R/W	RIGHT-OF-WAY
SWPP	STORMWATER POLLUTION PREVENTION PLAN
TYP	TYPICAL
UV	ULTRA VIOLET
YD	YARD

LEGEND

- RAILROAD RIGHT-OF-WAY
- APPROXIMATE LIMIT OF REFUSE
- EXISTING FENCE
- RAILROAD TRACKS
- 424 EXISTING GRADE INTERMEDIATE CONTOUR
- 424 FINISHED GRADE INTERMEDIATE CONTOUR
- 440 EXISTING GRADE INDEX CONTOUR
- 430 FINISHED GRADE INDEX CONTOUR
- 430 GB GRADE BREAK CONTOUR
- GB TOP/TOE OF SLOPE OR GRADE BREAKS
- SF SILT FENCE
- FLOWLINE
- CUT/FILL LINE
- RIDGE LINE
- DRAINAGE HEADER
- DRAINAGE LATERAL
- LIMIT OF LLDPE GEOMEMBRANE
- LANDFILL GAS VENT LOCATION
- CUT
- FILL
- 2% DIRECTION AND RATE OF SLOPE
- TEMPORARY EQUIPMENT STAGING AREA
- TURF REINFORCEMENT MAT
- EROSION CONTROL BLANKET
- ACCESS ROAD
- LF01-MW12D GROUNDWATER MONITORING WELL
- LF01-GW42 LFG MONITORING WELL
- SETTLEMENT MARKER
- SURVEY BENCHMARKS
- DETAIL NUMBER
- SHEET NUMBER
- DETAIL SHEET NUMBER



SITE PLAN

SCALE: 1"=120'

NORTH

GRAPHIC SCALE

120 60 0 120 240

(IN FEET)

CONSTRUCTION NOTES

1. CLEAR AND GRUB
2. ABANDON WELL
3. EXCAVATE OR FILL TO CONTOURS SHOWN
4. PROTECT IN PLACE
5. PLACE 12 OZ NONWOVEN GEOTEXTILE 40 MIL DOUBLE SIDED TEXTURE LLDPE GEOMEMBRANE AND SUBSURFACE DRAINAGE GEOCOMPOSITE
6. CONSTRUCT DRAINAGE HEADER
7. CONSTRUCT DRAINAGE HEADER OUTLET
8. CONSTRUCT TOE DRAIN
9. DEMOLISH AND RECONSOLIDATE INTO WASTE PRISM
10. CONSTRUCT PROTECTIVE LAYER
11. PLACE TURF REINFORCEMENT MAT AND HYDROSEED
12. PLACE EROSION CONTROL BLANKET
13. HYDROSEED
14. CONSTRUCT SLOPE LINER TERMINATION
15. CONSTRUCT TOE DRAIN OUTLET
16. CONSTRUCT SILT FENCE
17. INSTALL SETTLEMENT MARKER
18. REFUSE REMOVAL
19. CONSTRUCT ACCESS ROAD
20. CONSTRUCT DRAINAGE LATERAL
21. CONSTRUCT IMPROVED COVER FOR EOD FOUNDATION

LFG CONSTRUCTION NOTES

1. INSTALL DECK HORIZONTAL GAS COLLECTOR TRENCH
2. INSTALL 4" HDPE CAP
3. INSTALL 4" HDPE TEE
4. INSTALL VENT RISER
5. INSTALL SLOPE HORIZONTAL GAS COLLECTOR TRENCH

LF01-MW17S
LF01-MW17D

SITE SURVEY BENCHMARKS:

Point #	Description	Elevation	Northing	Easting
100	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	432.0500	680757.0300	2391872.1500
101	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	432.3500	680597.8900	2391769.5900
102	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	432.1600	680392.2100	2391833.8200
103	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	433.8400	680131.5200	2392112.0900
104	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	433.3900	680011.6600	2391998.4300
105	5/8" Iron Rod with Orange plastic Cap stamped "ABNA TRAV PNT"	432.7700	679830.2400	2391704.9100

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

DEPARTMENT OF THE AIR FORCE

SCOTT AFB, IL

SCOTT AIR FORCE BASE LANDFILL

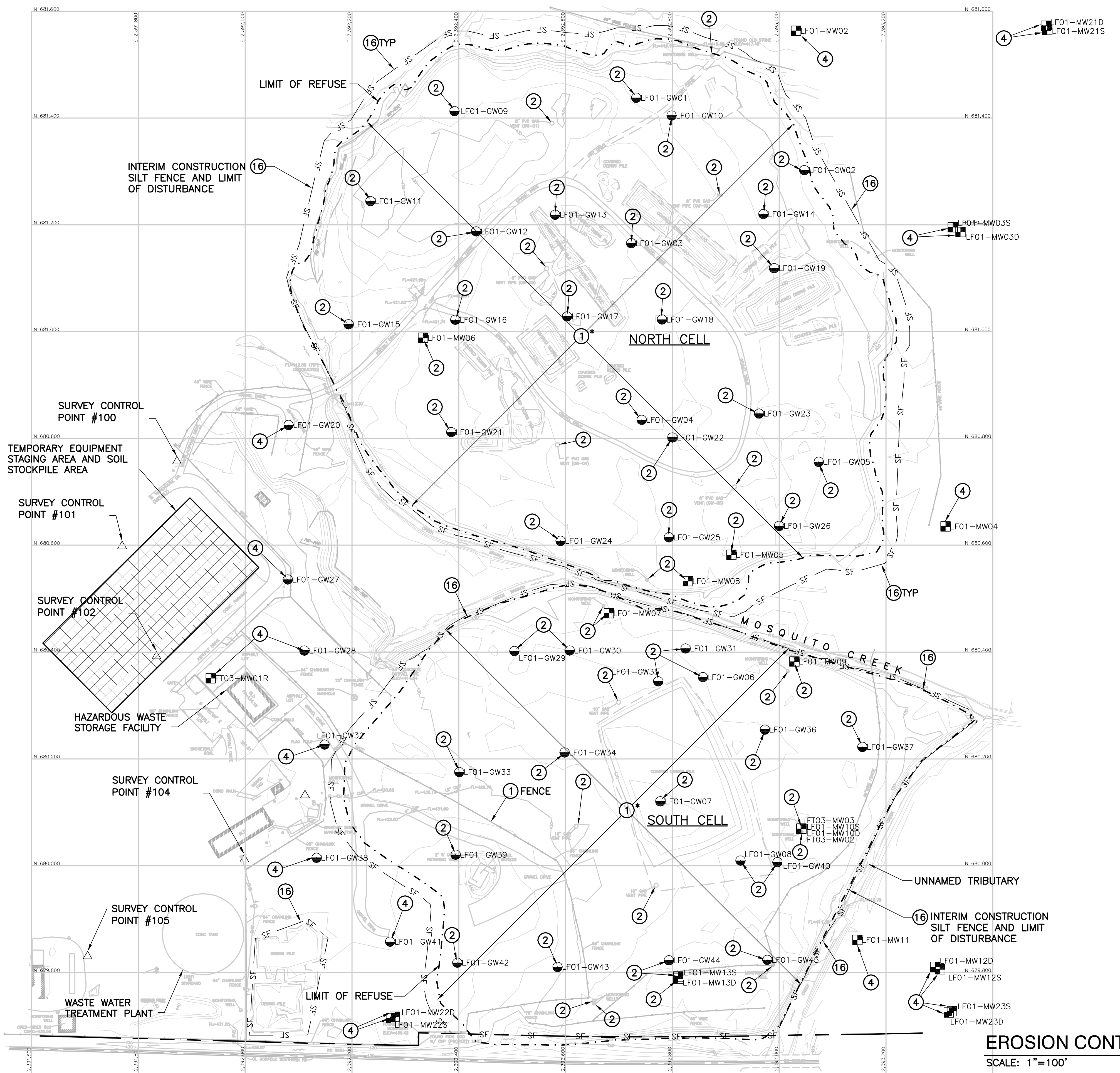
FINAL CLOSURE

SITE PLAN

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 2 OF 15	

C-1

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\04-0001XS.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:00pm



CONSTRUCTION NOTES

1. CLEAR AND GRUB (SEE NOTE)
2. ABANDON WELL — 5/C2 D3
4. PROTECT IN PLACE
16. CONSTRUCT SILT FENCE — 1/C2 D4

LEGEND

- RAILROAD RIGHT–OF–WAY
- APPROXIMATE LIMIT OF REFUSE
- 424 EXISTING GRADE INTERMEDIATE CONTOUR
- 440 EXISTING GRADE INDEX CONTOUR
- TEMPORARY EQUIPMENT STAGING AREA
- SF SILT FENCE
- LF01–MW12D GROUNDWATER MONITORING WELL
- LF01–GW42 LFG MONITORING WELL
- EXISTING SURVEY MONUMENT

NOTE:

EXISTING STOCKPILES TO BE USED AS SUBGRADE MATERIAL, TARPS AND OTHER PROTECTIVE COVER TO BE DISPOSED OF OFF SITE.

EROSION CONTROL NOTES:

THE CONTRACTOR IS RESPONSIBLE FOR THE SELECTION, INSTALLATION, INSPECTION, MAINTENANCE, AND REMOVAL OF ALL STORMWATER BMP'S AND THE IMPLEMENTATION OF APPROPRIATE STORMWATER MANAGEMENT PRACTICES TO PREVENT THE DISCHARGE OF SEDIMENT AND POLLUTANTS INTO PUBLIC WATERS IN ACCORDANCE WITH SWPPP FOR THE LANDFILL LF–01 CAP CONSTRUCTION ACTIVITIES.

ALL EXISTING MANHOLES, UTILITIES AND OTHER SURFACE FEATURES NOT CALLED OUT TO BE REMOVED SHALL BE PROTECTED FROM DAMAGE AS A RESULT OF THE CONSTRUCITON. THE CONTRACTOR SHALL PROVIDE ALL LABOR, EQUIPMENT, AND MATERIALS REQUIRED TO REPAIR DAMAGE RESULTING FROM CONSTRUCTION ACTIVITIES.

*CLEARING SHALL INCLUDE THE REMOVAL AND DISPOSAL/RECONSOLIDATION OF ALL STRUCTURES/DEBRIS THAT ARE NOT PART OF THE COVER SYSTEM (i.e. FENCING, CONCRETE, WOOD, PLASTIC, etc.)

GRAPHIC SCALE

100 50 0 100 200

(IN FEET)

DRAFT FINAL - NOT FOR CONSTRUCTION

EROSION CONTROL / CLEAR AND GRUB PLAN

SCALE: 1"=100'

NORTH

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860–7777

HGL HydroGeoLogic, Inc

DES C.H.M. DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

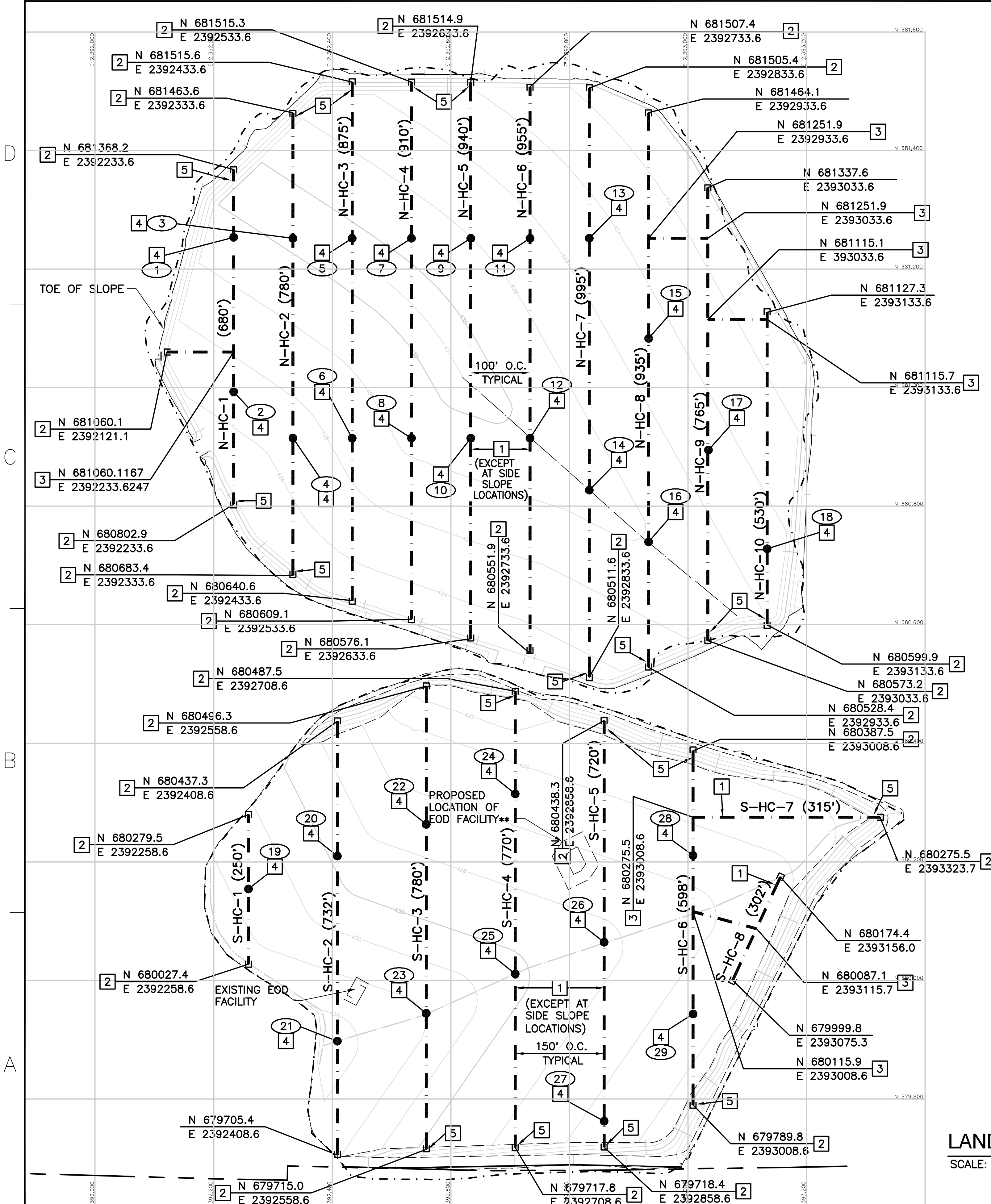
DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

FINAL CLOSURE
EROSION CONTROL / CLEAR AND GRUB PLAN

CODE ID. NO. 80091 SIZE D
SCALE: AS SHOWN
MAXIMO NO. TRACKING#
STA. PROJ. NO. #
WORK ORDER NO. #
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#
DRAWING NO. #
SHEET 3 OF 15
C-2

SYM DESCRIPTION DATE APPR

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\10-0001SITE.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Thursday, June 30, 2011 - 4:11pm



VENT COORDINATE TABLE

POINT NUMBER	POINT DESCRIPTION	COORDINATES NORTHING	COORDINATES EASTING
1	N-HC-1 VENT #1	681253.6	2392233.6
2	N-HC-1 VENT #2	680993.2	2392233.6
3	N-HC-2 VENT #3	681251.9	2392333.6
4	N-HC-2 VENT #4	680914.7	2392333.6
5	N-HC-3 VENT #5	681251.9	2392433.6
6	N-HC-3 VENT #6	680914.7	2392433.6
7	N-HC-4 VENT #7	681251.9	2392533.6
8	N-HC-4 VENT #8	680914.7	2392533.6
9	N-HC-5 VENT #9	681251.9	2392633.6
10	N-HC-5 VENT #10	680914.7	2392633.6
11	N-HC-6 VENT #11	681251.9	2392733.6
12	N-HC-6 VENT #12	680914.7	2392733.6
13	N-HC-7 VENT #13	681251.9	2392833.6
14	N-HC-7 VENT #14	680827.1	2392833.6
15	N-HC-8 VENT #15	681083.1	2392933.6
16	N-HC-8 VENT #16	680739.9	2392933.6
17	N-HC-9 VENT #17	680894.7	2393033.6
18	N-HC-10 VENT#18	680728.2	2393133.6
19	S-HC-1 VENT #19	680154.1	2392258.6
20	S-HC-2 VENT #20	680209.8	2392408.6
21	S-HC-2 VENT #21	679897.4	2392408.6
22	S-HC-3 VENT #22	680263.1	2392558.6
23	S-HC-3 VENT #23	679944.2	2392558.6
24	S-HC-4 VENT #24	680314.7	2392708.6
25	S-HC-4 VENT #25	680011.0	2392708.6
26	S-HC-5 VENT #26	680064.1	2392858.6
27	S-HC-5 VENT #27	679762.6	2392858.6
28	S-HC-6 VENT #28	680210.5	2393008.6
29	S-HC-6 VENT #29	679943.3	2393008.6

LFG CONSTRUCTION NOTES

1. INSTALL DECK HORIZONTAL GAS COLLECTOR TRENCH (1, C3 D3)
2. INSTALL 4" HDPE CAP (3, C3 D3)
3. INSTALL 4" HDPE TEE
4. INSTALL VENT RISER (2, C3 D3)
5. INSTALL SLOPE HORIZONTAL GAS COLLECTOR TRENCH (4, C3 D3)

**PROPOSED EOD FACILITY LOCATION IS BASED ON THE 500' SET BACK FROM THE RAIL ROAD RIGHT-OF-WAY AND FACILITIES.

LEGEND

- RAILROAD RIGHT-OF-WAY
- APPROXIMATE LIMIT OF REFUSE
- 424 PROPOSED SUBGRADE INTERMEDIATE CONTOUR
- 440 PROPOSED SUBGRADE INDEX CONTOUR
- LANDFILL GAS VENT LOCATION
- HORIZONTAL GAS COLLECTOR TRENCH
- END CAP, HORIZONTAL GAS COLLECTOR

GRAPHIC SCALE
100 50 0 100 200
(IN FEET)

LANDFILL GAS SYSTEM PASSIVE VENTING PLAN

SCALE: 1"=100'

NORTH

DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES C.H.M. DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

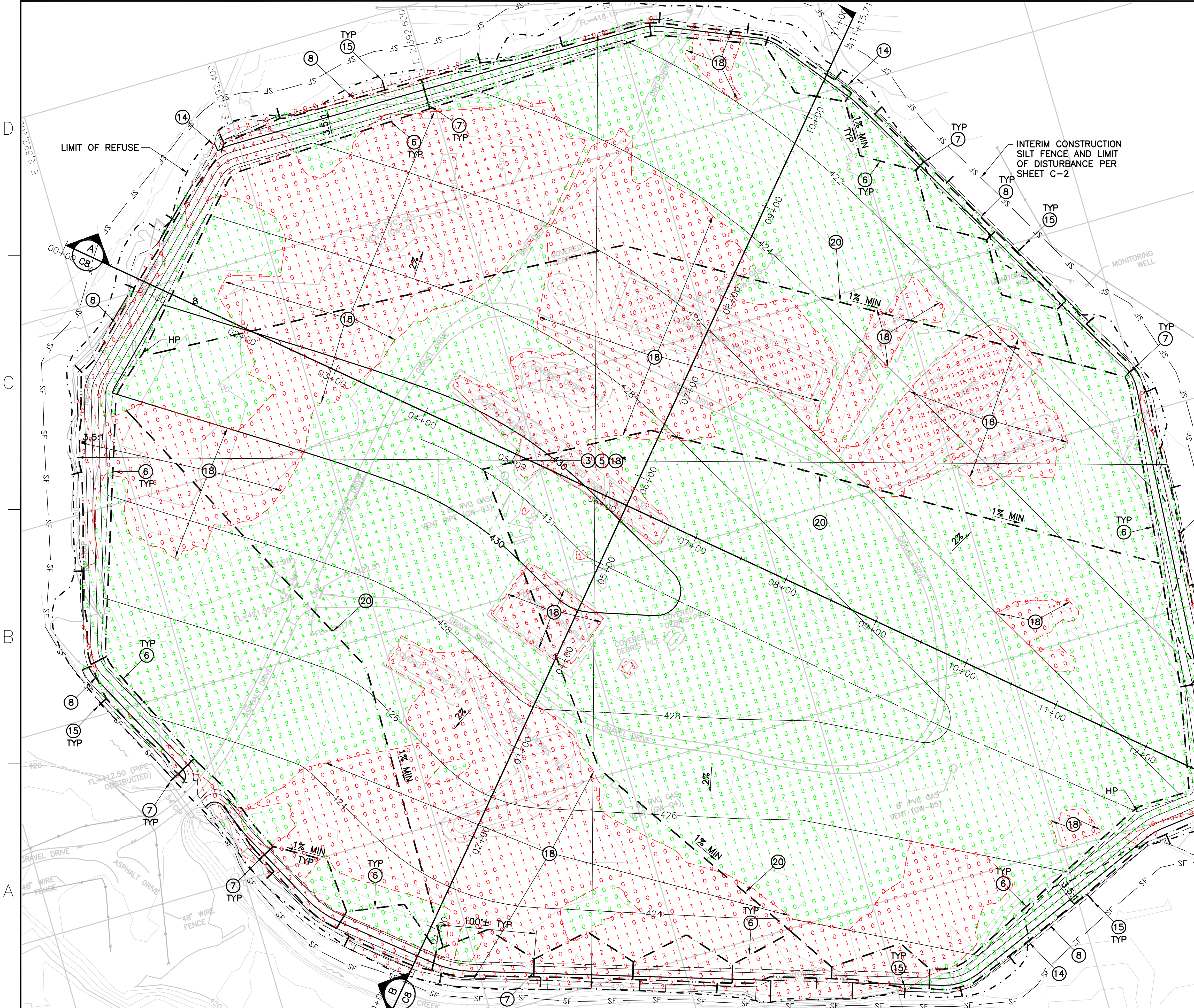
DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
LANDFILL GAS SYSTEM PASSIVE VENTING PLAN

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 4 OF 15	

C-3

SYM | DESCRIPTION | DATE | APPR

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\86-0001GSP.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:02pm



CONSTRUCTION NOTES

- (3) EXCAVATE TO CONTOURS SHOWN (NOTES 1–3) 1/C4 D1, 2/C4 D1
- (5) PLACE 12 OZ NONWOVEN GEOTEXTILE 40 MIL DOUBLE SIDED TEXTURE LLDPE GEOMEMBRANE AND SUBSURFACE DRAINAGE GEOCOMPOSITE 1/C4 D1, 2/C4 D1
- (6) CONSTRUCT DRAINAGE HEADER 3/C4 D1
- (7) CONSTRUCT DRAINAGE HEADER OUTLET 4/C4 D1
- (8) CONSTRUCT TOE DRAIN 6/C4 D1
- (14) CONSTRUCT SLOPE LINER TERMINATION 5/C4 D1
- (15) CONSTRUCT TOE DRAIN OUTLET 6/C4 D1
- (18) REFUSE REMOVAL
- (20) CONSTRUCT DRAINAGE LATERAL 7/C4 D1

NOTES:
1. GRADES SHOWN ARE GEOMEMBRANE GRADES.
2. DUE TO UNCERTAINTIES IN SUBSURFACE CONDITIONS, A DETAILED SITE SPECIFIC RECONSOLIDATION PLAN IS UNABLE TO BE PREPARED. CONTRACTOR SHALL CONFORM WITH SPECIFICATION SECTION 3.2 OF 31 00 00.
3. RECONSOLIDATION WORK SHALL NOT COMMENCE UNTIL A WASTE RECONSOLIDATION PLAN IS APPROVED BY THE HGL PM OR DESIGN ENGINEER.

LEGEND

- LIMIT OF REFUSE
- 424 EXISTING GRADE INTERMEDIATE CONTOUR
- 428 FINISHED GRADE INTERMEDIATE CONTOUR
- 440 EXISTING GRADE INDEX CONTOUR
- 430 FINISHED GRADE INDEX CONTOUR
- FLOW LINE
- 2% DIRECTION AND RATE OF SLOPE
- TOP/TOE OF SLOPE OR GRADE BREAKS
- RIDGE LINE
- DRAINAGE HEADER
- DRAINAGE LATERAL
- SF SILT FENCE
- CUT/FILL LINE
- 1 CUT
- 1 FILL

VOLUME SUMMARY
CUT: 20,300 CY
FILL: 32,600 CY

*CONTRACTOR TO LOCATE AND REMOVE REFUSE/DEBRIS WITHIN LIMITS OF DISTURBANCE

GRAPHIC SCALE
50 25 0 50 100
(IN FEET)

DRAFT FINAL - NOT FOR CONSTRUCTION

SUBGRADE PLAN - NORTH

SCALE: 1"=50'

NORTH

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

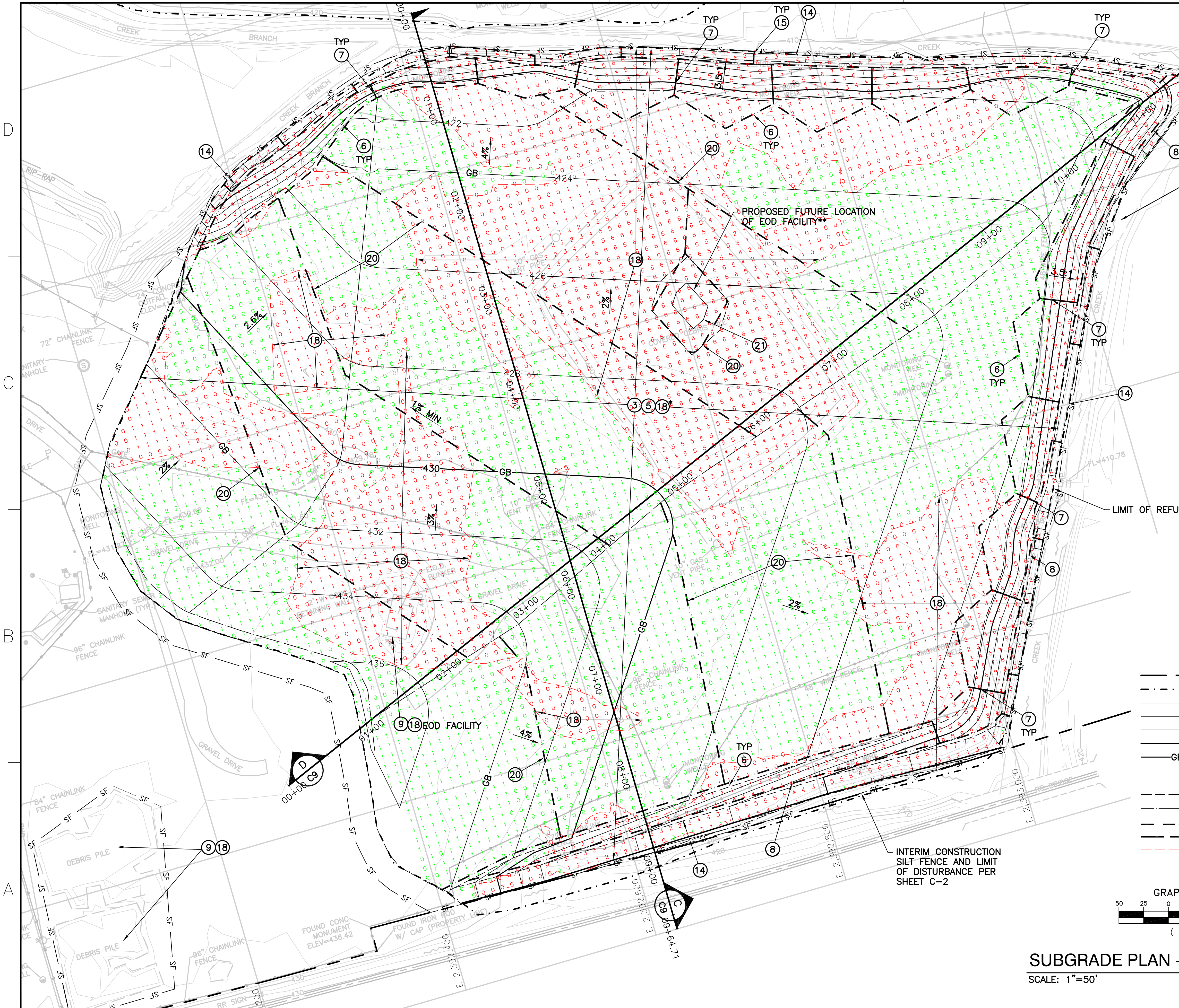
DES C.H.M. DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
SUBGRADE PLAN - NORTH CELL

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO.	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 5 OF 15	
C-4	

SYM | DESCRIPTION | DATE | APPR

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\12-00040RF.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:03pm



CONSTRUCTION NOTES

3. EXCAVATE TO CONTOURS SHOWN (NOTES 1–3) (1/C5 D1) (2/C5 D1)
5. PLACE 12 OZ NONWOVEN GEOTEXTILE 40 MIL DOUBLE SIDED TEXTURE LLDPE GEOMEMBRANE AND SUBSURFACE DRAINAGE GEOCOMPOSITE (1/C5 D1) (2/C5 D1)
6. CONSTRUCT DRAINAGE HEADER (3/C5 D1)
7. CONSTRUCT DRAINAGE HEADER OUTLET (4/C5 D1)
8. CONSTRUCT TOE DRAIN (6/C5 D1)
9. DEMOLISH AND RECONSOLIDATE INTO WASTE PRISM
14. CONSTRUCT SLOPE LINER TERMINATION (5/C5 D1)
15. CONSTRUCT TOE DRAIN OUTLET (6/C5 D1)
18. REFUSE REMOVAL
20. CONSTRUCT DRAINAGE LATERAL (7/C5 D1)
21. CONSTRUCT IMPROVED COVER FOR EOD FOUNDATION (1/C5 C10)

NOTES:
1. GRADES SHOWN ARE GEOMEMBRANE GRADES.
2. DUE TO UNCERTAINTIES IN SUBSURFACE CONDITIONS, A DETAILED SITE SPECIFIC RECONSOLIDATION PLAN IS UNABLE TO BE PREPARED. CONTRACTOR SHALL CONFORM WITH SPECIFICATION SECTION 3.2 OF 31 00 00.
3. RECONSOLIDATION SHALL NOT COMMENCE UNTIL A WASTE RECONSOLIDATION PLAN IS APPROVED BY THE HGL PM OR DESIGN ENGINEER.

*CONTRACTOR TO LOCATE AND REMOVE REFUSE/DEBRIS WITHIN LIMITS OF DISTURBANCE

**PROPOSED EOD FACILITY LOCATION IS BASED ON THE 500' SET BACK FROM THE RAIL ROAD RIGHT-OF-WAY AND FACILITIES.

LEGEND

	RR RIGHT-OF-WAY
	LIMIT OF REFUSE
424	EXISTING GRADE INTERMEDIATE CONTOUR
428	FINISHED GRADE INTERMEDIATE CONTOUR
440	EXISTING GRADE INDEX CONTOUR
430	FINISHED GRADE INDEX CONTOUR
GB 430	GRADE BREAK CONTOUR
2%	DIRECTION AND RATE OF SLOPE
	TOP/TOE OF SLOPE OR GRADE BREAKS
	RIDGE LINE
	DRAINAGE HEADER
	DRAINAGE LATERAL
	CUT/FILL LINE
1	CUT
1	FILL

GRAPHIC SCALE
50 25 0 50 100
(IN FEET)

VOLUME SUMMARY
CUT: 25,000 CY
FILL: 11,600 CY

SUBGRADE PLAN - SOUTH
SCALE: 1"=50'

NORTH

DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc

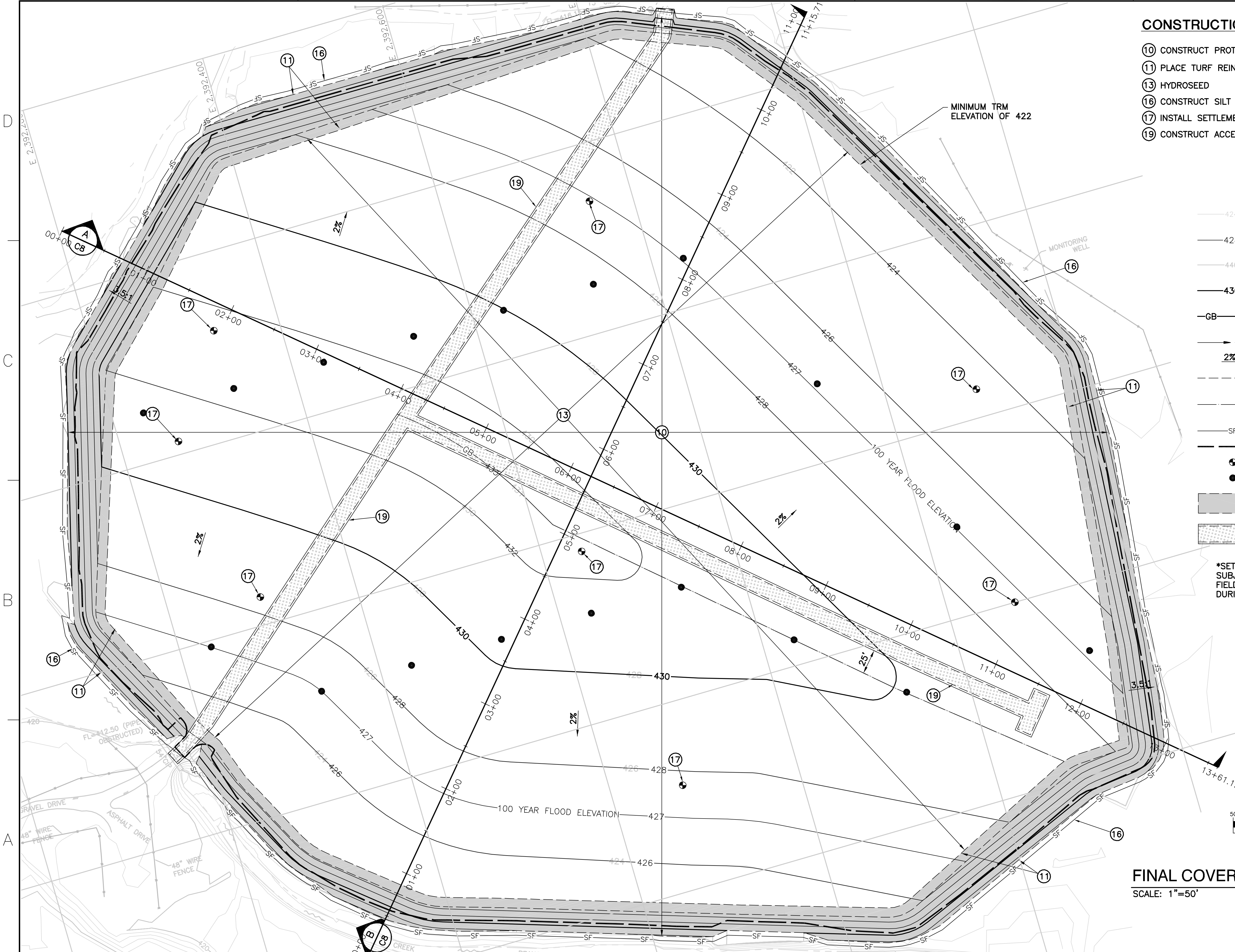
DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
SUBGRADE PLAN - SOUTH CELL

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 6 OF 15	

C-5

REVDATE: FILE NAME: C:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\12-0006RF.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Thursday, June 30, 2011 - 8:55am



CONSTRUCTION NOTES

- (10) CONSTRUCT PROTECTIVE LAYER (1 C6 D1) (2 C6 D1)
- (11) PLACE TURF REINFORCEMENT MAT AND HYDROSEED (1 C6 D2)
- (13) HYDROSEED
- (16) CONSTRUCT SILT FENCE (1 C6 D4)
- (17) INSTALL SETTLEMENT MARKER* (6 C6 D3)
- (19) CONSTRUCT ACCESS ROAD (7 C6 D3)

LEGEND

- EXISTING GRADE INTERMEDIATE CONTOUR (424)
- FINISHED GRADE INTERMEDIATE CONTOUR (428)
- EXISTING GRADE INDEX CONTOUR (440)
- FINISHED GRADE INDEX CONTOUR (430)
- GB 430 GRADE BREAK CONTOUR
- FLOW LINE
- 2% DIRECTION AND RATE OF SLOPE
- TOP/TOE OF SLOPE OR GRADE BREAKS
- RIDGE LINE
- SF SILT FENCE
- LIMIT OF LLDPE GEOMEMBRANE
- SETTLEMENT MARKER
- LANDFILL GAS VENT LOCATION
- TURF REINFORCEMENT MAT
- ACCESS ROAD

*SETTLEMENT MARKER LOCATION SUBJECT TO CHANGE BASED ON FIELD CONDITIONS ENCOUNTERED DURING CONSTRUCTION

GRAPHIC SCALE 50 25 0 50 100 (IN FEET)

DRAFT FINAL - NOT FOR CONSTRUCTION

FINAL COVER PLAN - NORTH

SCALE: 1"=50'

NORTH

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

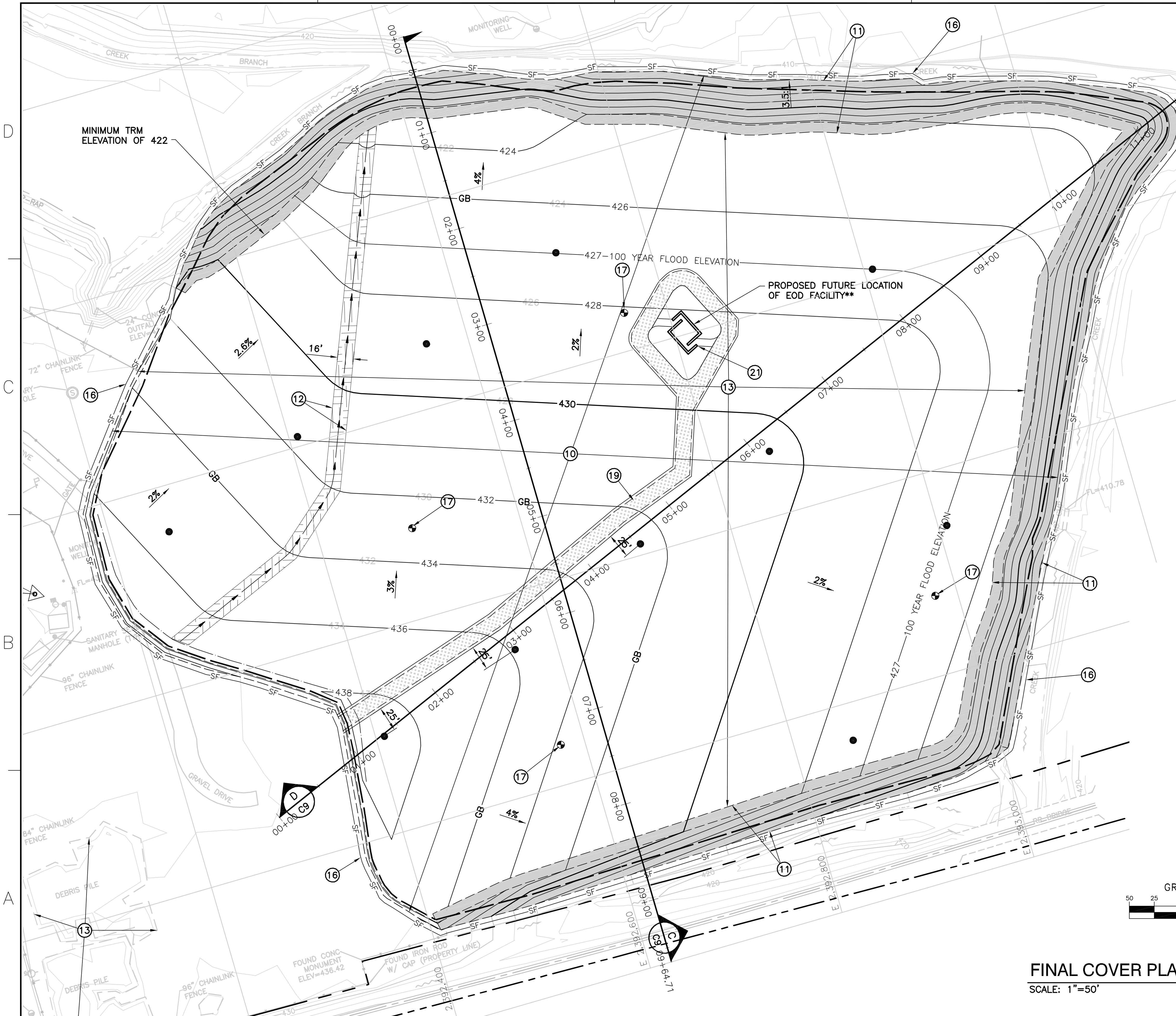
DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
FINAL COVER PLAN - NORTH CELL

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 7 OF 15	

C-6

SYM | DESCRIPTION | DATE | APPR

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\12-0001GRF.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:42pm



FINAL COVER PLAN - SOUTH

SCALE: 1"=50'

NORTH

GRAPHIC SCALE

50 25 0 50 100

(IN FEET)

CONSTRUCTION NOTES

(10) CONSTRUCT PROTECTIVE LAYER — 1/C7 D1, 2/C7 D1
(11) PLACE TURF REINFORCEMENT MAT AND HYDROSEED — 1/C7 D2
(12) PLACE EROSION CONTROL BLANKET — 2/C7 D2
(13) HYDROSEED
(16) CONSTRUCT SILT FENCE — 1/C7 D4
(17) INSTALL SETTLEMENT MARKER* — 6/C7 D3
(19) CONSTRUCT ACCESS ROAD — 7/C7 D3
(21) CONSTRUCT IMPROVED COVER FOR EOD FOUNDATION — 1/C7 C10

*SETTLEMENT MARKER LOCATION SUBJECT TO CHANGE BASED ON FIELD CONDITIONS ENCOUNTERED DURING CONSTRUCTION

**PROPOSED EOD FACILITY LOCATION IS BASED ON THE 500' SET BACK FROM THE RAIL ROAD RIGHT-OF-WAY AND FACILITIES.

LEGEND

SCOTT AFB BOUNDARY
RAILROAD RIGHT-OF-WAY
EXISTING GRADE INTERMEDIATE CONTOUR
FINISHED GRADE INTERMEDIATE CONTOUR
EXISTING GRADE INDEX CONTOUR
FINISHED GRADE INDEX CONTOUR
GRADE BREAK CONTOUR
FLOW LINE
TOP/TOE OF SLOPE OR GRADE BREAKS
SILT FENCE
LIMIT OF LLDPE GEOMEMBRANE
SETTLEMENT MARKER
LANDFILL GAS VENT LOCATION
DIRECTION AND RATE OF SLOPE
TURF REINFORCEMENT MAT
EROSION CONTROL BLANKET
ACCESS ROAD

DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

FINAL CLOSURE

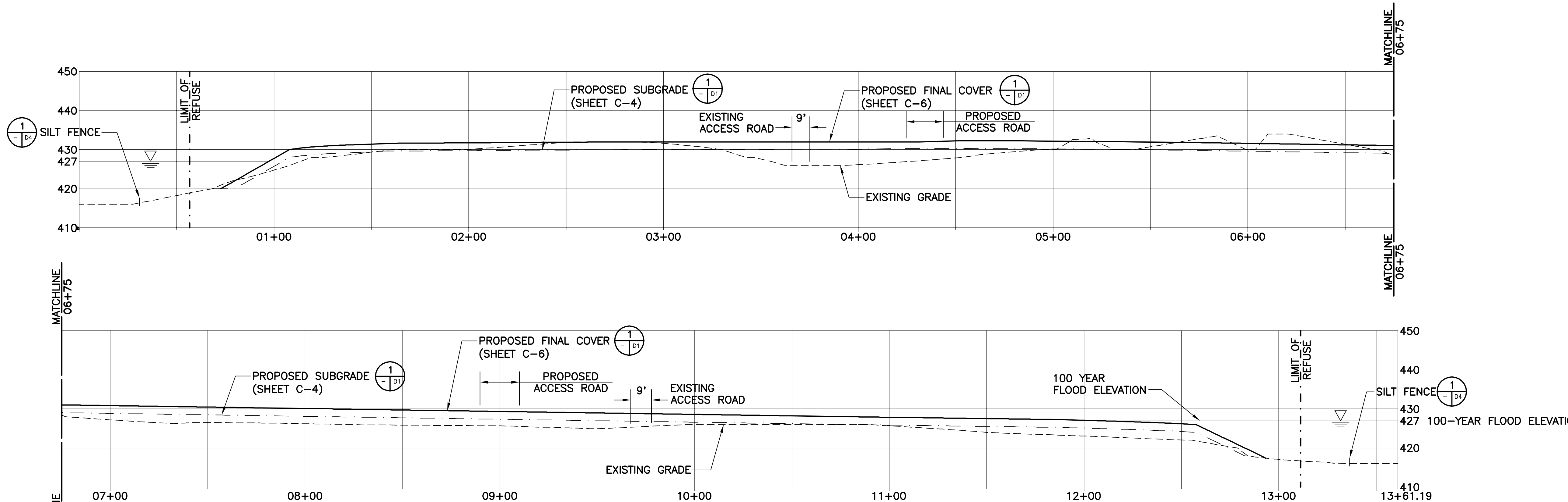
FINAL COVER PLAN - SOUTH CELL

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 8 OF 15	

C-7

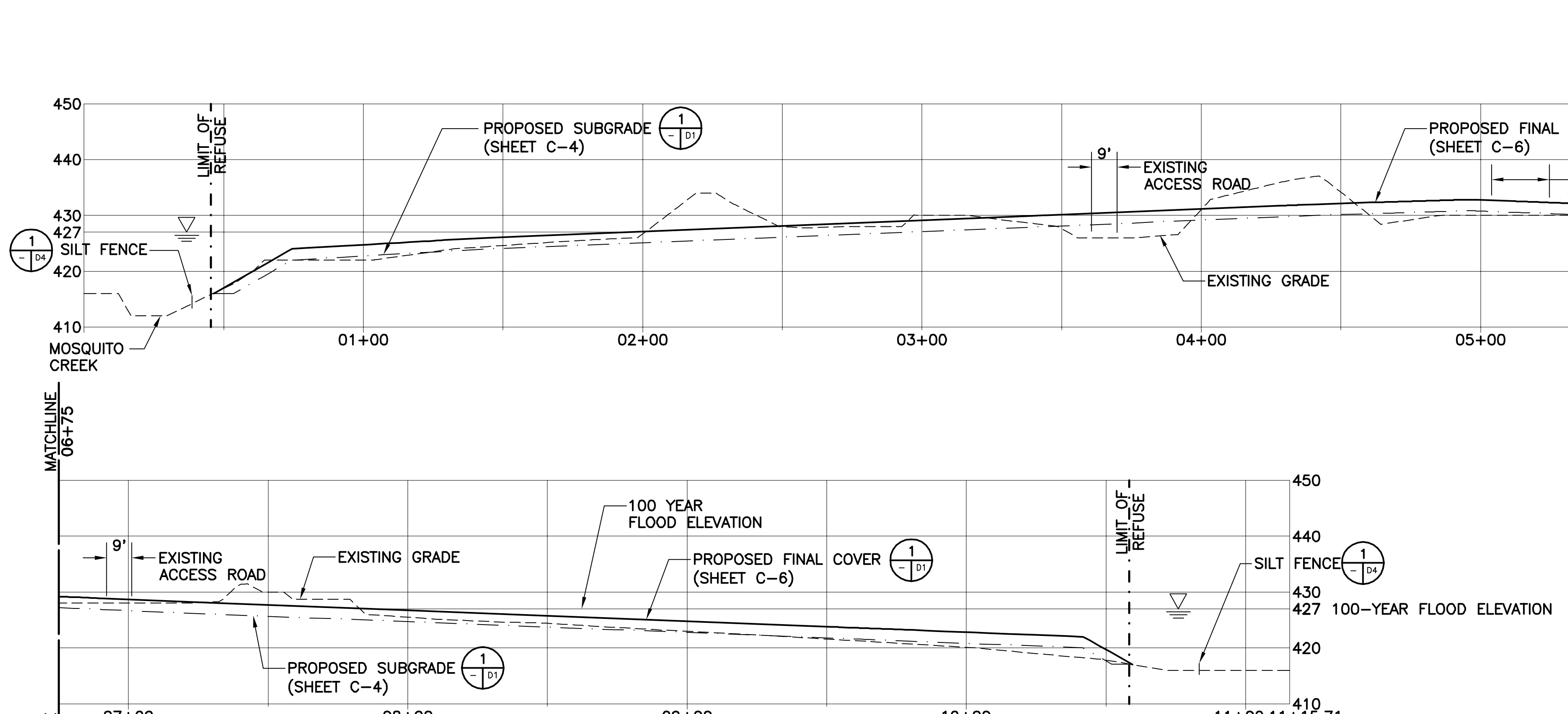
SYM | DESCRIPTION | DATE | APPR

REVDATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\12-0003DRF.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:43pm



SECTION A (C4 C8) A (C6 C8)

SCALE: HORIZONTAL: 1"=30'
VERTICAL: 1"=15'



SECTION B (C4 C8) B (C6 C8)

SCALE: HORIZONTAL: 1"=30'
VERTICAL: 1"=15'

DRAFT FINAL - NOT FOR CONSTRUCTION

SYM	DESCRIPTION	DATE	APPR

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

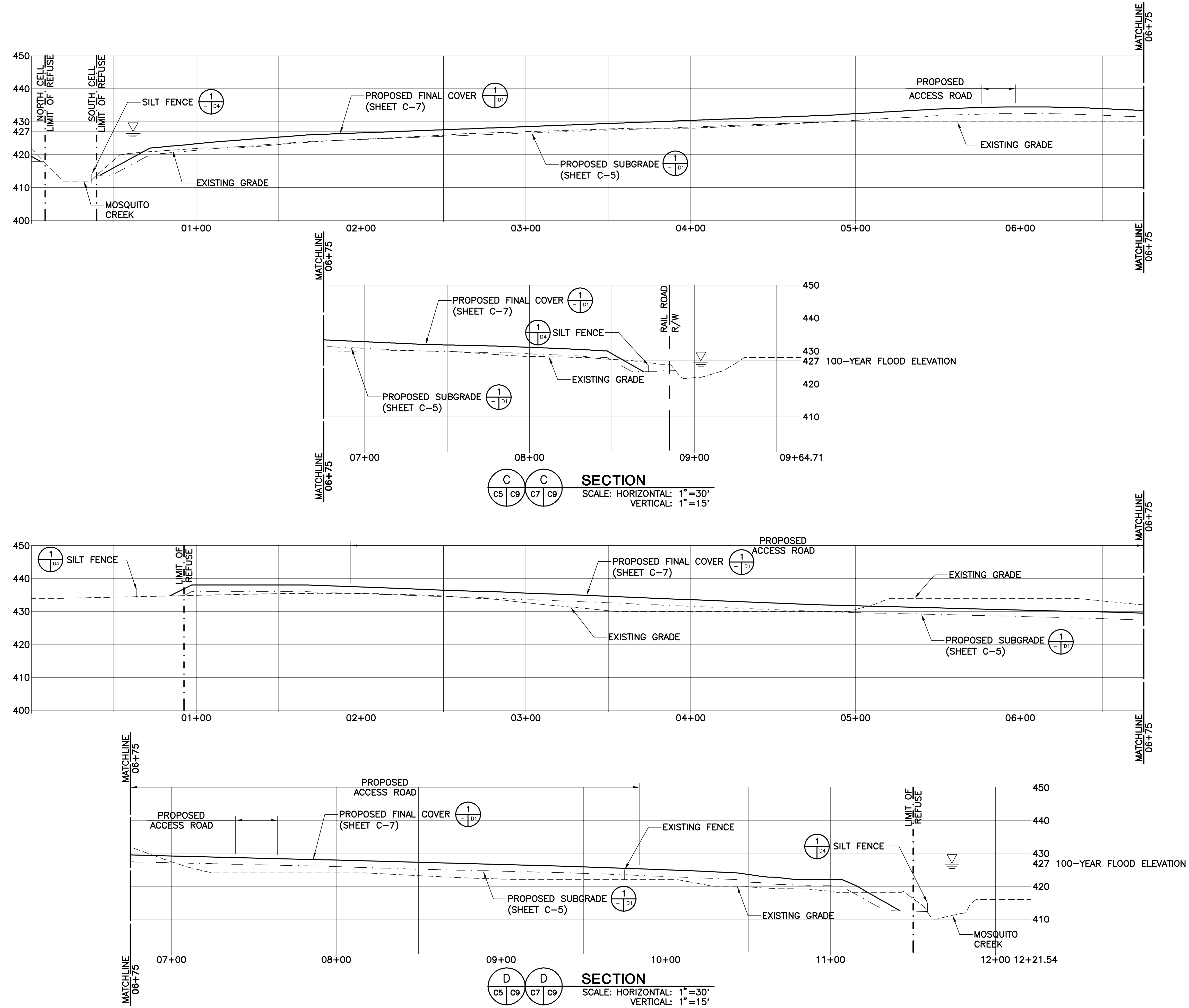
FINAL CLOSURE

LANDFILL SECTIONS - NORTH CELL

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 9 OF 15	

C-8

REV/DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\33-0001XSEC.dwg LAYOUT NAME: TAB PLOTTED: Wednesday, June 29, 2011 - 5:10pm



DRAFT FINAL - NOT FOR CONSTRUCTION

SYM	DESCRIPTION	DATE	APPR

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES	C.H.M.	DRW	J.M.B.
REVIEWED BY	C.H.M.		
PM/DM	C.H.M.		
CHIEF ENG	B.A.S.		

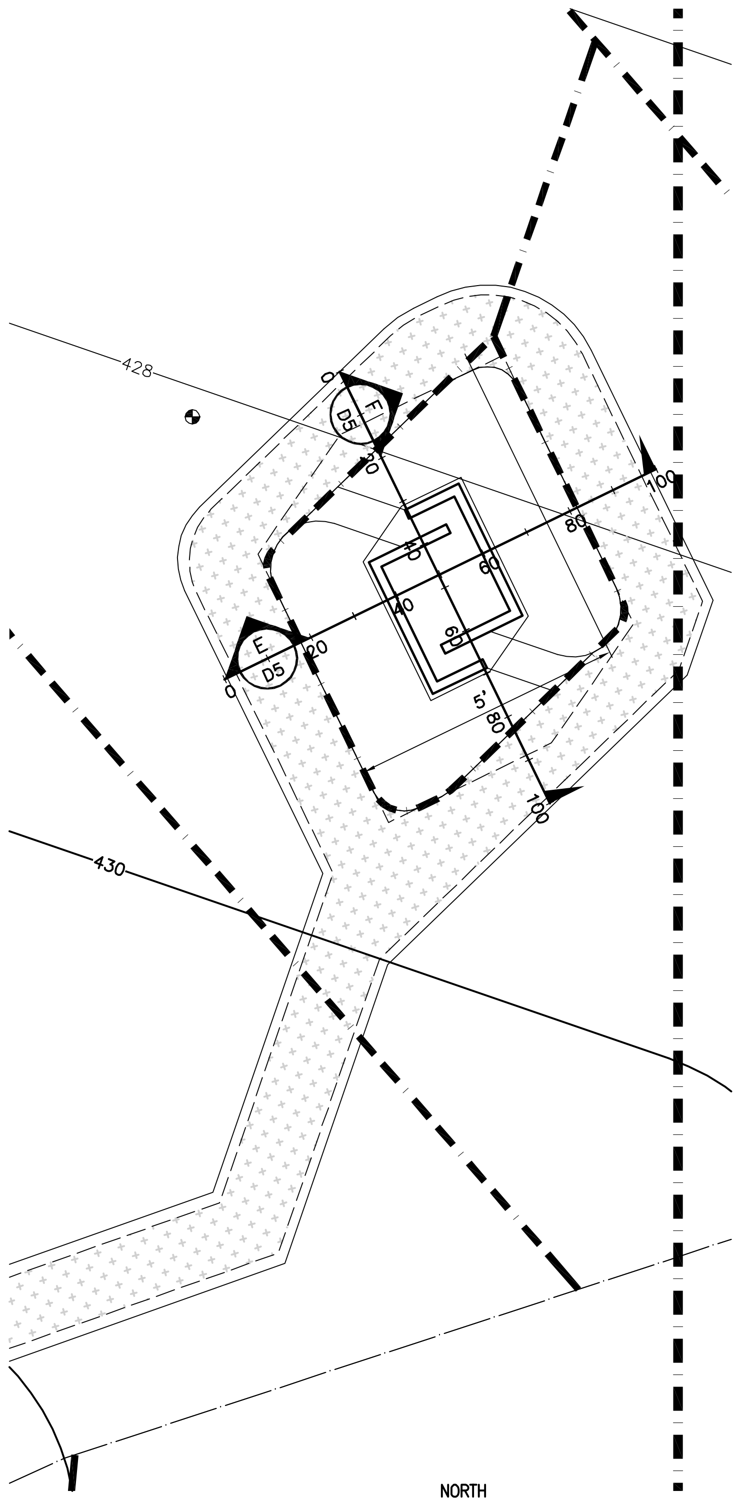
DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

FINAL CLOSURE
LANDFILL SECTIONS - SOUTH CELL

CODE ID. NO. 80091	SIZE D
SCALE:	AS SHOWN
MAXIMO NO.	TRACKING#
STA. PROJ. NO.	#
WORK ORDER NO.	#
CONSTR. CONTR. NO.	CONSTRUCTION CONTRACT#
DRAWING NO.	#
SHEET 10 OF 15	

C-9

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\33-0002XSEC.dwg LAYOUT NAME: TAB PLOTTED: Wednesday, June 29, 2011 - 5:11pm

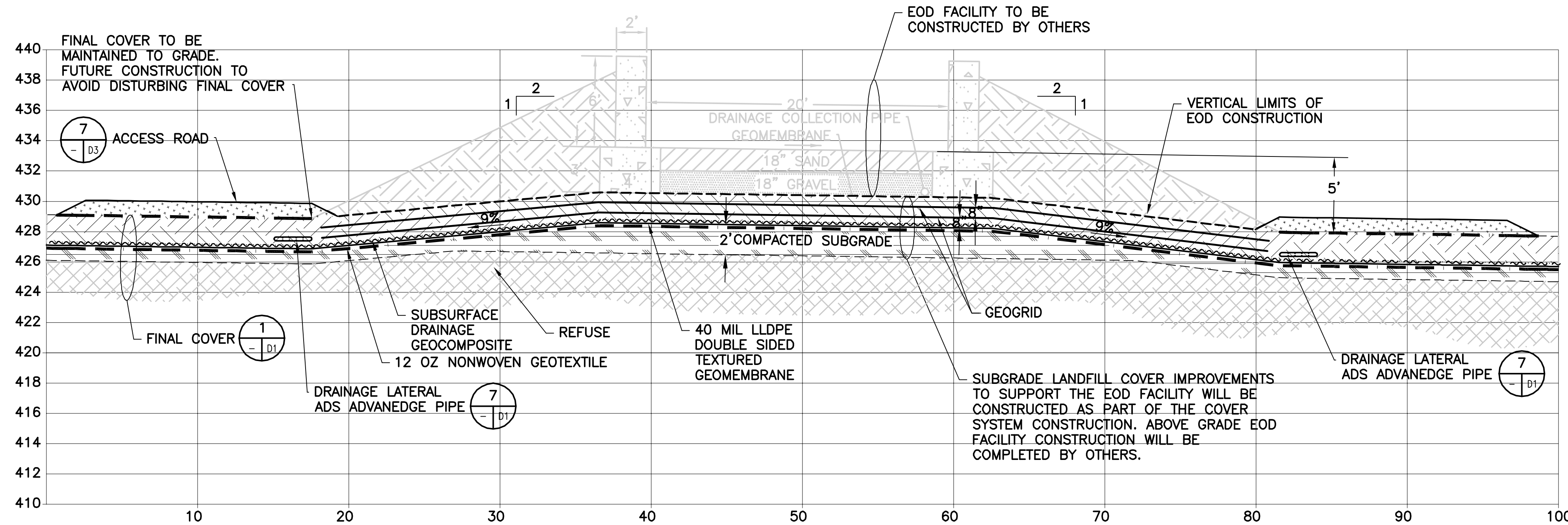


PLAN VIEW

SCALE: 1"=20'

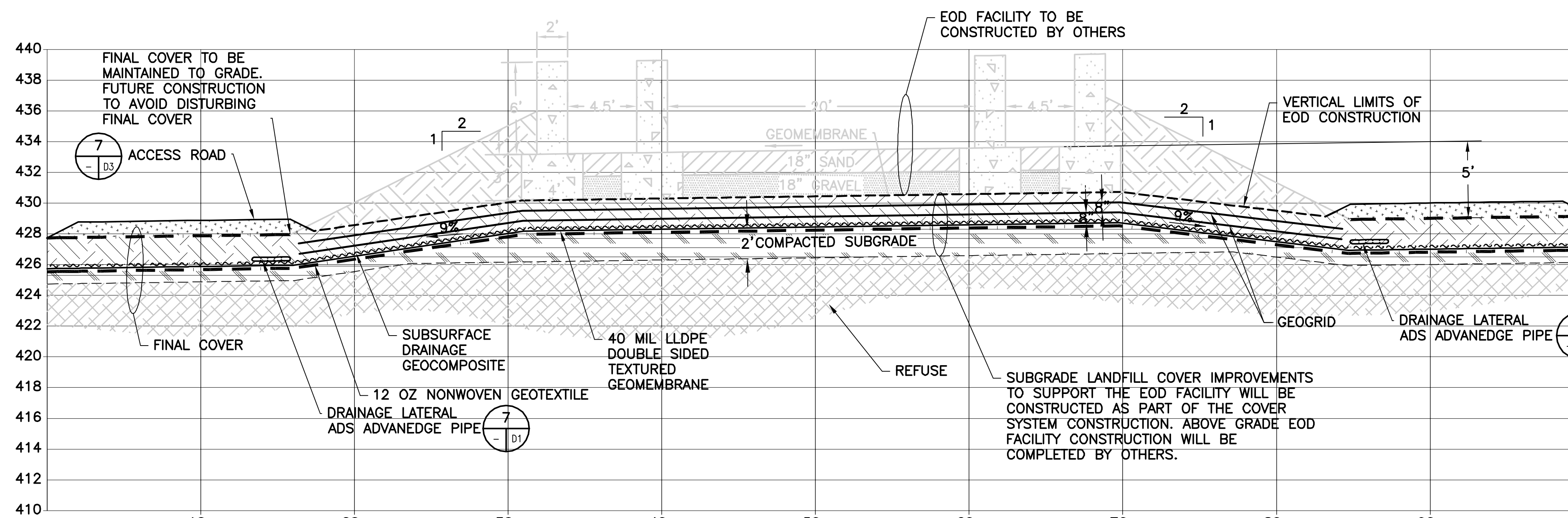
LEGEND

- 430 — FINISHED GRADE INDEX CONTOUR
- 428 — FINISHED GRADE INTERMEDIATE CONTOUR
- DRAINAGE LATERAL
- HORIZONTAL GAS COLLECTOR TRENCH
- RIDGE LINE
- SETTLEMENT MARKER
- ACCESS ROAD



E SECTION (C7 D5)

SCALE: HORIZONTAL: 1"=5'
VERTICAL: 1"=5'



F SECTION (C7 D5)

SCALE: HORIZONTAL: 1"=5'
VERTICAL: 1"=5'

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES	C.H.M.	DRW	J.M.B.
REVIEWED BY	C.H.M.		
PM/DM	C.H.M.		
CHIEF ENG	B.A.S.		

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL

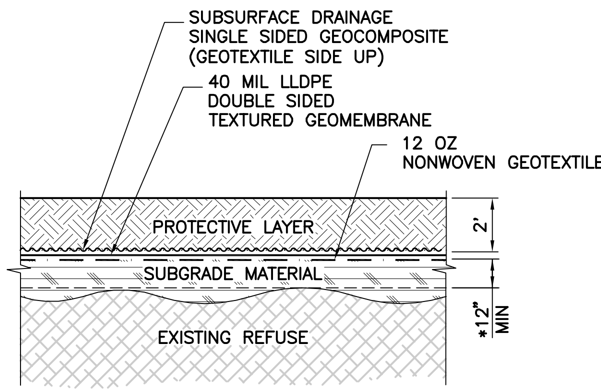
FINAL CLOSURE

EOD SECTIONS

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO. TRACKING#	
STA. PROJ. NO. #	
WORK ORDER NO. #	
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO. #	
SHEET 15 OF 15	

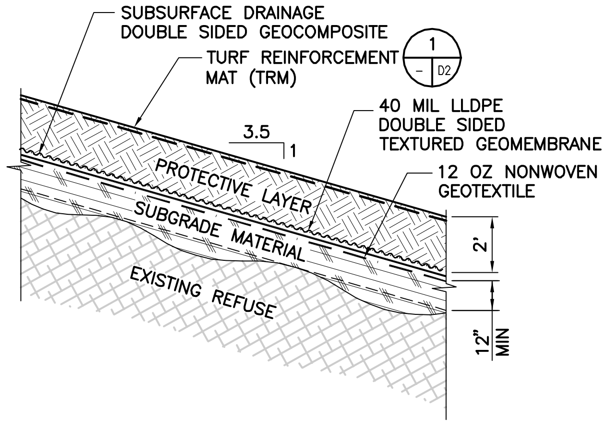
C-10

REV/DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\33-0023XSEC.dwg LAYOUT NAME: 22X34 PLOTTED: Wednesday, June 29, 2011 - 5:44pm



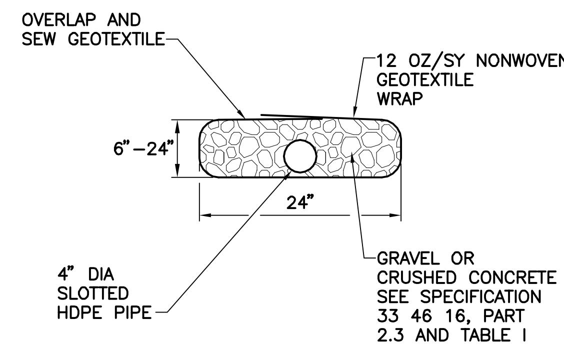
* THE FINAL COVER GRADING AND MATERIAL PROPERTIES SHALL CONFORM TO SECTION OF 31 00 00 EARTHWORK OF THE SPECIFICATIONS. THE SUBGRADE SHALL NOT CONTAIN ROCK GREATER THAN 1" AT THE SURFACE OR ROCKS GREATER THAN 3" IN THE UPPER 12"

DECK FINAL COVER
NTS
1 | C4,C5,C6,C7,C8,C9,D2,D3 | D1

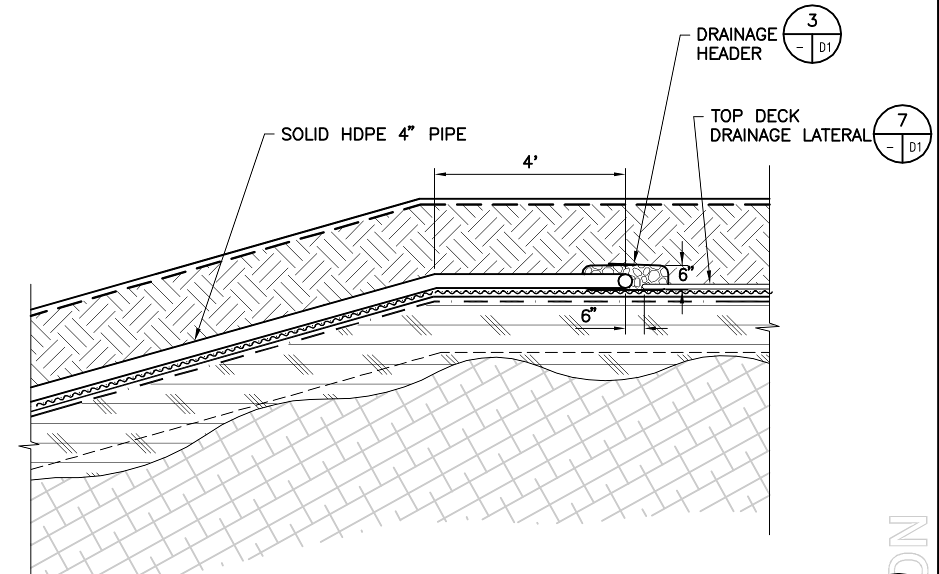


* THE FINAL COVER GRADING AND MATERIAL PROPERTIES SHALL CONFORM TO SECTION OF 31 00 00 EARTHWORK OF THE SPECIFICATIONS. THE SUBGRADE SHALL NOT CONTAIN ROCK GREATER THAN 1" AT THE SURFACE OR ROCKS GREATER THAN 3" IN THE UPPER 12"

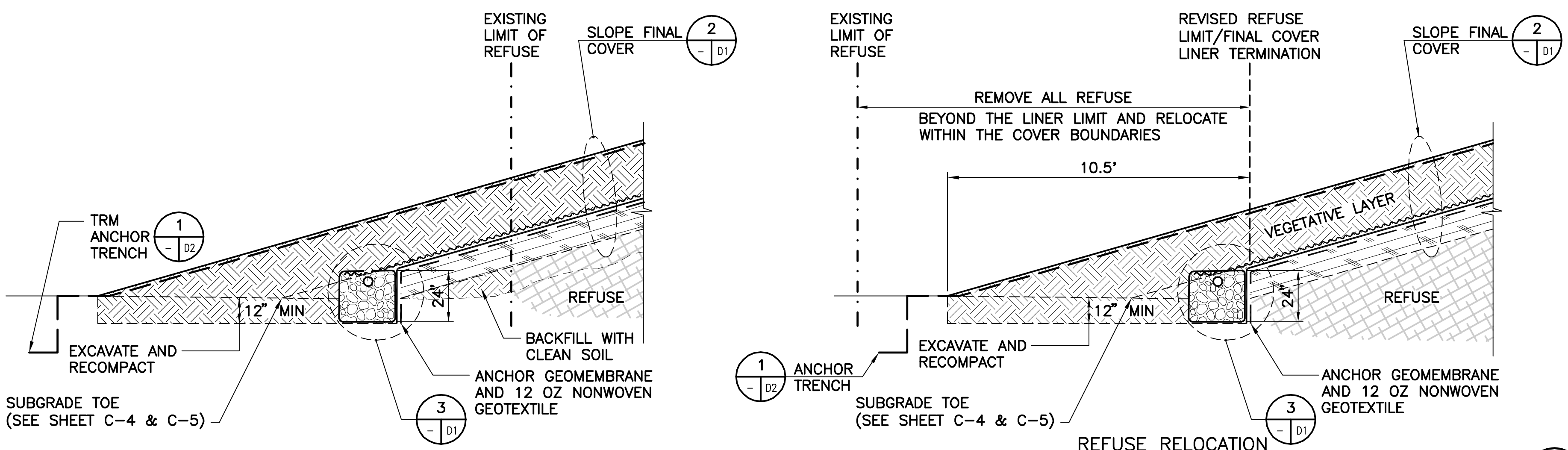
SLOPE FINAL COVER
NTS
2 | C4,C5,C6,C7,D1,D2,D3 | D1



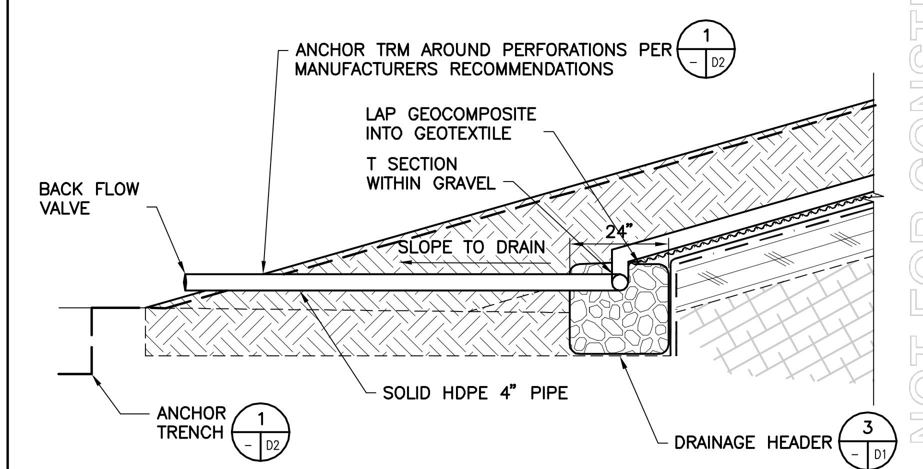
DRAINAGE HEADER
NTS
3 | C4,C5 | D1



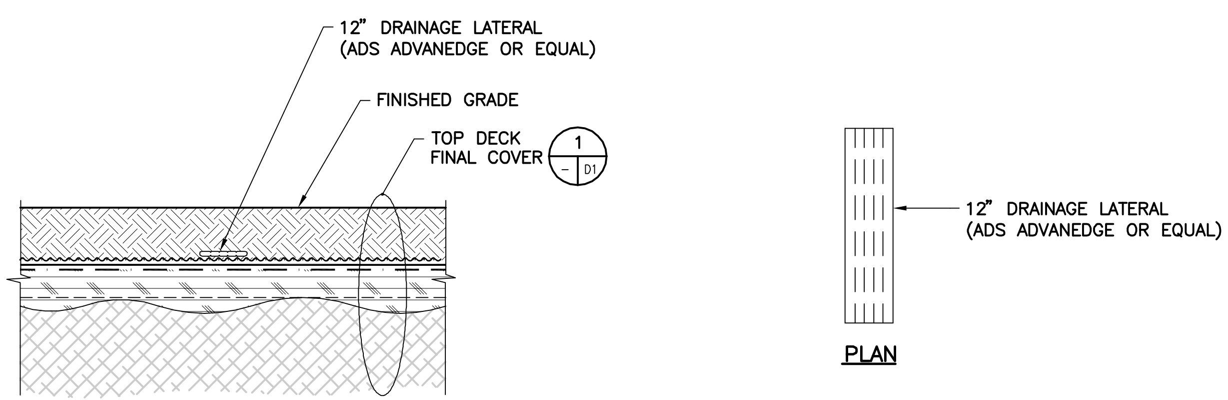
DRAINAGE HEADER OUTLET
NTS
4 | C4,C5 | D1



SLOPE LINER TERMINATION
NTS
5 | C4,C5 | D1



TOE DRAIN
NTS
6 | C4,C5 | D1



DRAINAGE LATERAL
NTS
7 | - | D1

SYM | DESCRIPTION | DATE | APPR

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc

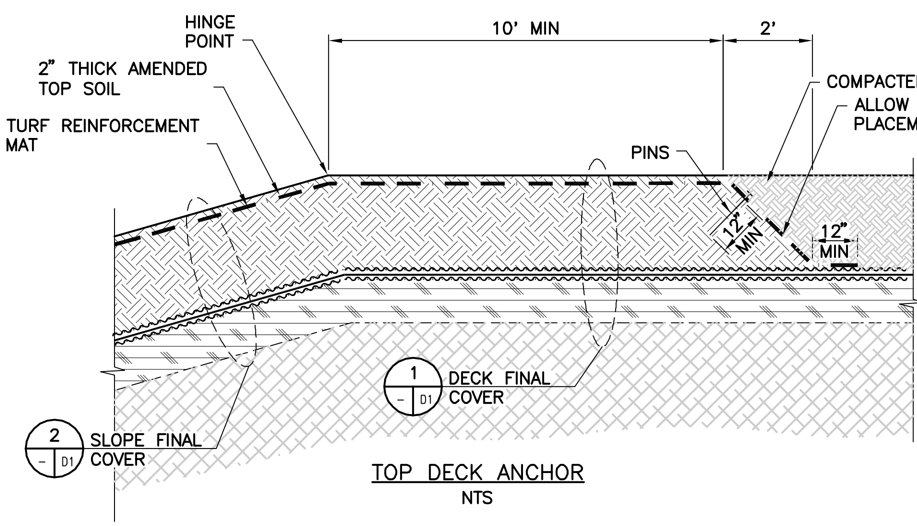
DES C.H.M. | DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
DETAILS

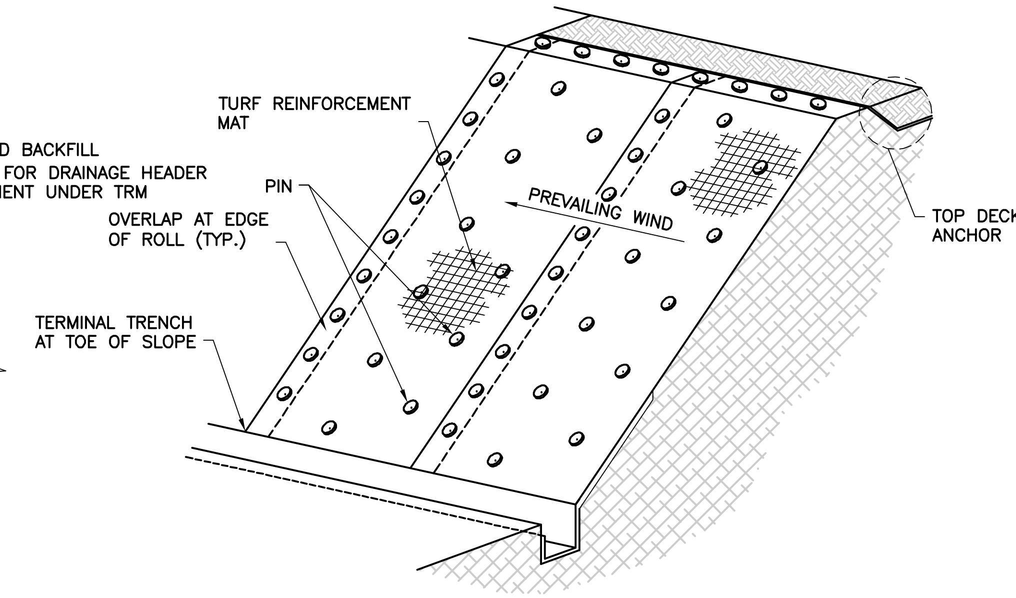
DRAFT FINAL - NOT FOR CONSTRUCTION

CODE ID. NO. 80091 | SIZE D
SCALE: AS SHOWN
MAXIMO NO. TRACKING#
STA. PROJ. NO. #
WORK ORDER NO. #
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#
DRAWING NO. #
SHEET 11 OF 15
D-1

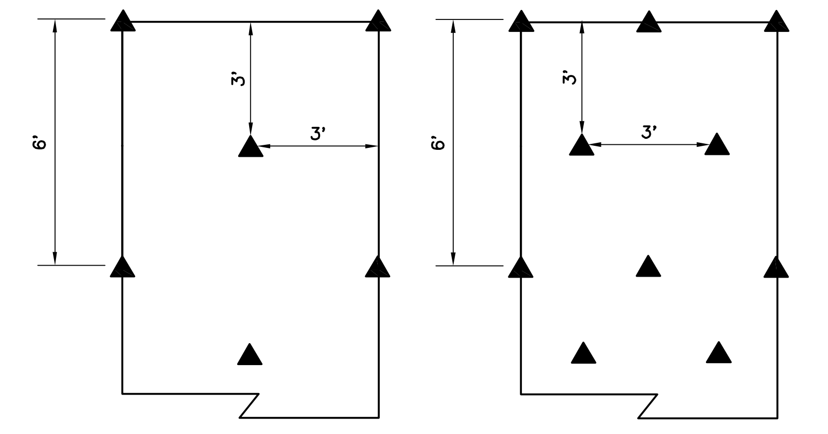
REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\31-0001DTL.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:14pm



TOP DECK ANCHOR
NTS



TURF REINFORCEMENT ON SLOPE
NTS

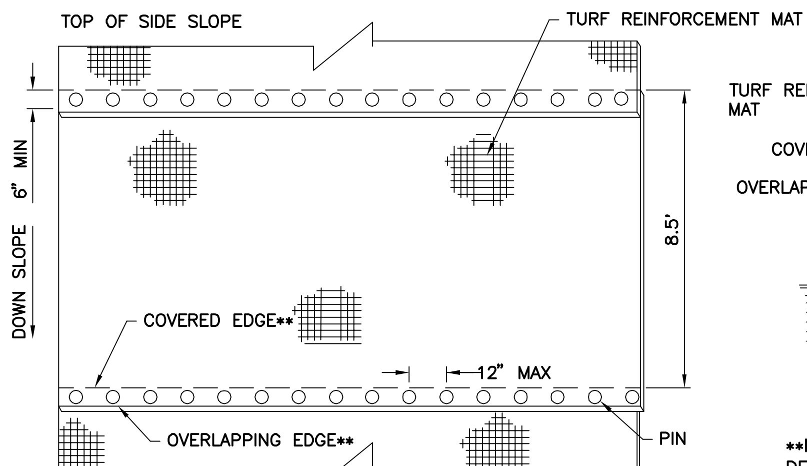


1 PINS/yd~2
TOP DECK PIN PATTERN

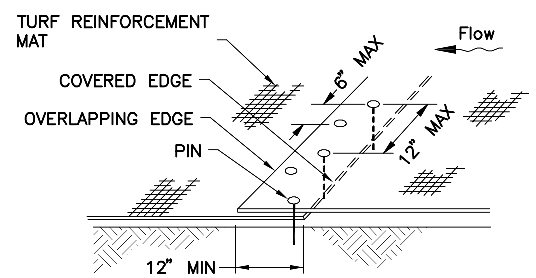
1.5 PINS/yd~2*
3.5:1 SLOPE PIN PATTERN

* WHERE PERFORATIONS FOR UNDER DRAIN PIPE OCCUR ANCHORING PER MANUFACTURERS RECOMMENTATIONS

PIN PATTERN
NTS

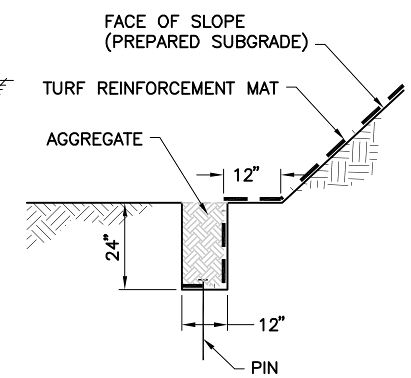


OVERLAP AT ROLL EDGE
NTS

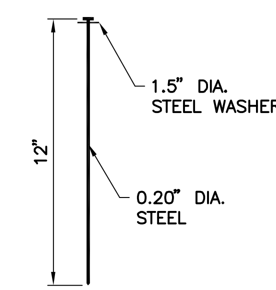


**NOTE: EDGE SHINGLE/OVERLAP PLACEMENT DEPENDS ON DOWN SLOPE DIRECTION (I.E SHINGLE IN THE DIRECTION OF THE DOWN SLOPE)

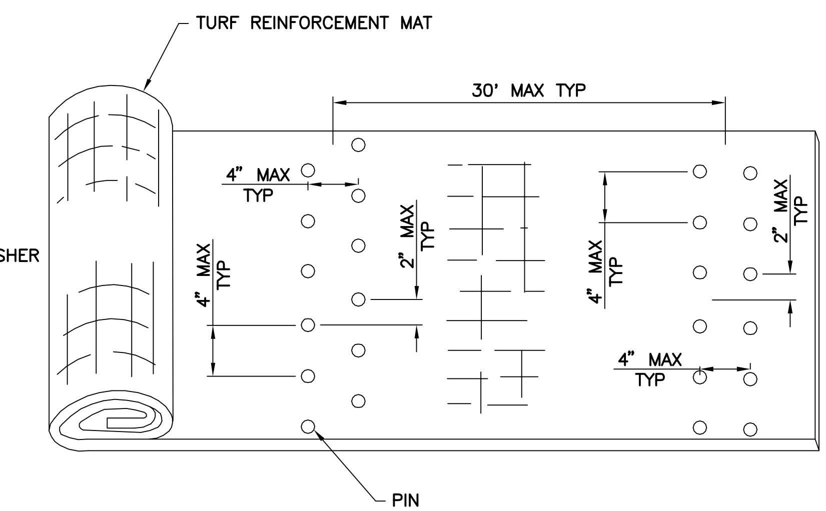
OVERLAP AT ROLL END
NTS



ANCHOR TRENCH AT TOE OF SLOPE
NTS



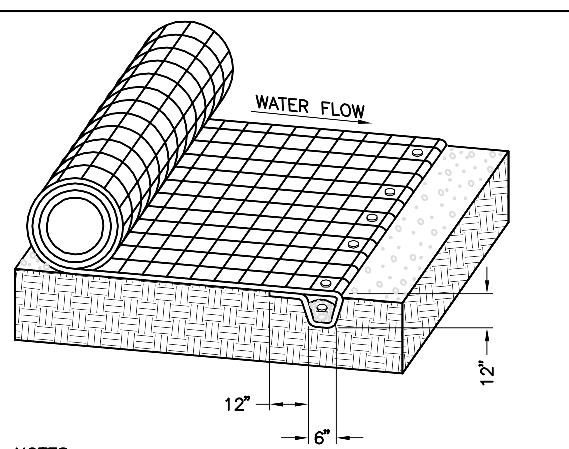
PIN
NTS



SIMULATED CHECK SLOT
NTS

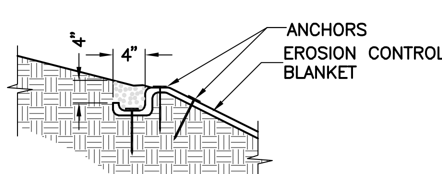
TURF REINFORCEMENT MAT
NTS

1 C6,C7,D1 D2



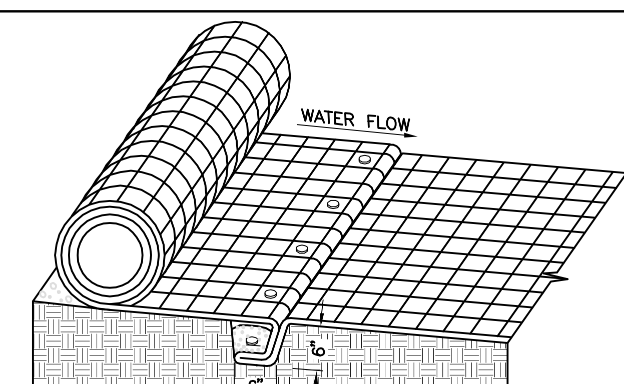
NOTES:
1. SECURE AT 12" INTERVALS, BACKFILL AND COMPACT SOIL

TERMINAL ANCHOR TRENCH



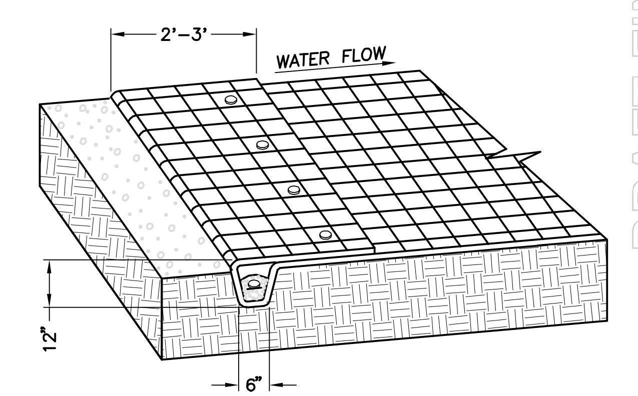
NOTES:
1. INSTALL EROSION CONTROL BLANKET AND SOIL FILL.
2. SECURE LONGITUDINAL ANCHOR TRENCH AT 12" INTERVALS, BACKFILL AND COMPACT SOIL

EDGE ANCHOR



NOTES:
1. SECURE AT 12" VERTICAL INTERVALS, BACKFILL AND COMPACT SOIL

INTERMITTENT CHECK SLOT



NOTES:
1. SECURE AT 12" VERTICAL INTERVALS, BACKFILL AND COMPACT SOIL

INITIAL ANCHOR TRENCH

EROSION CONTROL BLANKET

2 C6,C7 D2

DRAFT FINAL - NOT FOR CONSTRUCTION

SYM | DESCRIPTION | DATE | APPR

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc

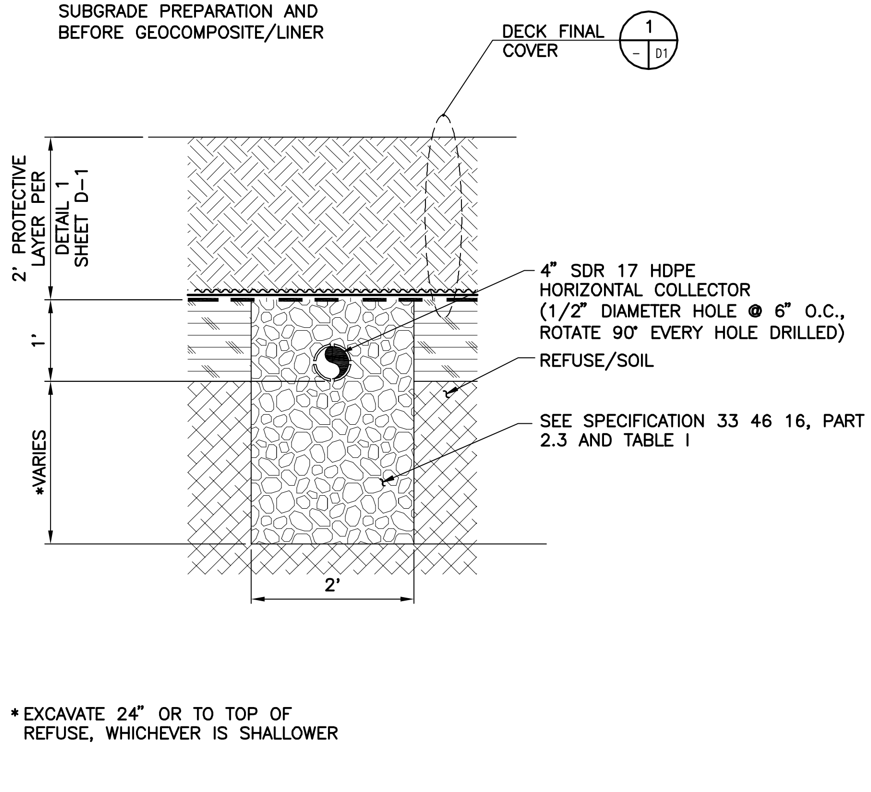
DES C.H.M. DRW J.M.B.
REVIEWED BY C.H.M.
PM/DM C.H.M.
CHIEF ENG B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
DETAILS

CODE ID. NO. 80091 SIZE D
SCALE: AS SHOWN
MAXIMO NO. TRACKING#
STA. PROJ. NO. #
WORK ORDER NO. #
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#
DRAWING NO. #
SHEET 12 OF 15
D-2

REV/DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\31-00020T1.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:15pm

NOTE:
COLLECTOR INSTALLED AFTER
SUBGRADE PREPARATION AND
BEFORE GEOCOMPOSITE/LINER

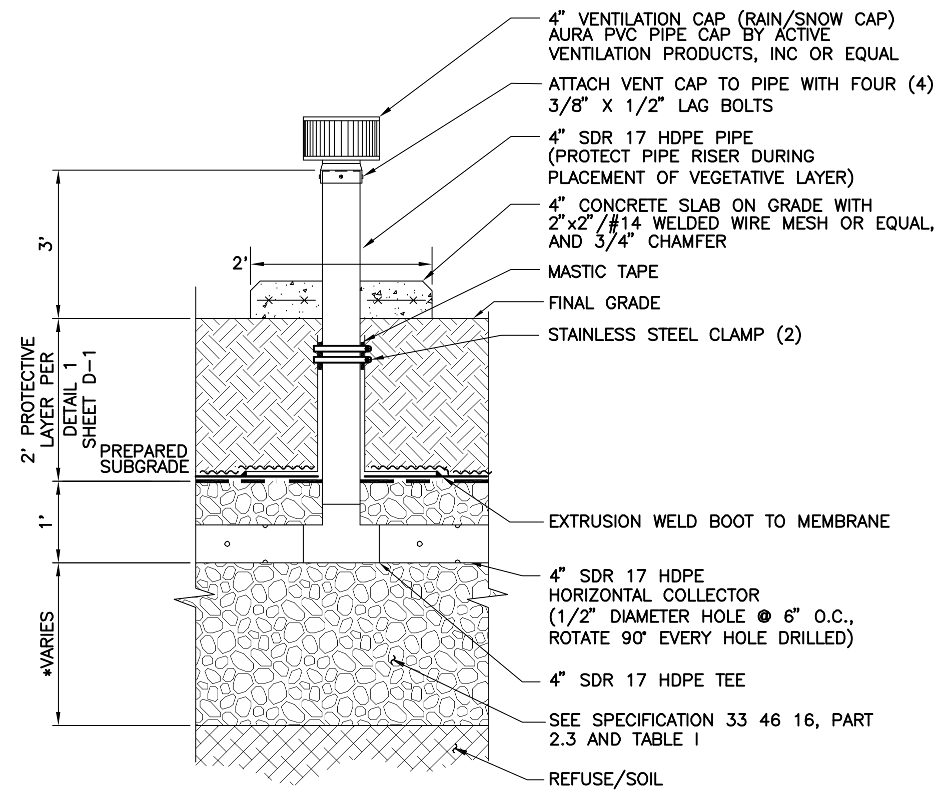


*EXCAVATE 24" OR TO TOP OF
REFUSE, WHICHEVER IS SHALLOWER

DECK HORIZONTAL GAS COLLECTOR TRENCH

NTS

1 C3 D3

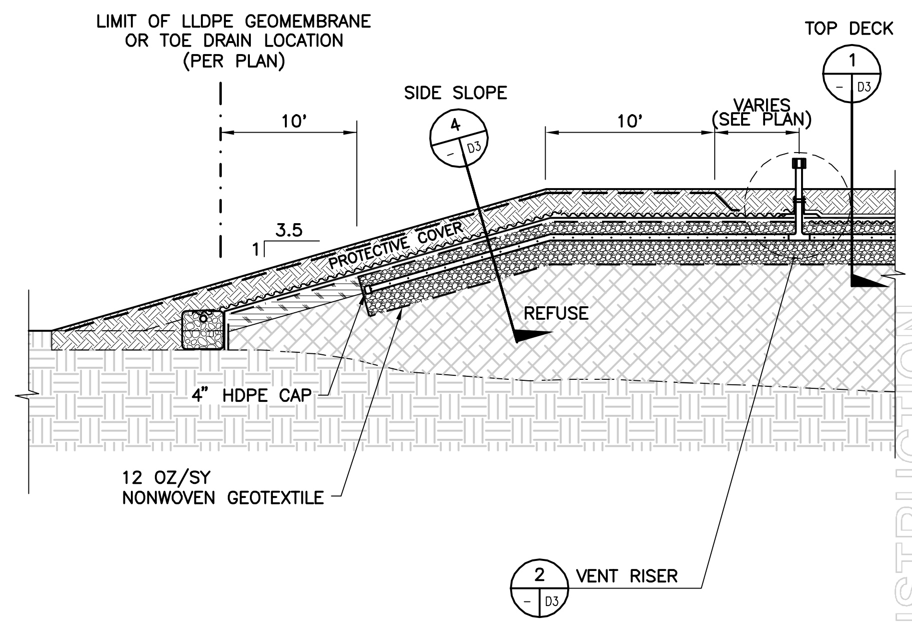


*EXCAVATE 24" OR TO TOP OF
REFUSE, WHICHEVER IS SHALLOWER

VENT RISER

NTS

2 C3 D3

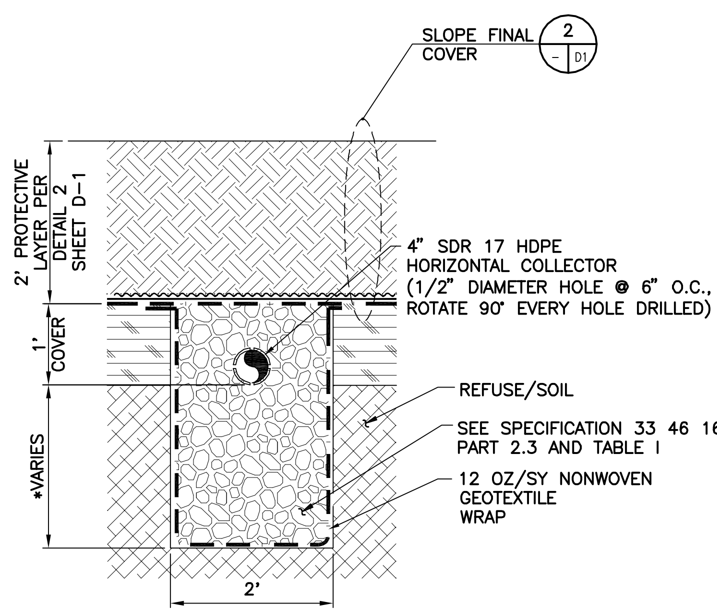


TYPICAL GAS COLLECTOR TERMINATION DETAIL

NTS

3 C3 D3

NOTE:
COLLECTOR INSTALLED AFTER
SUBGRADE PREPARATION AND
BEFORE GEOCOMPOSITE/LINER



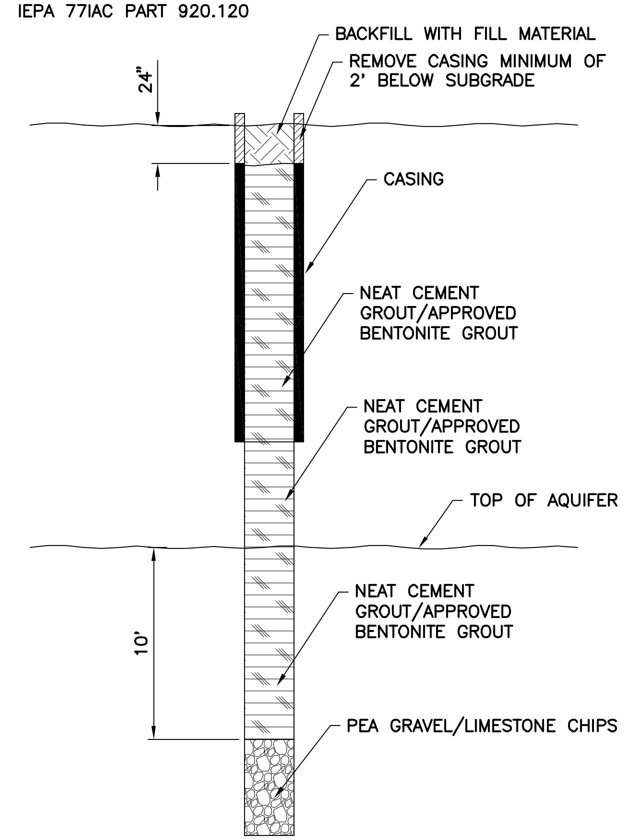
*EXCAVATE 24" OR TO TOP OF
REFUSE, WHICHEVER IS SHALLOWER

SLOPE HORIZONTAL GAS COLLECTOR TRENCH

NTS

4 C3 D3

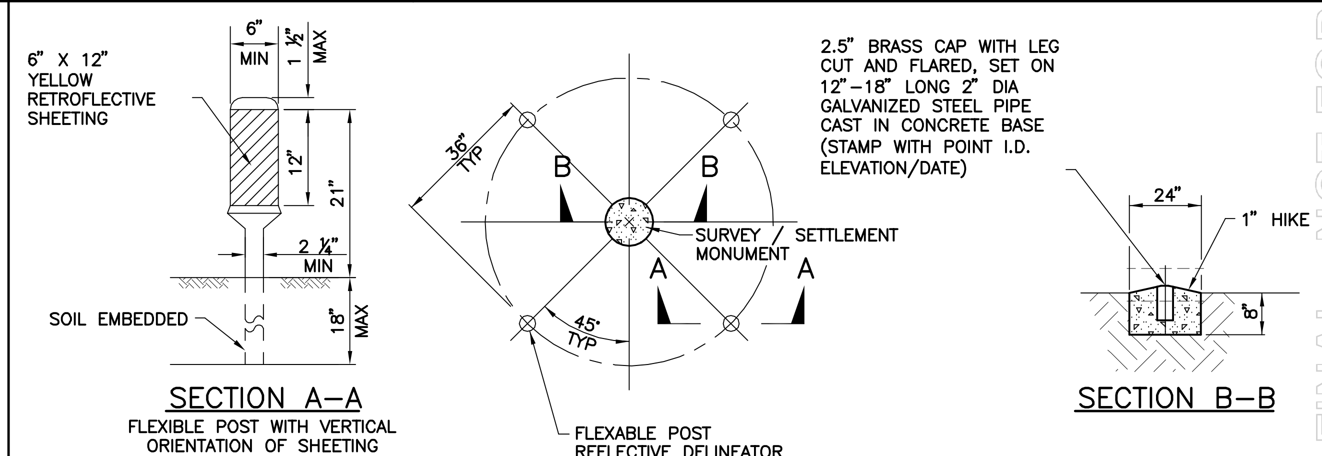
NOTE:
MUST BE ABANDONED PER
IEPA 77IAC PART 920.120



GROUNDWATER WELL ABANDONMENT

NTS

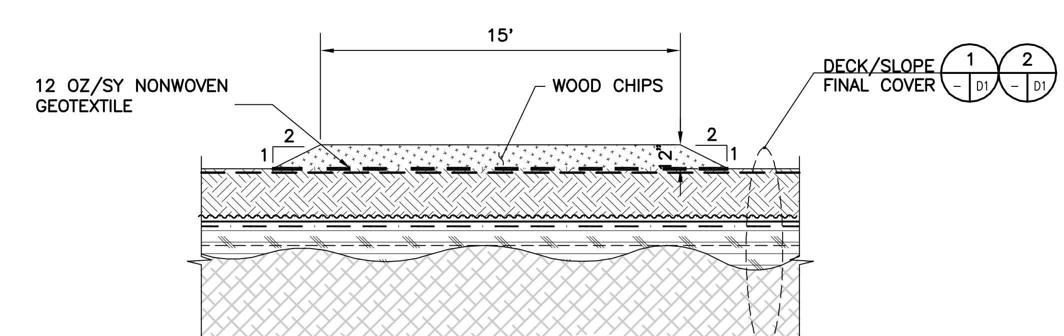
5 C2 D3



SETTLEMENT MARKER

NTS

6 C6,C9 D3



ACCESS ROAD

NTS

7 C6,C7 D3

DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc.

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

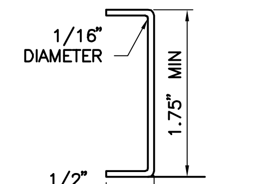
DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL 13
SCOTT AFB, IL
FINAL CLOSURE
DETAILS

CODE ID. NO. 80091 SIZE D
SCALE: AS SHOWN
MAXIMO NO. TRACKING#
STA. PROJ. NO. #
WORK ORDER NO. #
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#
DRAWING NO. #
SHEET 13 OF 15
D-3

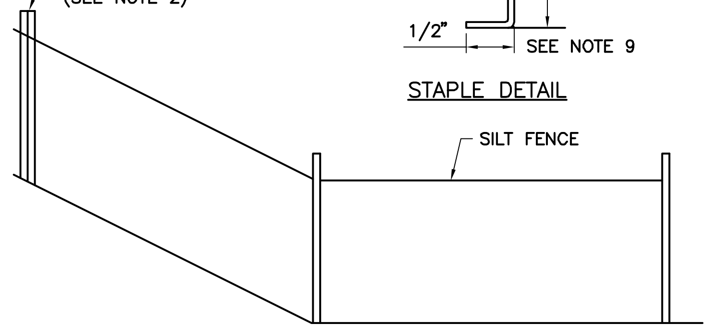
REV:DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\85-0001GSD.dwg LAYOUT NAME: 22x34 TAB PLOTTED: Wednesday, June 29, 2011 - 5:16pm

NOTES:

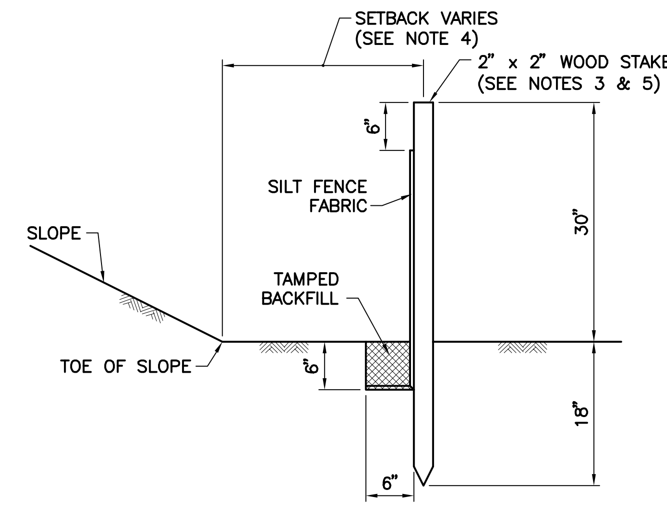
1. CONSTRUCT THE LENGTH OF EACH REACH SO THAT THE CHANGE IN BASE ELEVATION ALONG THE REACH DOES NOT EXCEED 1/3 THE HEIGHT OF THE LINEAR BARRIER, IN NO CASE SHALL THE REACH LENGTH EXCEED 490'.
2. THE LAST 8'-0" OF FENCE SHALL BE TURNED UP SLOPE.
3. STAKE DIMENSIONS ARE NOMINAL.
4. DIMENSIONS MAY VARY TO FIT FIELD CONDITION.
5. STAKES SHALL BE SPACED AT 8'-0" MAXIMUM AND SHALL BE POSITIONED ON DOWNSTREAM SIDE OF FENCE.
6. STAKES TO OVERLAP AND FENCE FABRIC TO FOLD AROUND EACH STAKE ONE FULL TURN. SECURE FABRIC TO STAKE WITH 4 STAPLES MINIMUM.
7. STAKES SHALL BE DRIVEN TIGHTLY TOGETHER TO PREVENT POTENTIAL FLOW-THROUGH OF SEDIMENT AT JOINT. THE TOPS OF THE STAKES SHALL BE SECURED WITH WIRE.
8. FOR END STAKE, FENCE FABRIC SHALL BE FOLDED AROUND TWO STAKES ON FULL TURN AND SECURED WITH 4 STAPLES.
9. MINIMUM 4 STAPLES PER STAKE. DIMENSIONS SHOWN ARE TYPICAL.
10. CROSS BARRIERS SHALL BE A MINIMUM OF 1/3 AND A MAXIMUM OF 1/2 THE HEIGHT OF THE LINEAR BARRIER.
11. MAINTENANCE OPENINGS SHALL BE CONSTRUCTED IN A MANNER TO ENSURE SEDIMENT REMAINS BEHIND SILT FENCE.
12. JOINING SECTIONS SHALL NOT BE PLACED AT SUMP LOCATIONS.
13. SANDBAG ROWS AND LAYERS SHALL BE OFFSET TO ELIMINATE GAPS.
14. ALL EROSION CONTROL MEASURES SHALL BE IN ACCORDANCE WITH THE SITE SWPPP.



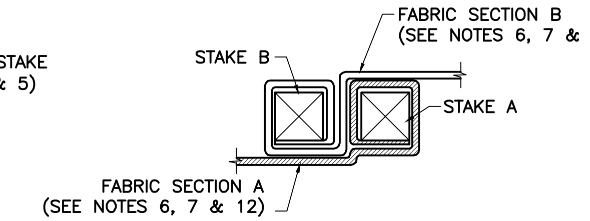
STAPLE DETAIL



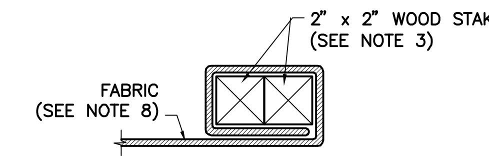
END DETAIL



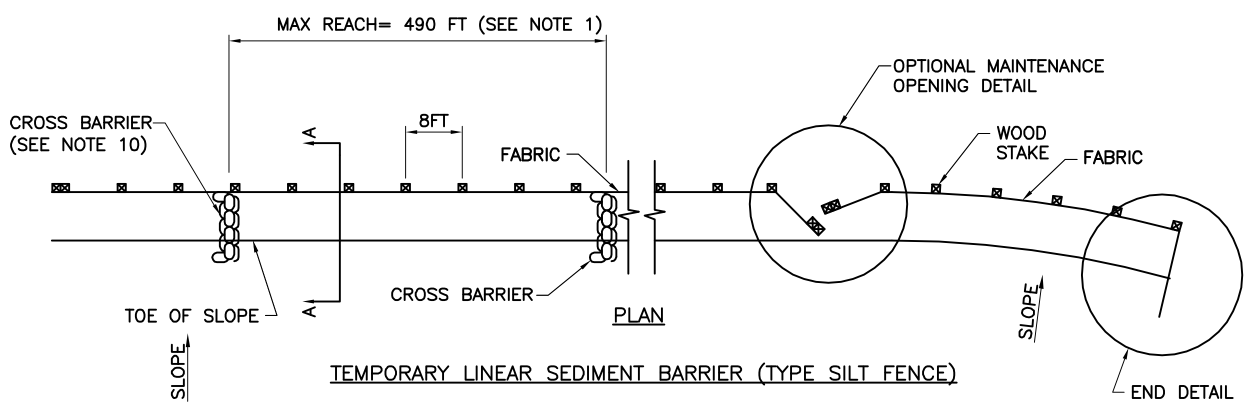
SECTION A-A
(SETBACK)



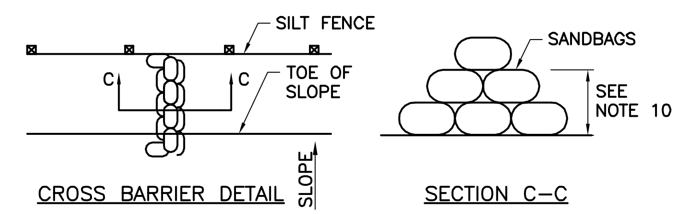
JOINING SECTION DETAIL
(TOP VIEW)



JOINING SECTION DETAIL
(TOP VIEW)



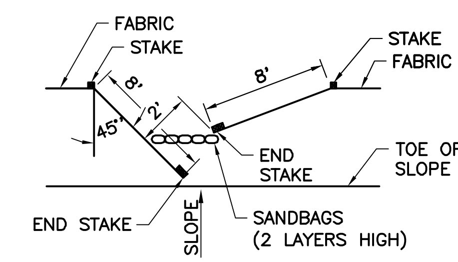
TEMPORARY LINEAR SEDIMENT BARRIER (TYPE SILT FENCE)



CROSS BARRIER DETAIL

SECTION C-C

TEMPORARY LINEAR SEDIMENT BARRIER (TYPE SILT FENCE)



OPTIONAL MAINTENANCE OPENING DETAIL
(SEE NOTE 11)

SILT FENCE
NTS

1 / D4 — C2,C6,C7,C8,C9

DETAILS
NTS
2 / – D4

DETAILS
NTS
3 / – D4

DETAILS
NTS
4 / – D4

DRAFT FINAL - NOT FOR CONSTRUCTION

TETRA TECH, INC.
CIVIL AND ENVIRONMENTAL ENGINEERS
1360 VALLEY VISTA DRIVE
DIAMOND BAR, CA. 91765
(909) 860-7777

HGL HydroGeoLogic, Inc

DES C.H.M.	DRW J.M.B.
REVIEWED BY	C.H.M.
PM/DM	C.H.M.
CHIEF ENG	B.A.S.

DEPARTMENT OF THE AIR FORCE
SCOTT AIR FORCE BASE LANDFILL
SCOTT AFB, IL
FINAL CLOSURE
DETAILS

CODE ID. NO. 80091	SIZE D
SCALE: AS SHOWN	
MAXIMO NO.	TRACKING#
STA. PROJ. NO.	#
WORK ORDER NO.	#
CONSTR. CONTR. NO. CONSTRUCTION CONTRACT#	
DRAWING NO.	#
SHEET 14 OF 15	

D-4

REV.DATE: FILE NAME: G:\dwg\SCOTT AFB\FINAL CLOSURE\PLAN SETS\31-0014DTL.dwg LAYOUT NAME: 22X34 PLOTTED: Wednesday, June 29, 2011 - 5:17pm

APPENDIX B

CONSTRUCTION DETAILS FOR MONITORING WELLS, GAS WELLS, AND COVER SETTLEMENT MONUMENTS

This page was intentionally left blank.

APPENDIX B

CONSTRUCTION DETAILS FOR MONITORING WELLS, GAS WELLS, AND COVER SETTLEMENT MONUMENTS

This page was intentionally left blank.

Table B.1
Monitoring Well Construction Summary
Landfill LF-01
Scott AFB, IL

Well Identification	Date Installed	Well Coordinates		Well Material	Top of Casing Elevation	Ground Surface Elevation	Borehole Depth	Well Stickup	Screening Interval					Soil Material in Screen Interval	
		Northing	Easting						Depth to Top of Screen	Top Elevation	Depth to Bottom of Screen	Bottom Elevation	Length	USCS Symbol	Description
					ft amsl	ft amsl	ft bgs	ft	ft bgs	ft amsl	ft bgs	ft amsl	ft		
LF-01 Monitoring Wells															
LF01-MW01S	10/19/1988	681807.85	2392284.96	2" Sch 40 PVC	418.58	416.53	30.50	2.05	17.97	398.56	27.95	388.58	9.98	SW	Fine SAND
LF01-MW01D	10/18/1988	681804.56	2392295.48	2" Sch 40 PVC	418.80	416.89	62.30	1.91	39.60	377.29	59.60	357.29	20.00	CL-ML; SP	Silty Clay; Fine-Coarse SAND
LF01-MW02	10/20/1988	681563.43	2393031.48	2" Sch 40 PVC	419.74	417.51	26.90	2.23	14.65	402.86	24.65	392.86	10.00	SM	Silty SAND
LF01-MW03S	10/23/1988	681195.52	2393324.78	2" Sch 40 PVC	419.59	417.27	32.30	2.32	19.90	397.37	29.90	387.37	10.00	SP	Fine-Coarse SAND
LF01-MW03D	10/22/1988	681186.67	2393339.45	2" Sch 40 PVC	418.91	416.94	62.60	1.97	40.90	376.04	60.90	356.04	20.00	SM; SW	Silty Fine SAND; Fine SAND
LF01-MW11	11/02/1988	679861.01	2393146.53	2" Sch 40 PVC	427.65	425.23	45.50	2.42	33.60	391.63	43.60	381.63	10.00	SM	Silty Fine-Coarse SAND; Sandy SILT
LF01-MW12S	5/10/2001	679805.86	2393301.55	2" Sch 40 PVC	421.50	418.14	23.80	3.36	8.00	410.14	18.00	400.14	10.00	ML; CL	SILT; CLAY
LF01-MW12D	5/09/2001	679810.42	2393292.10	2" Sch 40 PVC	421.50	419.29	92.00	2.21	55.30	363.99	60.30	358.99	5.00	SM	Silty Fine-Medium SAND
LF01-MW12SR	8/24/2010	679764.76	2393140.74	2" Sch 40 PVC	419.55	416.80	34.00	2.95	19.00	397.80	34.00	382.80	15.00	GC; CH; SC	Clayey gravel, Clay high plasticity, Clayey sand
LF01-MW17S	5/10/2001	679902.57	2393669.48	2" Sch 40 PVC	421.93	418.86	24.00	3.07	10.00	408.86	20.00	398.86	10.00	ML; CL	SILT; CLAY
LF01-MW17D	5/10/2001	679893.73	2393658.96	2" Sch 40 PVC	421.72	418.52	95.00	3.20	54.80	363.72	59.80	358.72	5.00	SP	Fine-Medium SAND; Some Silt
LF01-MW17SR	8/25/2010	679851.24	2393499.60	2" Sch 40 PVC	419.99	417.00	39.50	2.50	23.00	394.00	38.00	379.00	15.00	ML; CL	SILT; CLAY
LF01-MW18S	2/08/2001	680278.95	2394359.46	2" Sch 40 PVC	421.13	417.37	20.20	3.76	6.00	411.37	16.00	401.37	10.00	SC; CL	Sandy CLAY; CLAY
LF01-MW18D	2/08/2001	680281.31	2394364.02	2" Sch 40 PVC	421.11	417.32	40.20	3.79	31.00	386.32	36.00	381.32	5.00	SP; SC	Fine-Coarse SAND; Clayey SAND
LF01-MW19S	3/19/2001	681009.96	2394655.16	2" Sch 40 PVC	422.56	419.31	19.30	3.25	5.50	413.81	15.50	403.81	10.00	MCL-ML; SC	Silty CLAY; Sandy CLAY
LF01-MW19D	3/19/2001	681004.44	2394661.50	2" Sch 40 PVC	422.72	419.31	41.30	3.41	33.00	386.31	38.00	381.31	5.00	SP	Fine-Coarse SAND
LF01-MW20S	2/07/2001	681373.23	2394353.36	2" Sch 40 PVC	422.33	418.37	22.40	3.96	8.00	410.37	18.00	400.37	10.00	SC	Sandy CLAY; Clayey SAND
LF01-MW20D	2/07/2001	681376.34	2394358.31	2" Sch 40 PVC	422.38	418.46	31.80	3.92	32.00	386.46	37.00	381.46	5.00	SP	Fine-Coarse SAND
LF01-MW21S	3/20/2001	681565.20	2393501.95	2" Sch 40 PVC	419.94	416.37	19.10	3.57	5.50	410.87	15.50	400.87	10.00	CL-ML; SC	Silty CLAY; Sandy CLAY
LF01-MW21D	3/20/2001	681573.16	2393499.48	2" Sch 40 PVC	419.88	416.38	39.90	3.50	31.00	385.38	36.00	380.38	5.00	CL; SW	CLAY; Coarse SAND
LF01-MW22S	9/23/2009	679714.43	2392272.83	2" Sch 40 PVC	437.60	434.78	34.00	2.82	24.00	410.78	34.00	400.78	10.00	SW; SP	Coarse SAND; Fine-Coarse SAND
LF01-MW22D	9/22/2009	679717.13	2392280.24	2" Sch 40 PVC	437.85	434.87	89.30	2.98	53.00	381.87	63.00	371.87	10.00	CL; SP	CLAY; Fine-Coarse SAND
LF01-MW23S	9/16/2009	679727.53	2393323.27	2" Sch 40 PVC	418.64	415.84	37.50	2.80	24.00	391.84	34.00	381.84	10.00	SW; CL	Coarse SAND; CLAY
LF01-MW23D	9/17/2009	679724.81	2393317.48	2" Sch 40 PVC	418.86	416.07	59.00	2.79	49.00	367.07	59.00	357.07	10.00	ML; SW	SILT; Coarse SAND
LF01-MW24S	6/10/2011	679522.00	2393040.54	2" Sch 40 PVC	420.89	417.70	33.00	3.19	23.00	394.70	33.00	384.70	10.00	SW; SP	Fine SAND; Fine-Medium SAND; Some Silt
LF01-MW24D	6/10/2011	679529.18	2393038.17	2" Sch 40 PVC	420.52	417.60	78.00	2.92	60.00	357.60	70.00	347.60	10.00	SW; SP	Coarse SAND; Fine-Coarse SAND
LF01-MW25S	7/12/2011	679561.82	2392927.24	2" Sch 40 PVC	420.56	417.60	38.00	2.96	24.70	392.90	34.70	382.90	10.00	SP; SW	Fine-Coarse SAND; Fine SAND
LF01-MW25D	7/13/2011	679555.28	2392925.74	2" Sch 40 PVC	420.56	417.50	83.00	3.06	72.00	345.50	82.00	335.50	10.00	ML; SW	SILT; Fine-Coarse SAND
LF01-MW26S	6/10/2011	679345.30	2392891.17	2" Sch 40 PVC	420.33	417.50	33.00	2.83	21.50	396.00	31.50	386.00	10.00	SP	Fine SAND; Medium SAND; Some Silt; Fine Gravel
LF01-MW26D	6/13/2011	679341.88	2392903.63	2" Sch 40 PVC	420.43	417.50	81.90	2.93	70.00	347.50	80.00	337.50	10.00	SP	Medium SAND; Some Silt; Fine Gravel
LF01-MW27	8/31/2011	678998.13	2392332.67	2" Sch 40 PVC	420.04	417.20	30.50	2.84	20.00	397.20	30.00	387.20	10.00	SC; SP	Sandy CLAY; Fine-Medium SAND
LF01-MW28	8/30/2011	679173.42	2393714.84	2" Sch 40 PVC	418.66	415.80	34.50	2.86	24.00	391.80	34.00	381.80	10.00	SP	Fine-Coarse SAND
LF01-MW29	8/31/2011	678545.41	2392803.11	2" Sch 40 PVC	418.89	416.00	30.50	2.89	20.00	396.00	30.00	386.00	10.00	SP; ML; CL	Fine-Coarse SAND; Sandy Silt; Clay

" = inch.
amsl = above mean sea level
bgs = below ground surface
D = deep well
ft = feet.
NA = not available
PVC = polyvinyl chloride
S = shallow well
Sch = schedule
SR = shallow replacement well
USCS = Unified Soil Classification System

This page was intentionally left blank.

Table B.2
Proposed Monitoring Well Construction
Landfill LF-01
Scott AFB, IL

Well Identification	Proposed Well Coordinates		Screened Transmissive Zone	Proposed Screening Interval			Description
	Northing	Easting		Depth to Top of Screen feet bgs	Depth to Bottom of feet bgs	Length feet	
Proposed Long Term Monitoring Wells for Site LF-01							
LF01-MW30D	2393261.69	680572.30	Deep Sand Unit	52	62	10	2" Sch 40 PVC
LF01-MW30S	2393253.24	680584.50	Shallow Sand Unit	18	28	10	2" Sch 40 PVC
LF01-MW31D	2393364.15	680142.84	Deep Sand Unit	53	63	10	2" Sch 40 PVC
LF01-MW31S	2393350.44	680167.96	Shallow Sand Unit	22	32	10	2" Sch 40 PVC
LF01-MW32D	2393241.64	2393241.64	Deep Sand Unit	53	63	10	2" Sch 40 PVC
LF01-MW32S	2393216.37	679966.75	Shallow Sand Unit	22	32	10	2" Sch 40 PVC
LF01-MW33D	2392306.60	679594.65	Deep Sand Unit	60	70	10	2" Sch 40 PVC
LF01-MW33S	2392292.90	679594.65	Shallow Sand Unit	40	50	10	2" Sch 40 PVC
LF01-MW34D	2391602.23	681733.89	Deep Sand Unit	52	62	10	2" Sch 40 PVC
LF01-MW34S	2391587.05	681746.10	Shallow Sand Unit	18	28	10	2" Sch 40 PVC
LF01-MW35D	2391791.27	680992.12	Deep Sand Unit	52	62	10	2" Sch 40 PVC
LF01-MW35S	2391805.11	680977.85	Shallow Sand Unit	18	28	10	2" Sch 40 PVC
LF01-MW36D	2392806.09	681656.86	Deep Sand Unit	50	60	10	2" Sch 40 PVC
LF01-MW36S	2392792.26	681671.13	Shallow Sand Unit	15	25	10	2" Sch 40 PVC
LF01-MW37D	2393127.97	681400.29	Deep Sand Unit	50	60	10	2" Sch 40 PVC
LF01-MW37S	2393108.29	681400.89	Shallow Sand Unit	15	25	10	2" Sch 40 PVC
LF01-MW38D	2393297.99	680849.73	Deep Sand Unit	52	62	10	2" Sch 40 PVC
LF01-MW38S	2393295.69	680867.35	Shallow Sand Unit	18	28	10	2" Sch 40 PVC
LF01-MW39D	2392705.27	679672.27	Deep Sand Unit	72	82	10	2" Sch 40 PVC
LF01-MW39S	2392668.55	679675.44	Shallow Sand Unit	25	35	10	2" Sch 40 PVC

Notes:
1. Wells will be constructed from 2-inch Schedule 40 PVC per Basewide FSP.
2. Proposed locations may change based on field conditions per the field geologist.

" = inch.
bgs = below ground surface
D = deep well
COCs = contaminant of concern
PVC = polyvinyl chloride
S = shallow well
Sch = schedule

Table B.3
Landfill Gas Well Construction
Landfill LF-01
Scott AFB, IL

Gas Well Identification	Date Installed	Coordinates		Ground Surface Elevation	Borehole Depth	Screen Interval					Soil Material in Screen Interval	
		North	East	(feet amsl)	(feet bgs)	Top of Screen (feet)	Top Elevation (feet amsl)	Bottom of Screen (feet)	Bottom Elevation (feet amsl)	Length (feet)	USCS Symbol	Description
LF-01 Landfill Gas Wells												
LF01-GW20	09/15/09	680825.17	2392081.38	422.6	14	4	418.6	14	408.6	10	ML	SILT
LF01-GW27	09/02/09	680535.89	2392079.76	432.6	24	3	429.6	13	419.6	10	ML	Clay-SILT
LF01-GW32	09/02/09	680226.57	2392148.68	433.4	24	12.5	420.9	22.5	410.9	10	ML	Clayey-SILT
LF01-GW38	09/02/09	680014.71	2392133.67	433.1	22.5	12.5	420.6	22.5	410.6	10	ML	Clayey-SILT
LF01-GW41	09/02/09	679857.61	2392271.38	433.8	24	13.5	420.3	23.5	410.3	10	ML	Clayey-SILT

Wells constructed from 1-inch schedule 40 PVC.
amsl = above mean sea level.
ft = feet.
bgs = below ground surface.
USCS = Unified Soil Classification System.
PVC = polyvinyl chloride.

Table B.4
Settlement Marker Survey
Landfill LF-01
Scott AFB, IL

Settlement Marker Identification	Easting (X), feet	Northing (Y), feet	Elevation (Z), feet amsl
North Cell			
LF01-NC-01	2392336.33	681279.36	433.91
LF01-NC-02	2392235.34	681230.38	432.20
LF01-NC-03	2392267.43	681040.09	429.74
LF01-NC-04	2392550.59	681380.73	430.34
LF01-NC-05	2392553.01	681211.65	433.47
LF01-NC-06	2392436.66	680881.48	429.83
LF01-NC-07	2392755.22	681278.88	429.56
LF01-NC-08	2392650.17	680933.10	433.23
LF01-NC-09	2392632.38	680727.12	429.37
LF01-NC-10	2393001.81	681255.47	424.68
LF01-NC-11	2392397.44	680698.70	426.44
LF01-NC-12	2392863.02	680780.28	431.64
LF01-NC-13	2393097.06	681038.88	425.29
LF01-NC-14	2393069.86	680798.89	427.98
South Cell			
LF01-SC-01	2392332.03	680127.20	434.96
LF01-SC-02	2392561.30	680327.87	428.34
LF01-SC-03	2392458.33	680051.53	435.12
LF01-SC-04	2392553.38	679830.23	434.91
LF01-SC-05	2392748.72	680250.94	428.62
LF01-SC-06	2392781.52	680019.89	431.96
LF01-SC-07	2392834.10	679874.77	429.57
LF01-SC-08	2392892.49	680158.04	429.44
LF01-SC-09	2392974.75	680305.04	425.95
LF01-SC-10	2393041.83	680087.59	428.59

Table B.5
Seed Mixture
Landfill LF-01
Scott AFB, Illinois

Common Name	Botanical Name	Minimum Percent Pure Seed	Minimum Percent Germination and Hard Seed	Maximum Percent Weed Seed	Pounds per acre PLS	Percent (by weight)
Top Deck Seed Mix						
Creeping Red Fescue	*Festuca rubra*	97	83	0.25	25	30
Timothy Grass	*Phleum pratense*	95	85	0.25	16	20
Kentucky Bluegrass	*Poa pratensis*	95	85	0.25	20	24
White Ladino Clover	*Trifolium repens*	92	87	0.30	16	20
Black Eyed Susan	*Rudbeckia hirta*	98	85	0.10	5	6
Sideslope Application Seed Mix						
Creeping Red Fescue	*Festuca rubra*	97	83	0.25	30	45
Timothy Grass	*Phleum pratense*	95	85	0.25	16	25
Kentucky Bluegrass	*Poa pratensis*	95	85	0.25	20	30
Temporary Cover Crop Seed Mix						
Oat, Wheat, or Cereal Rye	--	--	--	--	90	100
Annual Oats	*Avena sativa*	92	88	0.50	--	--
Annual or Red Wheat	*Triticum*	92	89	0.50	--	--
Annual Ryegrass	*Lolium multiflorum*	97	85	0.50	--	--

Notes:
Seed mix in accordance Engineering Specification 31 32 11 of the LF-01 Remedial Action Work Plan.
PLS = pure live seed

Table B.6
North Cell Survey Features
Landfill LF-01
Scott AFB, Illinois

FID	Type	X Coordinates (feet)	Y Coordinates (feet)	Elevation (feet above mean sea level)
1300	Backflow Prevention Valve	2392164.43	680920.99	421.13
1303	Backflow Prevention Valve	2392108.48	681049.57	423.08
1306	Backflow Prevention Valve	2392592.67	681523.92	421.96
1309	Backflow Prevention Valve	2392969.81	681462.27	418.91
1312	Backflow Prevention Valve	2393023.30	681399.93	418.45
1315	Backflow Prevention Valve	2393060.48	681305.14	419.06
1318	Backflow Prevention Valve	2393107.94	681216.33	420.64
1321	Backflow Prevention Valve	2393153.03	681129.78	420.70
1324	Backflow Prevention Valve	2393209.11	681045.85	419.66
1327	Backflow Prevention Valve	2393197.77	680846.19	420.44
1330	Backflow Prevention Valve	2393188.57	680721.67	421.23
1333	Backflow Prevention Valve	2392835.40	680504.93	421.84
1336	Backflow Prevention Valve	2392736.79	680526.54	420.35
1339	Backflow Prevention Valve	2392638.55	680558.56	420.72
1342	Backflow Prevention Valve	2392545.23	680585.86	420.48
1345	Backflow Prevention Valve	2392449.19	680611.10	420.64
1348	Backflow Prevention Valve	2392348.89	680652.46	422.29
3073	Benchmark	2392336.33	681279.36	433.91
3076	Benchmark	2392235.34	681230.38	432.20
3079	Benchmark	2392267.43	681040.09	429.74
3082	Benchmark	2392436.66	680881.48	429.83
3085	Benchmark	2392632.38	680727.12	429.37
3088	Benchmark	2392863.02	680780.28	431.64
3091	Benchmark	2393069.86	680798.89	427.98
3094	Benchmark	2393097.06	681038.88	425.29
3097	Benchmark	2393001.81	681255.47	424.68
3100	Benchmark	2392755.22	681278.88	429.56
3103	Benchmark	2392550.59	681380.73	430.34
3106	Benchmark	2392553.01	681211.65	433.47
3109	Benchmark	2392650.17	680933.10	433.23
3110	Brass Cap in PVC Pipe	2392397.44	680698.70	426.44
3112	Gas Vent Riser	2392535.04	681251.63	432.71
3113	Gas Vent Riser	2392433.83	681251.88	433.85
3114	Gas Vent Riser	2392334.03	681252.05	433.62
3115	Gas Vent Riser	2392234.38	681252.11	432.11
3116	Gas Vent Riser	2392633.76	681251.95	431.61
3117	Gas Vent Riser	2392734.43	681251.46	430.13
3118	Gas Vent Riser	2392834.50	681250.49	428.26

Table B.6
North Cell Survey Features
Landfill LF-01
Scott AFB, Illinois

FID	Type	X Coordinates (feet)	Y Coordinates (feet)	Elevation (feet above mean sea level)
3119	Gas Vent Riser	2392934.03	681085.09	428.22
3120	Gas Vent Riser	2393034.48	680895.66	427.27
3121	Gas Vent Riser	2392933.96	680738.49	430.68
3122	Gas Vent Riser	2393133.64	680728.53	427.35
3123	Gas Vent Riser	2392833.95	680826.52	431.46
3124	Gas Vent Riser	2392733.46	680912.25	432.98
3125	Gas Vent Riser	2392634.06	680913.89	432.73
3126	Gas Vent Riser	2392534.39	680913.83	431.03
3127	Gas Vent Riser	2392434.29	680914.38	429.37
3128	Gas Vent Riser	2392334.58	680914.42	428.35
3129	Gas Vent Riser	2392232.90	680992.92	427.93

Table B.7
South Cell Survey Features
Landfill LF-01
Scott AFB, Illinois

FID	Type	X Coordinates (feet)	Y Coordinates (feet)
2409	Gas Vent Riser	2392408.80	680209.46
2410	Gas Vent Riser	2392258.50	680154.32
2411	Gas Vent Riser	2392408.72	679894.44
2412	Gas Vent Riser	2392558.60	679944.00
2413	Gas Vent Riser	2392708.88	680010.04
2414	Gas Vent Riser	2392558.87	680262.61
2415	Gas Vent Riser	2392709.18	680315.67
2416	Gas Vent Riser	2392859.28	680062.57
2417	Gas Vent Riser	2392859.07	679763.19
2418	Gas Vent Riser	2393009.11	679943.37
2419	Gas Vent Riser	2393009.91	680210.63
3033	Backflow Prevention Valve	2392998.33	679742.95
3034	Backflow Prevention Valve	2393048.43	679829.48
3035	Backflow Prevention Valve	2393092.69	679906.51
3036	Backflow Prevention Valve	2393137.75	680006.10
3037	Backflow Prevention Valve	2393183.97	680094.95
3038	Backflow Prevention Valve	2393226.76	680164.59
3039	Backflow Prevention Valve	2393281.74	680218.97
3040	Backflow Prevention Valve	2393319.91	680296.45
3041	Backflow Prevention Valve	2393248.18	680330.19
3042	Backflow Prevention Valve	2393150.33	680359.33
3043	Backflow Prevention Valve	2393054.78	680385.69
3044	Backflow Prevention Valve	2392959.16	680414.70
3045	Backflow Prevention Valve	2392864.04	680444.88
3046	Backflow Prevention Valve	2392769.54	680474.18
3047	Backflow Prevention Valve	2392674.09	680505.89
3048	Backflow Prevention Valve	2392564.92	680514.22
3049	Backflow Prevention Valve	2392490.17	680487.27
3050	Backflow Prevention Valve	2392426.28	680461.99
3132	Backflow Prevention Valve	2392929.84	679706.69

This page was intentionally left blank.

APPENDIX C

STANDARD OPERATING PROCEDURES

This page was intentionally left blank.

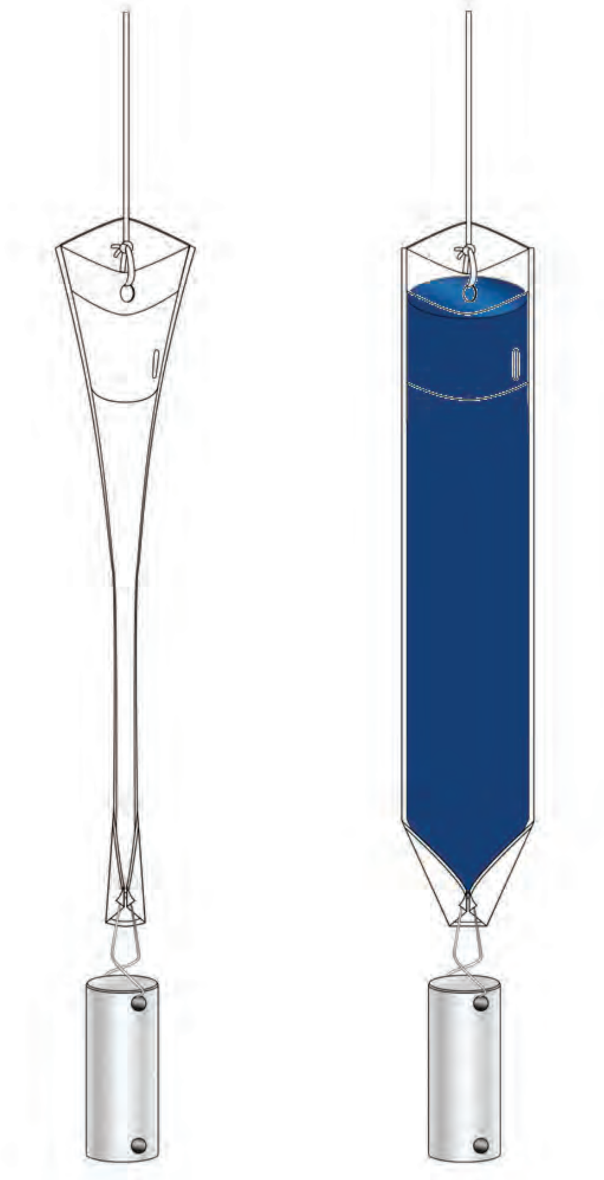
Appendix C

Standard Operating Procedures

HYDRASleeve™

Simple by Design US Patent No. 6,481,300; No. 6,837,120 others pending

Standard Operating Procedure: Sampling Ground Water with a HydraSleeve



This Guide should be used in addition to field manuals appropriate to sampling device (i.e., HydraSleeve or Super Sleeve).

Find the appropriate field manual on the HydraSleeve website at http://www.hydrasleeve.com.

For more information about the HydraSleeve, or if you have questions, contact:
GeoInsight, 2007 Glass Road, Las Cruces, NM 88005, 1-800-996-2225,
info@hydrasleeve.com.

Table of Contents

Introduction 1

Applications of the HydraSleeve 1

Description of the HydraSleeve 3

Selecting the HydraSleeve Size to Meet Site-Specific Sampling Objectives 4

HydraSleeve Deployment 5

Information Required Before Deploying a HydraSleeve 5

HydraSleeve Placement 6

Procedures for Sampling with the HydraSleeve 8

Measurement of Field Indicator Parameters 11

Alternate Deployment Strategies 11

Post-Sampling Activities 14

References 15

Introduction

The HydraSleeve is classified as a no-purge (passive) grab sampling device, meaning that it is used to collect ground-water samples directly from the screened interval of a well without having to purge the well prior to sample collection. When it is used as described in this Standard Operating Procedure (SOP), the HydraSleeve causes no drawdown in the well (until the sample is withdrawn from the water column) and only minimal disturbance of the water column, because it has a very thin cross section and it displaces very little water (<100 ml) during deployment in the well. The HydraSleeve collects a sample from within the screen only, and it excludes water from any other part of the water column in the well through the use of a self-sealing check valve at the top of the sampler. It is a single-use (disposable) sampler that is not intended for reuse, so there are no decontamination requirements for the sampler itself.

The use of no-purge sampling as a means of collecting representative ground-water samples depends on the natural movement of ground water (under ambient hydraulic head) from the formation adjacent to the well screen through the screen. Robin and Gillham (1987) demonstrated the existence of a dynamic equilibrium between the water in a formation and the water in a well screen installed in that formation, which results in formation-quality water being available in the well screen for sampling at all times. No-purge sampling devices like the HydraSleeve collect this formation-quality water as the sample, under undisturbed (non-pumping) natural flow conditions. Samples collected in this manner generally provide more conservative (i.e., higher concentration) values than samples collected using well-volume purging, and values equivalent to samples collected using low-flow purging and sampling (Parsons, 2005).

Applications of the HydraSleeve

The HydraSleeve can be used to collect representative samples of ground water for all analytes (volatile organic compounds [VOCs], semi-volatile organic compounds [SVOCs], common metals, trace metals, major cations and anions, dissolved gases, total dissolved solids, radionuclides, pesticides, PCBs, explosive compounds, and all other analytical parameters). Designs are available to collect samples from wells from 1” inside diameter and larger. The HydraSleeve can collect samples from wells of any yield, but it is especially well-suited to collecting samples from low-yield wells, where other sampling methods can’t be used reliably because their use results in dewatering of the well screen and alteration of sample chemistry (McAlary and Barker, 1987).

The HydraSleeve can collect samples from wells of any depth, and it can be used for single-event sampling or long-term ground-water monitoring programs. Because of its thin cross section and flexible construction, it can be used in narrow, constricted or damaged wells where rigid sampling devices may not fit. Using multiple HydraSleeves deployed in series along a single suspension line or tether, it is also possible to conduct in-well vertical profiling in wells in which contaminant concentrations are thought to be stratified.

As with all groundwater sampling devices, HydraSleeves should not be used to collect ground-water samples from wells in which separate (non-aqueous) phase hydrocarbons (i.e., gasoline, diesel fuel or jet fuel) are present because of the possibility of incorporating some of the separate-phase hydrocarbon into the sample.

Description of the HydraSleeve

The HydraSleeve (Figure 1) consists of the following basic components:

- A suspension line or tether (A.), attached to the spring clip or directly to the top of the sleeve to deploy the device into and recover the device from the well. Tethers with depth indicators marked in 1-foot intervals are available from the manufacturer.
- A long, flexible, 4-mil thick lay-flat polyethylene sample sleeve (C.) sealed at the bottom (this is the sample chamber), which comes in different sizes, as discussed below with a self-sealing reed-type flexible polyethylene check valve built into the top of the sleeve (B.) to prevent water from entering or exiting the sampler except during sample acquisition.
- A reusable stainless-steel weight with clip (D.), which is attached to the bottom of the sleeve to carry it down the well to its intended depth in the water column. Bottom weights available from the manufacturer are 0.75" OD and are available in three sizes: 5 oz. (2.5" long); 8 oz. (4" long); and 16 oz. (8" long). In lieu of a bottom weight, an optional top weight may be attached to the top of the HydraSleeve to carry it to depth and to compress it at the bottom of the well (not shown in Figure 1);
- A discharge tube that is used to puncture the HydraSleeve after it is recovered from the well so the sample can be decanted into sample bottles (not shown).
- Just above the self-sealing check valve at the top of the sleeve are two holes which provide attachment points for the spring clip and/or suspension line or tether. At the bottom of the sample sleeve are two holes which provide attachment points for the weight clip and weight.

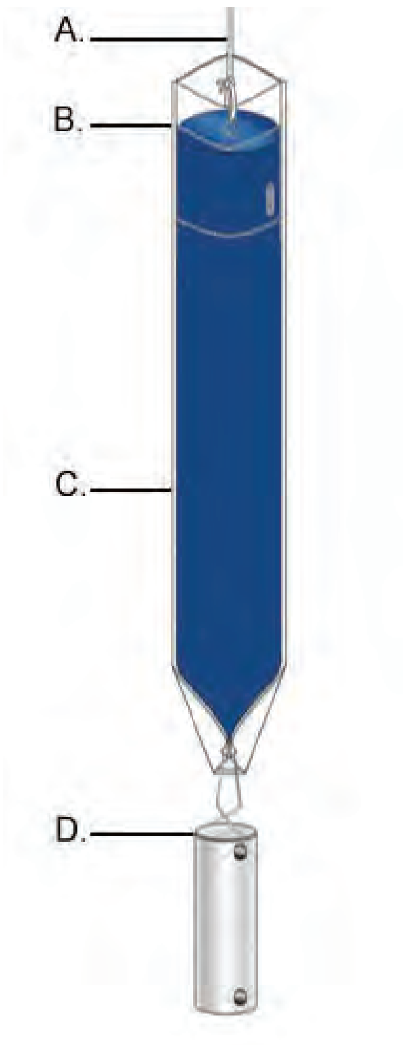


Figure 1. HydraSleeve components.

> **Note:** The sample sleeve and the discharge tube are designed for one-time use and are disposable. The spring clip, weight and weight clip may be reused after thorough cleaning. Suspension cord is generally disposed after one use although, if it is dedicated to the well, it may be reused at the discretion of the sampling personnel.

Selecting the HydraSleeve Size to Meet Site-Specific Sampling Objectives

It is important to understand that each HydraSleeve is able to collect a finite volume of sample because, after the HydraSleeve is deployed, you only get one chance to collect an undisturbed sample. Thus, the volume of sample required to meet your site-specific sampling and analytical requirements will dictate the size of HydraSleeve you need to meet these requirements.

The volume of sample collected by the HydraSleeve varies with the diameter and length of the HydraSleeve. Dimensions and volumes of available HydraSleeve models are detailed in Table 1.

Table 1. Dimensions and volumes of HydraSleeve models.

Diameter	Volume	Length	Lay-Flat Width	Filled Dia.
2-Inch HydraSleeves				
Standard 625-ml HydraSleeve	625 ml	< 30”	2.5”	1.4”
Standard 1-Liter HydraSleeve	1 Liter	38”	3”	1.9”
1-Liter HydraSleeve SS	1 Liter	36”	3”	1.9”
2-Liter HydraSleeve SS	2 Liters	60”	3”	1.9”
4-Inch HydraSleeves				
Standard 1.6-Liter HydraSleeve	1.6 Liters	30”	3.8”	2.3”
Custom 2-Liter HydraSleeve	2 Liters	36”	4”	2.7”

HydraSleeves can be custom-fabricated by the manufacturer in varying diameters and lengths to meet specific volume requirements. HydraSleeves can also be deployed in series (i.e., multiple HydraSleeves attached to one tether) to collect additional sample to meet specific volume requirements, as described below.

If you have questions regarding the availability of sufficient volume of sample to satisfy laboratory requirements for analysis, it is recommended that you contact the laboratory to discuss the minimum volumes needed for each suite of analytes. Laboratories often require only 10% to 25% of the volume they specify to complete analysis for specific suites of analytes, so they can often work with much smaller sample volumes that can easily be supplied by a HydraSleeve.

HydraSleeve Deployment

Information Required Before Deploying a HydraSleeve

Before installing a HydraSleeve in any well, you will need to know the following:

- The inside diameter of the well
- The length of the well screen
- The water level in the well
- The position of the well screen in the well
- The total depth of the well

The inside diameter of the well is used to determine the appropriate HydraSleeve diameter for use in the well. The other information is used to determine the proper placement of the HydraSleeve in the well to collect a representative sample from the screen (see HydraSleeve Placement, below), and to determine the appropriate length of tether to attach to the HydraSleeve to deploy it at the appropriate position in the well.

Most of this information (with the exception of the water level) should be available from the well log; if not, it will have to be collected by some other means. The inside diameter of the well can be measured at the top of the well casing, and the total depth of the well can be measured by sounding the bottom of the well with a weighted tape. The position and length of the well screen may have to be determined using a down-hole camera if a well log is not available. The water level in the well can be measured using any commonly available water-level gauge.

HydraSleeve Placement

The HydraSleeve is designed to collect a sample directly from the well screen, and it fills by pulling it up through the screen a distance equivalent to 1 to 1.5 times its length. This upward motion causes the top check valve to open, which allows the device to fill. To optimize sample recovery, it is recommended that the HydraSleeve be placed in the well so that the bottom weight rests on the bottom of the well and the top of the HydraSleeve is as close to the bottom of the well screen as possible. This should allow the sampler to fill before the top of the device reaches the top of the screen as it is pulled up through the water column, and ensure that only water from the screen is collected as the sample. In short-screen wells, or wells with a short water column, it may be necessary to use a top-weight on the HydraSleeve to compress it in the bottom of the well so that, when it is recovered, it has room to fill before it reaches the top of the screen.

Example

2” ID PVC well, 50’ total depth, 10’ screen at the bottom of the well, with water level above the screen (the entire screen contains water).

Correct Placement (figure 2): Using a standard HydraSleeve for a 2” well (2.6” flat width/1.5” filled OD x 30” long, 650 ml volume), deploy the sampler so the weight (an 8 oz., 4”-long weight with a 2”-long clip) rests at the bottom of the well. The top of the sleeve is thus set at about 36” above the bottom of the well. When the sampler is recovered, it will be pulled upward approximately 30” to 45” before it is filled; therefore, it is full (and the top check valve closes) at approximately 66” (5 ½ feet) to 81” (6 ¾ feet) above the bottom of the well, which is well before the sampler reaches the top of the screen. In this example, only water from the screen is collected as a sample.

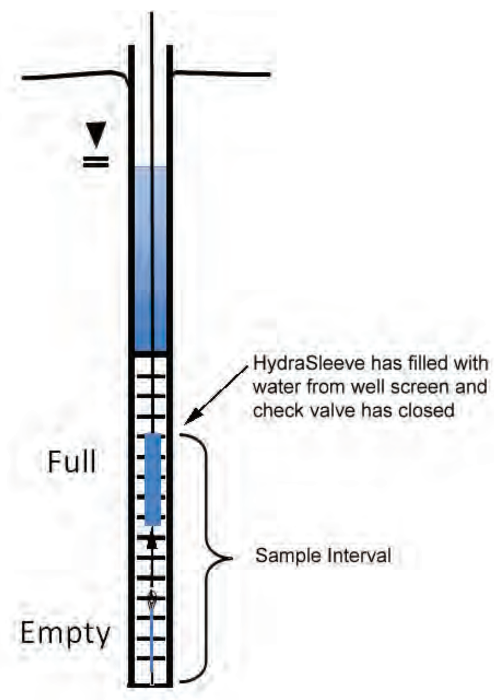


Figure 2. Correct placement of HydraSleeve.

Incorrect Placement (figure 3): If the well screen in this example was only 5' long, and the HydraSleeve was placed as above, it would not fill before the top of the device reached the top of the well screen, so the sample would include water from above the screen, which may not have the same chemistry.

The solution? Deploy the HydraSleeve with a top weight, so that it is collapsed to within 6" to 9" of the bottom of the well. When the HydraSleeve is recovered, it will fill within 39" (3 ¼ feet) to 54" (4 ½ feet) above the bottom of the well, or just before the sampler reaches the top of the screen, so it collects only water from the screen as the sample.

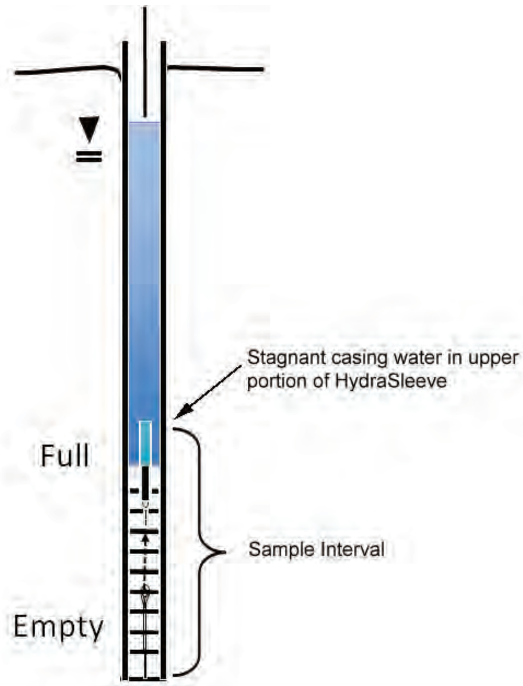


Figure 3. Incorrect placement of HydraSleeve.

This example illustrates one of many types of HydraSleeve placements. More complex placements are discussed in a later section.

Procedures for Sampling with the HydraSleeve

Collecting a ground-water sample with a HydraSleeve is a simple one-person operation.

Note: Before deploying the HydraSleeve in the well, collect the depth-to-water measurement that you will use to determine the preferred position of the HydraSleeve in the well. This measurement may also be used with measurements from other wells to create a ground-water contour map. If necessary, also measure the depth to the bottom of the well to verify actual well depth to confirm your decision on placement of the HydraSleeve in the water column.

Measure the correct amount of tether needed to suspend the HydraSleeve in the well so that the weight will rest on the bottom of the well (or at your preferred position in the well). Make sure to account for the need to leave a few feet of tether at the top of the well to allow recovery of the sleeve

Note: Always wear sterile gloves when handling and discharging the HydraSleeve.

I. Assembling the HydraSleeve

1. Remove the HydraSleeve from its packaging, unfold it, and hold it by its top.

2. Crimp the top of the HydraSleeve by folding the hard polyethylene reinforcing strips at the holes.

3. Attach the spring clip to the holes to ensure that the top will remain open until the sampler is retrieved.

4. Attach the tether to the spring clip by tying a knot in the tether.

Note: Alternatively, attach the tether to one (NOT both) of the holes at the top of the Hydrasleeve by tying a knot in the tether.

5. Fold the flaps with the two holes at the bottom of the HydraSleeve together and slide the weight clip through the holes.

6. Attach a weight to the bottom of the weight clip to ensure that the HydraSleeve will descend to the bottom of the well.

II. Deploying the HydraSleeve

1. Using the tether, carefully lower the HydraSleeve to the bottom of the well, or to your preferred depth in the water column

   During installation, hydrostatic pressure in the water column will keep the self-sealing check valve at the top of the HydraSleeve closed, and ensure that it retains its flat, empty profile for an indefinite period prior to recovery.

Note: Make sure that it is not pulled upward at any time during its descent. If the HydraSleeve is pulled upward at a rate greater than 0.5'/second at any time prior to recovery, the top check valve will open and water will enter the HydraSleeve prematurely.

2. Secure the tether at the top of the well by placing the well cap on the top of the well casing and over the tether.

Note: Alternatively, you can tie the tether to a hook on the bottom of the well cap (you will need to leave a few inches of slack in the line to avoid pulling the sampler up as the cap is removed at the next sampling event).

III. Equilibrating the Well

The equilibration time is the time it takes for conditions in the water column (primarily flow dynamics and contaminant distribution) to restabilize after vertical mixing occurs (caused by installation of a sampling device in the well).

- Situation: The HydraSleeve is deployed for the first time or for only one time in a well

  The HydraSleeve is very thin in cross section and displaces very little water (<100 ml) during deployment so, unlike most other sampling devices, it does not disturb the water column to the point at which long equilibration times are necessary to ensure recovery of a representative sample.

  In most cases, the HydraSleeve can be recovered immediately (with no equilibration time) or within a few hours. In regulatory jurisdictions that impose specific requirements for equilibration times prior to recovery of no-purge sampling devices, these requirements should be followed.

- Situation: The HydraSleeve is being deployed for recovery during a future sampling event

  In periodic (i.e., quarterly or semi-annual) sampling programs, the sampler for the current sampling event can be recovered and a new sampler (for the next sampling event) deployed immediately thereafter, so the new sampler remains in the well until the next sampling event.

Thus, a long equilibration time is ensured and, at the next sampling event, the sampler can be recovered immediately. This means that separate mobilizations, to deploy and then to recover the sampler, are not required. HydraSleeves can be left in a well for an indefinite period of time without concern.

IV. HydraSleeve Recovery and Sample Collection

1. Hold on to the tether while removing the well cap.

2. Secure the tether at the top of the well while maintaining tension on the tether (but without pulling the tether upwards)

3. Measure the water level in the well.

4. In one smooth motion, pull the tether up between 30” to 45” (36” to 54” for the longer HydraSleeve) at a rate of about 1’ per second (or faster).

   The motion will open the top check valve and allow the HydraSleeve to fill (it should fill in about 1 to 1.5 times the length of the HydraSleeve). This is analogous to coring the water column in the well from the bottom up.

   When the HydraSleeve is full, the top check valve will close. You should begin to feel the weight of the HydraSleeve on the tether and it will begin to displace water. The closed check valve prevents loss of sample and entry of water from zones above the well screen as the HydraSleeve is recovered.

5. Continue pulling the tether upward until the HydraSleeve is at the top of the well.

6. Decant and discard the small volume of water trapped in the Hydrasleeve above the check valve by turning the sleeve over.

V. Sample Collection

Note: Sample collection should be done immediately after the HydraSleeve has been brought to the surface to preserve sample integrity.

1. Remove the discharge tube from its sleeve.

2. Hold the HydraSleeve at the check valve.

3. Puncture the HydraSleeve just below the check valve with the pointed end of the discharge tube

4. Discharge water from the HydraSleeve into your sample containers.

Control the discharge from the HydraSleeve by either raising the bottom of the sleeve, by squeezing it like a tube of toothpaste, or both.

5. Continue filling sample containers until all are full.

Measurement of Field Indicator Parameters

Field indicator parameter measurement is generally done during well purging and sampling to confirm when parameters are stable and sampling can begin. Because no-purge sampling does not require purging, field indicator parameter measurement is not necessary for the purpose of confirming when purging is complete.

If field indicator parameter measurement is required to meet a specific non-purging regulatory requirement, it can be done by taking measurements from water within a HydraSleeve that is not used for collecting a sample to submit for laboratory analysis (i.e., a second HydraSleeve installed in conjunction with the primary sample collection HydraSleeve [see Multiple Sampler Deployment below]).

Alternate Deployment Strategies

Deployment in Wells with Limited Water Columns

For wells in which only a limited water column exists to be sampled, the HydraSleeve can be deployed with an optional top weight instead of a bottom weight, which collapses the HydraSleeve to a very short (approximately 6” to 9”) length, and allows the HydraSleeve to fill in a water column only 36” to 45” in height.

Multiple Sampler Deployment

Multiple sampler deployment in a single well screen can accomplish two purposes:

- It can collect additional sample volume to satisfy site or laboratory-specific sample volume requirements.
- It can accommodate the need for collecting field indicator parameter measurements.
- It can be used to collect samples from multiple intervals in the screen to allow identification of possible contaminant stratification.

It is possible to use up to 3 standard 30" HydraSleeves deployed in series along a single tether to collect samples from a 10' long well screen without collecting water from the interval above the screen.

The samplers must be attached to the tether at both the top and bottom of the sleeve. Attach the tether at the top with a stainless-steel clip (available from the manufacturer). Attach the tether at the bottom using a cable tie. The samplers must be attached as follows (figure 4):

- The first (attached to the tether as described above, with the weight at the bottom) at the bottom of the screen
- The second attached immediately above the first
- The third (attached the same as the second) immediately above the second

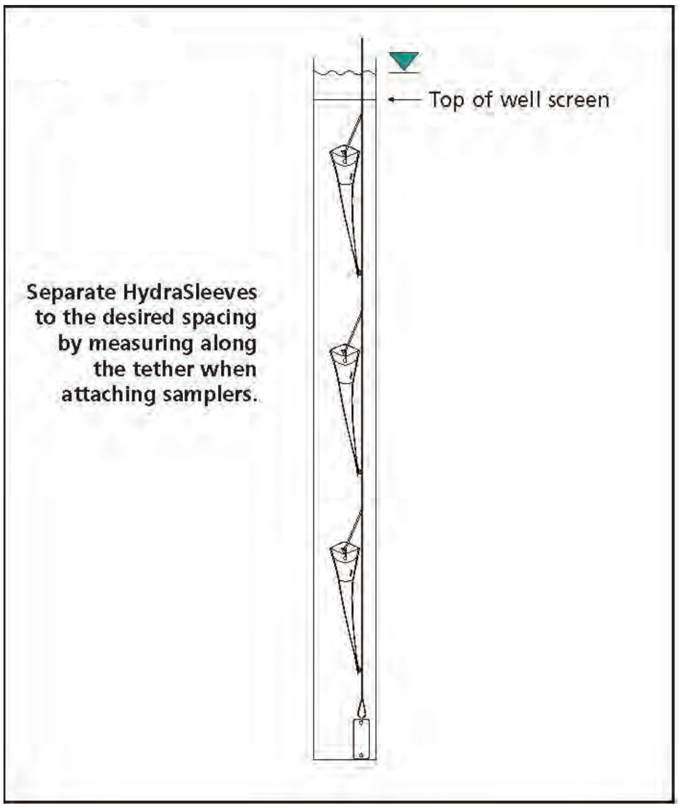


Figure 4. Multiple HydraSleeve deployment.

Alternately, the first sampler can be attached to the tether as described above, a second attached to the bottom of the first using a short length of tether (in place of the weight), and the third attached to the bottom of the second in the same manner, with the weight attached to the bottom of the third sampler (figure 5).

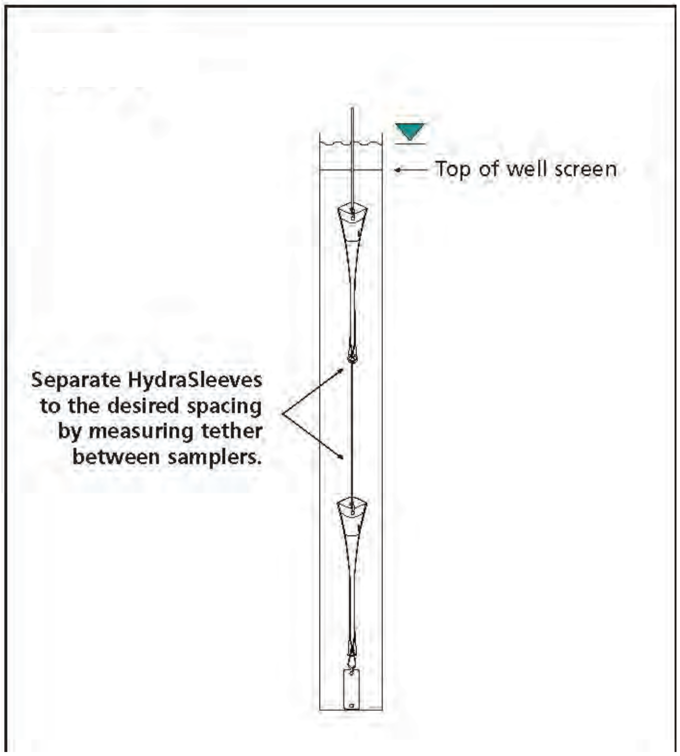


Figure 5. Alternative method for deploying multiple HydraSleeves.

In either case, when attaching multiple HydraSleeves in series, more weight may be required to hold the samplers in place in the well than would be required with a single sampler. Recovery of multiple samplers and collection of samples is done in the same manner as for single sampler deployments.

Post-Sampling Activities

The recovered HydraSleeve and the sample discharge tubing should be disposed as per the solid waste management plan for the site. To prepare for the next sampling event, a new HydraSleeve can be deployed in the well (as described previously) and left in the well until the next sampling event, at which time it can be recovered.

The weight and weight clip can be reused on this sampler after they have been thoroughly cleaned as per the site equipment decontamination plan. The tether may be dedicated to the well and reused or discarded at the discretion of sampling personnel.

References

McAlary, T. A. and J. F. Barker, 1987, Volatilization Losses of Organics During Ground-Water Sampling From Low-Permeability Materials, Ground-Water Monitoring Review, Vol. 7, No. 4, pp. 63-68

Parsons, 2005, Results Report for the Demonstration of No-Purge Ground-Water Sampling Devices at Former McClellan Air Force Base, California; Contract F44650-99-D-0005, Delivery Order DKO1, U.S. Army Corps of Engineers (Omaha District), U.S. Air Force Center for Environmental Excellence, and U.S. Air Force Real Property Agency

Robin, M. J. L. and R. W. Gillham, 1987, Field Evaluation of Well Purging Procedures, Ground-Water Monitoring Review, Vol. 7, No. 4, pp. 85-93

HYDRASleeve™

Simple by Design

US Patents No. 6,481,300; 6,837,120; others pending

SuperSleeve Assembly Instructions

Introduction

The most common practical limitation for no-purge sampling is sample volume. The HydraSleeve SuperSleeve was designed for maximum fill rate and maximum volume to address this issue. The HydraSleeve SS consists of a special 2-piece top weight, a slightly wider sleeve with a wider check valve.

Please use these assembly instructions in conjunction with the standard HydraSleeve Field Manual.

GeoInsight

(800) 996-2225 • www.hydrasleeve.com • www.nopurgesampling.com
P.O. Box 1266 • Mesilla Park, NM 88047

Assembly

Attach the Bottom Weight
Fold bottom of sleeve in half vertically to align holes. Open prongs of bottom weight clip by squeezing. Insert reusable weight clip through holes and attach the bottom weight.

Attaching the Sleeve to the Top Weight
Insert the open (check valve) end of the HydraSleeve SS through the bottom of the stainless steel portion of top weight until about 1/2 inch of the open sleeve protrudes above the female threads.

Thread stainless steel weight (female thread) onto PVC top piece (male thread) locking the top of the HydraSleeve SS between the threads.

Assembled

Assembled 1-liter HydraSleeve SS

2-Inch HydraSleeve SS Sizes

- 1-liter HydraSleeve SS 3-feet in length
- 2-Liter HydraSleeve SS 5-feet in length

Allow Weight to Settle

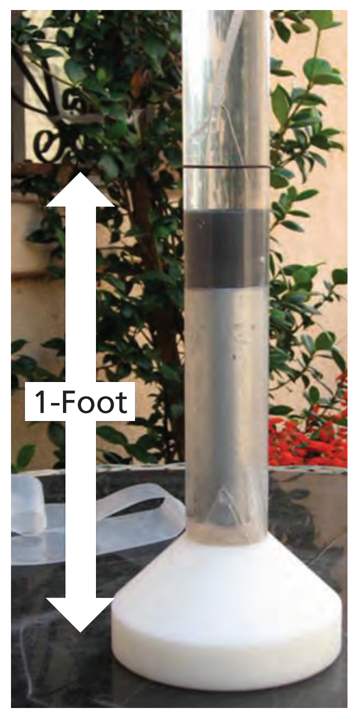
1. Lower the HydraSleeve SS into the well until the bottom weight touches the bottom.

2. Depending on the length of the HydraSleeve, provide enough slack tether to allow the top weight to fully compress the sampler into the bottom of the well.
Example – At least an additional 3 feet of tether for a 1-liter (36-inches long) HydraSleeve SS.

Compression and Fill Rates

1-liter HydraSleeve SS (3-feet long) will compress to within 1 foot of bottom of 2-inch well screen in about 2 hours.
1-liter HydraSleeve SS requires about 3 feet of water over the top of the sampler to completely fill.

2-liter HydraSleeve SS (5-feet long) will compress to within 2 feet of bottom of 2-inch well screen in about 4 hours.
2-liter HydraSleeve SS requires about 5 feet of water over the top of the sampler to completely fill.



Surface Water Sampling	SOP No.: 2.16 SOP Category: HTRW Revision No.: 0 Date: December 2010

1.0 PURPOSE

The purpose of this procedure is to describe the methods for surface water sampling. It describes the procedures and equipment to be used to obtain representative surface water samples that are capable of producing accurate quantification of water quality.

2.0 SCOPE AND APPLICATIONS

This procedure provides guidance for routine field operations on environmental projects. Procedures are included for collecting grab and composite surface water samples from flowing water bodies (streams, rivers, and creeks) and from standing water bodies (lakes, lagoons, and ponds).

3.0 GENERAL REQUIREMENTS

All work will be performed in a manner that is consistent with Occupational Safety and Health Administration established standards and requirements. Refer to the site- or project-specific health and safety plan for relevant health and safety requirements. All activities will be conducted in conformance with the Site Health and Safety Plan.

Personnel who use this procedure must provide documented evidence to the program manager or project manager that they have been trained on the procedure. This documentation will be retained in the project file.

Any deviations from specified requirements will be justified to and authorized by the project manager and/or the relevant program manager and discussed in the approved project plans. Deviations from requirements will be sufficiently documented to re-create the modified process.

4.0 DEFINITIONS

Aliquot: Fractional amount.

Composite Samples: Samples composed of more than one aliquot collected at various sampling sites and/or at different times.

Epilimnetic zone: The uppermost layer of water in a lake, characterized by an essentially uniform temperature that is generally warmer than elsewhere in the lake and by a relatively uniform mixing caused by wind and wave action. Specifically, the epilimnetic zone is the light (less dense), oxygen-rich layer of water in a thermally stratified lake.

Grab Samples: Samples that are collected at one particular point and time.

Hypolimnetic zone: The lowermost layer of water in a lake, characterized by an essentially uniform temperature (except during turnover) that is generally colder than elsewhere in the lake and is often characterized by relatively stagnant or oxygen-deficient water.

Rinsate: Wastewater generated as a result of rinsing sampling equipment during decontamination procedures.

Surface water samples: Samples of water collected from streams, ponds, rivers, lakes, or other impoundments open to the atmosphere.

5.0 PROCEDURE

5.1 INTRODUCTION

The objective of surface water sampling is to evaluate the surface water quality entering and/or leaving a site. It is also used to obtain data on waste loads, water quality and characteristics that will permit prediction or modeling of the water system (to describe probable water quality), and effects on uses under a variety of conditions.

5.2 SAMPLING EQUIPMENT

Sampling equipment includes all sampling devices and containers that are used to collect or contain a sample prior to final sample analysis. All surface water sampling equipment shall have a design that will maintain sample integrity and to provide the desired level of quality in achieving desired analytical results. There is a variety of equipment available for surface water sampling. Because each site may contain varied surface water conditions, collection of a representative sample may be difficult. In general, a sampling device will include the following characteristics:

- Be constructed of disposable or non-reactive material (Teflon® or stainless steel); and
- Have a minimum capacity of 500 milliliters (ml) to minimize sample disturbance.

5.3 DECONTAMINATION

Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination*, after use and between sampling if dedicated disposable equipment is not used.

5.4 SAMPLING LOCATION/SITE SELECTION

Before sampling, consideration must be given to the specific sampling locations to provide a representative sample. This and other considerations are detailed in the project-specific Work Plan.

The general determining factors in the selection of a sampling device for sampling liquids in lakes, ponds, lagoons, and surface impoundments are listed below:

- Accessibility:
    - Boat: If the water is navigable, any sampling location is accessible by boat.
    - Bridges: Provide ready access, are readily identifiable, and permit water sampling at any point across the width of the water body.
    - Wading: Personnel safety must be paramount. Wading is not recommended in areas where bottom deposits are easily disturbed, thereby increasing the possibility of increased sediment in the samples.
- Rivers, streams, and creeks:
    - Sampling stations will be located wherever a marked physical change occurs in the stream channel. For example, between rapids/deep water transitions, as well as at both ends of a reach.
    - Sampling stations will be located short distances above and below dams and weirs, to determine the artificial increase in dissolved oxygen.
    - A minimum of three sampling locations will be established between any two points of major change in a stream.
    - Sampling stations will be located upstream and downstream of any waste discharge site. Since the inflow frequently hugs the stream bank with very little lateral mixing, care must be taken to establish the sampling station after complete mixing with the main stream.
    - A tributary sampling station will be established near the mouth and upstream of any effects from the main stream. The station on the main stream will be just upstream from the confluence.
    - Sample as close as is practical to areas or points of important water uses.
    - At stations where wastes and tributary waters are well-mixed, one sampling point near mid-channel is usually adequate. At stations where mixing is inadequate, the station will be sampled at quarter points across the width of the station.

- Lakes, ponds, lagoons, and impoundments:
  - A single station at the deepest point may be sufficient for naturally formed ponds (near the center) and for impoundments (near the dam or spillway).
  - A sampling grid is the most representative for lakes and large impoundments.
  - In lakes with irregular shapes and with several bays and coves that are protected from the wind, sampling stations should be established in these areas.
  - A control station above a waste source is usually necessary to compare background water quality. It should be carefully selected and it may be necessary to have two or three control stations to establish the rate at which unstable material is changing. The time of travel between stations should be sufficient to permit accurate measurement of the change in the constituents under consideration.

5.5 SAMPLING METHODS

5.5.1 General

The specific sampling method utilized will depend on the accessibility to, the size, and the depth of the water body, as well as the type of samples being collected.

In most ambient water quality studies, grab samples will be collected. However, the objectives of the study will dictate the sampling method and will be specified in the project specific work plan.

For rivers, streams and creeks, the type of samples collected will be dependent upon the size and the amount of turbulence in the water body. Approximate the depth and location of samples in order to assure consistency. Flow rates will be measured in accordance with SOP 2.08: *Stream Flow Measurement*.

- With small streams less than 20 feet wide, a single grab sample collected at mid-depth in the center of the channel is usually adequate to represent the entire cross-section. In small streams and creeks less than 10 feet wide, a single grab sample can be collected by immersing the bottle directly under the surface of the water as close to the center of the channel as possible.
- For slightly larger streams, a vertical composite sample in the center of the channel may be required. The composite sample consists of samples taken just below the surface, at mid-depth, and just above the bottom.
- For rivers, several vertical composite samples are collected across the water body. The vertical composite samples will be collected at points in the cross section

approximately proportional to flow. The number of vertical composites required and the number of depths sampled for each are usually determined in the field. This determination is based on a reasonable balance between two considerations:

- o The larger the number of subsamples, the more nearly the composite sample will represent the water body; but
- o Taking many subsamples is time-consuming and increases the chance of cross-contamination.

- For lakes, ponds, lagoons, and impoundments, the greater tendency to stratify and the relative lack of adequate mixing usually requires that more subsamples be collected.
    - o In ponds, lagoons, and small impoundments, a single vertical composite sample at the deepest point is usually adequate.
    - o In lakes and larger impoundments, several vertical composites should be combined into a single sample. In some cases, it may be useful to form several composites of the epilimnetic and hypolimnetic zones. Normally, however, a composite consists of several verticals with subsamples collected at various depths.

5.5.2 Sample Bottle

Collecting a representative sample from small streams and creeks less than 10 feet wide, a single grab sample can be collected by immersing the sample bottle directly under the surface of the water as close to the center of the channel as possible. This method reduces the potential for cross contamination as it does not require the decontamination of equipment. The following procedures will be followed when sampling with a sample bottle:

- If sampling from within the stream, always sample from the furthest down stream location and work up stream;
- When wading to location, move slowly to minimize the amount of sediment stirred up from the bottom:
- Stand facing up stream;
- A single grab sample can be collected by immersing the bottle directly under the surface of the water as close to the center of the channel as possible ensuring the sample is collected up stream of the sampler.
- If preservatives are required for sample preservation, add preservative after samples have been collected. If pre-preserved bottleware is to be used, first collect the sample in an unpreserved bottle and decant the water to the pre-preserved bottle.

- Raise the sampler, seal, wipe clean, label or identify and prepare the bottle for transport in accordance with project guidelines;
- Identify or label samples carefully and clearly, addressing all the categories or parameters;
- After labeling of the sample bottles has been completed, place the filled sample containers on ice immediately;
- All field crew members handling the sampling equipment and/or sample bottles shall don new clean gloves prior to beginning sampling activities and between each successive sample point.
- Complete all chain of custody documents and record information in the field logbook (see the project-specific Quality Assurance Project Plan [QAPP] for sample custody procedures).
- Mark sample location and approximate depth, if possible, and note on maps and in field log book.

One additional grab sample from each location will be collected and a water quality probe will be used to collect pH, conductivity, temperature, turbidity, and salinity (where applicable) data. This sample will not be submitted for laboratory analysis. Other odors and significant characteristics will also be documented on the field sampling sheet and in the field logbook.

5.5.3 Weighted Bottle Sampler

Collecting a representative sample from a larger body of water requires the gathering of samples from various depths and locations. A weighted bottle sampler is typically utilized for this type of sampling. The sampler consists of a Teflon® bottle, a weighted sinker, a bottle stopper and a wire cord used to raise, lower and open the samples. This type of sampler can be fabricated or purchased. The following procedures will be followed when sampling with a weighted bottle sampler (Attachment 1, Weighted Bottle Sampler):

- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination*;
- Assemble the weighted bottle sampler in accordance with the sampler instruction manual;
- Gently lower the sampler to the desired depth so as not to remove the stopper prematurely. Do not let sampler disturb bottom sediments;
- Pull out the stopper with a sharp jerk of the sampler line;
- Allow the bottle to fill completely, as evidenced by the cessation of air bubbles;

- Raise the sampler, seal, wipe clean, label or identify and prepare the bottle for transport in accordance with project guidelines;
- Identify or label samples carefully and clearly, addressing all the categories or parameters;
- After labeling of the sample bottles has been completed, place the filled sample containers on ice immediately;
- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination*;
- All field crew members handling the sampling equipment and/or sample bottles shall don new clean gloves prior to beginning sampling activities and between each successive sample point.
- Complete all chain of custody documents and record information in the field logbook (see the project-specific QAPP for sample custody procedures).
- Mark sample location and approximate depth, if possible, and note on maps and in field logbook.

One additional grab sample from each location will be collected and a water quality probe will be used to collect pH, conductivity, temperature, turbidity, and salinity (where applicable) data. This sample will not be submitted for laboratory analysis. Other odors and significant characteristics will also be documented on the field sampling sheet and in the field log book.

5.5.4 Pond Sampler

The pond or dip sampler (Attachment 2, Pond Sampler) consists of a scoop or container attached to the end of a telescoping or solid pole. The sampler will be of non-reactive material such as wood, plastic, or stainless steel. The sample will be collected in a jar or beaker made of stainless steel or Teflon®. Preferably, a disposable beaker that can be replaced before each sampling will be used at each station. Liquid wastes from water courses, ponds, pits, lagoons or open vessels will be ladled into the sample container.

Perform the following procedures when sampling with a pond sampler:

- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination*;
- Assemble pond sampler in accordance with manufacturer's instructions;
- Extend pole to length that will allow safe access to desired sample location;
- Submerge pond sampler to desired sample depth. Submerge the sampler very slowly to minimize surface disturbance;

- Allow the sampler to fill very slowly;
- Retrieve the sampling device with minimal surface water disturbance;
- Remove the cap from the sample bottle and slightly tilt the mouth of the bottle below the sampler edge;
- Empty the sampler slowly, allowing the sample stream to flow gently down the side of the bottle with minimal entry turbulence. Fill sample bottle to appropriate headspace, if any;
- Identify or label samples carefully and clearly, addressing all the categories or parameters;
- After labeling of the sample bottles has been completed, place the filled sample containers on ice immediately;
- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination*.
- All field crew members handling the sampling equipment and/or sample bottles shall don new clean gloves before beginning sampling activities and between each successive sample point.
- Complete all chain of custody documents and record information in the field logbook (see the project specific QAPP for sample custody procedures).
- Mark sample location and approximate depth, if possible, and note on maps and in field log book.
- Collect additional grab samples to acquire field measurements such as temperature, pH, conductivity, turbidity, salinity (where applicable) and other significant characteristics;

5.5.5 Manual Hand Pumps

Manual pumps are available in various sizes and configurations. Manual hand pumps are commonly operated by peristaltic, bellows or diaphragm, and siphon action. Manual hand pumps that operate by a bellows or diaphragm, and siphon action should not be used to collect samples that will be analyzed for volatile organics (Attachment 3, Manual Hand Pump). These types of pumps should be constructed of inert materials, such as Teflon® or stainless steel.

Perform the following procedures when collecting surface water samples with a manual hand pump:

- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination;*
- Assemble and operate the pump in accordance with the manufacturer's instructions;

- The inlet hose and any surface of the pump used for sampling will be constructed of materials that are operable and non-reactive;
- To avoid agitation, insert the sampling tube into the liquid sample prior to pump activation;
- Insert a liquid trap (preferably the sample container) into the sample inlet hose to collect the sample and to prevent pump contamination;
- Identify or label samples carefully and clearly, addressing all the categories or parameters;
- After labeling of the sample bottles has been completed, place the filled sample containers on ice immediately;
- Decontaminate the sampling equipment in accordance with SOP 2.01: *Sampling Equipment Cleaning and Decontamination.*
- All field crew members handling the sampling equipment and/or sample bottles shall don new clean gloves prior to beginning sampling activities and between each successive sample point.
- Complete all chain of custody documents and record information in the field logbook (see the project-specific QAPP for sample custody procedures).
- Mark sample location and approximate depth, if possible, and note on maps and in the field logbook. Record applicable data in the field logbook such as color, turbidity, pH, temperature, turbidity (where appropriate), degree of turbulence, and weather conditions).

5.5.6 Peristaltic Pump

Gathering surface water samples with the assistance of a peristaltic pump is another commonly used sampling technique. In this method the sample is drawn through heavy-walled tubing and pumped directly into the sample container. This system allows the operator to extend into the liquid body to sample from depth, or sweep the width of narrow streams. Medical-grade silicon tubing is often used in the peristaltic pump and the system is suitable for sampling almost any parameter, including most organics (Attachment 4, Peristaltic Pump).

Peristaltic pumps are available with a range of power sources. For field use the battery operated units have proven most convenient and very reliable.

Perform the following procedures when sampling with a peristaltic pump:

- Prepare the peristaltic pump in accordance with manufacturer's instructions. When using a battery-operated pump, be sure battery is fully charged prior to entering the field.

- In most situations, it is necessary to change the Teflon® suction line and the silicon pump tubing between sample locations to avoid cross-contamination.
- Gently lower the pump intake tube to the desired sample depth. Avoid unnecessary agitation (aeration) of the liquid to be sampled and bottom sediments.
- Prior to activating the pump, note in which direction the pump will be rotating. (Most peristaltic pumps are capable of rotating in two directions.) Accidental reverse rotation of the pump will cause aeration of the liquid to be sampled.
- Run the pump until no air bubbles are noted in the discharge.
- Discharge water shall be released down stream from sampling area during sampling event.
- To prevent excess agitation and/or aeration of the sampler, fill the sample containers by tilting the container and flow the sample water down the side of sampling container.
- Identify or label samples carefully and clearly, addressing all the categories or parameters;
- After labeling of the sample bottles has been completed, place the filled sample containers on ice immediately;
- In most cases, no specific decontamination procedures are required due to the use of disposable tubing. However, site-specific sample procedures may require additional decontamination and will be specified in the project-specific work plan.
- All field crew members handling the sampling equipment and/or sample bottles shall don new clean gloves prior to beginning sampling activities and between each successive sample point.
- Complete all chain of custody documents and record information in the field logbook (see the project-specific QAPP for sample custody procedures).
- Mark sample location and approximate depth, if possible, and note on maps and in field log book. Record applicable data in the field log book such as color, turbidity, pH, salinity (where applicable), degree of turbulence, and weather conditions).

When medical grade silicon tubing is not available for analytical requirements, the system can be altered as illustrated in Attachment 5, Peristaltic Pump - Modified. In this configuration, the sample volume accumulates in the vacuum flask and does not enter the pump. This system will provide excellent sample integrity for most analyses; however, the potential for losing volatile fractions to the reduced pressure of the vacuum flask renders this method unacceptable for sampling of volatiles.

It may sometimes be necessary to sample large bodies of water where a near-surface sample will not sufficiently characterize the body as a whole. In this instance, the above-mentioned pump is appropriate. It is capable of lifting water from slightly deeper than six meters. It should be noted that this lift ability decreases somewhat with higher density fluids and with increased wear on the silicone pump tubing. Similarly, increases in altitude will decrease the ability of the pump to lift from depth. When sampling a liquid stream that exhibits a considerable flow rate, it may be necessary to weight the bottom of the suction line.

5.5.7 Optional Sampling Methods

The above-mentioned methods of surface water sampling will be used most often on HGL environmental projects; however, choice of sampling equipment depends on site specific conditions. Additional types of samplers available are:

- Kemmerer sampler;
- Wheaton sampler;
- Bacon Bomb sampler;
- Open tube sampler;
- D.O. Punker sampler; and
- Bailer.

Prior to any field work, the Project Manager or designee will review the available sampling equipment and choose the sampling device that will best suit the project requirements. The sampling device to be used will be specified in the project-specific work plan.

6.0 RECORDS

Documentation generated as a result of this procedure is collected and maintained in accordance with requirements specified in the project-specific work plan.

- Document all daily field activities on a daily field activity report in accordance with procedures listed in SOP 4.01: *Documentation – Daily Field Activity Reports*.
- The Surface Water Sampling Data form contained in Attachment 6 must be filled out for each surface water sample collected; and
- Complete the field logbook in accordance with procedures listed in SOP 4.07: *Field Logbook Use and Maintenance*.

7.0 REFERENCES

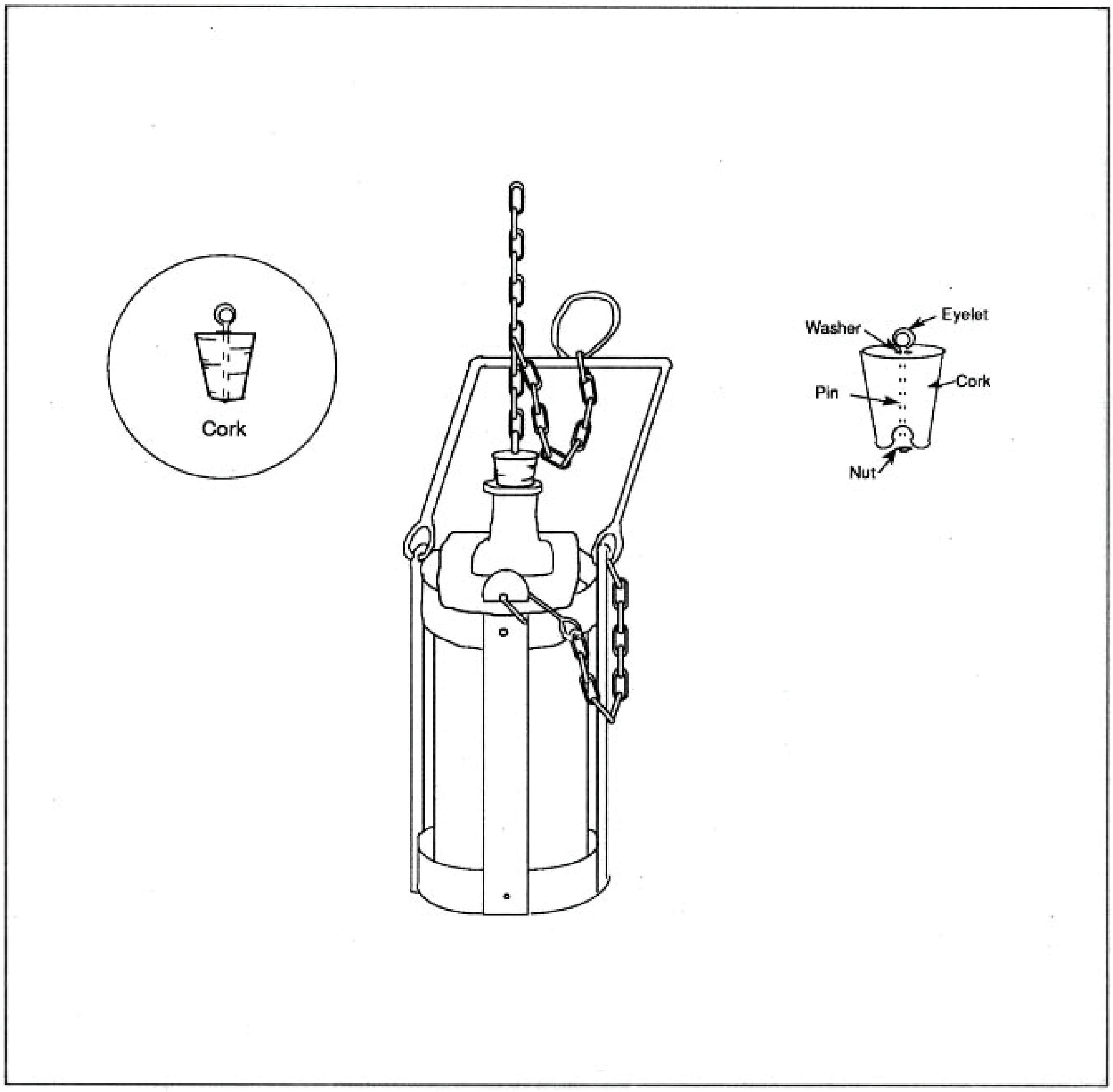
U.S. Army Corps of Engineers, 2001. *Requirements for the Preparation of Sampling and Analysis Plans* (EM200-1-3). Appendix C.3.

8.0 ATTACHMENTS

Attachment 1	Weighted Bottle Sampler
Attachment 2	Pond Sampler
Attachment 3	Manual Hand Pump
Attachment 4	Peristaltic Pump
Attachment 5	Peristaltic Pump - Modified
Attachment 6	Surface Water Sampling Data

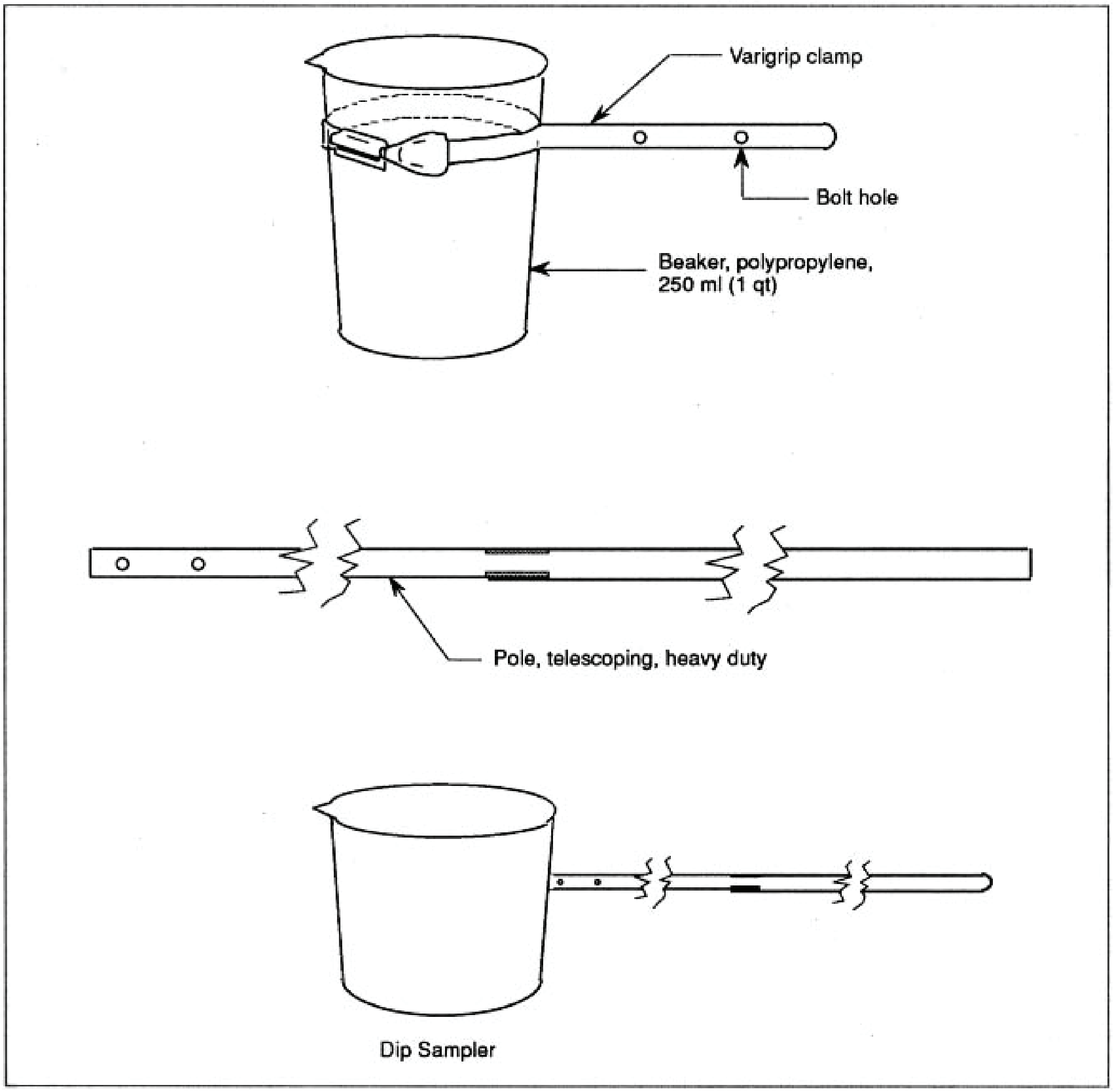
ATTACHMENT 1
Weighted Bottle Sampler

WEIGHTED-BOTTLE SAMPLER



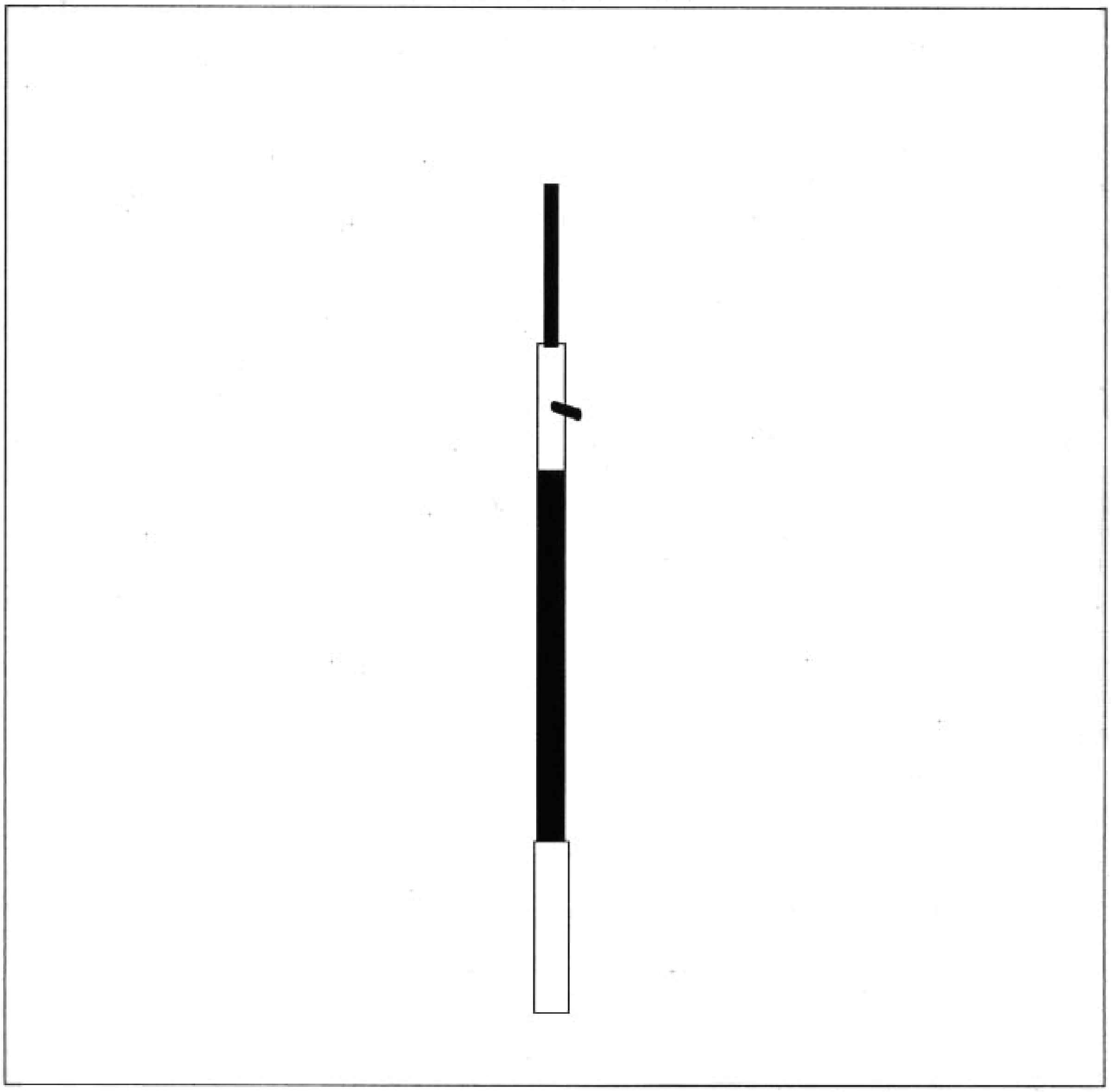
ATTACHMENT 2
Pond Sampler

POND SAMPLER



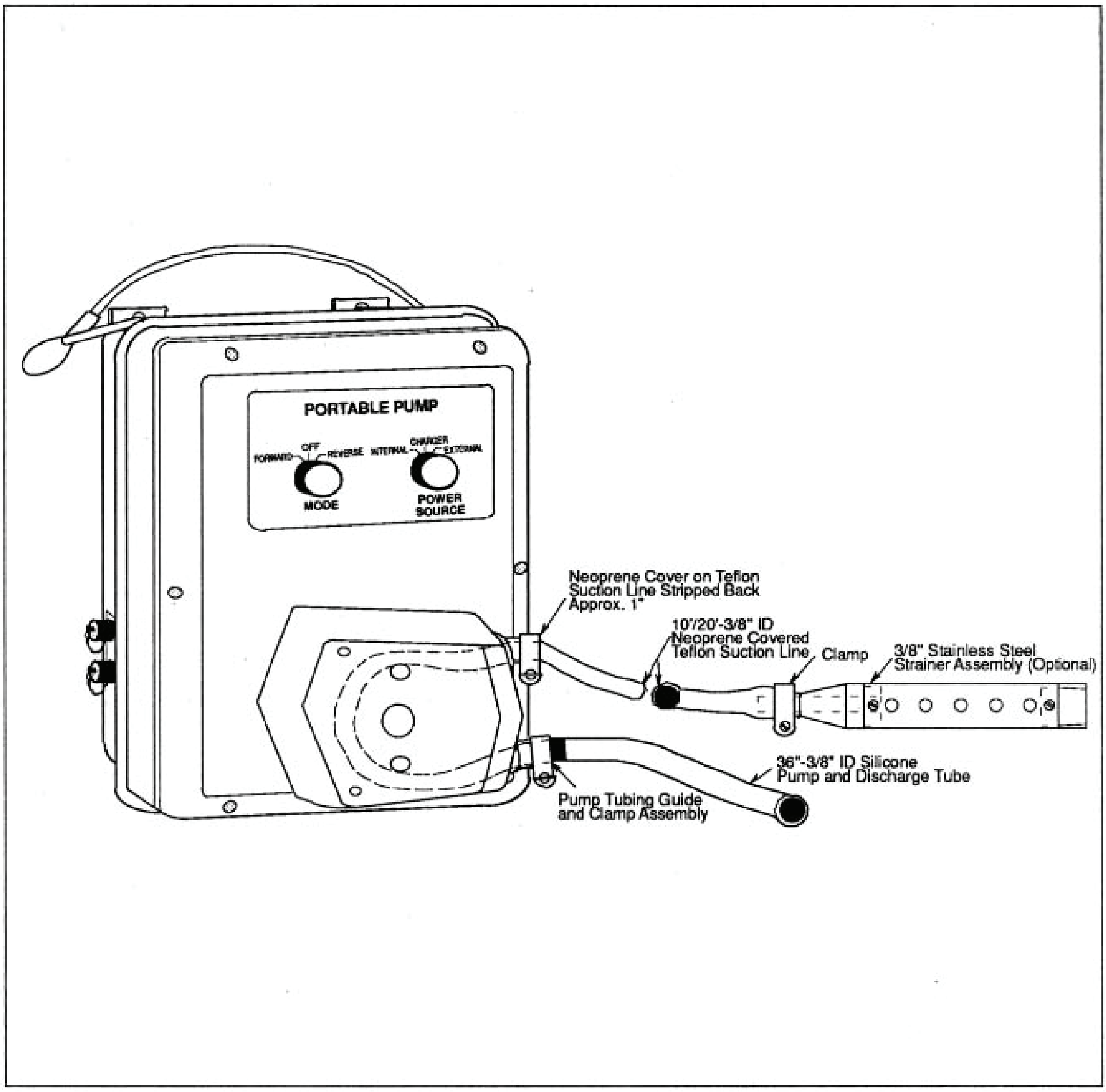
ATTACHMENT 3
Manual Hand Pump

MANUAL HAND PUMP



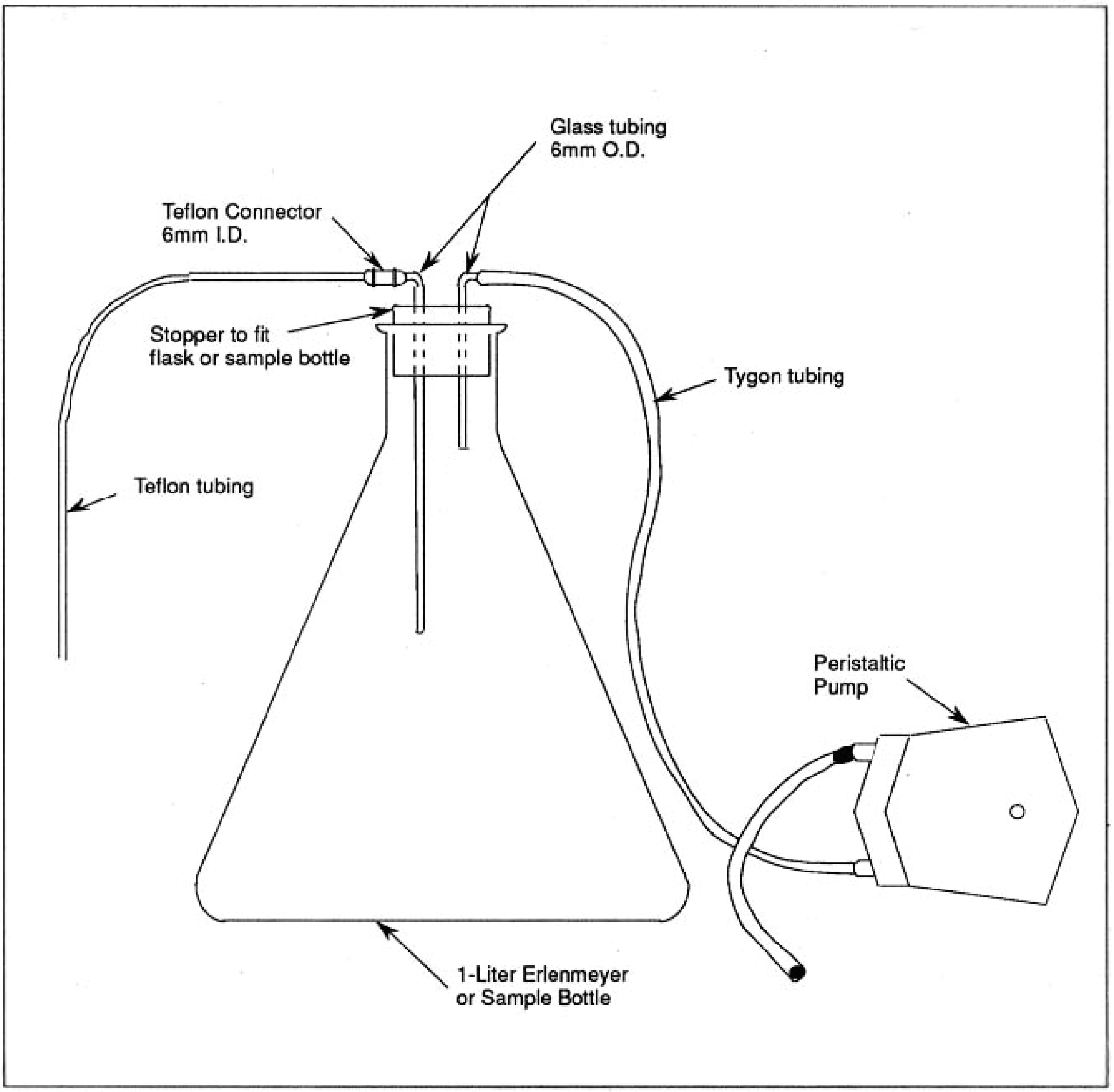
ATTACHMENT 4
Peristaltic Pump

PERISTALTIC PUMP



ATTACHMENT 5
Peristaltic Pump - Modified

PERISTALTIC PUMP - MODIFIED



ATTACHMENT 6
Surface Water Sampling Data

HGL HydroGeoLogic, Inc. Exceeding Expectations

Surface Water Sampling Data

Records Management Data

Project Number	Project Name	Page
		____ of ____

Time/Date: ______________________ Elevation: ______________________

Sample No.: ______________________ Weather: ______________________

Location: ______________________ Amb. Temp (°F): ______________________

Sampling Method: ______________________

WATER SAMPLE DATA

Water Temp: ______________________ °C Method of Measurement: ______________________

Specific Conductance: ______________ micromhos Method of Measurement: ______________________

pH: ______________________ Method of Measurement: ______________________

Containers Used (VOA Vial, 1 liter jar, etc.): ______________________

Physical Appearance: ______________________

Contamination Observed: ______________________

Remarks: ______________________

____

____

____

____

____

____

Location Sketch

Recorded By:	Date	Checked By:	Date:

STANDARD OPERATING PROCEDURE

Field Logbook Use and Maintenance	SOP No.: 4.07 SOP Category: HTRW Revision No.: 1 Date: December 2010

1.0 PURPOSE

The purpose of this standard operating procedure (SOP) is to describe the methods for use and maintenance of field logbooks. This procedure outlines methods, lists examples for proper data entry into a field logbook, and provides the standardized HydroGeoLogic, Inc. (HGL) format.

2.0 SCOPE AND APPLICATIONS

This procedure provides guidance for routine field operations on environmental projects. Site-specific deviations from the methods presented herein must be approved by the assigned HGL project manager and the HGL project quality assurance/quality control officer. Consult the project-specific planning documents for other documentation requirements that apply to the project.

3.0 GENERAL REQUIREMENTS

All work will be performed in a manner that is consistent with Occupational Safety and Health Administration established standards and requirements. Refer to the site- or project-specific health and safety plan for relevant health and safety requirements.

Personnel who use this procedure must provide documented evidence to the program manager or project manager that they have been trained on the procedure. This documentation will be retained in the project file.

Any deviations from specified requirements will be justified to and authorized by the project manager and/or the relevant program manager and documented in the planning documents. Deviations from requirements will be sufficiently documented to re-create the modified process.

All field personnel who travel to a site to conduct work related to environmental projects are responsible for documenting field investigation activities in project field logbooks in a legible manner and maintaining field logbooks over the course of the project in accordance with this SOP. Daily logs will be kept during field activities by the HGL field team leader, or approved designee, to provide daily records of significant events, observations, and measurements taken in the field.

The project manager or an approved designee is responsible for checking the field logbooks and verifying that they have been completed in accordance with this SOP.

4.0 PROCEDURE

4.1 INTRODUCTION

Field logbooks provide a means for recording observations and activities at a site. Field logbooks are intended to provide sufficient data and observation notes to enable participants to reconstruct events that occurred while performing field activities and to refresh the memory of field personnel when writing reports or giving testimony during legal proceedings. As such, all entries will be as factual, detailed, and as descriptive as possible so that a particular situation can be reconstructed without reliance on the collector's memory. Field logbooks are not intended to be used as the sole source of project or sampling information. A sufficient number of logbooks will be assigned to a project to ensure that each field team has a logbook at all times.

4.2 FIELD LOGBOOK IDENTIFICATION

Field logbooks shall be bound books with consecutively numbered pages. Logbooks will be permanently assigned to field personnel for the duration of a project, or sampling event. When not in use, the field logbooks are to be stored in site project files. If site activities stop for an extended period of time (2 weeks or more), field logbooks will be stored in the project files in the appropriate HGL office.

The cover of each logbook will contain the following information:

- Organization to which the book is assigned (HGL)
- Project number (if different than site number)
- Book number
- Site name

4.3 LOGBOOK ENTRY PROCEDURES

Every field team will have a logbook, and each field activity will be recorded in the logbook by a designated field team member to provide daily records of significant events, observations, and measurements during field operations. Beginning on the first blank page and extending through as many pages as necessary, the following list provides examples of useful and pertinent information that may be recorded (optional).

- Serial numbers and model numbers for equipment that will be used for the project duration
- Formulas, constants, and example calculations
- Useful telephone numbers

- County, state, and site address

Entries into the logbook may contain a variety of information. At a minimum, logbook entries must include the following information at the beginning of each day:

- Date
- Site name, site location, and project number
- Start time
- Weather
- All field personnel and subcontractors present and directly involved
- Level of personal protective equipment being used on the site
- Equipment used and calibration procedures followed
- Any field calculations

In addition, information recorded in the field logbook during the day will include, but is not limited to, the following:

- Sample description including sample numbers, time, depth, volume, containers, preservative, and media sampled
- Information on field quality control samples (e.g., duplicates)
- Sample courier airbill numbers and associated chains-of-custody
- Observations about site and samples (odors, appearances, etc.)
- Information about any activities, extraneous to sampling activities, that may affect the integrity of the samples
- Any public involvement, visitors, or press interest, comments, or questions; as well as times present at site
- Equipment used on site including time and date of calibration along with calibration gas/fluid lot numbers and expiration dates
- Background levels of each instrument and possible background interferences
- Instrument readings for the borehole, cuttings, or samples in the breathing zone and from the specified depth of the borehole, etc.
- Field parameters (pH, specific conductivity, etc.)
- Unusual observances, irregularities, or problems noted on site or with instrumentation used

- Maps or photographs acquired or taken at the sampling site, including photograph numbers and descriptions
- A description of the investigation-derived waste (IDW) generated, the quantity generated, and the manner of IDW storage employed
- A photograph log that lists subject and persons, distance to subject, person taking photograph, distance, direction, time, photograph number, and noteworthy items for each photograph
- Forms numbers and any information contained therein used during sampling (Note that a form does not take the place of the field logbook.)

All logbook entries will be made in indelible black or blue ink. No erasures are permitted. If an incorrect entry is made, the data will be crossed out with a single strike mark and initialed and dated by the originator. Entries will be organized into easily understandable tables if possible. A sample format is shown in Attachment 1.

All logbook pages will be initialed and dated at the top of each page. Times will be recorded next to each entry. No pages or spaces will be left blank. If the last entry for a day is not at the end of a page, a diagonal line will be drawn through the remaining space and the line will be initialed and dated.

Logbooks can become contaminated when used in the field. Every effort should be made by the field team to avoid contaminating the logbook. Logbooks can be kept in seal-top poly bags or temporary plastic covers may be used.

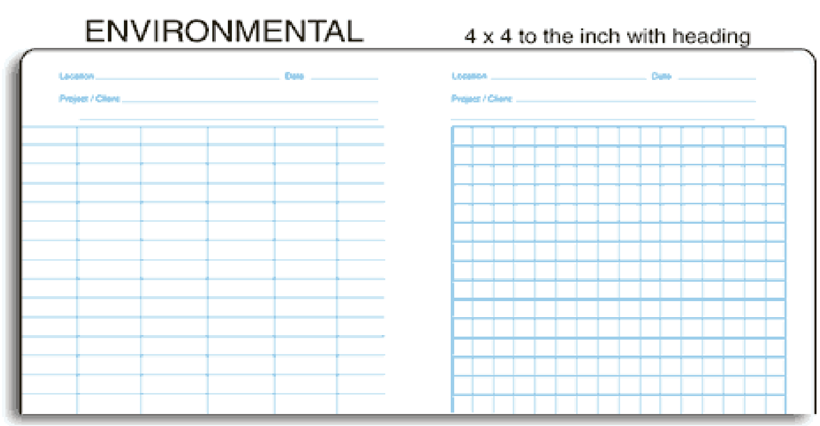
4.4 REVIEW

The assigned project leader or an approved designee will check field logbooks for completeness and accuracy on an appropriate site specific schedule determined by the project leader. Any discrepancies in these documents will be noted and returned to the originator for correction. The reviewer will acknowledge that these review comments have been incorporated by signing and dating the applicable reviewed documents.

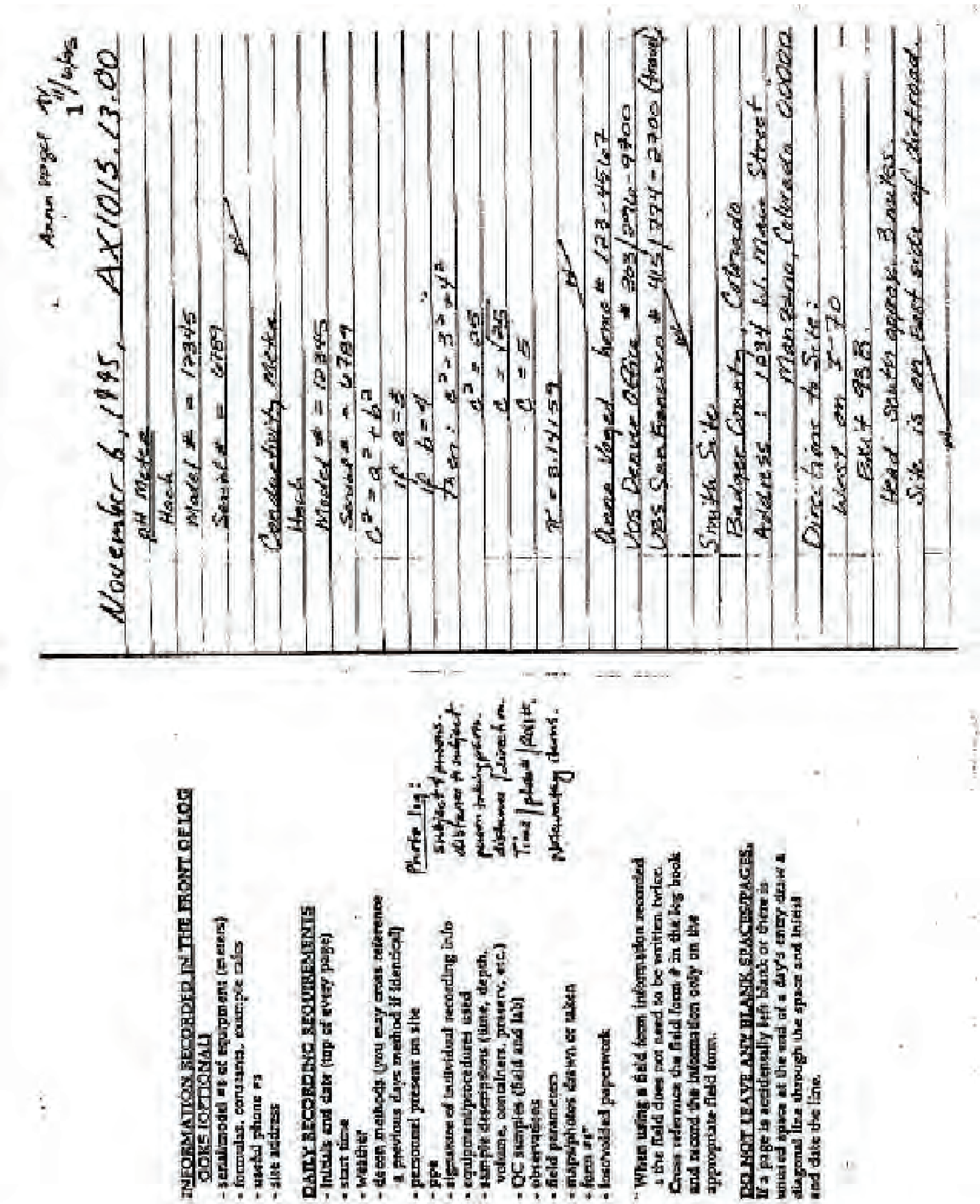
5.0 ATTACHMENTS

Attachment 1 Example Field Logbook

ATTACHMENT 1
Example Field Logbook



ATTACHMENT 1 (continued)
Example Field Logbook



INFORMATION RECORDED IN THE FRONT OF LOG BOOKS (OPTIONAL)
- serial/model #s of equipment (meters)
- formulas, constants, example calcs
- useful phone #s
- site address

DAILY RECORDING REQUIREMENTS
- initials and date (top of every page)
- start time
- weather
- decon methods (you may cross reference a previous days method if identical)
- personnel present on site
- ppe
- signature of individual recording info
- equipment/procedures used
- sample descriptions (time, depth, volume, containers, preserv, etc.)
- QC samples (field and lab)
- observations
- field parameters
- maps/photos drawn or taken
- form #s*
- lost/voided paperwork

Photo log:
subject of photo.
distance to subject.
person taking photo.
distance / direction.
Time / photo# / roll#.
photo-voucher doc#.

* When using a field form information recorded on the field does not need to be written twice. Cross reference the field form # in the log book and record the information only on the appropriate field form.

DO NOT LEAVE ANY BLANK SPACES/PAGES. If a page is accidentally left blank or there is unused space at the end of a day's entry draw a diagonal line through the space and initial and date the line.

Anna Vogel AV 11/6/95
1

November 6, 1995, AX1015. 13:00
pH Meter
Hach
Model # = 12345
Serial # = 6789
AV

Conductivity Meter
Hach
Model # = 12345
Serial # = 6789
$C^2 = a^2 + b^2$
if $a = 3$
if $b = 4$
Then: $c^2 = 3^2 + 4^2$
$c^2 = 25$
$c = \sqrt{25}$
$c = 5$
$\pi = 3.14159$
AV

Anna Vogel home # 123-4567
VOS Denver Office # 303/296-9700
VOS San Francisco # 415/774-2700 (travel)
AV

Smith Site
Badger County, Colorado
Address: 1234 W. Main Street
Manzano, Colorado 00000
Directions to Site:
West on I-70
Exit 93B.
Head South approx. 3 miles.
Site is on East side of dirt road.
AV

ATTACHMENT 1 (continued)
Example Field Logbook

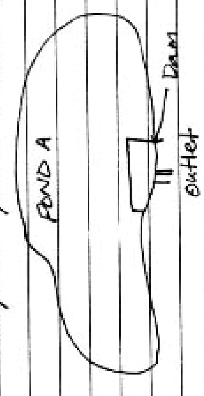
AV 11/6/95
2

November 6, 1995 Site Visit
0700 Arrive on site
Weather: 80°, sunny, slight breeze (~5 mph) from southwest.
UOS Field Team: EPA OSC:
M.R. Smith J.P. Scarton
K.W. Wagner
P.R. Lane
PRP representative, L.M. Stein, will be accompanying the UOS Field Team.
Personal Protective Equipment – LEVEL D will be used on-site (refer to site-specific health & safety plan).
All equipment will be deconed as follows:
- Brush equipment scrub brush to remove gross particulates.
- Scrub thoroughly with Alconox/water solution.
- Rinse with reagent-grade distilled water.
- Rinse with reagent-grade Methanol.
- Rinse with reagent-grade distilled water.

Allow equipment to gravity drain
Wrap equipment in tinfoil if not immediately used.
Sample procedure:
All surface water samples will be taken using a clean decontaminated TEFLON scoop; stainless steel spoon and stainless steel bowl will be used for sediment samples.

AV 11/6/95
3

The samples will be taken from the ponds at the center of the dam opposite the outlets. (see below; refer to sample plan).
All total suspended solids (TSS) samples will be collected in a 500 ml polystyrene bottle – No preservative is necessary.
All VOA samples will be collected in two 40-ml amber glass vials and will be collected first. Preservation will be 4°C (ice).
→ Meters (pH) Decon = Rinse with reagent-grade distilled water.



0730: leave trailer. Go to sample location SS-1 @ Pond A.
0745: Arrive @ POND A.
Decon. equipment as described on page 2 of this logbook.
Calibrate pH meter – Rinse probe

Time	STD	Reading	
0753	7.00	7.00	Rinse probe
0754	4.00	4.00	Rinse probe

0756 Calibrate Conductivity Meter using 10000 STD – Rinse probe

ATTACHMENT 1 (continued)
Example Field Logbook

AV 11/6/95
4

Time Sample Sample# Label #
0802 VOA 81088 VOAA 101
0803 TSS 8108 TSSA 103*
Decon equipment (scoop only).
*label 102 fell in mud - destroyed it.
Field Parameters
Time pH Conductivity
0815 6.35 610
Decon equipment (meters only)
Fill out surface water quality Sheet.
Note – wind speed is picking up –
The ponds become turbulent.
0829 – Leave Pond A – go to Pond B.
AV

0840 – Arrive at Pond B
Pond B sampling procedure.
0842 – Decon equipment.
Calibrate pH meter
Time STD Reading
0844 4.00 4.00 Rinse probe
0845 7.00 7.00 Rinse probe
0847 Calibrate conductivity meter
using 10000 STD – Rinse probe.
Decon sampling equipment (scoop).
Time Sample Sample# Label #
0902 VOA 81088 VOA BD 106
0903 TSS 81088 TSS BD 107
0903 Decon scoop.
AV

Rinsate Samples
Time Sample Sample# Label#
0920 VOA 81088 VOAR AV ~~107~~ 108

11/6/95 AV
5

FIELD PARAMETERS
TIME PH Conductivity
0924 6.00 590
Decon meters as noted on page 3
of this logbook.
Fill out surface water quality sheet.
AV

0940 – Leave Pond B – head back
to trailer to pack samples for
shipment.
0952 – Arrive at Trailer.
0959 – Complete chain-of-custody
forms for samples to be shipped.
Wrap samples according to VOS
TSOB.
1020 – Seal Cooler and attach
Custody seals.
1030 – Take cooler to Federal Express
for shipping.
COC # 1234567.
1035 – Leave Federal express.
Sampling complete.

AV 11/6/95

STANDARD OPERATING PROCEDURE

LANDFILL GAS MEASUREMENTS

1.0 PURPOSE

The purpose of this standard operating procedure (SOP) is to describe the equipment and operations used for sampling landfill gas from a subsurface gas monitoring well. This procedure outlines the methods for landfill gas sampling for routine field operations on environmental projects. Regardless of application, any sampling program should include a review of the following items: sampling program development, sample documentation techniques, analytical instrumentation, sample analysis, and final data interpretation. Site-specific deviations from the methods presented herein must be approved by the Project Leader, and the HydroGeoLogic, Inc. (HGL) Quality Assurance Manager.

2.0 DEFINITIONS AND ABBREVIATIONS

2.1 Definitions

Landfill Gas: A general term used to describe the gas produced by the bacterial breakdown of waste materials within a landfill. The gas consists primarily of methane and carbon dioxide, with traces of other constituents such as volatile organic compounds (VOCs) and other hazardous air pollutants.

Landfill Gas Meter: A device used to obtain real time measurements of various properties of landfill gas, such as volume percent methane, carbon dioxide, oxygen, and lower explosive limit (LEL).

2.2 Abbreviations

LEL	Lower explosive limit
SOP	Standard Operating Procedure
VOCs	Volatile organic compounds

3.0 RESPONSIBILITIES

Sampling personnel are responsible for performing the applicable tasks and procedures outlined herein when conducting work related to environmental projects.

The Project Leader or an approved designee is responsible for checking all work performance and verifying that the work satisfies the applicable tasks required by this procedure. This will be accomplished by reviewing all documents and data produced during work performance.

4.0 PROCEDURES

4.1 Introduction

Overall objectives of landfill gas sampling can vary, but generally is performed to assess the presence and concentration of landfill gas in the subsurface so that decisions may be made on whether the gas presents a concern, and if so, how the gas may best be managed. Landfill gas monitoring may be used to define:

- Whether landfill gas is being produced from historical buried waste.
- The magnitude of methane concentrations and lower explosive limit of the landfill gas, which can dictate whether a landfill gas management and/or monitoring system is required.
- Whether landfill gas is migrating beyond the limits of the landfill, which could require mitigation if the gas could potentially migrate into inhabited areas and create safety and health issues.

Landfill gas sampling described in this SOP is for qualitative screening purposes only to help determine the presence and general magnitude of landfill gas concentrations. Based on the results obtained from this qualitative screening, it may be necessary to collect and submit landfill gas samples for offsite laboratory analysis.

Landfill gas presents several health and safety concerns that should be fully understood before undertaking any sampling effort. Key concerns associated with landfill gas are listed below.

- Landfill gas is typically composed primarily of methane (typically 45 to 60 percent) and carbon dioxide (40 to 60 percent).
- Methane, the primary component of landfill gas is a colorless, odorless, highly flammable and potentially explosive gas.
- The flammable range of methane is approximately 5 to 15 percent by volume in air.

- Landfill gas can migrate through long distances in subsurface soil, conduits, permeable backfill, and into confined spaces.
- Landfill gas that accumulates within structures or enclosed or confined spaces can create explosive environments.

Based on these properties of landfill gas, care must be taken to avoid ignition sources in locations where landfill gas may be present. Intrinsically safe devices should be used, and no open flames or other ignition sources are allowed within these areas.

4.2 Sampling Equipment

Landfill gas sampling equipment includes:

- Logbook or log sheets for recording landfill gas measurements
- Barometer for barometric pressure measurements
- Thermometer for measurement of ambient temperatures
- Monitoring well construction details
- Monitoring well cap with sampling port/valve capable of being tightly sealed
- Unused, clean tygon or other inert tubing for sampling
- Cap, fittings, hose barbs, and/or clamps for connecting tubing to well sampling ports or valves to provide a gas tight seal

- Landfill gas meter capable of measuring and recording concentrations of methane, oxygen, and carbon dioxide, pressure, flowrate, temperature, and LEL (Landtec GEM 500 or 2000, or similar)

Depending on the wellhead arrangement at a particular well, not all equipment (fittings, valves, clamps, etc.) may be necessary for sampling.

4.3 Decontamination

Sample tubing, caps, and other fittings will be dedicated to each sample location, so no decontamination should be required. Used tubing will be disposed as municipal waste.

4.4 Sampling Location/Site Selection

The sample locations are shown on location maps within the site specific work plan. The sample locations were selected to provide the data necessary to achieve the objectives of the sampling.

4.5 Measurement Procedure

Procedures to be followed for measurement of landfill gas properties are listed below.

- Inspect the wellhead to be sampled and, if necessary, install a cap, sample port, hose barb, and/or valve to allow connection of sample tubing to the wellhead and provide a gas tight seal for sample collection.
- Based on well construction data, calculate the casing volume of the well. As necessary, convert the volume units to units displayed by the landfill gas meter.
- Record ambient weather conditions in the logbook or on the log sheet.
- Calibrate and zero the landfill gas meter per manufacturer's instructions. An operations manual for a Landtec GEM 500 landfill gas meter is attached as an example set of instructions. If the meter will not calibrate or zero appropriately, make meter adjustments or replace the meter until it can be properly calibrated and zeroed.
- Connect the sample collection tubing to the gas monitoring wellhead using inert flexible tubing and gas tight fittings. If a valve is installed at the wellhead, make sure the valve is open.
- The landfill gas meter should begin purging gas from the wellhead. Based on the well casing volume calculated earlier and the gas flowrate measured by the landfill gas meter, purge at least one well casing volume from the well.
- If readings have not stabilized, continue purging. Readings are considered stable when they vary by less than 0.5 percent by volume.
- Landfill gas typically contains little oxygen (less than 2 percent). If oxygen levels are at or above this level, this indicates an air leak in the system. Check tubing, fittings, connections, and wellhead for leakage.
- Record meter data as required.

5.0 Data Recording

At a minimum, the following data shall be recorded in the field logbook or on data log sheets for each measurement event.

- Name of person(s) taking the measurements
- Date and time of measurements
- Measurement location identification
- Ambient temperature, barometric pressure, and weather conditions
- Landfill gas meter make and model
- Documentation of meter calibration
- Percent methane, oxygen, carbon dioxide, temperature, pressure, gas flowrate, LEL
- Duration of purging required to achieve stable readings

6.0 Attachments

Attachment A: Landtec GEM 500 Landfill Gas Meter Operations Manual

Attachment A

Attachment A: Landtec GEM 500 Landfill Gas Meter Operations Manual

This page was intentionally left blank.

GEM-500

OPERATION MANUAL

CES LANDTEC

TABLE OF CONTENTS:

Chapter 1 - Getting Started....1
- Unpacking the GEM-500™....1
- Attaching the Hose Assembly....1
- GEM-500™ Keyboard and Port Descriptions....2
- Must Do's Before Using the GEM-500™....3
- Calibration Gases....3
- Special Key Functions....3
  - Entering an ID code with Letters and Numbers....3
  - Backspace Function....3
  - Contrast Adjustment....3

Chapter 2 - Using Menu Screens....4
- Starting Up the GEM....4
- GEM-500 Menu Tree....5
- Review of the Main Menu and Sub-Menu Screens....5
  - General Utilities....5
  - Read Gas Levels....6
  - View/Print Data....6
  - Download Data....6

Chapter 3 - Field Calibration....7
- Calibration Gas/Span Gases....7
- Zero Methane....7
- Span Methane....8
- Span Carbon Dioxide....8
- Zero Oxygen....8
- Span Oxygen....8
- Equipment....9
- Setting Up the Equipment....10
- General Utilities KEY 5-Gas Calibration....11
- Methane (CH_4) Calibration - Zero CH_4....12
- Methane (CH4) Calibration....13
- Carbon Dioxide (CO_2) Calibration....15
- Oxygen (O_2) Calibration - Zero O_2....17
- Factory Setting Calibrations....21
- After Completing Gas Calibrations....22

Chapter 4 - General Utilities Functions....23
- General Utilities Screen Tree Diagram....23
- General Utilities Menu....23
- General Utilities Functions....24
  - Check Time/Date....24
  - Battery Status....26
  - Zero Pressures....27
  - Memory....28
  - USA/Metric Units....30
  - Gas Calibration....31
  - Gas Alarms....31
  - ID Maintenance....32

Chapter 5 - Read Gas Levels....36
- GEM-500™ Hose and Wellhead....36
- CES-LANDTEC Horizontal Accu-Flo Wellhead....37
- CES-LANDTEC Vertical Accu-Flo Wellhead....38
- Read Gas Levels Screen Tree Diagram....38
- Read Gas Levels Menu – Read Using ID? No....39

Read Gas Levels Menu – Read Using ID? Yes 43
Chapter 6 - View Data....53
View Data 53
Chapter 7 - Communications....55
Installing The Gas Analyzer Communications Program....55
Starting The Gas Analyzer Communications Program 56
Main Menu....57
Configure 58
User Defined Comments....59
Well ID Functions....62
GEM Data Functions 66
Show GEM memory statistics....70
Clear Functions....70
Chapter 8 - Maintenance....72
Servicing....72
Cleaning 72
Sunlight and Heat....72
Dust Cap....72
Filters....73
Travel and Storage....73
Battery Charging 74
Battery Shut-Off....74
Battery Low Symbol 74
Automatic Power-Off....75
Emergency Battery Power....75
Chapter 9 - Troubleshooting....76
Chapter 10 - 77
Measurement Units & Specifications....77
Measurement Units 77
Operating Temperature....77
Range and Resolution....78
Accuracy....78
Chapter 11 - Field Operations 79
Landfill Gas Generation 79
Subsurface Fires....79
Techniques for Controlling Landfill Gas....80
Controlling by Wellhead Valve Position 80
Controlling by Wellhead Vacuum 80
Controlling by Gas Composition....80
Controlling by Flow Rate 81
Well field Monitoring....81
Typical Field Readings 81
Abbreviated Field Readings 82
Well field Adjustment Criteria....83
Establishing Target Flows....84
Well field Optimization....84
Migration Control—Dealing with Poor Methane Quality 84
Well field Adjustment—Purpose and Objectives 85
CES-LANDTEC Technical Tips 85

Chapter 1 - Getting Started

Unpacking the GEM-500™

The GEM-500™ unit is normally shipped in a special protective Styrofoam shipping unit. An optional protective hard case with a foam interior offers additional protection, transportation convenience and component hardware storage. When properly sealed, the hard case is watertight. The hard case is equipped with a pressure relief valve (located under the handle on the case) that is normally kept closed. If there is a change in elevation, the hard case may not open until internal pressure is equalized by turning the pressure relief valve. When shipping a GEM-500™ back to CES-LANDTEC for calibration or service, always ship it in the original packaging to protect unit from damage.

Carefully unpack the contents of the GEM-500™, inspect and inventory them. The following items should be contained in your package:

- The GEM-500™ unit
- GEM-500™ Operation Manual
- Registration/Warranty Card and other instructional information
- Soft carrying case with replaceable protective window and carrying strap
- External (clear vinyl) sampling hose assembly (5 ft.) with external water trap filter assembly
- Blue ¼" vinyl pressure tubing sampling hose (5 ft.)
- Spare internal particulate filter element
- Polypropylene male connector (hose barb) connects to blue vinyl tubing
- Spare external water trap filter element
- 110-volt Nickel-Cadmium battery charger
- GEM-500™ download software disk on 3 ½ inch floppy disk
- RS-232 serial cable for computer/printer data downloading
- Temperature probe (optional)
- Hard carrying case (optional)

Immediately notify shipper if the GEM-500™ unit or accessories are damaged due to shipping. Contact CES-LANDTEC if any items are missing. If you have any questions, please contact CES/ LANDTEC technical support at (800) 526-3832 or (800) LANDTEC. Complete the Registration/Warranty Card and return it to CES-LANDTEC. The model and serial numbers are located on the back of the GEM-500™ unit.

Attaching the Hose Assembly

The GEM-500™ hose assembly comes fully assembled but it needs to be connected to the GEM-500™. Connect the clear tubing with the external filter/water trap assembly to the static pressure/ sampling port (top left corner) on the GEM-500™ (See Figure 1.1). The shorter piece of tubing (from the water trap filter hosing) should be connected to the GEM-500™. This allows you to see any liquid entering the hose and shut the unit off before the liquid reaches the GEM-500™. Always connect the hose in the same direction. Connect the blue tubing to the impact pressure port on the GEM-500™ (See Figure 1.1). This port is located on the bottom left corner of the GEM-500™. **DO NOT** block the exhaust port (See Figure 1.1).

Quick Connect Fittings

The quick connect fittings will simplify taking well field readings. They are easy to install on your landfill gas extraction system and on perimeter probes. Many different types are available. CES-LANDTEC maintains a stock of fittings used on its equipment for your convenience.

The GEM-500™ comes with quick connect fittings for the AccuFlo™ wellhead. Insert the hose barb end of the male connector into the end of the clear and blue tubing.

GEM-500™ Keyboard and Port Descriptions

1. **Red On/Off Key**—Turns unit on or off.
2. **Blue Number/Letter Toggle Key**—Enables well ID code to be entered by toggling between number and letter mode and toggles contrast on the gas read screens.
3. **Receptacle Port**—Used for battery recharging, RS232 serial communications, temperature probe or gas pod.
4. **Backspace/Exit Key**—Acts as backspace key when pressed and held for one second, to correct entry of wrong number/letter, returns to previous procedure or steps back one layer of menus (similar to pressing the ESCAPE key in many computer programs).

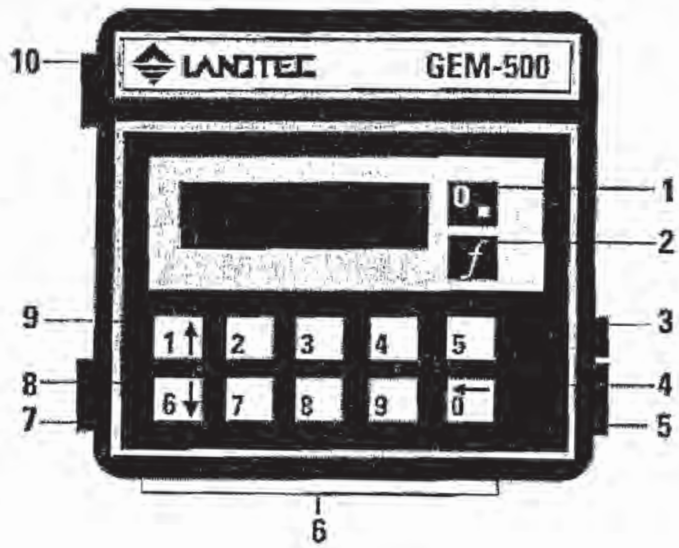


FIGURE 1.1

5. **Exhaust Port**—This port must be kept clear. If blocked while operating, over-pressurization may occur causing damage to internal components and case.
6. **Number Keys**—Enter numbers 0 through 9.
7. **Impact Pressure Port**—Measures impact pressure when connected to wellhead impact pressure port, pitot tube or orifice plate.
8. **Cursor-Down Key**—Enters number 6, scrolls down lines of information on display screens, and also scrolls down alphabetic character list.
9. **Cursor-Up Key**—Enters number 1, scrolls up lines of information on display screens, and also scrolls up alphabetic character list.
10. **Static Pressure/Sampling Port**—Measures static pressure and is inlet for gas sampling.

Must Do's Before Using the GEM-500™

Read Chapter 2 – *Using Menu Screens*.

Proper operation of the GEM-500™ requires the following functions to be completed before proceeding.

- Charge the unit with the battery charger
- Check the Time/Date
- Field Calibrate the unit

Calibration Gases

Calibration gases are required to field calibrate the GEM-500™. Portable Calibration Gas Kits and 4-unit or 12-unit cylinder cases are available from CES-LANDTEC. (See Chapter 2 -- *Field Calibration*)

Special Key Functions

Entering an ID code with Letters and Numbers

Use the blue toggle key (*f*) to shift back and forth between number mode and letter mode. When in number mode, use number keys to enter numbers. When switched to letter mode, use the **1** KEY (**UP ARROW**) or the **6** KEY (**DOWN ARROW**) to scroll to desired letter, press **0** KEY to enter the letter on the display. Repeat this process for all letters. After entry, the first four characters will remain as a default for ease in entering the next ID. If different characters are desired, replace the defaults by using the backspace function described below.

Backspace Function

To change or correct an entry, use the **0** KEY (**BACK ARROW**) as a backspace key by holding it down for one second. In normal use, this key is quickly pressed and released.

Contrast Adjustment

Contrast can be adjusted when the unit is either first turned on or while taking a reading. While taking a reading, use the Blue *f* KEY to enter the contrast adjustment screen. To adjust, use 1 KEY (UP ARROW) to darken the screen and the 6 KEY (DOWN ARROW) to lighten screen.

Starting Up the GEM

This procedure is the same each time the GEM-500™ is turned on by pressing the RED On/Off KEY. The following steps will allow you to proceed to the Main Menu Screen of the GEM-500™.

1. Turn unit on by pressing the RED **On/Off** KEY (see Figure 1.1)

   Note: If the GEM is turned on and no additional keys are pressed within 15 minutes, the unit will automatically shut off.

2. The Warning screen appears for five seconds. This is a reminder that the GEM-500™ is not to be used in areas such as vaults, excavations or other confined spaces. An explosion could result causing serious injury or death.

FIGURE 2.1

Warning! --Do not use
in confined spaces.
Unit NOT certified
intrinsically safe.

3. The Service Contract screen may appear for five seconds if activated by CES-LANDTEC. Otherwise, the Not Covered screen is displayed. The GEM-500™ is a portable, scientific, field instrument that does require factory maintenance and calibration at recommended six-month intervals under normal landfill usage.

FIGURE 2.2

This analyzer has a
Service Contract
Next service due:-
dd/mm/yy

FIGURE 2.3

Unit not covered by
Service Contract
Next service due:-
dd/mm/yy

4. The CES-LANDTEC/Contrast screen allows the user to adjust the contrast of the characters on the liquid crystal display screen. Press and hold the **1** KEY (**UP ARROW**) to increase contrast. Press and hold the **6** KEY (**DOWN ARROW**) to decrease the contrast. Adjust the contrast as necessary (contrast levels are **NOT** saved when the unit is turned off). Press the **0** KEY to proceed to the Main Menu screen.

FIGURE 2.4

CES-LANDTEC
GEM 500
(800) 821-0496
↑↓–Contrast 0-Cont

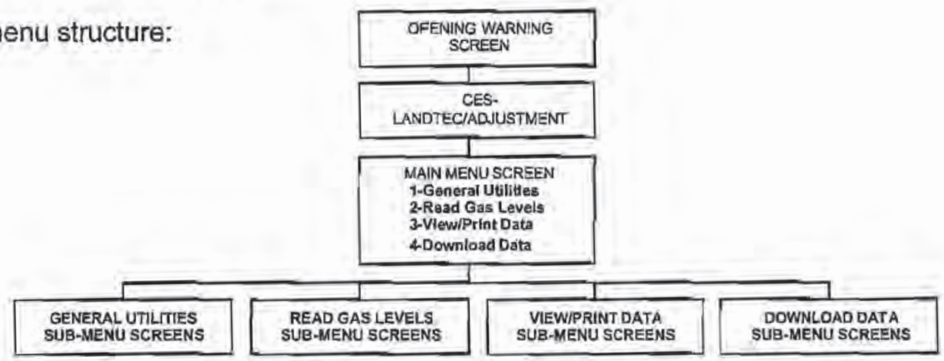
5. **The Main Menu Screen.** All the GEM-500™ functions are accessed from the Main Menu Screen. All subsequent instructions about the GEM-500™ functions will start from this screen.

FIGURE 2.5

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

GEM-500 Menu Tree

The overall menu structure:



Review of the Main Menu and Sub-Menu Screens

General Utilities
Refer to Chapter 4 for further information. The General Utilities function has sub-menu screens that allow housekeeping and other maintenance including:

1. CHECK TIME/DATE: Set or check time and date.
2. BATTERY STATUS: Graphic display of remaining power in batteries.
3. ZERO PRESSURES: Zero pressure transducers.
4. MEMORY: Check memory available or clear all data and ID information.
5. USA/METRIC UNITS: Select either USA standard or metric measurement units.
6. GAS CALIBRATION: Allow methane, Carbon Dioxide and oxygen to be field calibrated by the user with calibration gas mixtures for increased accuracy (see Chapter 3).
7. GAS ALARM: Set gas alarm levels.
8. ID MAINTENANCE: View, enter, edit or delete ID information.

Read Gas Levels

Refer to Chapter 5 for further information. Read Gas Levels function allows gas, pressure, flow and BTU readings to be viewed and recorded. Sub-menu screens include:

1. Read GAS with Existing ID code.
2. Read GAS without ID code.

View/Print Data

For further information, refer to Chapter 6. The View/Print Data function allows previously stored data to be scanned on the GEM-500™ display screen, individually displayed, or printed via the RS-232 cable to a serial printer.

Download Data

The Download Data function allows stored data to be downloaded via the RS-232 cable to a computer in a format that can be uploaded into DataField (CES-LANDTEC database management program) or onto spreadsheets. See Chapter 7 for further information.

> **Note: The 0 KEY (BACKSPACE) acts as an exit or ESCAPE key at the end of each sub-menu by returning to the Main Menu.**

Chapter 3 - Field Calibration

Field Calibration is menu guided and can be completed in about ten minutes. To streamline the procedure, the pump remains running during field calibration. The GEM-500™ contains a calibration map accessed by its microprocessor for baseline reference data. This reference data was programmed into the GEM-500™ during factory calibration using various traceable gas mixtures in an environmental chamber. At any time, the GEM-500™ can be reset to factory settings which clears any user calibration settings and restores the GEM-500™ to its original factory calibration.

The factory calibration has been designed to give the best possible results over a wide range of conditions. However, the instrument's accuracy can be improved in specific operating ranges by performing a field calibration. Most field instruments are calibrated or adjusted prior to taking a series of gas or pressure readings. They may also be checked for calibration during and after readings in order to verify the accuracy of the data collected.

It is important to field calibrate the GEM-500™ on-site after the instrument has stabilized at working temperature. For this reason, a GEM-500™ that was calibrated in the cool of the morning may not read as accurately during the hottest part of the day.

Note: Field calibration of the GEM-500™ will improve the data collected in the range of the calibration gases used. Less accurate readings of concentrations outside the calibrated range may occur. For example, a GEM-500™ that was field calibrated using 50% CH_4 and 35% CO_2 will give improved readings for most gas extraction systems. Recommended gas mixtures for reading migration probes are 15% methane, 15% Carbon Dioxide with balance nitrogen. A 4.0% oxygen with balance nitrogen mixture may be used for both types of testing.

Calibration Gas/Span Gases

Field calibration requires two calibration gas mixtures. One gas mixture is used to span oxygen and zero methane. The other is used to span methane, Carbon Dioxide and zero oxygen. The oxygen has two curves: 0-5% and 0-25%. The zero point is the same for both curves; however, the span is different. The user need only span the instrument using calibration gas below 5% for the 0-5% range or calibration gas below 25% for the 0-25% range. Regardless of the ranges used, the instrument **must** be zeroed. Various calibration gas mixtures are available from CES-LANDTEC.

Zero Methane

Calibration of the GEM-500™ starts by establishing the bottom point of the methane gas curve. The methane (CH_4) is zeroed prior to taking readings at the start of each day. This function significantly improves the GEM-500™'s CH_4 accuracy over the entire range. **It is essential that the gas analyzer be clear of CH_4 when zeroed.** Care must be taken if the GEM-500™ is to be zeroed using air near a landfill site because there are situations where methane could be in the atmosphere.

Span Methane

A field calibration spans the methane range prior to taking readings at the start of each day. The best results are obtained after the instrument has stabilized at its working temperature. This procedure alters the methane calibration at all concentrations and stores the revised data in protected memory.

Note: Methane zero must be performed before setting the Methane Span.

Span Carbon Dioxide

Field calibration of CO_2 should be performed prior to taking readings at the start of each day after the instrument has stabilized at its working temperature. This procedure alters the calibration at all concentrations and stores the revised data in protected memory.

Zero Oxygen

This function is essential where low concentrations of oxygen are expected (below 5%). This establishes the zero point of an oxygen curve that is stored in the GEM-500™ protected memory.

Span Oxygen

The oxygen calibration map contains two span curves, one for oxygen below 5% and one for oxygen above 5%. The proper curve is automatically selected. If a calibration gas with less than 5% oxygen is used, the lower span curve is set. If the calibration gas has more than 5% oxygen, the higher calibration curve is set.

Note: The Oxygen zero must be set before setting the Oxygen Span.

Equipment

The following items are required to perform a field calibration:

1. Cylinder of methane and Carbon Dioxide span gas
2. Cylinder of 4/96 (4% O_2 and 96% N_2) calibration gas
3. Pressure regulators for the above cylinders capable of regulating in the range of 0 - 2 psig fitted with connectors suitable for ¼" tubing
4. CES-LANDTEC regulator and flow meter preset to deliver the required flow of 399-500 cc per minute at 2 psig. (See Figure 3.1a)
5. Interconnecting lengths of ¼" tubing

This equipment is available from CES-LANDTEC. The calibration equipment set up is shown in Figure 3.1.a and 3.1.b.

FIGURE 3.1.a Pressure/Flow Regulator

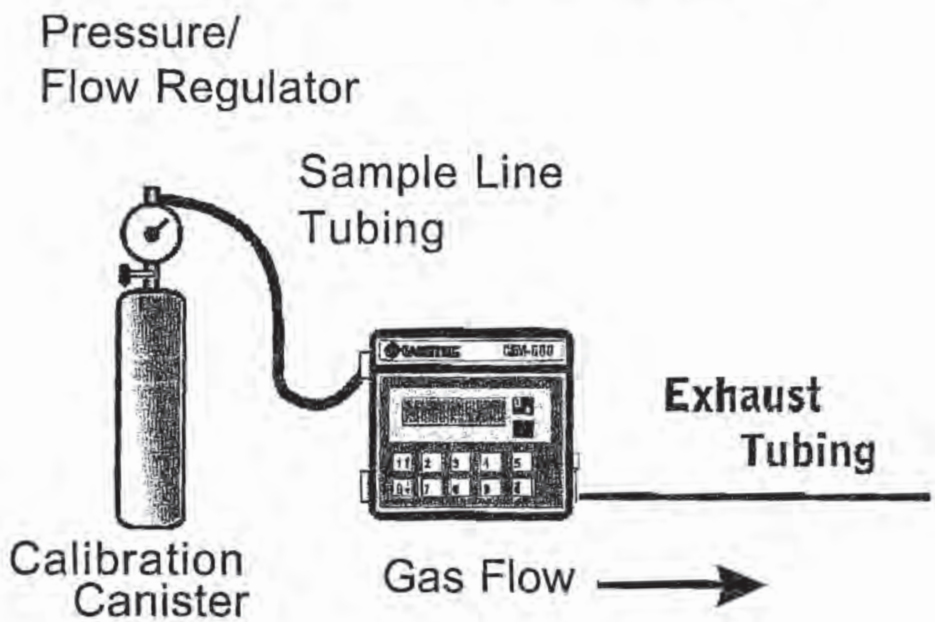
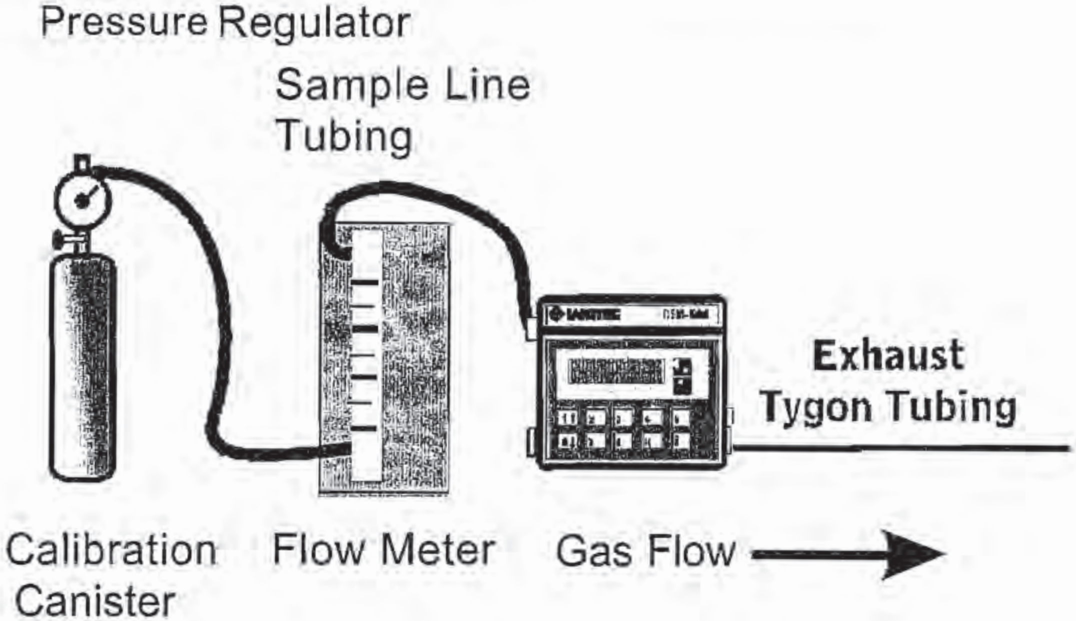


FIGURE 3.1.b Pressure Regulator with Flow Meter



Setting Up the Equipment

1. Connect the calibration gas cylinder to the pressure regulator.
2. Connect the sample input line to the regulator and to the GEM-500™.
3. Connect the second 24" section of ¼" tubing to the exhaust nozzle of the GEM-500™. Direct exhaust away from you and out of the immediate area.
4. If using a CES-LANDTEC regulator, no flow meter is required.
5. If **not** using the CES-LANDTEC regulator, adjust the regulator discharge pressure to 2 psig and the flow meter to 500 cc per minute. Pinch the gas supply hose that will attach to the GEM-500™ and verify the regulator discharge pressure does not exceed 5 psig. Turn off the cylinder valve.

Note: This procedure will be duplicated for the second span gas when oxygen is calibrated. The Oxygen/N_2 calibration gas cylinder will be substituted for the Methane/Carbon Dioxide calibration gas.

General Utilities KEY 5-Gas Calibration

The GEM-500™ is factory calibrated. To improve accuracy, all standard landfill gas instruments should be field calibrated, zeroed, or in other ways adjusted prior to every use. Field calibration is performed from the General Utilities Menu of the GEM-500™.

1. Press **1** KEY for **General Utilities** on the Main Menu Screen (See Figure 3.2.).

FIGURE 3.2

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

2. The General Utilities Screen appears as shown in Figure 3.3.

FIGURE 3.3

1-Check Time/Date	
2-Battery Status	
3-Zero Pressures	
9-More	**0-Exit**

3. The gas calibration function is not on the first General Utilities screen. To reach this screen, press **9** KEY for **More** and **6** KEY for **Gas Calibration** (Figure 3.4). You may also press the **6** KEY while at the first General Utilities Screen to proceed directly to the Gas Calibration screen.

FIGURE 3.4

4-Memory	
5-USA/Metric Units	
6-Gas Calibration	
9-More	**0-Exit**

4. Pressing the **6** KEY for **Gas Calibration** on the General Utilities Sub-Menu screen, the first Gas Calibration screen is displayed as shown in Figure 3.5.

FIGURE 3.5

Calibrate Gas Type		
1-CH_4	**2-$C0_2$**	**3-0_2**
5-Factory Settings		

Methane (CH_4) Calibration - Zero CH_4

1. Press **1** KEY, **CH_4 Calibration,** to start the calibration procedure. Pressing the **0** KEY will exit the screen without changing the previous calibration.

FIGURE 3.6

1-Zero CH_4
2-Calibrate CH_4 Span
0-Previous Menu

2. Pressing **1** KEY, **Zero CH4**, initializes the Zero Methane procedure (Figure 3.7). A methane percentage will not display until the Infrared (IR) Bench warms up. A plus or minus sign may appear on the far left of the display. This symbol may be ignored.

DO NOT PERFORM THIS PROCEDURE IN THE PRESENCE OF METHANE.

FIGURE 3.7

+**00.0**%	**CH_4 Gas**	**1-Zero Level** **5-Pump** **0-Exit**

3. If using air to zero methane, press the **5** KEY, **Pump,** to turn on the GEM-500™ sample pump. Calibration gas hoses should not be attached to the GEM-500™ during this procedure. Allow the pump to run for two minutes or until gas reading stabilizes. If using oxygen calibration gas to zero methane, see step 3-6 in the Span Methane section then return to step 2 of this section.

4. Press **1** KEY, **Zero level.** One of the following screens (Figure 3.8 or Figure 3.9) will be displayed for three seconds before returning to the Zero Methane Screen shown in Figure 3.7.

CH_4 Not Zeroed

FIGURE 3.8

CH_4 Zeroed

FIGURE 3.9

5. If the CH_4 Not Zeroed screen (Figure 3.8) is displayed, return to the Gas Calibration screen by pressing the **0** KEY, **Exit.** Verify methane is not present and re-zero the methane. If the problem continues, proceed to instructions contained in this section for Factory Settings.

6. If the CH_4 Zeroed OK screen (Figure 3.9) is displayed, press **0** KEY, **Exit,** to return to the Methane Calibration Screen (Figure 3.10). Press **2** KEY, **Calibrate CH_4 Span**, to proceed to the next section.

FIGURE 3.10

1-Zero CH_4
2-Calibrate CH_4 Span
0-Previous Menu

Methane (CH4) Calibration

1. Read the warning below before proceeding with the next steps.

> **WARNING! The GEM-500™ is not certified intrinsically safe. The following procedure MUST NOT be performed in a confined area (such as well vaults, underground or indoors) or where there is any possibility of sparking or ignition. Ensure that the exhaust port is not blocked and is properly vented away from you. Ensure that no leaks are present. Unless all above conditions are maintained, an explosion could occur resulting in serious injury or death.**

2. After selecting the **2** KEY, **Calibrate CH_4 Span,** on the Methane Calibration screen, the following CH_4 Span screen appears (Figure 3.11).

FIGURE 3.11

+**00.0**%	CH_4 Gas	**1-Enter** Gas Con 5-Pump 0-Exit

3. Connect the ¼" tubing from the calibration gas regulator/flow meter to the GEM-500™ gas sample/impact port (Figure 1.1). It is **NOT** recommended to use the water trap sample tubing for calibration. Attach tubing to the exhaust port of the GEM-500™ and direct the exhaust flow away from you and out of the immediate area.

4. Press the **5** KEY, **Pump**, on the CH_4 Span screen to turn on the sample pump.

5. If not using CES-LANDTEC supplied regulator, make sure the calibration gas flow is 500 cc and pressure is no greater than 2 psig.

6. Allow the calibration gas to flow into the GEM-500™ for one minute or until instrument gas reading stabilizes.

7. After one minute, read the methane gas concentration on the screen. It should be stable and not changing more than a few tenths of one percent at the 15% gas level or 2% at the higher gas level.

8. Press the **1** KEY, **Enter Gas Con**, and input the methane concentration of the calibration gas using the keyboard of the GEM-500™ (Figure 3.12). Enter the percentage as three digits XX.X%. (50% methane would be input as 500.) The GEM-500™ will automatically place a decimal point in the proper position. After the percentage is entered, press **0** KEY, **Exit**.

FIGURE 3.12

Enter Concentration
of Calibration Gas
CH_4 ?__._ %
0-Exit

9. The next screen is the Caution Re-Calibrate Screen (Figure 3.13).

FIGURE 3.13

CAUTION Re-Calibrate
Are you sure ?

1-Yes 2-No

10. Press **1** KEY, **Yes**, and one of the two following messages will appear (Figure 3.14 or Figure 3.15). .

Calibration Gas NOT
Accepted. Refer to
Operating manual.
Retry? 1-Yes 2-No

FIGURE 3.14

Calibration OK

FIGURE 3.15

11. If the Calibration OK screen flashes (Figure 3.15), proceed to Step 13.

12. If the Calibration Gas Not Accepted screen appears (Figure 3.14), press the **1** KEY, **Yes**, and re-enter the methane percentage. If the Calibration Gas Not Accepted screen still appears, press **0** KEY, **No**, and start procedure again from zero methane. If problem persists, proceed to Factory Settings, discussed later in this chapter.

13. If required, proceed to $C0_2$ calibration Step 1.

14. Press the **0** KEY twice. Turn off the calibration gas cylinder. Remove the calibration gas hose attached to the gas sample/static pressure port on the GEM-500™. Leave the exhaust port hose connected and turn on the pump and allow it to purge the GEM-500™ with air for 60 seconds. Press the **5** KEY, **Pump**, again. The pump turns off and automatically returns to the Calibrate Methane screen.

15. If there is no further calibration, press the **0** KEY, **Exit**, to return to the Gas Calibration screen. Field calibration has successfully been completed.

Carbon Dioxide (CO_2) Calibration

1. Because the cylinder used in this calibration contains methane, the following warning must be adhered to before proceeding with the steps below.

> **WARNING!** **The GEM-500™ is not certified intrinsically safe. The following procedure MUST NOT be done in a confined area (such as well vaults, underground or indoors) or where there is any chance of sparking or ignition. No smoking, exposed lighting, or other sources of ignition should be in the area. On the GEM-500™, ensure that exhaust port is not blocked and properly vented away from you. Ensure that no leaks are present. Unless all above conditions are maintained, an explosion could occur resulting in serious injury or death.**

2. Press the **2** KEY, **CO_2 Calibration**, on the Gas Calibration screen (Figure 3.16).

FIGURE 3.16

Calibrate Gas Type
1-CH_4 2-$C0_2$ 3-0_2

5-Factory Settings

3. There is no Zero CO_2 function as there is in the methane or oxygen calibration procedures. The following CO_2 Span screen appears (Figure 3.17).

FIGURE 3.17

+00.0%	**CO_2 Gas**	**1-Enter Gas Con** **5-Pump** **0-Exit**

4. Press the **1** KEY, **Enter Gas Con**, to access the Enter Concentration screen (Figure 3.18). Input the percentage of Carbon Dioxide concentration of the calibration gas as three digits XX.X%. (40% Carbon Dioxide would be input as 400) The GEM-500™ will automatically place a decimal point in the proper position. After the percentage is entered, press the **0** KEY, **Exit**.

FIGURE 3.18

Enter Concentration
of Calibration Gas
CO_2 ?__._ %
0-Exit

5. The next screen is the Caution Re-Calibrate screen (Figure 3.19).

FIGURE 3.19

CAUTION Re-Calibrate
Are you sure?

1-Yes 2-No

6. Press the **1** KEY, **Yes**, and one of the two following messages will appear (Figure 3.20 or Figure 3.21). If the Calibration OK screen appears, go to step 9 below.

Calibration gas NOT
Accepted. Refer to
Operating manual.
Retry ? 1-Yes / 2-No

FIGURE 3.21

Calibration OK

FIGURE 3.22

7. If the Calibration gas NOT Accepted screen appears, several things could have happened. Press the **1** KEY, **Yes,** and enter the percentage of Carbon Dioxide in the calibration gas. It is possible that the wrong percentage was input. If on a second attempt this does not work, press the **0** KEY, **No**, to return to the Gas Calibration screen and turn to the Factory Settings section for additional instructions.

8. If O_2 is to be zeroed, proceed to O_2 Calibration, step 1.

9. If no further calibration is needed, press the **0** KEY to **Exit** and return to the CO_2 Calibration screen shown on the prior page.

10. Turn off the calibration gas. Remove the calibration gas hose attached to the gas sample/static pressure port on the GEM-500™. Leave the exhaust port hose connected. Allow the GEM-500™ to purge with air for 60 seconds. Press the **5** KEY, **Pump**, to turn off the pump; then press the **0** KEY, **Exit**, to return to the Gas Calibration screen.

11. You have successfully completed a Carbon Dioxide Field Calibration. Immediately proceed to the next function, O_2 Calibration.

Oxygen (O_2) Calibration - Zero O_2

1. There are two calibration gas mixtures used for the calibration of oxygen. The methane/ Carbon Dioxide calibration gas previously used to calibrate the methane and Carbon Dioxide is used to Zero oxygen. A second calibration gas with a mixture of oxygen and nitrogen is used to set the oxygen level in the next section. Because the calibration gas used contains methane, the warning below must be followed before proceeding with the following steps.

> **WARNING! The GEM-500™ is not certified as intrinsically safe. The following procedure MUST NOT be done in a confined area (such as well vaults, underground and indoors) or where there is any chance of sparking or ignition. No smoking, exposed lighting, or other sources of ignition should be in the area . On the GEM-500™, ensure that exhaust gas port is not blocked and properly vented away from you. Ensure that no leaks are present. Unless all above conditions are maintained, an explosion could occur resulting in serious injury or death.**

2. Press Key 3-O_2 Calibration on the Gas Calibration Screen (Figure 3.22).

FIGURE 3.22

Calibrate Gas Type
1-CH_4 2-$C0_2$ 3-0_2

5-Factory Settings

3. The Oxygen Calibration Screen will appear (Figure 2.23).

FIGURE 3.23

1-Zero O_2
2-Calibrate O_2 Span
0-Previous Menu

4. Pressing the **1** KEY, **Zero O_2**, will bring up the Zero Oxygen screen (Figure 3.24).

FIGURE 3.24

+**00.0**% O_2 Gas	**1-Zero Level** **5-Pump** **0-Exit**

5. Read the oxygen Gas Concentration on the screen. It should be very near 00.0% and not changing more than a few tenths of one percent.

   Note: Even if the screen displays 00.0% oxygen, proceed with step 6 below, the Oxygen must be zeroed anyway.

6. Press the 1 KEY, Zero level, and one of the following screens (Figure 3.25 or Figure 3.26) is displayed for three seconds before returning to the Zero Oxygen screen shown above.

O_2 Zeroed

FIGURE 3.25

Level NOT Zeroed **Please refer to** **Operating Manual**

FIGURE 3.26

7. If the O_2 Zeroed screen displays, proceed to step 9 below.

8. If the Oxygen NOT Zeroed screen displays, return to the Oxygen Calibration screen. Check that the calibration gas contains no oxygen. Connect the correct gas and re-zero the oxygen. If the problem continues, proceed to instructions contained in this section for Factory Settings.

9. If the Oxygen Zeroed OK screen appears, turn off the calibration gas.

10. Remove the hose from the flow regulator to the GEM-500™. Let the pump run for at least 60 seconds to purge the instrument with air. Press the **5** KEY, **Pump**, to turn off the pump.

11. Press the **0** KEY, **Exit**, to return to the Oxygen Calibration screen and proceed to oxygen span.

O_2 Calibration - O_2 Span

1. From the Gas Calibration screen, press the 3 KEY, zero O_2 and the Oxygen Calibration screen (Figure 3.27) will appear.

FIGURE 3.27

1-Zero O_2
2-Calibrate O_2 Span
0-Previous Menu

2. Press the **2** KEY, **Calibrate O_2 Span**, on the Oxygen Calibration screen will appear (Figure 3.28).

FIGURE 3.28

+00.0% O_2 Gas	**1-Enter** **Gas con** **5-Pump** **0-Exit**

Note: The calibration gas used in this procedure is a mixture of oxygen and nitrogen. The oxygen concentration by volume can be 2-5% with the remainder N_2.

3. Change the calibration gas mixture to Oxygen/Nitrogen. Install the regulator/flow meter on the new calibration gas mixture as directed previously in *Setting Up the Equipment*, page 4. Check and adjust the gas flow to 500 cc and pressure to 2 psig. Turn off the gas.

4. Connect the ¼" tubing from the calibration gas regulator/flow meter to the GEM-500™ gas sample/static pressure port (Figure 1.1). Attach tubing to the exhaust port of the GEM-500™, if not already attached, and direct the exhaust away from you and out of the immediate area.

5. Press the **5** KEY, **Pump**, displayed on the O_2 Span screen shown above. (Figure 3.28)

6. Turn on the calibration gas mixture of oxygen and nitrogen.

7. Allow the calibration gas to flow into the GEM-500™ for 60 seconds.

8. After 60 seconds, read the Oxygen Gas Concentration on the screen. It should be stable and not changing more than a few tenths of one percent.

9. Press the **1** KEY, **Enter Gas Con**, and input the oxygen concentration of the calibration gas (typically 4%) using the keyboard of the GEM-500™ (Figure 3.29). Enter the percentage as three digits XX.X%. (4% O_2 would be input as 040) The GEM-500™ will automatically place a decimal point in the proper position. After the percentage is entered, press the **0** KEY to **Exit**.

FIGURE 3.29

Enter Concentration
of Calibration Gas
O_2 ?__._ %
0-Exit

10. The next screen to appear, Figure 3.30, is the Caution Re-Calibrate Screen.

FIGURE 3.30

CAUTION Re-Calibrate
Are you sure?
1-Yes 2-No

11. Press the **1** KEY, **Yes**, and one of two screens will appear (Figure 3.31 or Figure 3.32).

12. If the Calibration OK Screen appears proceed to step 15.

Calibration gas NOT Accepted. Refer to Operating manual.
Retry ? 1-Yes / 2-No

FIGURE 3.31

Re-Calibrated OK

FIGURE 3.32

13. If the Calibration Gas NOT Accepted screen appears, press the **1** KEY, **Yes,** and re-enter the percentage of oxygen in the calibration gas. It is possible the wrong percentage was input. If, on a second attempt, this has not worked, press the **0** KEY, **No**, and return to the Oxygen Calibration Menu. Start the procedure over again. Zero and then calibrate the oxygen. If there are still problems, proceed to Factory Settings in this section.

14. Press the **0** KEY, **Exit**, to return to the Oxygen Calibration screen shown on the following page.

15. Turn off the calibration gas. Remove the calibration gas hose attached to the gas sample/static pressure port on the GEM-500™.

Factory Setting Calibrations

As previously mentioned, it is sometimes necessary to return the GEM-500™ to factory settings before trying to field calibrate the unit. If for some reason sampling conditions change radically, overall accuracy of the GEM-500™ can be improved by returning to factory settings and then re-calibrating. This procedure will overwrite previous field calibrations.

1. From the Gas Calibration screen, Figure 3.34, press the **5** KEY, **Factory Settings**.

FIGURE 3.34

Calibrate Gas Type
1-CH_4 2-$C0_2$ 3-0_2

5-Factory Settings

2. The Caution screen, Figure 3.35, shown below will be displayed. If the **0** KEY, **No**, is pressed, the Not Set screen (Figure 3.36) appears for two seconds, then the Gas Calibration screen returns.

CAUTION Return to
factory setting of
calibration & zero ?
1-Yes 0-No

FIGURE 3.35

Factory Settings
NOT Set

FIGURE 3.36

3. Press **1** KEY, **Yes**, and the Factory Setting Set OK screen (Figure 3.37) is displayed for three seconds before returning to the Gas Calibration screen shown in step 1.

FIGURE 3.37

Factory Settings
Set OK

4. After loading the factory settings, the methane and oxygen calibration MUST BE RE-ZEROED PRIOR TO USE. After completing the gas calibrations, the GEM -500™ is ready to read gas levels. Go to Chapter 5 of this manual, Read Gas Levels.

After Completing Gas Calibrations

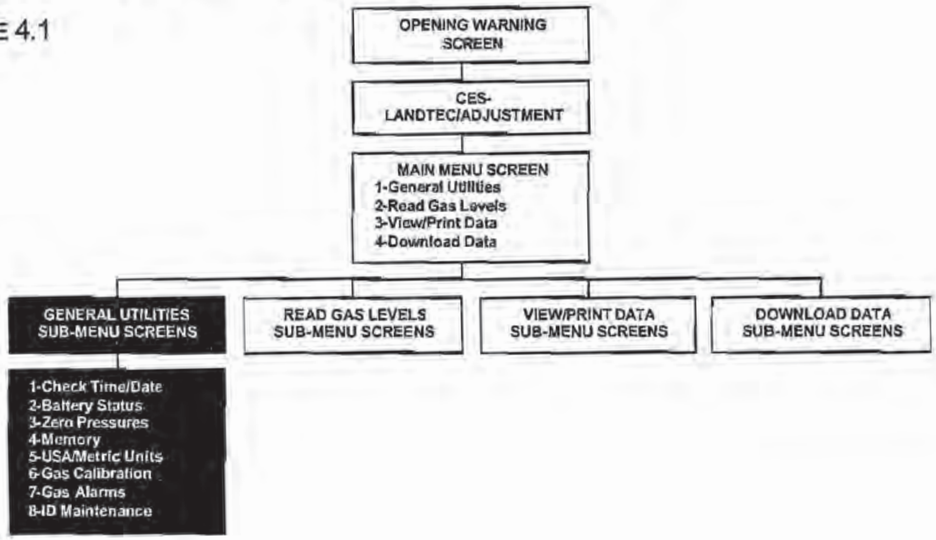
Additional general utilities functions should be addressed after the GEM-500™ is field calibrated. These functions are available on the General Utilities menu and include:

- **Check Time/Date** Assures that the data collected is properly time/date stamped.
- **Check Memory** Assures that there is enough memory space in the GEM-500™ to store the readings you plan to take. Otherwise the memory will need to be cleared. (See page 28, chapter 4)
- **Set Gas Alarms** Alerts the user to unusual gas conditions.

Chapter 4 - General Utilities Functions

General Utilities Screen Tree Diagram

FIGURE 4.1



General Utilities Menu

The General Utilities functions are displayed on three subsequent screens. (Figures 4.2, 4.3, & 4.4) Any of the functions may be selected while any of the three screens is displayed. Use the **9** KEY, **More**, to move from one screen forward to the next. Press the **0** KEY, **Exit**, to return to the main menu.

1-Check Time/Date
2-Battery Status
3- Zero Pressures
9-More 0-Exit

FIGURE 4.2

4- Memory
5- USA/Metric Units
6- Gas Calibration
9-More 0-Exit

FIGURE 4.3

7- Gas Alarms
8- ID Maintenance
9-More 0-Exit

FIGURE 4.4

General Utilities Functions

1. **CHECK TIME/DATE:** Set or check time and date.
2. **BATTERY STATUS:** Graphic display of the percentage of power remaining in the batteries.
3. **ZERO PRESSURES:** Zero pressure transducers.
4. **MEMORY:** Check available memory and facilitate clearing of all data and ID information.
5. **USA/METRIC UNITS:** Select either USA standard (Imperial) or metric (SI) measurement units.
6. **GAS CALIBRATION:** Field calibrate methane, Carbon Dioxide and oxygen with special gas mixtures for increased accuracy.
7. **GAS ALARMS:** Set gas alarm levels.
8. **ID MAINTENANCE:** View, enter, edit and delete well ID information.

Check Time/Date
There is an internal clock and calendar in the GEM-500™ powered by a secondary battery that maintains the clock function when the GEM-500™ is turned off. As each reading is stored in the GEM-500™, it is time and date stamped. Both the clock and calendar are set by CES-LANDTEC, however, they should be reset to the local time zone and checked weekly thereafter.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.

FIGURE 4.5

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

2. (Figure 4.6). Press the **1** KEY, **Check Time/Date**, on the General Utilities Sub-Menu screen to access the Check Time/Date function.

FIGURE 4.6

1-Check Time/Date
2-Battery Status
3- Zero Pressures
9-More 0-Exit

3. Press the **1** KEY, **Set Time/Date**, to proceed. (Figure 4.7)

FIGURE 4.7

15:46:07 09/07/98
1-Set Time/Date
0-Exit

4. The time and date are displayed on the top line of the screen. A 24-hour clock or military time is used. If the time is after 12 noon, add 12 to the hour to convert it to the 24-hour format. Example: 3 p.m. is 12+3 = 15:00 hours. The time format is Hours: Minutes. Seconds. The date format used in the example is in U.S. calendar format with the month first and day second (mm/dd/yy).

5. If the time and date are accurate, end procedure by pressing the **0** KEY to **Exit** to the General Utilities Sub-Menu screen. If the time or date, or both, is wrong, press the **1** KEY, **Set Time/Date**.

6. Set the time and date by entering numbers from the GEM-500™ keyboard. For setting the time hh = hours, mm = minutes, and ss = seconds. The date is entered in the U.S. calendar format where mm = months, dd = days, and yy = years. When finished, press the **0** KEY to **Set**.

FIGURE 4.8

?h:mm:ss mm/dd/yy
Enter New Time/Date
0-Set

NOTE: If it is necessary to correct an entry error, use the 0 KEY as a Backspace Key by holding it down for 1 second. In normal use, the 0 KEY is quickly pressed and released.

7. After setting, one of two screens displays. If the date is valid, Figure 4.9 displays for three seconds. If the time or date is invalid, Figure 4.10 displays. The time or date is invalid when impossible numbers are entered into the field. For example, mm=15 is an invalid month. Return to step 5 above and reenter the correct time or date as instructed.

Time/Date Set OK

FIGURE 4.9

Invalid Time/Date
Please enter again

FIGURE 4.10

Battery Status

A Fast charger came with the GEM-500™. The Fast charger takes approximately 2.5 hours for a 90% charge, the charger will automatically switch to a slow charge after this time to prevent damage to the batteries. The fast charger may be left connected overnight without damage to the NiCad batteries. When the GEM-500™ is fully charged, it should be able to operate continuously for 6-8 hours depending upon the battery used and how it was charged. The GEM-500™ may be operated with Alkaline batteries, however, **ONLY** the NiCad batteries can be recharged. If the GEM-500™ is without batteries for more than 30-45 minutes, memory/data loss will occur and the internal back backup battery will run down. If this occurs, the unit will need to be returned to the lab for service.

WARNING! DO NOT TRY TO RECHARGE ALKALINE BATTERIES – DAMAGE TO THE UNIT WILL OCCUR. (Figure 4.11)

Figure 4.11

◆ WARNING ◆

THE BATTERIES MUST BE IN THE UNIT AND MAKING CONTACT WHEN RECHARGING UNIT. CHECK FOR CONTACT BY TURNING ON UNIT AFTER BATTERIES ARE CONNECTED. IF NOTHING APPEARS ON THE SCREEN, THE BATTERIES ARE NOT IN CONTACT OR DEFECTIVE. CLEAN CONTACTS AND RETRY. IF STILL NO CONTACT, REPLACE BATTERY STICKS.
FAILURE TO COMPLY CAN CAUSE DAMAGE TO YOUR GEM

1. From the Main Menu, press the **1** KEY, **General Utilities,** for the General Utilities Sub-Menu screen.

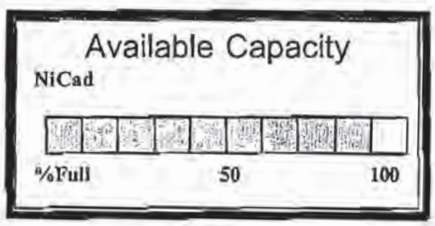
2. Press the **2** KEY, **Battery Status**, on the General Utilities Sub-Menu screen as shown in Figure 4.12.

FIGURE 4.12

1-Check Time/Date
2-Battery Status
3- Zero Pressures
9-More 0-Exit

3. The battery status graph displays the percentage of power remaining in the battery. When the graph reads 20-30%, a battery symbol indicator displays on the upper right of the screen, indicating approximately 1 hour of use remaining. When finished viewing the screen, press the **0** KEY to **Exit**.

FIGURE 4.13



Zero Pressures

The GEM-500™ measures atmospheric pressure as part of the LFG flow calculation. To properly measure pressure and the vacuum used in landfill gas extraction systems, the pressure transducers must be reset to zero each time before taking a pressure or vacuum reading.

This procedure may also be done prior to doing any **Read Gas Levels** because the **Zero Pressures** function is also contained on the **Read Gas Levels** Sub-Menu screen as shown in Chapter 5.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.

2. Press the **3** KEY, **Zero Pressures**, on the General Utilities Sub-Menu screen as shown in Figure 4.14.

FIGURE 4.14

```
1-Check Time/Date
2-Battery Status
3- Zero Pressures
9-More       0-Exit
```

3. Figure 4.15 displays the current readings of the static and differential pressure transducers. If both pressures do not read 00.0 **DISCONNECT ANY HOSES** ATTACHED TO THE GEM-500™ and press **1** KEY, **Zero Pressures**.

FIGURE 4.15

```
        Static  +00.2"H2O
Differential  -0.30"H2O
1-Zero Pressures
                    0-Exit
```

Note: Units displayed are inches of water column or (millibar) MB depending on measurement unit selected (USA or metric).

4. After the pressures have been zeroed, Figure 4.16 appears for three seconds. The Zero Pressures screen (Figure 4.15) then redisplays.

FIGURE 4.16

```
Zeroed OK
```

5. Press the **0** KEY, **Exit**, to return to the General Utilities Sub-Menu screen.

Memory

> **CAUTION: THIS FUNCTION CAN ERASE ALL STORED DATA. ONCE CLEARED, THE DATA CANNOT BE RECOVERED.**

All well ID's and readings are stored in the GEM-500™'s memory. Eventually the memory becomes full. After each day's readings are completed, the remaining amount of memory should be checked. Normally, the readings for the day are downloaded to a computer. Downloading copies the information but does not clear it out of memory. That must be done manually, as described below. If the memory becomes full, a **MEMORY FULL** message displays. When this happens, the memory must also be manually cleared.

The GEM-500™ can store many well ID's. It is, therefore, possible to use it on several landfills.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.

2. Since the **Memory** function is not on the first General Utilities screen, press the **9** KEY, **More**, for the next screen, or enter **4** at this screen.

3. Press the **4** KEY, **Memory**, on the General Utilities Sub-Menu screen (Figure 4.17).

FIGURE 4.17

```
4- Memory
5- USA/Metric Units
6- Gas Calibration
9-More        0-Exit
```

4. The amount of available memory left in the GEM-500™ displays on The Number of Free Readings screen. (Figure 4.18) Three choices may be made on this screen. Press the **1** KEY, **Clear Readings**, to erase all gas/data readings but leave the ID's. Press the **2** KEY, **Clear ID Info**, to erase all ID numbers and the associated readings that have accumulated in the GEM-500™ from the ID MAINTENANCE and READ GAS functions. Press the **0** KEY, **EXIT**, to ESCAPE from the procedure and return to the General Utilities Sub-Menu screen.

FIGURE 4.18

```
 0479 Free Readings
1-Clear Readings
2-Clear ID Info
               0-Exit
```

> **CAUTION: THIS STEP ERASES STORED DATA. YOU MAY WANT TO DOWNLOAD THE DATA FIRST SO IT IS NOT LOST.**

5. After making your choice from the screen above, the Caution screen displays (Figure 4.19). As a final safety check, the code **0102** must be input from the GEM-500™ keyboard to clear the memory. **IF YOU DECIDE NOT TO CLEAR THE MEMORY AT THIS POINT, TURN THE GEM-500™ OFF BY PRESSING THE RED ON/OFF KEY** or enter an incorrect code then press the **0** KEY, **EXIT**, to return to the Memory screen. Do not input 0102 unless you want to clear the memory.

FIGURE 4.19

CAUTION! **DATA WILL BE LOST** **Enter 0102 to clear** **?**

6. Enter 0102 from the keyboard and press the **0** KEY, Exit. The Clearing Memory screen displays for 3 seconds if the memory was erased.

FIGURE 4.20

Clearing Memory

After the Clearing Memory screen displays, the Number of Free Readings screen (Figure 4.18) redisplays. Press the **0** KEY to **Exit** to the General Utilities Menu screen.

USA/Metric Units

The GEM-500™ can store and display data in 2 units of measure, Metric (SI) or Imperial (USA). This function allows setting the unit of measure.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.

2. Since the **USA/Metric Units** function is not on the first General Utilities screen, press the **9** KEY, **More**, for the next screen, or enter **5** at this screen.

3. Press the **5** KEY, **USA/Metric Units**, on the General Utilities Sub-Menu screen (Figure 4.21).

FIGURE 4.21

```
4- Memory
5- USA/Metric Units
6- Gas Calibration
9-More        0-Exit
```

4. The Measurement Units screen (Figure 4.22) displays how the GEM-500™ is currently set (Set to USA Std or Set to Metric). Press the **1** KEY to change from USA Std to Metric. (This setting acts as a toggle switching from one to the other.) If the GEM-500™ is currently displaying USA Std measurement units (Imperial — Btu's, Standard Cubic Feet, Fahrenheit temperatures, etc.) it switches to Metric and vice versa. When the GEM-500™ is set to the correct measurement unit, press the **0** KEY, **Exit**, to return to the General Utilities Sub-Menu screen.

FIGURE 4.22

```
  Measurement Units
Set to USA Std
1-Change to Metric
              0-Exit
```

Gas Calibration

Please refer to Chapter 2 - Field Calibration for all information and instructions relating to the Gas Calibration function.

Gas Alarms

The GEM-500™ has two alarm options that can warn the operator if a gas sample contains concentrations of **Methane below** established levels or **Oxygen above** preset levels. If the alarms are activated, there is a beeping and the affected gas blinks when displayed on the Read Gas Levels screen.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.
2. Since the **Gas Alarms** function is not on the first General Utilities screen, press the **9** KEY, **More**, for the next screen, or enter **7** at this screen. (Figure 4.23)

FIGURE 4.23

7- Gas Alarms
8- ID Maintenance
9-More 0-Exit

3. The Gas Alarm Set screen displays the alarm set point of both methane and oxygen and presents the functions to change them. (Figure 4.24). Chose one to change the methane alarm set point or two to change the oxygen alarm set point. If no change in alarm set points is required, press the **0** KEY, **Exit**, to return to the General Utilities Sub-Menu screen.

FIGURE 4.24

Methane Below 00.0%
Oxygen Above 25.0%
1-Change Methane
2-Change Oxygen

NOTE: To turn off alarms, set methane alarm to 00.0% and oxygen alarm to 25.0%.

4. If the **1** KEY, **Change Methane**, is pressed, the Methane Alarm Set Point screen displays. (Figure 4.25) Using the numbered keys on the GEM-500™ keyboard, input the new alarm level for methane (CH_4). All three digits must be entered (XX.X%). The decimal point is automatically inserted. Press the **0** KEY to save and **Exit**.

FIGURE 4.25

CH_4 Level is 00.0%
Level Required ? --.-%
Enter level 0-Exit

Note: If the GEM-500™ receives CH_4 at or below this set point during the Read Gas Levels procedure, an audible alarm sounds to alert the operator.

5. If the **2** KEY, **Change Oxygen**, is pressed, the Oxygen Alarm Set Point screen displays. (Figure 4.26). Using the numbered keys on the GEM-500™ keyboard input the new alarm level for oxygen (O_2). All three digits must be entered (XX.X%). The decimal point is automatically inserted. Press the **0** KEY to save and **Exit.**

FIGURE 4.26

O_2 Level is 00.0%
Level Required ? --.-%
Enter level 0-Exit

Note: If the GEM-500™ receives O_2 at or above this set point during the Read Gas Levels procedure, an audible alarm sounds to alert the operator.

6. Press the **0** KEY, **Exit,** to return to the General Utilities Sub Menu screen.

ID Maintenance

Each monitoring point on a site can be assigned a unique ID code using the ID Maintenance function. This code **must** be eight characters long. The characters can be any combination of letters and numbers. Typically, the landfill name or an abbreviation is used for the first four characters. After an ID code is entered (Step 4), the type of flow device (Accu-Flo, pitot tube, orifice plate or user defined) used at that ID location must also be entered (Step 5). Depending on the flow device selected, either no data, pipe ID (inner diameter), or both orifice and pipe ID size must also be entered.

1. From the Main Menu, press the **1** KEY, **General Utilities**, for the General Utilities Sub-Menu screen.

2. Since the **ID Maintenance** function is not on the first or second General Utilities screens, press the **9** KEY, **More**, twice or enter **8** at this screen. (Figure 4.27)

FIGURE 4.27

7-Gas Alarms
8-ID Maintenance
9-More 0-Exit

The ID Maintenance screen presents two options. If a well already has an ID number, option one, View/Edit/Delete should be accessed (Step , *Figure 4.*) If a well has no ID assigned in the GEM-500™, press the **2** KEY to **Enter New ID.**

FIGURE 4.28

ID MAINTENANCE
1-View/Edit/Delete
2-Enter New ID
0-Exit

Both numbers and letters can be input on the Enter ID screen. Use the **BLUE** f KEY to switch (toggle) between numbers and letters. See Keyboard Information in the Getting Started Section at the beginning of this Manual.

- For numbers, press the keypad **Number** KEYS (0-9). (Figure 4.29)
- For letters, press the **1** KEY (UP ARROW) or **6** KEY (DOWN ARROW) to scroll through the alphabet until the letter of choice appears. Press the **0** KEY to select the letter. (Figure 4.30)

After the final character is entered, the unit displays **0-Cont**. Press the **0** KEY if the ID is correct and ready to enter; otherwise, press and **hold** the **0** KEY to backspace and make corrections.

Please Enter ID
f- Numbers/ Letters
ID 0000 ?---

FIGURE 4.29

Please Enter ID
f- Numbers/ Letters
ID 0000 ?---
↑↓-Step 0-Enter

FIGURE 4.30

Note: When entering the very first well ID, four zeros will hold the first four places in the ID number. To replace these zeros with the well ID, use the 0 KEY to backspace over them by holding it down for more than a second. Then enter the first four characters of the well ID. These first four characters default to the second well ID entered, saving the user the time of reentering for each well.

5. If an existing code is entered, the unit will ask if you want to overwrite. (Figure 4.31) If so, press the **1** KEY, **Yes**, to overwrite. If a mistake was made and an overwrite is not desired, press the **2** KEY, **No**, to return to the ID Maintenance screen. (Figure 4.28)

FIGURE 4.31

ID 00000000 Already
Exists, overwrite ?

1-Yes 2-No

6. A well flow device is selected in the Flow Device screen. This selection is necessary for the GEM-500™ to be able to calculate flow when readings are taken. Use the **1** KEY (UP ARROW) or the **6** KEY (DOWN ARROW) to scroll through the choices listed in Figure 4.33. Once the desired flow device is located in the shaded selection window, press **0** to select and Continue. (Figure 4.32)

FIGURE 4.32

Select Flow Device
Accuflo-1.5V
↑-Scroll Up
↓-Scroll Down 0-Cont

Note: Selection of User Input allows the entry of flow in SCFM, if known. Those without a flow device may wish to use this selection to record velocity or other relevant data. (For example, using a Kurz meter.)

FIGURE 4.33

Accuflo–1.5V (1½" Accu-Flo Model 150 Vertical Wellhead)
Accuflo–1.5H (1½" Accu-Flo Model 150 Horizontal Wellhead)
Accuflo–2V (2" Accu-Flo Model 200 Vertical Wellhead)
Accuflo–2H (2" Accu-Flo Model 200 Horizontal Wellhead)
Accuflo–3V (3" Accu-Flo Model 300 Vertical Wellhead)
Accuflo–3H (3" Accu-Flo Model 300 Horizontal Wellhead)
Orifice Plate (Orifice diameter and pipe inner diameter required)
Pitot Tube (Pipe ID required)
User Input (Pipe ID required)

7. If an Orifice Plate, Pitot Tube or User Input flow device is selected, additional information is required. If the pipe or orifice diameter screen appears, input the required size as necessary. Insert inches or centimeters (depending on whether US or Metric Units were selected on the USA/Metric Units screen). The unit uses XX.XX as the format and automatically enters the decimal point. Press the **0** KEY to enter and Continue.

FIGURE 4.34

Enter Pipe ID
?-.-- Inches

8. The ID Stored OK Screen displays for three seconds (Figure 4.35) then the ID Maintenance displays so the next ID can be entered (Figure 4.36).

FIGURE 4.35

ID Stored OK

9. To View/Edit/Delete ID information, select the **1** KEY. (Figure 4.36)

FIGURE 4.36

ID MAINTENANCE
1-View/Edit/Delete
2-Enter New ID
0-Exit

10. The Well ID screen displays the Well ID and the associated flow device. The orifice and pipe data is also displayed if associated with a flow device. To scroll through the ID's stored in memory, use the **1** KEY, (UP ARROW) or the **6** KEY (DOWN ARROW). Press the **0** KEY to **Exit.**

FIGURE 4.37

```
ID 00000000   | 2-Edit
[flow device] | 3-Del
[  orifice  ] | ↑↓-Scan
[  pipe id  ] | 0-Exit
```

11. To edit the chosen Well ID information, press the **2** KEY, **Edit.** Use the **1** KEY (UP ARROW) or the **6** KEY (DOWN ARROW) to scroll through the choices listed in Figure 4.33. Once the desired flow device is located in the shaded selection window, press **0** to select and **Continue.** Press the **0** KEY, **Exit**, to return to the ID Maintenance screen. (Figure 4,38)

NOTE: Once the 2 KEY is pressed, the original flow device is erased and new data must be entered.

FIGURE 4.38

```
Select Flow Device
     Accuflo-1.5V
↑-Scroll Up
↓-Scroll Down  0-Cont
```

12. To delete a Well ID and the associated flow device, press the **3** KEY, **Delete.**

FIGURE 4.39

```
ID 00000000   | 2-Edit
[flow device] | 3-Del
[  orifice  ] | ↑↓-Scan
[  pipe id  ] | 0-Exit
```

13. The Delete ID screen displays the Well ID chosen and asks for confirmation of the delete. If this Well ID ***should not*** be deleted, press the **2** KEY, **No**, (Figure 4.40). The unit cancels the delete command and returns to the Well ID screen (Figure 4.39). If this Well ID ***should*** be deleted, press the **1** KEY, **Yes.** The ID is deleted and the unit returns to the Well ID screen (Figure 4.39). Press the **0** KEY to **Exit** to the ID Maintenance screen.

FIGURE 4.40

```
     DELETE ID ?
      00000000

1-Yes         2-No
```

14. Before leaving this section, store at least three ID's. These will be necessary for use in the following chapters.

Chapter 5 - Read Gas Levels

This section instructs the operator in how to use the GEM-500™ to collect data from LFG extraction system wells and other monitoring points. Several things should be done prior to beginning to collect data readings with the GEM-500™.

The operator should have performed the following:

- Check the TIME/DATE. (See Chapter 4 - General Utilities)
- Charge the unit's factory provided nickel cadmium batteries. (See Chapter 8 - Maintenance)
- Perform a Field Calibration on the unit. (See Section 3 - Field Calibration)

The GEM-500™ is a sensitive measuring instrument. Vibration, shock, and great temperature changes can alter the field calibration. It is suggested that a field calibration be performed just prior to using the instrument at the site. Additional calibration is sometimes necessary in the field during the day.

WARNING! Review the warnings given in the beginning of this manual. The GEM-500™ is NOT to be used in dangerous, explosive or confined atmospheres. Do not use the GEM-500™ inside vaults, manholes, trenches or indoors. Do NOT block the exhaust port. If the exhaust port is blocked while the pump is operating, the pressure could force the unit to over-pressurize and damage the internal components and the case.

GEM-500™ Hose and Wellhead

The proper hoses must be connected from the GEM-500™ to the wellhead in order to collect data. As mentioned in the Getting Started Chapter, the clear tubing with the external filter/water trap assembly is attached to the static pressure port on the GEM-500™ (Figure 5.1). The almond colored male quick connect goes on the end of this tubing to read the static pressure on the Accu-Flo Wellhead and the blue hose is connected to the impact port of the GEM-500™.

On the following pages are examples of the Accu-Flo Wellhead, both vertical and horizontal models. Note the locations of the Static Pressure Port, Impact Pressure Port, Temperature Gauge, and Gas Sample Port.

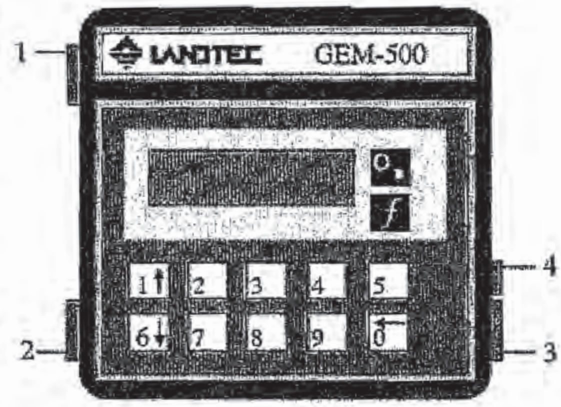
Note: Five O-rings for quick disconnect fittings are supplied with Unit. Replace O-Rings when necessary because oxygen will be drawn into sample if O-rings are damaged. The GEM-500™ pump will pull up to 80" of vacuum.

FIGURE 5.1

1. Static Pressure/Sampling Port—Measures static pressure when connected to wellhead static pressure port by tubing. Always use water trap assembly.

2. Impact Pressure Port—Measures impact pressure when connected to wellhead impact pressure port by tubing.

3. Exhaust Port—This port must be kept clear. If blocked while operating, over-pressurization and damage to internal components and case could occur.



4. Data Port - Used for Temperature Probe, POD, downloading data, and battery recharging.

CES-LANDTEC Horizontal Accu-Flo Wellhead

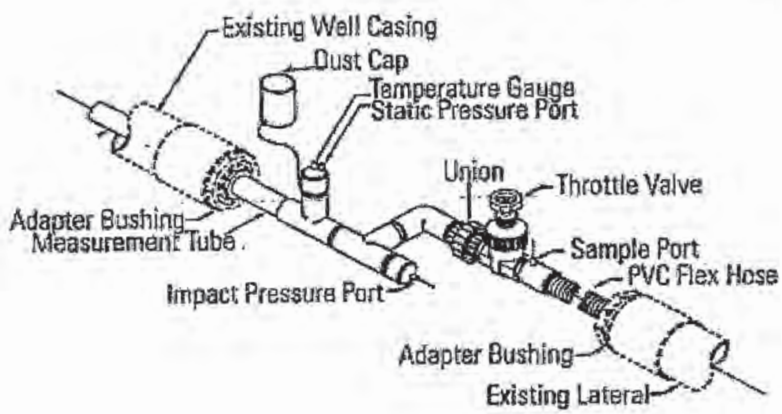


FIGURE 5.2

CES-LANDTEC Vertical Accu-Flo Wellhead

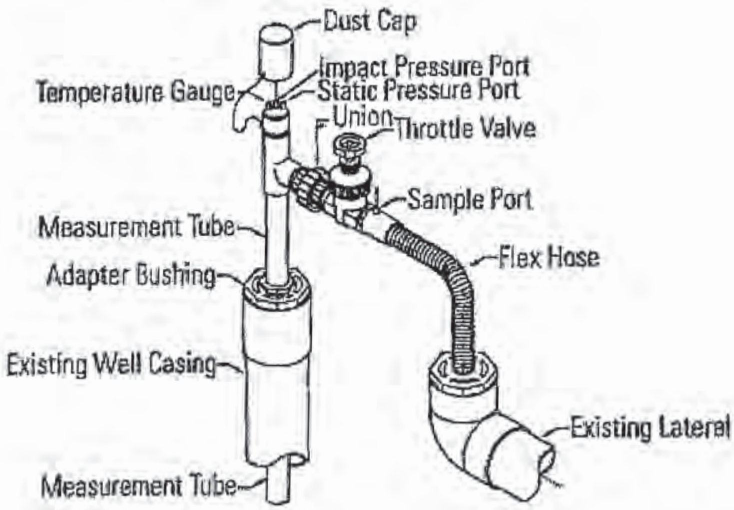
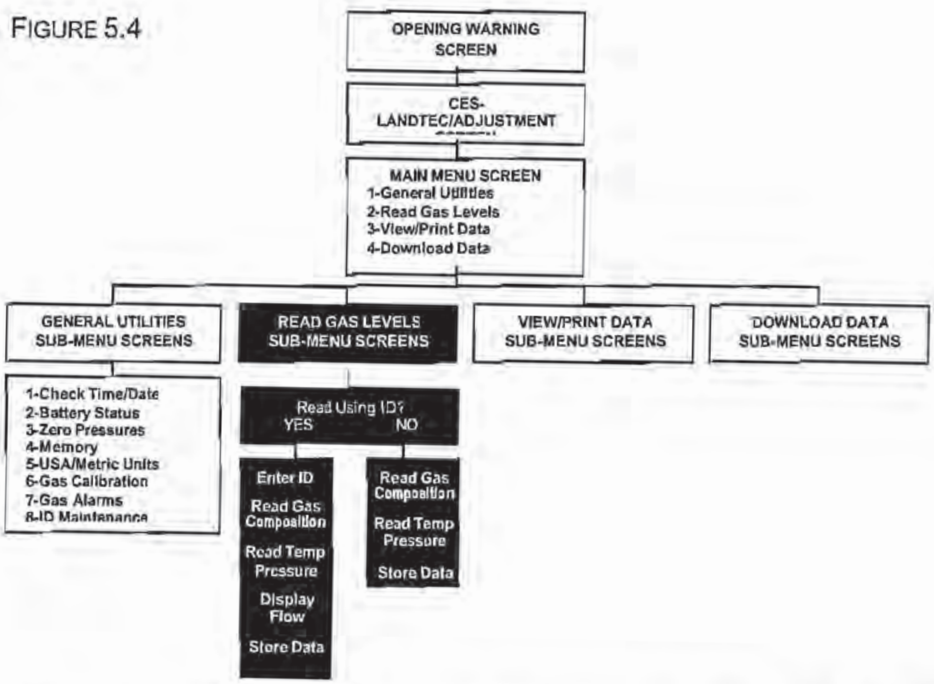


FIGURE 5.3

Read Gas Levels Screen Tree Diagram

FIGURE 5.4



As shown in the screen tree diagram above (Figure 5.4), there are two menu paths that can be followed through the Read Gas Levels function. The path taken depends upon whether or not a well ID has been defined and stored in the GEM-500™. Well IDs can be added at several points during this procedure.

Read Gas Levels Menu – Read Using ID? No

1. On the Main Menu screen press the **2** KEY (Figure 5.5) to initialize the Read Using ID screen.

FIGURE 5.5

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

2. The Read Using ID screen is displayed as shown in Figure 5.6. A choice needs to be made whether or not to read the well using an ID. If yes is chosen, a well ID is selected by scrolling through a list, or by entering the well ID manually. Typically, **NO** is chosen when a well only needs to be monitored or a well ID is not stored in memory in the unit.

FIGURE 5.6

	Read Using ID?	
1-Yes	**2-No**	**0-Exit**

3. Select the **2** KEY, **No.**

4. Connect the GEM-500™ to the wellhead with supplied tubing—Static Pressure/Sampling Port to the Static Pressure Port and Impact Pressure Port to Impact Pressure Port. The Gas Levels screen is divided into two parts. The left side of the screen displays current percentages of CH_4, CO_2, O_2, and BAL, which is the balance of all other gases excluding the CH_4, CO_2, and O_2. The right side of the screen displays functional choices for this reading (Lower Explosive Limit, Continue to temperature and pressure data, Pump On/Off, Last Data, Exit) and POD reminder.

FIGURE 5.7

CH_4 00.0%	**1-LEL 2-Cont**
CO_2 00.0%	**5-Pump OFF**
O_2 00.0%	**6-Store**
BAL 00.0%	**POD? 0-Exit**

POD refers to interchangeable electrochemical gas pods that are used to extend the measurement capabilities of the GEM-500™ (Figure 5.8). These pods are available in seven different gases with nine different ranges and easily plug into the data port. The reminder lets the user know that if a pod were attached at this time, additional gas readings could be taken.

FIGURE 5.8

Interchangeable Electrochemical Gas Pods

Gas	Range (ppm)	Resolution (ppm)
H_2S	0 – 50	0.1
	0 – 200	1.0
CO	0 – 1000	1.0
SO_2	0 – 20	0.1
	0 – 100	1.0
NO_2	0 – 20	0.1
Cl_2	0 – 20	0.1
H_2	0 – 1000	1.0
HCN	0 – 100	1.0

5. Press the **5** KEY, **Pump**, to start the pump and draw a gas sample into the GEM-500™. The **5** KEY, **Pump**, works as a toggle switch to turn the pump on and off. Once the pump is turned on, the readings are not considered to be accurate until the percentages on the left side of the display stabilize, typically within 30-45 seconds. A timer displays on the screen to monitor the pump running time.

FIGURE 5.9

CH_4 00.0%	**1-LEL 2-Cont**
CO_2 00.0%	**5-Pump OFF**
O_2 00.0%	**9-Last Data**
BAL 00.0%	**POD? 0-Exit**

Note: The GEM-500™ may sound an alarm (beeping) while taking gas readings. The alarm means that gas levels set in General Utilities Gas Alarms (page 31) have been reached or exceeded. In addition to the alarm, the screen display of the gas that set off the alarm will also blink.

6. To monitor the LEL (Lower Explosive Limit), press the 1 KEY (Figure 5.9), to enter the LEL screen. If the LEL needs to be continuously monitored, just leave the pump running. When the LEL no longer needs to be monitored, press the 5 KEY again to toggle off the pump. Press the 0 KEY to Exit back to the Gas Levels screen.

FIGURE 5.10

00.0	CH_4 %Gas	5-Pump OFF
000.	CH_4 %LEL	0-Exit

7. To store the current readings press the **6** KEY, **Store.**

FIGURE 5.11

CH_4 00.0%	1-LEL 2-Cont
CO_2 00.0%	5-Pump OFF
O_2 00.0%	6-Store
BAL 00.0%	POD? 0-Exit

Since a well ID was not entered to begin with, one must now be entered. Use the **BLUE** *f* KEY to switch (toggle) between numbers and letters. See Keyboard Information in the Getting Started Section at the beginning of this Manual.

- For numbers, press the keypad **Number** KEYS (1-0). (Figure 5.12)
- For letters, press the **1** KEY (UP ARROW) or **6** KEY (DOWN ARROW) to step through the alphabet until the letter of choice appears. Press the **0** KEY to select the letter. (Figure 5.13)

After the final character is entered, the unit displays **0-Cont**. Press the **0** KEY if the ID is correct and ready to enter; otherwise, press and **hold** the **0** KEY to backspace and make corrections.

Please Enter ID
f-Numbers/ Letters
ID AAAA ?---

FIGURE 5.12

Please Enter ID
f- Numbers/ Letters
ID AAAA ?---
↑↓-Step 0-Enter

FIGURE 5.13

Note: Four letters from the previously entered ID default into the first four places of the ID number. To replace these letters, if needed, use the 0 KEY to backspace over them by holding it down for more than a second. Change the characters as needed.

9. On the Select Comments screen, use the **1** KEY to **Scroll Up** or the **6** KEY to **Scroll Down** through the comment list (Figure 5.15). When the correct comment appears in the comment display area, press the **2** KEY, to **Select**, then the **0** KEY to Store the readings.

Select Comments	
[comment display]	
↑-Scroll Up	2-Select
↓-Scroll Dn	0-Store

FIGURE 5.14

COMMENTS	
Flex Hose	Labeling
Valve	Air Leakage
Casing Height	Water Blockage
Well Bore Seal	Other

FIGURE 5.15

10. If the readings are successfully stored, the Readings Stored screen displays for three seconds (Figure 5.16). The Gas Levels screen (Figure 5.11) redisplays. The unit is now ready to accept another reading.

FIGURE 5.16

Readings Stored

Read Gas Levels Menu – Read Using ID? Yes

1. On the Main Menu screen, press the **2** KEY (Figure 5.17) to initialize the Read Using ID screen.

FIGURE 5.17

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

2. A choice needs to be made at the Read Using ID screen (Figure 5.18) whether or not to read the well using an ID. If **YES** is chosen, the well ID is selected by scrolling through a list or by entering the well ID manually. Typically, **NO** is chosen when a well only needs to be monitored or a well ID is not yet in memory.

FIGURE 5.18

	Read Using ID?	
1-Yes	**2-No**	**0-Exit**

3. Select the **2** KEY **Yes.**

4. The Manually Enter ID screen allows the user to either scroll through a list of ID's that are already in memory or enter an ID manually. (Figure 5.19) If there are few ID's stored, it might be faster to use the scroll option; whereas the manual entry option might be faster to use if there are many ID's stored.

FIGURE 5.19

Manually enter ID or Scroll through ?	
1-scroll	**2-Manual**

5. To scroll through the existing list of well ID's, press the **1** KEY, **scroll**. The Select ID screen displays a single ID at a time. (Figure 5.20) Use the **1** KEY (**UP ARROW**) or the **6** KEY (**DOWN ARROW**) to scroll through the list. When the ID of choice is displayed, press the **2** KEY, **Select.**

FIGURE 5.20

16) ID *number*	
↑↓-Scroll	
2-Select	**0-Exit**

6. The chosen well ID and the type of flow device (Accu-Flo, Orifice Plate, Pitot Tube, or User Input) display on the left side of the screen and the Read, Retry, Edit, and Abort options display on the right. (Figure 5.21) Press the **1** KEY, **Read**, to continue to the Read Gas screen. (Skip to Step 14.)

FIGURE 5.21

ID [*number*]	1-Read
[*flow device*]	2-Retry
	3-Edit
[*pipe id*]	0-Abort

7. To select manual entry press the 2 KEY, **Manual.**

FIGURE 5.22

Manually enter ID or
Scroll through ?

1-scroll 2-Manual

8. Enter the well ID of choice. Use the **BLUE** *f* KEY to switch (toggle) between numbers and letters.

   - For numbers, press the keypad **Number** KEYS (0-9). (Figure 5.23)
   - For letters, press the **1** KEY (UP ARROW) or **6** KEY (DOWN ARROW) to step through the alphabet until the letter of choice appears. Press the **0** KEY to select the letter. (Figure 5.24)

After the final character is entered, the unit displays **0-Cont**. Press the **0** KEY if the ID is correct and ready to enter; otherwise, press and **hold** the **0** KEY to backspace and make corrections.

Please Enter ID
f- Numbers/ Letters
ID AAAA ?---

FIGURE 5.23

Please Enter ID
f- Numbers/ Letters
ID AAAA ?---
↑↓-Step 0-Enter

FIGURE 5.24

Note: Four letters from the previously entered ID default into the first four places of the ID number. To replace these letters, if needed, use the 0 KEY to backspace over them by holding it down for more than a second. Change the characters as needed.

9. If the well ID is not yet entered into the unit, a No Reading screen appears. This screen provides three options. If the number is correct but has never been previously added, Enter ID Info. allows entry and flow device definition. If the well ID entered was in error, Retry takes the user back to the Manually Enter ID screen (Figure 5.22). The Abort command takes the user completely back out to the Main Menu (Figure 5.17). Press the 1 KEY, **Enter ID Info.**

FIGURE 5.25

No Reading for ID
XXXX XXXX
1-Enter ID Info.
2-Retry 0-Abort

10. A well flow device is selected in the Flow Device screen. This selection is necessary for the GEM-500™ to be able to calculate flow when readings are taken. Use the **1** KEY (**UP ARROW**) or the **6** KEY (**DOWN ARROW**) to scroll through the choices listed in Figure 5.27. Once the desired flow device is located in the shaded selection window, press **0** to select and continue.

FIGURE 5.26

Select Flow Device
Accuflo-1.5V
↑-Scroll Up
↓-Scroll Down 0-Cont

Note: Selection of User Input allows the eventual entry of flow in SCFM, if known (see Step 11). Those without flow devices may wish to use this selection to record velocity or other relevant data. (For example, when using a Kurz meter.)

FIGURE 5.27

Accuflo–1.5V (1½" Accu-Flo Model 150 Vertical Wellhead)
Accuflo–1.5H (1½" Accu-Flo Model 150 Horizontal Wellhead)
Accuflo–2V (2" Accu-Flo Model 200 Vertical Wellhead)
Accuflo–2H (2" Accu-Flo Model 200 Horizontal Wellhead)
Accuflo–3V (3" Accu-Flo Model 300 Vertical Wellhead)
Accuflo–3H (3" Accu-Flo Model 300 Horizontal Wellhead)
Orifice Plate (Orifice diameter and pipe inner diameter required)
Pitot Tube (Pipe ID required)
User Input (Pipe ID required)

11. If an Orifice Plate, Pitot Tube or User Input flow device is selected, additional information is required. If the pipe or orifice diameter screen appears, input the required size as necessary. Insert inches or centimeters (depending on whether USA or Metric Units were selected on the General Utilities USA/Metric units screen). The unit uses XX.XX as the format and automatically enters the decimal point. Press the **0** KEY to enter and continue.

FIGURE 5.28

Enter Pipe ID

?-.-- Inches

12. The ID Stored OK Screen displays for three seconds (Figure 5.29)

FIGURE 5.29

ID Stored OK

13. The chosen well ID and the type of flow device (Accu-Flo, Orifice Plate, Pitot Tube, or User Input) display on the left side of the screen and the Read, Retry, Edit, and Abort options display on the right. (Figure 5.30) Press the **1** KEY, **Read**, to continue to the Read Gas screen.

Figure 5.30

ID [*number*]	**1-Read**
[*flow device*]	**2-Retry**
	3-Edit
[*pipe id*]	**0-Abort**

14. Connect the GEM-500™ to the wellhead with the supplied tubing—Static Pressure/Sampling Port to the Static Pressure Port and Impact Pressure Port to Impact Pressure Port. The Gas Levels screen is divided into two parts. The left side of the screen displays current percentages of CH_4, CO_2, O_2, and BAL, which is the balance of all other gases excluding the CH_4, CO_2, and O_2. The right side of the screen displays functional choices for this reading (Lower Explosive Limit, Continue to temperature and pressure data, Pump On/Off, Last Data, Exit) and a POD reminder.

FIGURE 5.31

CH_4 00.0%	**1-LEL 2-Cont**
CO_2 00.0%	**5-Pump OFF**
O_2 00.0%	**9-Last Data**
BAL 00.0%	**POD? 0-Exit**

POD refers to interchangeable electrochemical gas pods that are used to extend the measurement capabilities of the GEM-500™ (Figure 5.32). These pods are available in seven different gases with nine different ranges and easily plug into the data port. The reminder lets the user know that if a pod were attached at this time, additional gas readings could be taken.

FIGURE 5.32

Interchangeable Electrochemical Gas Pods

Gas	Range (ppm)	Resolution (ppm)
H_2S	0 – 50	0.1
	0 – 200	1.0
CO	0 – 1000	1.0
SO_2	0 – 20	0.1
	0 – 100	1.0
NO_2	0 – 20	0.1
Cl_2	0 – 20	0.1
H_2	0 – 1000	1.0
HCN	0 – 100	1.0

15. Press the **5** KEY, **Pump**, to start the pump and draw a gas sample into the GEM-500™. The **5** KEY, **Pump**, works as a toggle switch to turn the pump on and off. Once the pump is turned on, the readings are not considered to be accurate until the percentages on the left side of the display stabilize, typically within 30-45 seconds. A timer displays on the screen to monitor the pump running time.

FIGURE 5.33

CH_4 00.0%	**1-LEL 2-Cont**
CO_2 00.0%	**5-Pump OFF**
O_2 00.0%	**9-Last Data**
BAL 00.0%	**POD? 0-Exit**

Note: The GEM-500™ may sound an alarm (beeping) while taking gas readings. The alarm means that gas levels set in General Utilities Gas Alarms (page 31) have been reached or exceeded. In addition to the alarm, the screen display of the gas that set off the alarm also blinks.

16. To monitor the LEL (Lower Explosive Limit), press the 1 KEY (Figure 5.33) to enter the LEL screen. If the LEL needs to be continuously monitored, just leave the pump running. When the LEL no longer needs to be monitored, press the 5 KEY again to toggle off the pump. Press the 0 KEY to Exit back to the Gas Levels screen (Figure 5.33).

FIGURE 5.34

00.0	CH_4 %Gas	5-Pump OFF
000.	CH_4 %LEL	0-Exit

17. If at any time, while working on this screen, reference needs to be made to the prior reading on this wellhead, press the **9** KEY, **Last Data**. The unit displays the very last stored reading. The **9** KEY works as a toggle switch between the current reading and the last reading. From the Last Data (Figures 5.33 & 5.35) screen, the user can go directly into monitoring current LEL or continue directly to the current Temperature/Pressure screen.

FIGURE 5.35

CH_4 00.0%	1-LEL 2-Cont
CO_2 00.0%	
O_2 00.0%	9-Curr Data
BAL 00.0%	0-Exit

18. From the Read Gas Levels screen (Figure 5.33), press the **2** KEY, Continue, for the Temperature and Pressure screen (Figure 5.36). This screen shows temperature, static pressure, differential pressure, and a number of functional options. If a temperature probe is used, it should be connected from the data port on the right side of the GEM-500™ (Figure 5.1) to the temperature gauge port on the wellhead. When the temperature probe is connected to the GEM-500™, the functional option Enter TEMP disappears as it displays only for manual entry. The temperature probe needs to remain in the wellhead until the entire reading is completed and stored. If a temperature probe is not used, the temperature can be read on a thermometer placed in the temperature gauge port on the wellhead and entered manually into the GEM-500™. To enter the temperature manually, press the **1** KEY, **Enter TEMP**.

FIGURE 5.36

T >>>°F	1-Enter TEMP
SP+/-00.0"	3-Zero Pres
DP+/-0.00"	9-Last Data
0-Back	2-Continue

Note: If flow measurement is required, temperature must be entered either manually or with the optional temperature probe.

19. Temperature is entered in either Fahrenheit (F) or Celsius (C) depending on whether USA or Metric was chosen in the General Utilities USA/Metric Units screen (page 30). Number keys and leading zero(s) are used to enter a three digit temperature value, i.e., 78° is 078 and 5° is 005. Press the **0** KEY to Continue back to the Read Gas Levels screen (Figure 5.36).

FIGURE 5.37

Enter Temperature
?-- °F
0-Cont

20. From the Read Gas Levels screen (Figure 5.36), press the **3** KEY, **Zero Pressures**, before proceeding. This is necessary because by taking a LFG sample, the impact port pressure transducers have been pressurized. The **Warning Disconnect hoses** screen displays before the Zero Pressure screen. (Figure 5.38). Disconnect all the hoses from the GEM-500™ to the wellhead and press any key to continue.

FIGURE 5.38

WARNING: Disconnect
Hoses before zeroing
Press any key

21. The pressures displayed on the Static/Differential screen are displayed with either a positive or negative and need to stabilize (quit changing) before being zeroed. When stabilized, press the **1** KEY to **Zero Pressures**. (Figure 5.39)

FIGURE 5.39

Static +/-00.0"H_2O
Differential +/-0.00"H_2O
1-Zero Pressures
0-Exit

Note: The units displayed are in inches of water column or MB depending on whether USA or Metric was chosen in the General Utilities USA/Metric Units screen (page 30).

22. The Zeroed OK screen (Figure 5.40) displays for 3 seconds, then the Static/Differential screen (Figure 5.41) redisplays. Occasionally the pressures will not zero completely. If this happens, press the **1** KEY, **Zero Pressures**, again until the pressures are completely zeroed. Once the static and differential pressures are zeroed, press the **0** KEY to **Exit**. The Temperature/Pressure screen then redisplays (Figure 5.42).

Zeroed OK

FIGURE 5.40

Static +/-00.0"H_2O
Differential +/-0.00"H_2O
1-Zero Pressures
0-Exit

FIGURE 5.41

23. At the Temperature/Pressure screen, reconnect the tubing (Figure 5.43). Allow the instrument to read, then press the **2** KEY to Continue to the Flow/Btu screen.

FIGURE 5.42

T >>>°F	1-Enter TEMP
SP+/-00.0"	3-Zero Pres
DP+/-0.00"	9-Last Data
0-Back	2-Continue

FIGURE 5.43

CONNECTIONS

1. **On an Accu-Flo wellhead, connect the Static Pressure/Sampling port (top left) on the GEM-500™ to the Static Pressure Port on the wellhead, and the Impact Pressure Port on the wellhead. (Figures 5.1, 5.2, 5.3)**
2. **On a pitot tube, connect the hoses from the GEM-500™ to the Impact and static Pressure ports.**
3. **On an orifice plate, connect the hoses on the GEM-500™ to either side of the orifice plate.**

Note: To get a flow reading, the differential pressure (DP) must be positive with respect to the static pressure. If it is not, reverse the hoses.

The Flow screen displays the reference flow (REF) and adjusted flow (ADJ). The reference flow is static and will not change unless the Temperature/Pressure screen is re-accessed. Whereas the adjusted flow is active and is typically used to display changes in flow while adjustments are made. If an Accu-Flo, pitot tube, or orifice plate wellhead is in place, the GEM-500™ automatically calculates the reference and adjusted flow. Other functions available here are the ability to store the data, go to the next wellhead ID, view the last data on this wellhead, or go back to the Temperature/Pressure screen. The units displayed on this screen, Scfm (Standard cubic feet per minute) and BTU (British Thermal Units, in thousands) are the USA units that are selected in the General Utilities USA/Metric Units screen (page 30).

FIGURE 5.44

```
Scfm 000|BTU 000e3/h REF
Scfm 000|BTU 000e3/h ADJ
 1-Flow  6-Store  4-ID>
 9-Last Data     0-Back
```

If the flow needs to be changed, adjust the control valve on the wellhead. Within a few seconds the new flow is displayed on the GEM-500™ as the adjusted value. When satisfied with the flow adjustment, press the **6** KEY to **Store** the information and continue to the Select Comments screen.

24. On the Select Comments screen, use the **1** KEY to **Scroll Up** or the **6** KEY to **Scroll Down** through a comment list (Figure 5.45). When the correct comment appears in the comment display area, press the **2** KEY, to **Select**, then the **0** KEY to store the readings.

```
   Select Comments
  [comment display]
↑-Scroll Up   2-Select
↓-Scroll Dn   0-Store
```

FIGURE 5.45

```
            COMMENTS
Flex Hose        Labeling
Valve            Air Leakage
Casing Height    Water Blockage
Well Bore Seal   Other
```

FIGURE 5.46

25. If the readings are successfully stored, the Readings Stored screen displays for three seconds (Figure 5.47). The Flow screen (Figure 5.44) redisplays.

FIGURE 5.47

```
  Readings Stored
```

26. Press the **4** KEY, **ID** to advance to the next wellhead ID in memory. If the wellhead ID's were loaded into the GEM-500™ in order, the unit will purge and the next ID will display after purge is completed.

FIGURE 5.48

```
Scfm 000|BTU 000e3/h REF
Scfm 000|BTU 000e3/h ADJ
 1-Flow  6-Store  4-ID>
 9-Last Data     0-Back
```

27. The Purge Prompt Screen (Figure 5.20) reminds the user that the GEM-500™ needs to have any residual gases purged from its system to guarantee the accuracy of the next reading. Press the **5** KEY, **Purge,** to initialize the purge process.

FIGURE 5.49

```
  You should purge
  After each sample
1-Next ID
2-Purge        0-Back
```

28. The Warning screen displays to remind the user to disconnect the hoses from the wellhead before starting the purge process. Once the hoses are removed, press the 1 KEY to begin the purge (Figure 5.50).

FIGURE 5.50

```
WARNING! Disconnect
Hoses from the well
Before purging.
1-Begin Purge    0-Back
```

29. The pump starts and the Purge in Progress screen displays the number of seconds remaining in the purge (Figure 5.51). This purge process can be taking place while walking to the next wellhead.

FIGURE 5.51

```
  Purge in progress
60 seconds remaining

                 0-Back
```

30. The pump will run for 60 seconds and then shut off allowing the next well ID to be displayed. (Figure 5.52)

FIGURE 5.52

```
ID [ number ]     1-Read
[ flow device ]   2-Retry
                  3-Edit
[ pipe id ]       0-Abort
```

Chapter 6 - View Data

View Data

1. From the Main Menu, select the **3** KEY, **View Data** (Figure 6.1)

FIGURE 6.1

1-General Utilities
2-Read Gas Levels
3-View/Print Data
4-Download Data

From the View/Print Select screen, the user can choose to view data, print data or return to the main menu. Press the **1** KEY, **View Data** to view data in memory in the GEM-500™.

FIGURE 6.2

Please Select
1-View Data
2-Print Data
0-Main Menu

To view data from just one wellhead, press the **1** KEY, **With Specific ID**. This option allows manual entry of a single wellhead ID. If data from all wellheads needs to be viewed, press the **5** KEY, **All Data** and proceed to step 4.

FIGURE 6.3

View Data

1-With Specific ID
5-All Data

The Manually Enter ID screen allows the user to either scroll through a list of ID's that are already in memory or enter an ID manually. If there are few ID's stored, it might be faster to use the scroll option; whereas the manual entry option might be faster to use if there are many ID's stored (Refer to page 44 number 7). (Figure 6.4)

FIGURE 6.4

Manually enter ID or
Scroll through?

1-scroll 2-Manual

2. To scroll through the existing list of well ID's, press the **1** KEY, **Scroll**. The Select ID screen displays a single ID at a time. Use the **1** KEY (UP ARROW) or the **6** KEY (DOWN ARROW) to scroll through the list. When the ID of choice is displayed, press the **2** KEY, **Select.** (Figure 6.5)

FIGURE 6.5

```
16)   ID [number]

↑↓-Scroll
  2-Select   0-Exit
```

3. The following options are used to move around within the Data screens: (Figure 6.6)

   - ↑↓-Scan – Use the **1** KEY (UP ARROW) or the **6** KEY (DOWN ARROW) to scroll through the list.
   - 2-Go First – Use the 2 KEY to go to the first well ID in the list.
   - 5-Change Data Screen – Use the 5 KEY to toggle between the Gas Concentrations screen (Figure 6.7) and the Flow screen (Figure 6.8).
   - 7-Go Last – Use the 7 KEY to go to the last well ID in the list.
   - 0-Exit – Use the 0 KEY to exit at any time.

FIGURE 6.6

```
USE:   ↑↓-Scan    0-Exit
2-Go First   7-Go Last
5-Change data Screen
Any key to continue
```

```
      ID [number]
TAKEN 00:00 00/00/98
CH4  00.0%  |CO2  00.0%
O2   00.0%  |BAL  00.0%
```

FIGURE 6.7

```
SP+/-00.0"   |DP+/-0.00"
T  095°F
Scfm  000 |BTU 000e3/h REF
Scfm  000 |BTU 000e3/h ADJ
```

FIGURE 6.8

Chapter 7 - Communications

This chapter details how to:

- Install the GEM COMM software, contained on the disk that came with the GEM-500™ onto a PC.
- Connect the GEM-500™ to a PC with the special data interface cable (RS-232) provided with the GEM-500™ by CES-LANDTEC.
- Configure the GEM COMM software.
- Transfer information from and to the GEM-500™.
- Add, delete, and manipulate comments, IDs and well data with the GEM COMM software.

The special data interface cable (RS-232) attaches to the GEM with a specially manufactured plug. The temperature probe, battery charger and data cable all use the same input plug. **Do Not** try to substitute any other cables or plugs for the ones provided by CES-LANDTEC.

Installing The Gas Analyzer Communications Program

The CES-LANDTEC Gas Analyzer Communications software converts the binary data produced by the GEM-500™ to an ASCII format, comma delimited. The data in ASCII format can then be imported into Datafield™ (CES-LANDTEC proprietary software), LOTUS 1-2-3™, Excel™, and many other programs that accept comma delimited data files.

1. Turn on computer and go to the DOS (Disk Operating System) prompt.
2. Insert the CES-LANDTEC GEM COMM download disk into computer's floppy drive.
3. Install the GEM software into a subdirectory of your choice, using the drive of your choice, by typing the following instructions (assuming drive A contains source disk and C is the destination drive).

   a) C:\>MD GEM <ENTER>
   b) C:\>CD GEM <ENTER>
   c) C:\GEM> COPY A:*.* <ENTER>
   ```
        A:\GEM_COMM.EXE
            1 file(s) copied
        C:\GEM>
   ```
4. The GEM software is now installed on the computer in a directory named GEM.
5. Remove the diskette from drive A and store it in a safe place. One backup copy of the program disk may be made.

Starting The Communications Program

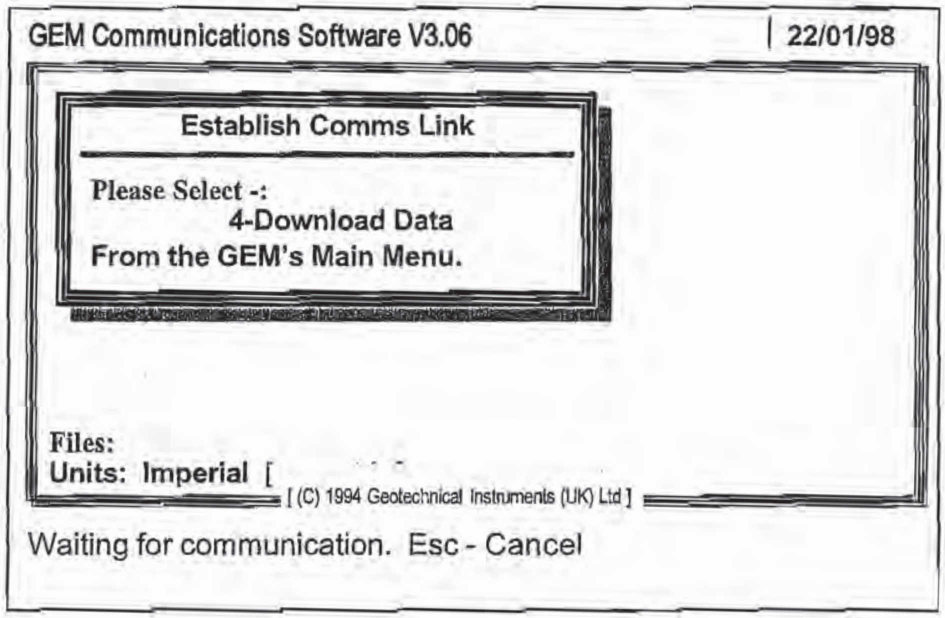
1. Make sure that the GEM-500™ is off.

2. Attach the RS-232 cable to **COM Port 1 or COM Port 2** on the computer and make a note of which COM Port is used, as this information is needed later. Plug the cable into the receptacle port (Figure 1.1) on the GEM-500™.

3. Access the GEM directory by typing the following at the **C:>** prompt (DOS prompt) :
   a) C:\> CD GEM <Enter>
   b) C:\> GEM_COMM<Enter>

4. The Com Link screen sets up the link between the computer and the GEM-500™. Now turn on the GEM-500™. Proceed to the Main Menu on the GEM-500™ (Figures 2.1 – 2.5). Press the **4** KEY, **Download Data**.

FIGURE 7.1



Note: While using the Gas Analyzer Communications Software - Version 3.06, brief function descriptions and instructions appear, on screen. Use the ↑ KEY and the ↓ KEY to move through the options, and the ESC (ESCAPE) key to Exit or abort.

5. Watch for the **C** to appear for a few seconds at the lower right of the Com Link screen. This is an indication of a successful communications link.

   - If this is the first time that the program has been run, the **C** may not appear because a COM port has not yet been specified. In this case, press the **ESCAPE** KEY and continue to **Main Menu – Configure** below.

     NOTE: The only time that the ESCAPE KEY is used to return to the Main Menu in the GEM COMM program is during the initial setting. At all other times press the 4 KEY, Download Data on the GEM-500™ keyboard.

   - If the "**C**" does not appear, and this is **NOT** the first time that the program has been run, the cable maybe plugged into the wrong COM port, the cable connections are not stable, there is a problem with the cable, or a Windows program is also running. In this case, exit out of the software, turn off the GEM-500™, check the cable connections, close any open Windows program, and start over from Step 1, above.

FIGURE 7.2

Establish Comms Link

Please Select -:
4-Download Data
From the GEM's Main Menu.

Main Menu

After the communication link is established, the Main Menu displays on the computer (Figure 7.3) and the RS-232 Mode screen (Figure 7.4) displays on the GEM-500™. RS-232 MODE means that the GEM-500™ is linked and awaiting instructions from computer.

Well ID functions ▸
User Defined Comments ▸
GEM Data Functions ▸
Clear functions ▸
Show GEM memory statistics ▸
Configure ▸
Exit ▸

FIGURE 7.3

RS-232 MODE

Awaiting Instruction
From computer

FIGURE 7.4

Configure

The Configure menu presents options that allow the user to set operating units, select a communications port, data directory, set well ID sorting, and set stored titles.

1. Using the up ↑ and down ↓ arrows on the computer keypad, highlight **Configure** on the Main Menu screen (Figure 7.3) and press the **Enter** key.

FIGURE 7.5

```
Well ID functions              ▸
User Defined Comments          ▸
GEM Data Functions             ▸
Clear functions                ▸
Show GEM memory statistics     ▸
Configure                      ▸
Exit
        Use Imperial Units                  :OFF/ON
        Set Data Directory
        Set Default COM Port
        Sort Well Ids                       :OFF/ON
        Separate Time/Date Fields           :OFF/ON
        Store CSV with or without Titles    :OFF/ON
```

2. **Set Default COM Port** by highlighting the option and pressing**enter**. This is one of the most important options because it tells the GEM-500™ software which computer communications port will be used when trading data with the GEM-500™. Enter **1** for COM1 and **2** for COM2. To verify the communication port setting, select Set Default COM Port again and view current setting, press **ESC** to exit.

FIGURE 7.6

```
        Set RS-232 port
   Press:  1) COM1
           2) COM2
   Current:    COM1
```

NOTE: If this is the initial setting of the COM port, once it is set correctly and verified, press the ESCAPE KEY to save the setting and return to the Main Menu. Use the ↓ to highlight the Exit option and press Enter. Re-access the GEM COMM software. This time at the Com Link screen (Figure 7.2) when the 4 KEY, Download Data, is pressed, the C should appear.

3. **Use Imperial Units.** The GEM-500™ was set with either USA Standard (Imperial) or Metric units under General Utilities – USA/Metric Units (page 30). The software needs to match this setting in order to store the well data properly. Pressing **Enter** at this option acts as a toggle switch, turning the use of Imperial units on or off.

4. **Set Data Directory.** This is the directory where the ID and comment files are located. Press **Enter** to access the option. The current data directory appears. In the example below, files are saved in the Desktop file on the C drive (hard disk). Change this path if necessary.

FIGURE 7.7

5. **Sort Well ID.** This function sorts the well IDs into alphabetic order after being received from the GEM-500™. Pressing **Enter** at this option acts as a toggle switch, turning the sort function on or off. **IDs remain in the same order as input with this option turned OFF.** The default for this function is OFF.

6. **Separate Time/Date Fields.** This function sets the file format to separate time and date into two columns. Pressing **Enter** at this option acts as a toggle switch, turning the separate function on or off. The default for this function is OFF.

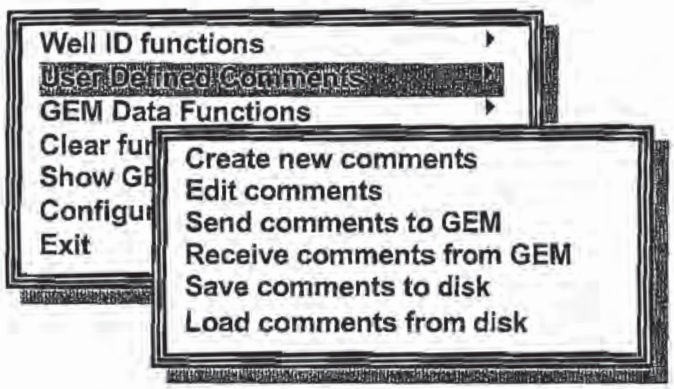
7. **Store CSV with or without Titles.** This function sets the file format to include titles at the top of the data columns. Pressing **Enter** at this option acts as a toggle switch, turning the addition of titles on and off. The default for this function is OFF.

8. Press the **ESCAPE** KEY to return to the main menu.

User Defined Comments

The Custom Comments menu presents options that allow the user to create, edit, send, receive, save, and load eight user defined comments that can be saved with the well data. (Chapter 5 – Read Gas Levels, Page 42, Step 10, Figures 5.14 and 5.15 and Page 51, Step 26, Figures 5.45 and 5.46)

Figure 7.8



1. **Create New Comments.** The Comment screen displays places for eight (8) comments, numbered one through eight. (Figure 7.9) Since space is so limited, it might be a good idea to use the preprogrammed comments for a while to get a feel for the additional comments that might be needed. To add a comment, position the cursor on the C of comment and type over the word COMMENT1. (Figure 7.9) There are 16 spaces available for the entry of a new comment. In order for the new comment to fit, words might need to be abbreviated. Use the **ESCAPE** KEY to back out of the option without saving or changing anything and the **F2** KEY to save the new comments.

Figure 7.9

PREPROGRAMMED COMMENTS	
Flex Hose	**Labeling**
Valve	**Air Leakage**
Casing Height	**Water Blockage**
Well Bore Seal	**Other**

```
COMMENT1
COMMENT2
COMMENT3
COMMENT4
COMMENT5
COMMENT6
COMMENT7
COMMENT8
```

FIGURE 7.10

```
WELLHEAD BROKEN
COMMENT2
COMMENT3
COMMENT4
COMMENT5
COMMENT6
COMMENT7
COMMENT8
```

FIGURE 7.11

2. **Edit Comments.** Allows the user to add, change, or correct the custom comments that are already stored in the system. Before the comments can be edited, they must be received by the GEM COMM software from either the GEM-500™ (See step 4 below) or from disk (See step 6 below). If the custom comments have just been created using the Create New Comments option, they are already in the GEM COMM software and do not need to be loaded again. Use the **F2** KEY to store the new comments or the **ESCAPE** KEY to back out of the option without saving or changing anything.

3. **Send comments to GEM.** Once accessed, this option automatically sends the list of user-defined comments to the GEM-500™ for use while reading gas levels. When finished sending to the GEM-500™, the user is notified that the comments have transferred.

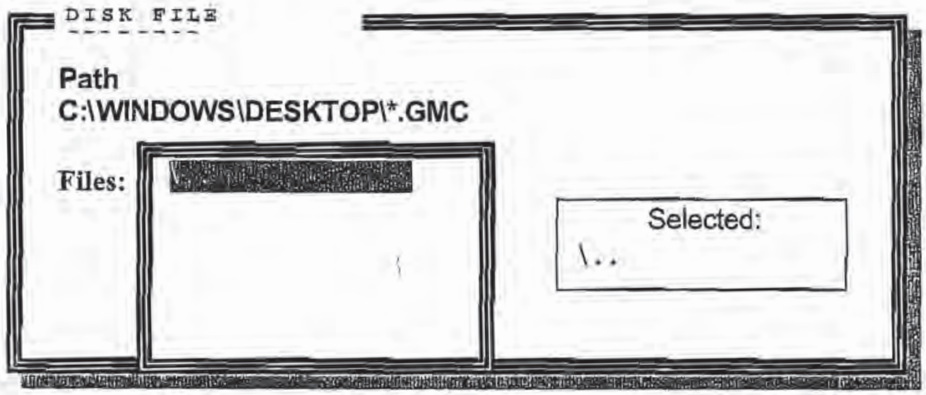
FIGURE 7.12

```
Send User Comments...
Comments transferred.
Press any key
```

4. **Receive comments from GEM.** This option allows the user to load the custom comments into the GEM COMM software that are stored in the GEM-500™. As in the Send Comments option (step 3, above), this takes place automatically once the option is accessed. Use the **F2** Key to store the new comments or the **ESCAPE** KEY to back out of the option without saving or changing anything.

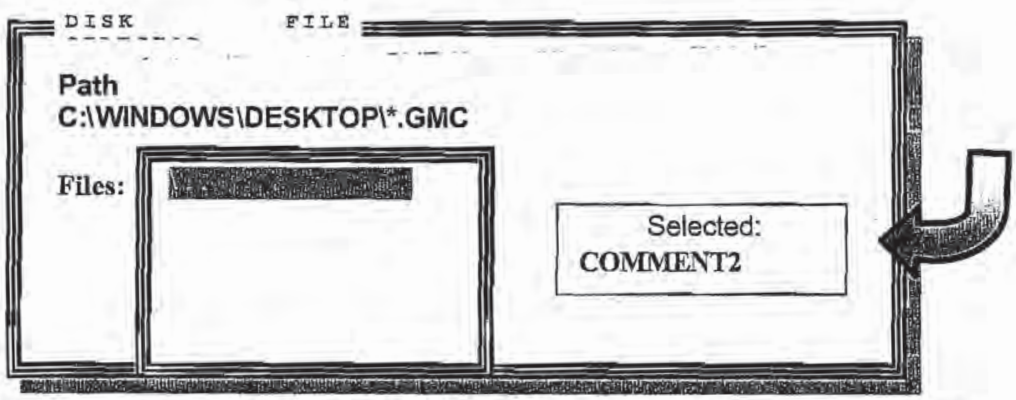
5. **Save comments to disk.** This option allows the user to store the custom comments on either a floppy disk or the hard drive. In the example below, documents are stored in the Desktop file on the C drive (hard disk) in the computer.

FIGURE 7.13



If the path is correct, enter a file name (eight character maximum) in which the custom comments are to be stored. Numerous sets of comments can be stored in this manner. Highlight The cursor displays in the **Selected:** window. Type the file name over the contents of the window and press **enter**. Do not type the .GMC file extension, it is automatically added by the program. The custom comments are stored in the newly named file and the system returns to the main Comments menu.

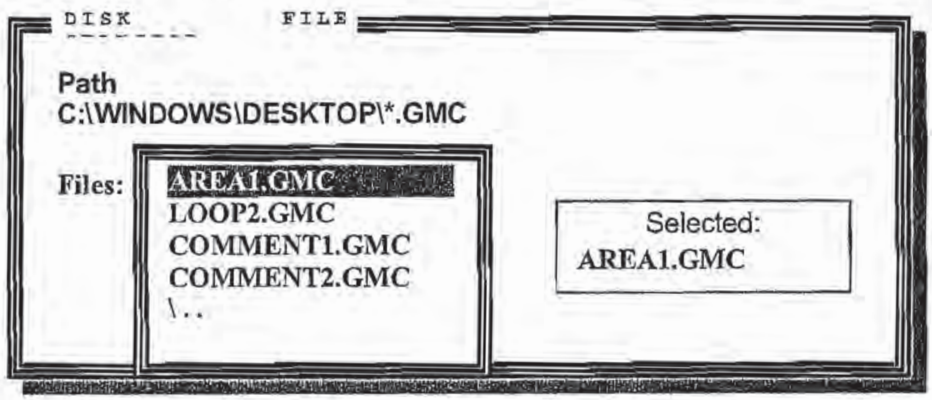
FIGURE 7.14



a) If the path is not correct, use the **ESCAPE** KEY to return to the Main menu (Figure 7.3). Access the **Set Data Directory** option under **Configure** (Figure 7.7) and change the path. When the path has been changed, return to the **Receive comments from GEM** option and type in the file name for the custom comments.

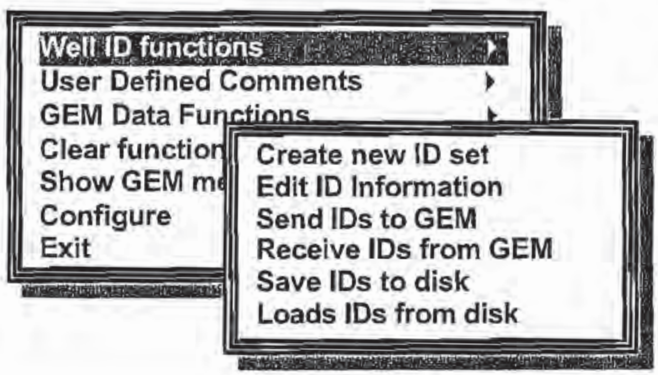
6. **Load comments from disk**. This option allows the user to load a set of custom comments from a computer file into the GEM COMM software. When this option is accessed, all of the files with a .GMC extension are displayed. Use the ↑ or ↓ arrow to highlight the wanted file name (so that it displays in the Selected: window) and press enter. The computer automatically loads the contents of the selected file into the GEM-COMM software and returns to the main Comment menu.

FIGURE 7.15



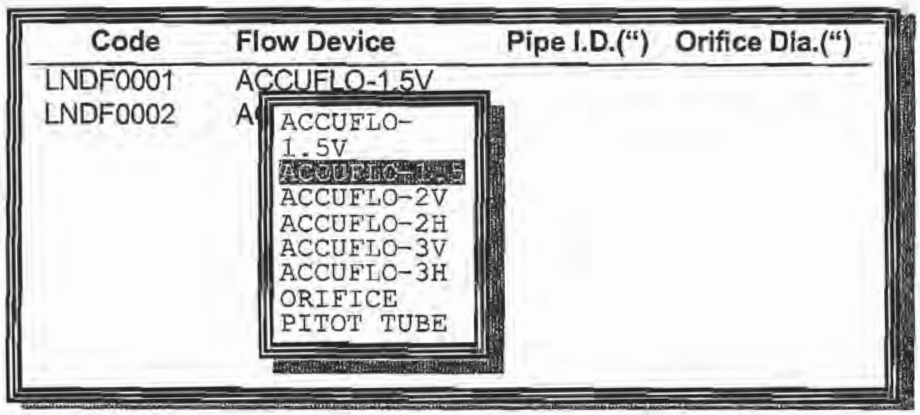
Well ID Functions
This menu contains options that enable the user to create, edit, receive, send, save and load well IDs.

FIGURE 7.16



1. **Create new ID set.** This option allows the easy addition of multiple well IDs. The Code column contains the 8-character, alpha/numeric well ID. The Flow Device column, when accessed, displays a complete table of available flow devices. Use the ↑ or ↓ arrow to highlight the appropriate device and press **enter**. If Pitot Tube or User Input is chosen, the Pipe Inner Diameter must be entered. If Orifice is chosen (Orifice Plate) both the Pipe Inner Diameter and Orifice Diameter must be entered. These diameters are needed by the GEM-500™ when calculating flow rates. When finished entering, press the **F2** KEY to store the new additions. Once the IDs are entered, they should be saved to disk (See Step 5 below.)

FIGURE 7.17



2. **Edit ID Information.** This option allows the editing of ID information. The IDs can be from the GEM-500™ or from a file saved to disk. Use the **F9** KEY to delete, **F10** KEY to insert, the **F2** KEY to store the changes, and the **ESCAPE** KEY to cancel.

3. **Send IDs to GEM**. Once accessed, this option automatically sends the list of IDs to the GEM-500™. When finished sending to the GEM-500™, the user is notified that the comments have transferred.

FIGURE 7.18

4. **Receive IDs from GEM.** This option allows the upload of IDs that are stored on the GEM-500™. Once the option is accessed, the well ID numbers are automatically loaded into the GEM COMM software where they may be viewed, edited, or saved to disk.

FIGURE 7.19

Receive ID Information . . .

[*well ID #s received* **]**

FIGURE 7.20

Code	Flow Device	Pipe I.D.(")	Orifice Dia.(")
LNDF0001	ACCUFLO-1.5V		
LNDF0002	ACCUFLO-1.5H		
LNDF0003	ACCUFLO-2V		
LNDF0004	ACCUFLO-2H		
LNDF0005	ACCUFLO-3V		
LNDF0006	ACCUFLO-3H		
LNDF0007	ORIFICE	2.00	.750
LNDF0008	PITOT TUBE	.250	
LNDF0009	USER INPUT	.250	

5. **Save IDs to disk.** This option allows the user to store the IDs on either a floppy disk or the hard drive. In the example below, documents are stored in the Desktop file on the C drive (hard disk).

FIGURE 7.21

DISK SELECTOR FILE

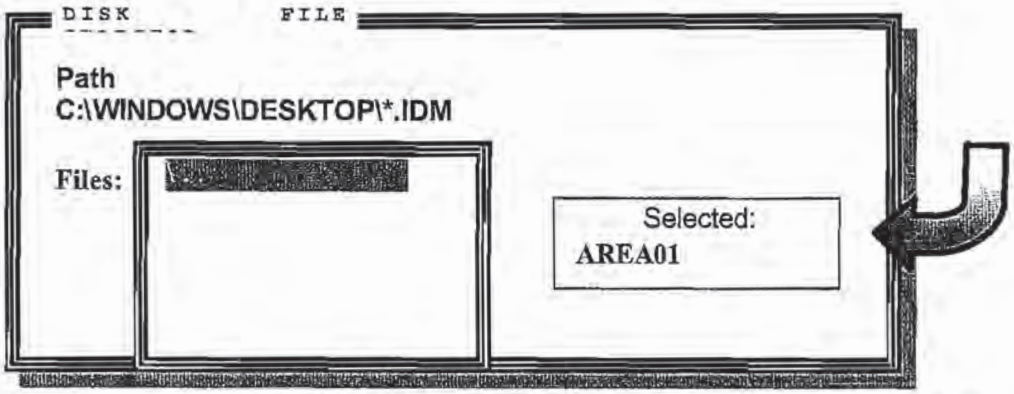
Path
C:\WINDOWS\DESKTOP*.IDM

Files:

Selected:
\..

a) If the path is correct, enter a file name (eight characters maximum) in which the IDs are to be stored. Numerous sets of IDs can be stored in this manner. The cursor displays in the **Selected:** window. Type the file name over the contents of the window and press **enter**. Do not type the .IDM file extension, it is automatically added by the program. The IDs are stored in the newly named file and the system returns to the main Comments menu.

FIGURE 7.22



b) If the path is not correct, use the **ESCAPE** KEY to return to the Main menu (Figure 7.3). Access the **Set Data Directory** option under **Configure** (Figure 7.7) and change the path. When the path has been changed, return to the **Receive comments from GEM** option and type in the file name for the custom comments.

6. **Load IDs from disk.** This option allows the user to move a selected set of IDs from the file where it is stored into the GEM COMM software. When this option is accessed, all of the files with an .IDM extension are displayed. Use the ↑ or ↓ arrow to highlight the wanted file name (so that it displays in the **Selected:** window) and press **enter**. The computer automatically loads the contents of the selected file into the GEM COMM software and returns to the main ID menu.

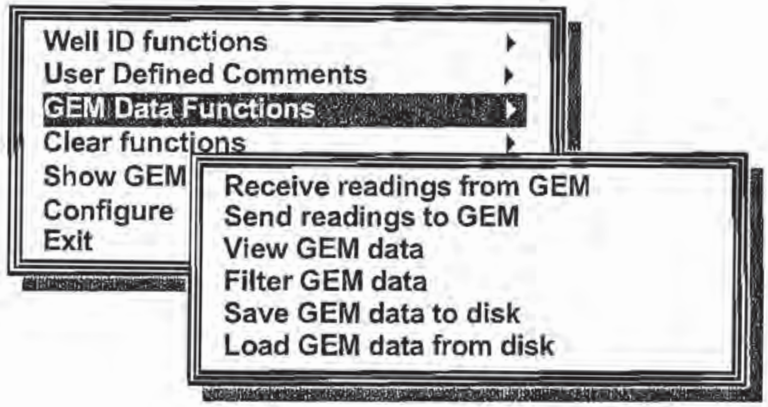
FIGURE 7.23

GEM Data Functions

This Menu provides options to upload readings from the GEM-500™ to the computer and download readings from the computer to the GEM-500™, view and filter the GEM-500™ data, and save the GEM-500™ data to disk or load data from disk to the GEM-500™.

1. From the Main Menu, highlight **GEM Data functions** and press enter.

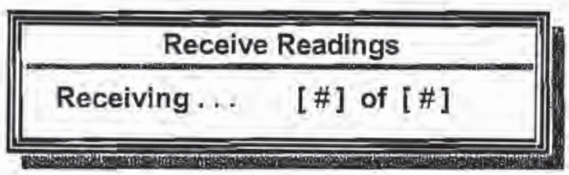
FIGURE 7.24



NOTE: There must be readings stored in the GEM-500™ to proceed with the following steps. Refer to Chapter 5 - Read Gas Levels if further information is needed.

2. **Receive readings from GEM.** The system does an automatic, temporary upload from the GEM-500™ to the GEM COMM software. The software must receive the readings in order for them to be viewed or manipulated. While this is happening, the Receive Readings screen displays how many readings, out of the total, the computer has received (i.e., 3 of 60, 4 of 60, etc.) until all of the readings have been received. (Figure 7.25) If the readings need to be viewed again after leaving the **Receive readings from GEM** option, use the **View GEM data** option.

FIGURE 7.25



a) Once the computer has received the readings, the View Data screen can be displayed for each reading. Use the ↑ or ↓ arrows to cycle through the data. Use the ESCAPE KEY to exit or the F3 KEY to change the filter (See Step 4, below).

View Data							4 of 5
ID:	[*Well ID*]	14:30	01/20/98				
CH4:	06.6 %	Static:	-00.2	"H2O	Ref Flow:	0000	Scf/m
CO2:	20.6 %	Diff'l:	+0.190	"H2O	Adj Flow:	0000	Scf/m
O2:	00.0 %				Ref Energy:	0000	BTU/hr
Bal:	72.8 %	Temp :	075	°F	Adj Energy:	0000	BUT/hr
		Gas Pod:	[*type fitted*]		Pod Value :	>>>	
Comments:							
[*comment or custom comment*]							

FIGURE 7.26

3. **View GEM data.** Allows the user the view the readings that were received from the GEM-500™ in the previous step. (Figure 7.20). A filter can be set so that specific data can be viewed. (See step 4, below.)

4. **Filter GEM data.** Allows the user to set parameters for viewing data. This option is used in conjunction with the **View GEM data** option above. If data is needed from one specific well, enter that well ID number into the Code field. If data is needed from a specific time forward, enter that time in the Time field. If data is needed from a specific date forward, enter that date in the Date field. Two or more of these fields can be used in combination.

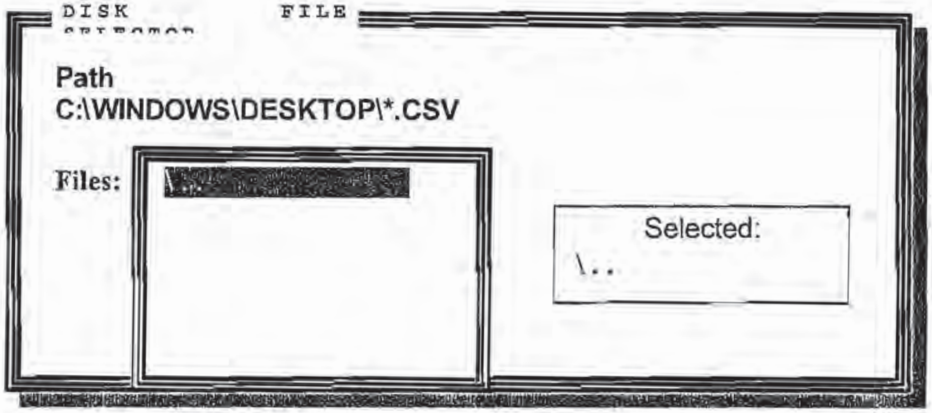
Note: An * instruction is a wildcard and can be used in place of a specific number or letter.

FIGURE 7.27

Set Data Filter		
Code	Time	Date
********	** **	** ** **

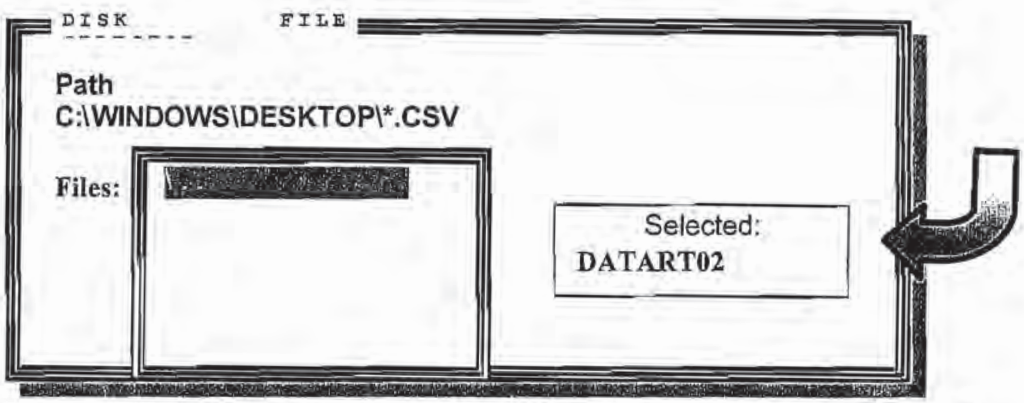
5. **Save GEM data to disk.** This option allows the user to store the well data on either a floppy disk or the hard drive. In the example below, documents are stored in the Desktop file on the C drive (hard disk) in the computer. CSV = Comma Separated Value

FIGURE 7.28



a) If the path is correct, enter a file name (eight character maximum) in which the well data is to be stored. Numerous sets of data can be stored in this manner. The cursor displays in the **Selected:** window. Type the file name over the contents of the window and press **enter**. Do not type the .CSV file extension, it is automatically added by the program. The well data is stored in the newly named file and the system returns to the main Data Functions menu.

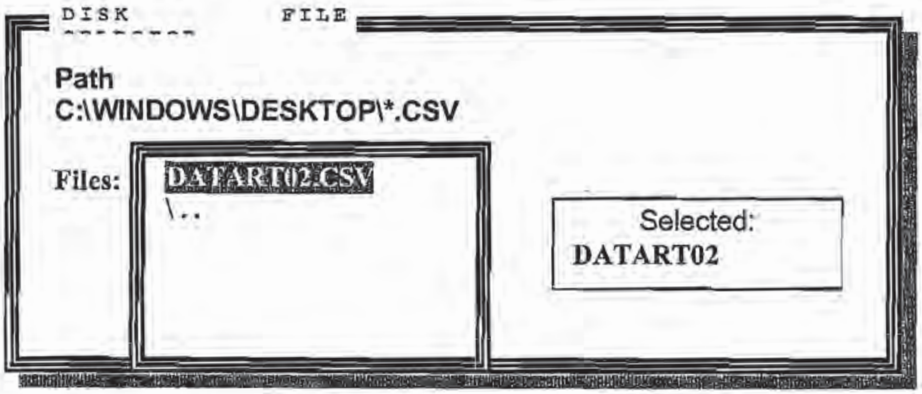
FIGURE 7.29



b) If the path is not correct, use the **ESCAPE** KEY to return to the Main menu (Figure 7.x). Access the **Set Data Directory** option under **Configure** (Figure 7.7) and change the path. When the path has been changed, return to the **Save GEM data to disk** option and type in the data file name.

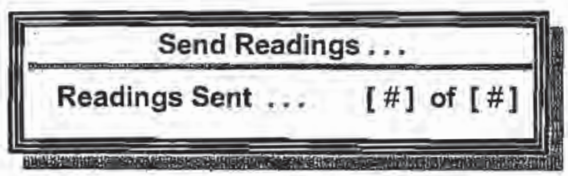
5. **Load GEM data from disk.** This option allows the user to load a data file from where it is stored in the computer into GEM COMM software so that it can then be edited or sent to the GEM-500™. When this option is accessed, all of the files with the chosen path are displayed. Data files have a .CSV extension, so be sure to only select one of those. Use the ↑ or ↓ arrows to highlight the wanted file name in the Files box (so that it displays in the **Selected:** window) and press **enter**. (Figure 7.30) The computer automatically loads the contents of the selected file into the GEM COMM software and returns to the main ID menu.

FIGURE 7.30



6. **Send readings to GEM.** Once accessed, this option automatically sends the list of well data readings to the GEM-500™. When finished sending to the GEM-500™, the user is notified that the data has been sent. (Figure 7.31)

FIGURE 7.31



Show GEM memory statistics

This option allows the user to view the Memory status of the GEM-500™.

1. On the main menu, highlight Show GEM memory statistics and press **enter**.

FIGURE 7.32

Well ID functions ▸
User Defined Comments ▸
GEM Data Functions ▸
Clear functions ▸
Show GEM memory statistics ▸
Configure ▸
Exit ▸

2. The GEM Memory Status screen displays the amount of free memory available and the amount of memory used by stored well Ids and readings. There are 699 blocks of memory available on the GEM-500™. Each block stores a well ID and the corresponding reading. If there are 50 wells in a field, as in the example shown below, the GEM-500™ should be able to store 13 readings of each well (650) before the need to clear the memory. Once the Memory Status is viewed, press the **ESCAPE** key to return to the Main Menu.

FIGURE 7.33

GEM Memory Status
Free Memory: 674
Ids: 50
Readings: 50
Version: GEM 3.17 – 21/01/98

Clear Functions

This Menu provides options to erase the IDs, readings, and user-defined comments in the GEM.

FIGURE 7.34

Well ID functions ▸
User Defined Comments ▸
GEM Data Functions ▸
Clear functions ▸
Show GEM memory statistics ▸
Configure
Exit

Clear IDs in GEM
Clear readings in GEM
Clear user-defined comments in GEM

1. **Clear IDs in GEM.** This option erases the well IDs from the GEM-500™. If the IDs to be cleared have readings attached, the readings are also erased.

FIGURE 7.35

Clear GEM

F1 – Clear IDs; Esc - Cancel

WARNING: Pressing F1 will clear memory!

2. **Clear readings in GEM.** This option clears the readings from the GEM-500™ and leaves the IDs in place, able to accept new readings. Of the three clear options, this is the option most frequently used. As the GEM-500™ becomes full of readings, the readings are uploaded into the computer and are then cleared from the GEM-500™ to make room for more readings.

FIGURE 7.36

Clear GEM

F1 – Clear Readings; Esc - Cancel

WARNING: Pressing F1 will clear memory!

3. **Clear user defined comments in GEM.** This option erases all of the user defined comments from the GEM-500™. It does leave the preprogrammed options in place.

FIGURE 7.37

Clear GEM

F1 – Restore default comments

WARNING: Pressing F1 will clear memory!

Note: Do not clear IDs or Readings without making copies or archiving the data to disk.

Chapter 8 - Maintenance

Servicing

The GEM-500™ has been electronically and functionally tested before leaving the factory. It is recommended that with normal usage, the unit should be serviced every **six months** for routine factory service and maintenance which includes:

- Replace all Filters and O-rings.
- Perform Bench Test with 10 Test Gases.
- Minor Adjustments.
- Check Overall Performance.
- If needed, run through Environmental Chamber.
- Check Charging Circuit and Battery Pack.
- Check Inlet Port Fittings.
- Calibrate Transducers.
- Check the Pump.
- Check the Flow Fail.
- Perform Leak Test.

Cleaning

Protect the GEM-500™ by keeping it in its protective soft case. The keypad (polycarbonate membrane) should be wiped clean with soapy water and a damp cloth. Other cleaning agents may damage the membrane.

Sunlight and Heat

The GEM-500™ should not be left out in direct sunlight for long periods of time as this raises the temperature inside the case and may cause damage to the components. The unit may not operate or may operate erratically if it gets too hot or cold. The operating temperature range may be extended by use of heat packs in extremely cold conditions or cold packs in extreme heat conditions, these packs may be placed in the rear pouch of the soft case.

Dust Cap

Always keep the protective dust cap in place when the data port is not in use.

Filters

The GEM-500™ is equipped with two filters:

Water Trap Filter — This filter is external to the GEM-500™ and is located in-line in the sample hose. Unscrewing the two halves of the filter holder gives easy access to the filter. This filter should be routinely changed every one hundred hours of use or when water is sucked through the filter. The filter should also be replaced when the sample pump has difficulty drawing a sample of gas through it and into the unit. When this happens, the GEM-500™ sounds a continuous audible warning and a **Flow Fail** message appears on the screen.

Particulate Filter — This filter is inside the GEM-500™ and is located just inside the Static Pressure/Sampling port. (Figure 1.1, Figure 5.1) This filter is accessed by unscrewing the port (counter-clockwise) using the wrench provided.

Both filter holders are sealed with o-rings. Periodically inspect the o-rings to check their condition. Replace the o-rings if they become nicked, cut, swollen, or otherwise damaged. The GEM-500™ unit is shipped with a spare filter of each type. Only genuine CES-LANDTEC filters should be used and can be purchased through the CES-LANDTEC Sales Department by dialing 1-800-LANDTEC or on our web page at CES-LANDTEC.COM.

Travel and Storage

Travel – The GEM-500™ is a delicate scientific instrument and should be stored in its optional protective hard case when carrying it from site to site. This case affords maximum protection for the unit and offers enough storage space to take along all of the required accessories for the GEM-500™.

Storage – If the unit is to be stored for a long period, the internal batteries should be charged prior to storage. Recharge the unit every two weeks during storage.

When loading the GEM-500™ into its protective hard case, place the unit with the keyboard facing out and the CES-LANDTEC logo (upside down) closest to the handle on the front of the case. This will assure that the unit is stored right-side up when the case is closed and standing with its handle up in the carry position.

Battery Charging

The internal battery pack of the GEM-500™ is designed to be recharged many times, but as with all nickel-cadmium cells, certain rules should be observed or the batteries might not provide their full power or operating time. Please follow these instructions carefully.

WARNING! ONLY CHARGE A GEM-500™ WITH A LANDTEC BATTERY CHARGER (PROVIDED WITH THE UNIT).

1. Let the batteries almost fully discharge before recharging.

2. Do not top off an almost full battery charge because memory patterns can be established and the battery may not provide its full capacity.

   Note: If the GEM-500™ is repeatedly given small "top-off" charges, the battery capacity can be reduced. To restore the battery to full capacity, totally discharge the unit and then charge it for a full 14 hour period.

   When charging the batteries, let them charge at least 12 to14 hours. If using the optional CES-LANDTEC Smart Charger, batteries can be completely recharged in approximately 3 hours.

3. Never let the batteries charge for more than three or four days.

4. Disconnect the charger from the GEM-500™ after the batteries have charged.

   Note: Heat from the battery compartment makes the front of the GEM-500™ (under the GEM-500™ label) warm to the touch while the batteries are charging.

Battery Shut-Off

A circuit within the GEM-500™ continuously monitors the battery voltage. If the battery voltage falls below a predetermined level, the unit automatically shuts itself off in order to prevent memory loss. If the unit shuts itself off, it requires a full charge of 14 hours (approximately 3 hours with the CES-LANDTEC Smart Charger) to restore the battery to its maximum level.

Battery Low Symbol

This Battery Symbol displays in the top right corner of the display screen. It displays as the battery capacity reaches about 10% of full charge. There are only about 30-45 minutes of full pump power left in the GEM-500™ when the symbol is displayed.

Automatic Power-Off

The GEM-500™ has an automatic power-off timer to conserve battery power. If no key is pressed for 15 minutes, the unit automatically switches itself off (no stored readings are lost).

Emergency Battery Power

In emergencies, the GEM-500™ may be operated with 6 "C" sized alkaline batteries. To use alkaline cells, remove the nickel-cadmium battery pack by using a Phillips screwdriver on the back battery compartment of the GEM-500™. Insert the alkaline battery "C" cells in the correct direction.

WARNING! DO NOT USE THE BATTERY CHARGER FOR STANDARD ALKALINE BATTERIES AS THEY MAY EXPLODE.

WARNING! BE SURE TO REPLACE BATTERIES IN CORRECT DIRECTION OR UNIT WILL BE DAMAGED.

Chapter 9 - Troubleshooting

Problem	Corrective Action/Reason
Unit does not turn on or operation is erratic	Battery charge is too low-recharge batteries. Unit is too hot - cool down unit and try again. Contact the factory.
"Flow Fail" is displayed and an audible alarm is heard	The inlet is blocked - remove blockage and retry. The particulate filter or water trap filter needs replacing – see chapter 8 Maintenance.
Readings taken are not what was expected	Unit may be cut out of calibration – calibrate unit with known gas concentration. Water trap filter or particulate filter are clogged – replace filter.
Unit displays***** or >>>>>	These symbols are substituted when the measured reading is out of range of the instruments capabilities in some fields or when a value needs to be entered manually such as temperature.
Oxygen reading is high on all wells	Check that the water trap housing is screwed on tight. Check or replace o-rings on the water trap and instrument inlet. Check the wellhead inset for cracks, replace o-ring on insert. Field calibrate oxygen channel.
Unit will not download readings or an error occurs while downloading.	Verify that the communications software is the right version for the instrument being used. Check that the proper serial port is selected in the software (see chapter 7). Contact the factory.
Methane and Carbon Dioxide readings drift	Perform a field calibration and check well again. Verify cal gas is flowing when regulator is turned on. Verify all connections are tight and filters are not clogged. Contact the factory.
Oxygen readings drift	Perform a field calibration – zero and span (see chapter 3) Contact the factory
Black screen displayed when unit turned on	Charge unit over night and try again. Unit too hot – cool down and try again. Try adjusting contrast level (see chapter 2) Contact factory
Nothing happens when the Gas Pod is installed	Remove and re-seat the Gas Pod. Contact the factory.
Temperature does not update when temperature probe is installed	Check the probe fitting is fully seated. Check the probe plug is screwed together tightly. Contact the factory.

Chapter 10 - Measurement Units & Specifications

Measurement Units

Screen 1

Type	Displayed As	USA (Imperial)	Metric (SI)
Methane	CH_4%	% by volume	% by volume
Carbon Dioxide	CO_2%	% by volume	% by volume
Oxygen	O_2%	% by volume	% by volume
Balance	BAL	% by volume	% by volume

Screen 2

Type	Displayed As	USA (Imperial)	Metric (SI)
Methane	CH_4%	% by volume	% by volume
Lower Explosive Limit	CH_4% LEL	% of 5% CH_4	% of 5%CH_4

Screen 3

Type	Displayed As	USA (Imperial)	Metric (SI)
Static Pressure	SP"	"w.c. (H_2O)	mb (millibar)
Differential Pressure	DP"	"w.c. (H_2O)	mb (millibar)
Temperature	T °F/°C	°F (degrees Fahrenheit)	°C (degrees Celsius)

Screen 4

Type	Displayed As	USA (Imperial)	Metric (SI)
Ref (past) BTU	BTU Ref	per cubic foot	
Ref (past) BTU	KJ Ref		per cubic meter
Ref (past) Gas Flow	scfm Ref	std. cubic feet per min.	
Ref (past) Gas Flow	scfm Ref		std. cubic meters per min.
Adj. (present) BTU	BTU ADJ		per hr.
Adj. (present) BTU	KJ ADJ		per hr.
Adj. (present) Gas Flow	scfm ADJ		std. cubic feet per min.
Adj. (present) Gas Flow	scfm ADJ		std. cubic meters per min.

Operating Temperature

10 to 104° F/-10 to 40° C

Range and Resolution

	Sensor Range Imperial	Resolution Imperial
Methane	0-100%	0.1
Carbon Dioxide	0-50%	0.1
Oxygen	0-25%	0.1
Pressure-Differential	0-10 "w.c.	0.01
Pressure-Static	0-100 "w.c.	0.1

Typical Accuracy

Gas	0 - 5% Volume	5% - 15% Volume	15% - Full Scale
Methane	0.3%	1.0%	3.0%
Carbon Dioxide	0.3%	1.0%	3.0%
Oxygen	0.5%	1.0%	1.0%

Chapter 11 - Field Operations

Landfill Gas Generation

A brief overview of the theory of landfill gas generation and methane recovery follows. Initially, when decomposable refuse is placed into a solid waste landfill, the refuse is entrained with air from the surrounding atmosphere. Through a natural process of bacterial decomposition, the oxygen from the air is consumed and an anaerobic (oxygen free) environment is created within the landfill. This anaerobic environment is one of several conditions necessary for the formation of methane-CH_4.

If oxygen is reintroduced into the landfill, those areas are returned to an aerobic (oxygen present) state and the methane producing bacteria population are destroyed. A period of time must pass before the productive capacity is returned to normal. Since there is some methane of a given quality within the landfill void space, a decline in methane quality is only gradually apparent depending upon the size of the landfill.

Carbon Dioxide is also produced under either an aerobic or anaerobic condition. Under static conditions, the landfill gas will be composed of roughly half methane and half Carbon Dioxide with a little nitrogen.

As air is introduced into the landfill, the oxygen is initially converted to Carbon Dioxide and residual nitrogen remains. Measurement of residual nitrogen is usually a good indicator of the anaerobic state of the landfill; however, it cannot be directly measured. It can, however, be assumed and estimated using a subtraction basis as the balance gas. Hence, the measurement of Carbon Dioxide is an intermediary step. Because Carbon Dioxide levels may fluctuate depending on the changing concentrations of the other constituent gases, Carbon Dioxide levels are not evaluated directly but are considered in light of other data.

In evaluation of residual nitrogen, allowances must be made if there has been any air leakage into the gas collection system or if there has been serious over pull. If enough air is drawn into the landfill, not all oxygen is converted into Carbon Dioxide and the oxygen is apparent in the sample. It is ideal to perform routine analysis of individual wells, as well as an overall well field composite sample, by a gas chromatography. This is not always practical at every landfill.

Under some conditions there may be a small amount of hydrogen in the LFG, (about 1 percent, usually much less). This may affect field monitoring response factors, but otherwise it can be ignored.

Subsurface Fires

If very large quantities of air are introduced into the landfill, either through natural occurrence or overly aggressive operation of the LFG system, a partly unsupported subsurface combustion of the buried refuse may be initiated. Subsurface fire situations are difficult to control or extinguish once started, present health and safety hazards, and can be quite costly. Therefore, prevention by good operation of the collection system and maintenance of the landfill cover, is the best course of action. The presence of Carbon Monoxide, Carbon Dioxide, and Hydrogen Sulfide are indicators of poorly supported combustion within the landfill.

Techniques for Controlling Landfill Gas

There are many techniques for controlling landfill gas extraction. These techniques represent tools which are used together to control landfill gas. The Accu-Flo™ wellhead is designed to work with all of these techniques. Below is a discussion of the individual techniques, how to use them, and their limitations. Reliance on only a few of the techniques discussed can lead to misinterpretation of field data and improper operation of the well field. Later the best use of these techniques to optimize landfill gas control will be discussed.

Controlling by Wellhead Valve Position

Unless the valve handle is calibrated for a given flow rate, this method is unreliable. The position of the valve handle alone does not provide sufficient information about the well to control it. It is useful to note the relative position of the valve, and essential to know which valves are fully open or fully closed.

Controlling by Wellhead Vacuum

This technique relies on the relationship of well pressure/vacuum to flow for a given well. Reliance upon this method, however, can be misleading. This is because the square root relationship between flow and pressure is difficult to affect while performing day-to-day well field adjustments. As decomposition, moisture, and other conditions change, this method shows itself to be inadequate and imprecise.

Controlling by Gas Composition

This method determines methane, nitrogen (balance gas) and other gas composition parameters at wellheads and at recovery facilities using portable field instruments and, sometimes, analytical laboratory equipment. Complete knowledge of gas composition (i.e., major fixed gases: Methane, Carbon Dioxide, Oxygen and Nitrogen) is desirable. It is also necessary to check other gas parameters, such as Carbon Monoxide, to fully evaluate the condition of the well field. Reliance on this information can lead to improper operation of the well field. Indications of excessive extraction often do not show up right away. This method often leads to a cycle of damage to the methane producing bacteria population and then to over-correction. This cycling of the well and producing area of the landfill is not a good practice. It leads to further misinterpretation of the condition of the well field and has a disruptive effect on the operation of the well field. The use of analytical laboratory instrumentation such as a gas chromatograph is a valuable supplementary tool to verify gas composition. This normally requires collection of samples at the wellhead and analysis at some fixed location where the equipment is located. The drawbacks of this method as a primary means of obtaining information for well field adjustment are the time expended, cost, and probably most important, responsiveness to the needs of the well field for timely adjustment. The laboratory equipment required is also very costly. Some analysis is recommended for verification of field readings from time to time. It is recommended a monthly sample of the composite gas be taken at the inlet to the flare or gas recovery facility.

Controlling by Flow Rate

This is a more exacting technique for determining and adjusting gas flow at individual wells. It requires using a fixed or portable flow measurement device at each wellhead to obtain the data needed to calculate volumetric (or mass) flow rates. It is normally convenient to use cubic feet per minute or per day, as a standard unit of measure for volumetric flow. It is important to distinguish between the volumetric quantity of landfill gas and the volumetric quantity of methane extracted from each well and the landfill in total. The two variables are somewhat independent of each other and it is the total quantity of methane extracted we are interested in. It is possible for the total quantity of landfill gas extracted to increase while the total quantity of methane extracted decreases. To monitor this, the quantity of methane extracted (LFG flow x percent methane) or the quantity of BTUs recovered per hour (LFG flow x percent methane x BTUs per cubic foot of methane x 60 minutes per hour) can be calculated. It is conventional to measure BTUs per hour as a unit of time. There are approximately 1012 BTUs of heat per cubic foot of pure methane (like natural gas), although this figure varies a little among reference texts.

Measuring flow is an essential part of monitoring and adjusting a well field. The well should be adjusted until the amount of methane recovered is maximized for the long term. A greater amount of methane or energy can usually be recovered over the short term; however, this ultimately leads to diminishing returns. This is seen in stages as increased CO_2 and gas temperature and later as increased oxygen from well over-pull. In time, the methane will also decline. This is the result of a portion of the landfill, usually at the surface, being driven aerobic. In this portion of the landfill, the methane producing bacteria will have been destroyed (due to the presence of oxygen). With the methane-producing capacity of the landfill reduced, the pore space in the area no longer producing may become filled with landfill gas equilibrating (moving in) from an unaffected producing area. This leaves the impression that more gas can be recovered from this area, and may lead to the operator opening the well or increasing flow.

Well field Monitoring

The frequency of LFG well field monitoring varies depending upon field requirements and conditions. Normal monitoring frequency for a complete field monitoring session with full field readings (suggested normal and abbreviated field readings list follows) will vary from typically once a month to once a week. Well field monitoring should not normally be extended beyond one month. The importance of regular, timely monitoring can not be overemphasized.

Typical Field Readings

- Name of person taking readings
- Date/time of each reading
- Methane (CH_4)
- Oxygen (O_2)
- Carbon Dioxide (CO_2)
- Balance Gas (primarily nitrogen N_2)
- Wellhead gas temperature (flowing)
- Ambient air temperature
- Static pressure (PS) (from GEM-500™ or magnehelic) or other device (anemometer/velometer)
- Velocity head (P or PT) (from GEM-500™ or pitot tube and magnehelic)

- Wellhead gas flow (from GEM-500™, or pitot tube & magnehelic, or anemometer/velometer)
- Wellhead adjustment valve position (initial and adjusted)
- New wellhead vacuum and flow information after adjustment
- Calculation of each well's LFG and methane flow and sum total
- Observations/comments

Additionally, Carbon Monoxide (CO) or Hydrogen Sulfide (H_2S) readings may be taken if problems are suspected. Supplementary monitoring once to several times a week may be performed using an abbreviated form of field readings.

Abbreviated Field Readings

- Name of person taking readings
- Date/time of each reading
- Methane (CH_4)
- Oxygen (O_2)
- Wellhead gas temperature (flowing)
- Ambient air temperature
- Static pressure (PS) (from GEM-500™, GA-90 or magnehelic)
- Velocity head (P or Pt) (from GEM-500™ or pitot tube and magnehelic)
- Wellhead gas flow (from GEM-500™, or pitot tube and magnehelic, or anemometer/velometer)
- Wellhead adjustment valve position (initial and adjusted)
- New wellhead vacuum and flow information after adjustment
- Observations/comments

Line vacuums and gas quality may be taken at key points along the main gas collection header and at subordinate branches. This helps to identify locations of poor performance, excessive pressure drop, or leakage. Perform systematic monitoring of the well field, taking and logging measurements at each wellhead and major branch junction in the collection system.

During monitoring, examine landfill and gas collection system for maintenance issues. Record needed maintenance or unusual conditions. Examples of unusual occurrences or conditions are unusual settlement, signs of subsurface fires, cracks and fissures, liquid ponding, condensate/leachate weeping from side slopes, surface emissions and hot spots, and liquid surging and blockage in the gas collection system. Field readings should be kept in a chronological log and submitted to management on a timely basis.

Well field Adjustment Criteria

There are several criteria used in well field adjustment. The primary criterion is methane quality. Methane quality is an indicator of the healthy anaerobic state of the landfill and thus proper operation of the LFG collection system. However, a decline in the healthy productive state of the landfill is usually not immediately apparent from methane quality. Due to this several criteria must be considered at once.

Following are well field adjustment criteria for consideration.

- Methane quality (ranging from 26 percent upwards)
- The degree to which conditions within the landfill favor methane production. Typical conditions include:
  - pH
  - temperature
  - general overall quality
  - moisture conditions
  - waste stream characteristics
  - placement chronology
  - Insulation characteristics
- Oxygen quality (ranging below 1 percent, preferably less then ½ percent)
- Landfill cover porosity and depth in the proximity of the well
- Landfill construction factors including:
  - type of fill
  - size and shape of refuse mass
  - depth of fill
  - compaction
  - leachate control methods
- Seasonal, climatic, geographical, and recent weather, or other considerations, including seasonally arid or wet conditions, precipitation, drainage, groundwater
- Surrounding topography and geologic conditions
- Proximity of the well to side slopes (within 150 to 200 feet and less may require conservative operation of the well)
- Nitrogen (typically 8 to 12 percent and less)
- Temperature (between ambient and about 130 °F)
- LFG and methane flow from the wellhead
- Design of the gas collection system
- Landfill perimeter gas migration and surface emission control, or energy recovery objectives
- Diurnal fluctuation (day to night) of atmospheric pressure

Establishing Target Flows

For a given individual well, a target flow is established which will likely support maintenance of methane and oxygen quality objectives while maximizing the recovery of landfill gas. Typically, small adjustments are made in flow to achieve and maintain quality objectives. The well must not be allowed to over pull. High well temperatures, (130° to 140° F and greater), are an indication of aerobic activity and, thus, well over-pull. These effects may not be immediately apparent.

Well adjustment should be made in as small an increment as possible, preferably an increment of ten percent of the existing flow or less. There may be obvious conditions when this is not appropriate, such as when first opening up a well or when serious over-pull is recognized. Every effort should be made to make adjustments and operations as smooth as possible. Dramatic adjustments, or operating while switching between a high flow mode and a well shutoff mode, should be avoided.

Well field Optimization

Every effort should be made to continuously locate and correct or eliminate conditions (e.g., gas condensate, surging and blockage, settlement, etc.) which inhibit efficient operation of the gas collection system. This allows well monitoring and adjustment to be significantly more effective.

Migration Control—Dealing with Poor Methane Quality

If methane and oxygen quality objectives cannot be maintained at a given well, such as a perimeter migration control well, then an attempt should be made to stabilize the well as closely as is practical, avoiding significant or rapid down-trending of methane or up-trending of oxygen.

It is not uncommon for perimeter migration control wells to be operated at less than 40 percent methane or greater than one-percent oxygen. It should be recognized that these wells are likely in a zone where some aerobic action is being induced, and that there is some risk of introducing or enhancing the spread of a subsurface fire. Sometimes a judicious compromise is necessary to achieve critical migration control objectives or because existing conditions do not allow otherwise. Such situations should be monitored closely.

Well field Adjustment—Purpose and Objectives

The objective of well field adjustment is to achieve a steady state of operation of the gas collection system by stabilizing the rate and quality of extracted LFG in order to achieve one or several goals. Typical reasons for recovery of LFG and close control of the well field are:

- Achieve and maintain effective subsurface gas migration control.
- Achieve and maintain effective surface gas emissions control.
- Assist with proper operation of control and recovery equipment.
- Avoidance of well over pull and maintenance of a healthy anaerobic state within the landfill.
- Optimize LFG recovery for energy recovery purposes.
- Control nuisance landfill gases odors.
- Prevent or control subsurface LFG fires.
- Protect structures on and near the landfill.
- Meet environmental and regulatory compliance requirements.

Well field adjustment is partly subjective and can be confusing because it involves judgment calls based on simultaneous evaluation of several variables, as well a general knowledge of site specific field conditions and historical trends. Well field evaluation and adjustment consist of a collection of techniques, which may be used, in combination, to achieve a steady state of well field operation.

CES-LANDTEC Technical Tips

Landfill Control Technologies regularly produces technical landfill related information and educational material. Please call CES-LANDTEC at (800) LANDTEC, to receive the current series of these Technical Tips.

AMBIENT AIR MONITORING LOG SHEET

Project Name, Number: ______________________________ Date: ____________________

Sampler(s): ____

Weather Conditions: ____

Ambient Temp.: ___________________ Barometric Pressure: ___________________

Time	Sample Location Coordinates	Methane, percent	Oxygen, percent	Carbon Monoxide, percent	Hydrogen Sulfide

Comments: ____

____

____

LANDFILL GAS MONITORING LOG SHEET

Project Name, Number: ______________________________

Sample Location: ____________ Sampler(s): ____________

Sample Number: ____________ Date: ____________

Weather Conditions: ______________________________

Ambient Temp.: ____________ Barometric Pressure: ____________

Time	Methane, percent	Oxygen, percent	Carbon Dioxide, percent	Pressure	Lower Explosive Limit (LEL), percent

APPENDIX D

SUPPORTING DOCUMENTS

- **IEPA Approval for LF-01 Groundwater Monitoring Network**
- **Technical Memorandums:**
  - **LTM Well Installation at LF-01 Technical Memorandum**
  - **Proposed LTM Well Locations at LF-01 Technical Memorandum**
- **Scott Air Force Base Basewide Groundwater Dataset Summary**

This page was intentionally left blank.

APPENDIX D

SUPPORTING DOCUMENTS

- **IEPA Approval for LF-01 Groundwater Monitoring Network**
- **Technical Memorandums:**
  - **LTM Well Installation at LF-01 Technical Memorandum**
  - **Proposed LTM Well Locations at LF-01 Technical Memorandum**
- **Scott Air Force Base Basewide Groundwater Dataset Summary**

This page was intentionally left blank.

IEPA APPROVAL FOR LF-01 GROUNDWATER MONITORING NETWORK

This page was intentionally left blank.

ILLINOIS ENVIRONMENTAL PROTECTION AGENCY

1021 NORTH GRAND AVENUE EAST, P.O. BOX 19276, SPRINGFIELD, ILLINOIS 62794-9276 • (217) 782-2829

PAT QUINN, GOVERNOR LISA BONNETT, DIRECTOR

217.785.7728

April 8, 2013

Ms. Richelle Neff Collingham
Environmental Restoration Program Manager
375th CES/CEAN
701 Hangar Road, Bldg. 531
Scott AFB, IL 62225-5035

Re: LTM Well Installation at LF-01
Technical Memorandum

1630105014 – St. Clair Co.
Scott Air Force Base
Superfund/Technical Reports

Dear Ms. Collingham:

The Illinois Environmental Protection Agency (Illinois EPA) is in receipt of an e-mail dated April 8, 2013 containing the *LTM Well Installation at LF-01 Technical Memorandum.* The latest Technical Memorandum selects modeling Scenario #6 from the *Proposed Long Term Monitoring (LTM) Well Locations at LF-01 Technical Memorandum* as the long-term groundwater monitoring well configuration for the base landfill, Landfill LF-01. Modeling Scenario #6 consists of 30 wells (5 existing well pairs and 10 new well pairs). At each well pair location, a shallow well will monitor Zone 2 groundwater (screened between 20 and 35 feet bgs) and a deeper well will monitor Zone 4 groundwater (screened between 50 and 60 feet bgs). It is anticipated that the new well pairs will be installed in two phases. The revised *Proposed LTM Well Locations Technical Memorandum* is included in this submission and will be appended to the Operations and Maintenance (O&M) Manual for LF-01. Illinois EPA approves the *LTM Well Installation at LF-01 Technical Memorandum.*

Should you have any question or require further assistance concerning this letter, do not hesitate to contact me at the number above or by e-mail at Paul.Lake@illinois.gov.

Sincerely,

Paul T. Lake, Remedial Project Manager
Federal Site Remediation Section
Bureau of Land

PTL:cah:rac:h:/site files/irp/Scott AFB/LF-01/LTM Well Installation Tech Memo_apr_040813.docx

Attachments:
1. Table 1, Model Scenario #6 LTM Well Locations
2. Figure 1, LF-01 LTM Well Installation

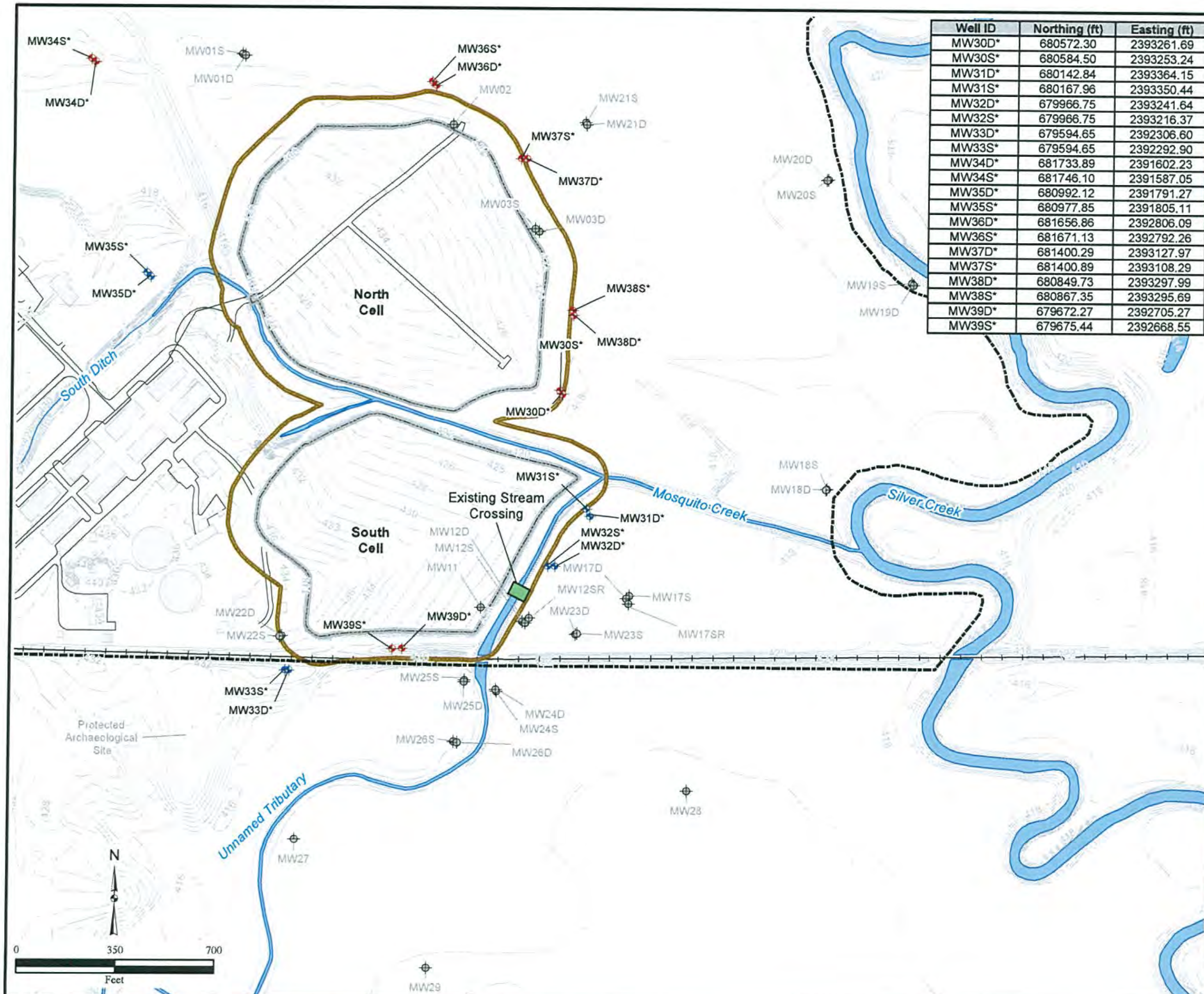
cc: Bruce Little, HGL
Caleb Moore, BAS
Brian Siegfried, AFCEC

Table 1
Model Scenario #6 Long Term Monitoring Well Locations
Landfill LF-01
Scott AFB, IL

Original Well I.D.	Revised Well I.D.
MW-01D	NC
MW-01S	NC
MW-03D	NC
MW-03S	NC
MW-12D	NC
MW-12SR	NC
MW-21D	NC
MW-21S	NC
MW-25D	NC
MW-25S	NC
MW-30D*	NC
MW-30S*	NC
MW-31D*	NC
MW-31S*	NC
MW-33D*	NC
MW-33S*	NC
MW-34D*	NC
MW-34S*	NC
MW-35D*	NC
MW-35S*	NC
MW-36D*	NC
MW-36S*	NC
IEPA-MW1*	MW-37D*
	MW-37S*
IEPA-MW2*	MW-38D*
	MW-38S*
IEPA-MW4*	MW-32D*
	MW-32S*
IEPA-MW5*	MW-39D*
	MW-39S*

Note:
1. * = wells are proposed but not yet existing.
2. NC = No Change

Figure 1
LF-01 Long Term Monitoring Well Installation



Well ID	Northing (ft)	Easting (ft)
MW30D*	680572.30	2393261.69
MW30S*	680584.50	2393253.24
MW31D*	680142.84	2393364.15
MW31S*	680167.96	2393350.44
MW32D*	679966.75	2393241.64
MW32S*	679966.75	2393216.37
MW33D*	679594.65	2392306.60
MW33S*	679594.65	2392292.90
MW34D*	681733.89	2391602.23
MW34S*	681746.10	2391587.05
MW35D*	680992.12	2391791.27
MW35S*	680977.85	2391805.11
MW36D*	681656.86	2392806.09
MW36S*	681671.13	2392792.26
MW37D*	681400.29	2393127.97
MW37S*	681400.89	2393108.29
MW38D*	680849.73	2393297.99
MW38S*	680867.35	2393295.69
MW39D*	679672.27	2392705.27
MW39S*	679675.44	2392668.55

Legend

- Existing Monitoring Well
- Proposed Phase 1 Monitoring Well Pair Location
- Proposed Phase 2 Monitoring Well Pair Location
- Topographic Contour (ft amsl)
- Rail Line
- Road
- Scott AFB Boundary
- Landfill Cover Boundary
- Attenuation Zone
- Existing Structure
- Surface Water

Notes:
1. The location of each well may vary based on field observations made by the field geologist.
2. Well pairs consist of a shallow well screened in Zone 2 groundwater and a deep well screened in Zone 4 groundwater.

*=proposed long term monitoring well
ft amsl=feet above mean sea level
D=deep well
IAC=Illinois Administrative Code
S=shallow well
SR=shallow replacement well

\\Gst-srv-01\HGLGIS\Scott_AFB\LF-01\Well_Installation_Techmemo\(1)LTM_Well_Installation.mxd
2/11/2013 ST, QC: 9/26/2012 PD
Source: HGL

HGL HydroGeoLogic, Inc.

LTM WELL INSTALLATION AT LF-01 TECHNICAL MEMORANDUM
APRIL 2013

This page was intentionally left blank.

LTM WELL INSTALLATION AT LF-01 TECHNICAL MEMORANDUM
SCOTT AIR FORCE BASE, ILLINOIS

TO:	Paul Lake, Illinois Environmental Protection Agency
FROM:	Robert Bird, HydroGeoLogic, Inc. (HGL) Task Manager
CC:	Doug Simpleman, Project Manager, U.S. Army Corps of Engineers-Omaha District Richelle Collingham, Program Manager, Scott Air Force Base Lynn Kessler, HGL, Project Manager
DATE:	April 5, 2013
SUBJECT:	LTM Well Installation at LF-01 Technical Memorandum, Scott Air Force Base, Illinois
CONTRACT NO:	W91238-06-D-0021-DK02
CLIN NO:	0001

HydroGeoLogic, Inc. (HGL) has reviewed the Illinois Environmental Protection Agency's (IEPA) letter dated January 28, 2013, which was prepared in response to the email submitted by HGL on November 20, 2012, regarding the *Proposed Long Term Monitoring (LTM) Well Locations at LF-01 Technical Memorandum*. In the response letter, the IEPA indicated that both original model Run #1 and model Scenario #6 meet the IEPA criterion of a 99 percent monitoring efficiency and concluded either well network is acceptable for LTM at LF-01. Based on this conclusion, HGL will implement the LTM network associated with Scenario #6, which consists of 30 monitoring wells. Table 1 summarizes the Scenario #6 well network. Please note that the well identifications for well pairs IEPA-MW01, IEPA-MW02, IEPA-MW04, and IEPA-MW05 referenced in the *Proposed LTM Well Locations at LF-01 Technical Memorandum* have been revised to retain numerical consistency with the existing wells at the site. These revisions are also summarized in Table 1.

As requested by the IEPA, the *Proposed LTM Well Locations at LF-01 Technical Memorandum* will be revised to select Scenario #6 and will incorporate a reference to the IEPA's efficiency goal found in Appendix C to its guidance document LPC-PA2 (IEPA, 2010). In addition, the revised *Proposed LTM Well Locations at LF-01 Technical Memorandum* is included as Attachment A of this technical memorandum and also will be appended to the Operation and Maintenance (O&M) Manual for LF-01.

MONITORING WELL INSTALLATION

As discussed above, model Scenario #6 consists of 30 wells. A total of 10 wells (5 well pairs) currently exist at the site and 20 wells (10 well pairs) are proposed for installation. The following wells are proposed for installation:

- MW30S/D
- MW31S/D
- MW32S/D
- MW33S/D
- MW34S/D
- MW35S/D
- MW36S/D
- MW37S/D
- MW38S/D
- MW39S/D

Figure 1 illustrates the location of the 20 proposed monitoring wells. HGL plans to proceed immediately with installing the proposed monitoring wells in order to initiate postclosure LTM at the landfill.

As depicted on Figure 1, the majority of these locations are located within or near wetland areas that frequently exhibit saturated conditions. Based on HGL's experience at LF-01, these conditions present logistical challenges during well installation activities (for example, equipment accessibility).

HGL conducted a site walk on February 1, 2013, to assess the current conditions at the landfill. A large rain event occurred the week before the site walk which heavily saturated the wetlands that surround the North and South Cells. Approximately 2 to 3 feet of standing water were observed at the proposed well locations on the north and east sides of the North Cell and approximately 1 to 2 feet of standing water were observed at the proposed well locations on the east side of the South Cell. As a result, these areas are inaccessible with drilling equipment.

Based on HGL's experience during construction of the landfill cap and other investigation activities at the site, the wetland areas to the north and east of the North Cell tend to remain saturated for longer periods of time as compared to the areas east of the South Cell. Based on these observations, HGL anticipates being able to access the proposed locations around the South Cell sooner than the North Cell locations. As a result, HGL anticipates completing the well installation activities in two phases. The specific wells installed during each phase will depend on the wetland conditions and accessibility. HGL will install as many wells as possible during each phase. However, HGL anticipates that Phase 1 will include the following eight monitoring wells (four well pairs):

- LF01-MW31S/D
- LF01-MW32S/D
- LF01-MW33S/D
- LF01-MW35S/D

Accordingly, HGL anticipates Phase 2 will include the remaining 12 monitoring wells (6 well pairs):

- LF01-MW30S/D
- LF01-MW34S/D
- LF01-MW36S/D
- LF01-MW37S/D
- LF01-MW38S/D
- LF01-MW39S/D

The wells included in each phase are depicted on Figure 1.

The proposed monitoring wells will be installed, developed, and surveyed in accordance with the Basewide Field Sampling Plan (HGL, 2009) and all federal, state, and local requirements. Each well will be installed using hollow stem auger drilling methods. Each well pair will consist of a shallow well screened in Zone 2 and a deep well screened in Zone 4. At each well pair location, continuous sampling (from ground surface to the total depth of the borehole) will be conducted at the Zone 4 (deep well) location using split spoon or 5-foot continuous sampling methodology. Soil cores will be provided to the HGL field geologist for field logging. The Zone 2 borehole will be blind augered and will be set based on the lithology observed in the Zone 4 well. The drill log of the Zone 4 well that was used to set the Zone 2 well will be referenced on the Zone 2 drill log. Monitoring well construction details are summarized below:

- The total depths of the Zone 2 wells will range from 30 to 35 feet below ground surface (bgs) and the total depths of the Zone 4 wells will range from 60 to 70 feet bgs. However, totals depths may vary based on observations made by the field geologist.
- Each monitoring well will be 2 inches in diameter and will be screened to yield representative groundwater samples from the water-bearing zone being monitored (that is, Zone 2 or Zone 4). Each screen (0.010-inch slot) and blank riser pipe will be composed of schedule 40 polyvinyl chloride (PVC). Screen lengths will be approximately 10 feet and will be sealed with a schedule 40 PVC end cap. The blank riser will extend from the top of the screen to approximately 3.5 feet above the ground surface.
- The sand pack will consist of No. 20-40 silica sand and will extend approximately 2 feet above the top of the screen.
- A buffer pack consisting of a 1-foot-thick seal of #100 fine silica sand will be placed above the sand pack.
- A seal consisting of 1/4-inch coated bentonite pellets will be placed on top of the buffer pack and will extend approximately 2 feet above the buffer pack.
- High-solids (30 percent) bentonite grout will then be installed from the top of the bentonite seal to ground surface using a tremie method to emplace the grout from the bottom up.
- To alleviate the concern of standing water surrounding the wells, well protectors that are 7 feet in length will be used. The bottom 3 feet of the protector will be set below ground surface into the grout seal with the remaining 4 feet of the protector above the ground surface. Each well also will be sealed with a locking, compression plug and surrounded by a 2-foot by 2-foot concrete pad. Additionally, three bollards will be set around the concrete pad at each individual well location.

Each monitoring well will be developed after construction activities are complete. In addition, the newly installed monitoring wells will be surveyed by a State of Illinois certified land surveyor.

In addition to the above mentioned well installation activities, HGL will refurbish three existing monitoring wells (MW-02, MW-03S, and MW-03D) to match the new wells being installed at LF-01.

SCHEDULE

Activities associated with each phase will begin as soon as HGL and the drilling subcontractor determine conditions are suitable. HGL anticipates mobilizing for Phase 1 in April 2013. The mobilization for Phase 2 is projected for late spring or the summer of 2013. The drilling subcontractor estimated that the well installation and development activities can be completed in approximately 17 days.

REFERENCES

HydroGeoLogic, Inc. (HGL), 2009. Final Basewide Field Sampling Plan, Scott Air Force Base, Illinois. October.

Illinois Environmental Protection Agency (IEPA), 2010. LPC-PA2: Landfill Development Permit.

TABLES

Table 1 - Model Scenario #6 Long Term Monitoring Well Locations

FIGURES

Figure 1 - LF-01 Long Term Monitoring Well Installation

ATTACHMENT

Attachment A - Proposed LTM Well Locations at LF-01 Technical Memorandum

TABLE

This page was intentionally left blank.

Table 1
Model Scenario #6 Long Term Monitoring Well Locations
Landfill LF-01
Scott AFB, IL

Original Well I.D.	Revised Well I.D.
MW-01D	NC
MW-01S	NC
MW-03D	NC
MW-03S	NC
MW-12D	NC
MW-12SR	NC
MW-21D	NC
MW-21S	NC
MW-25D	NC
MW-25S	NC
MW-30D*	NC
MW-30S*	NC
MW-31D*	NC
MW-31S*	NC
MW-33D*	NC
MW-33S*	NC
MW-34D*	NC
MW-34S*	NC
MW-35D*	NC
MW-35S*	NC
MW-36D*	NC
MW-36S*	NC
IEPA-MW1*	MW-37D*
	MW-37S*
IEPA-MW2*	MW-38D*
	MW-38S*
IEPA-MW4*	MW-32D*
	MW-32S*
IEPA-MW5*	MW-39D*
	MW-39S*

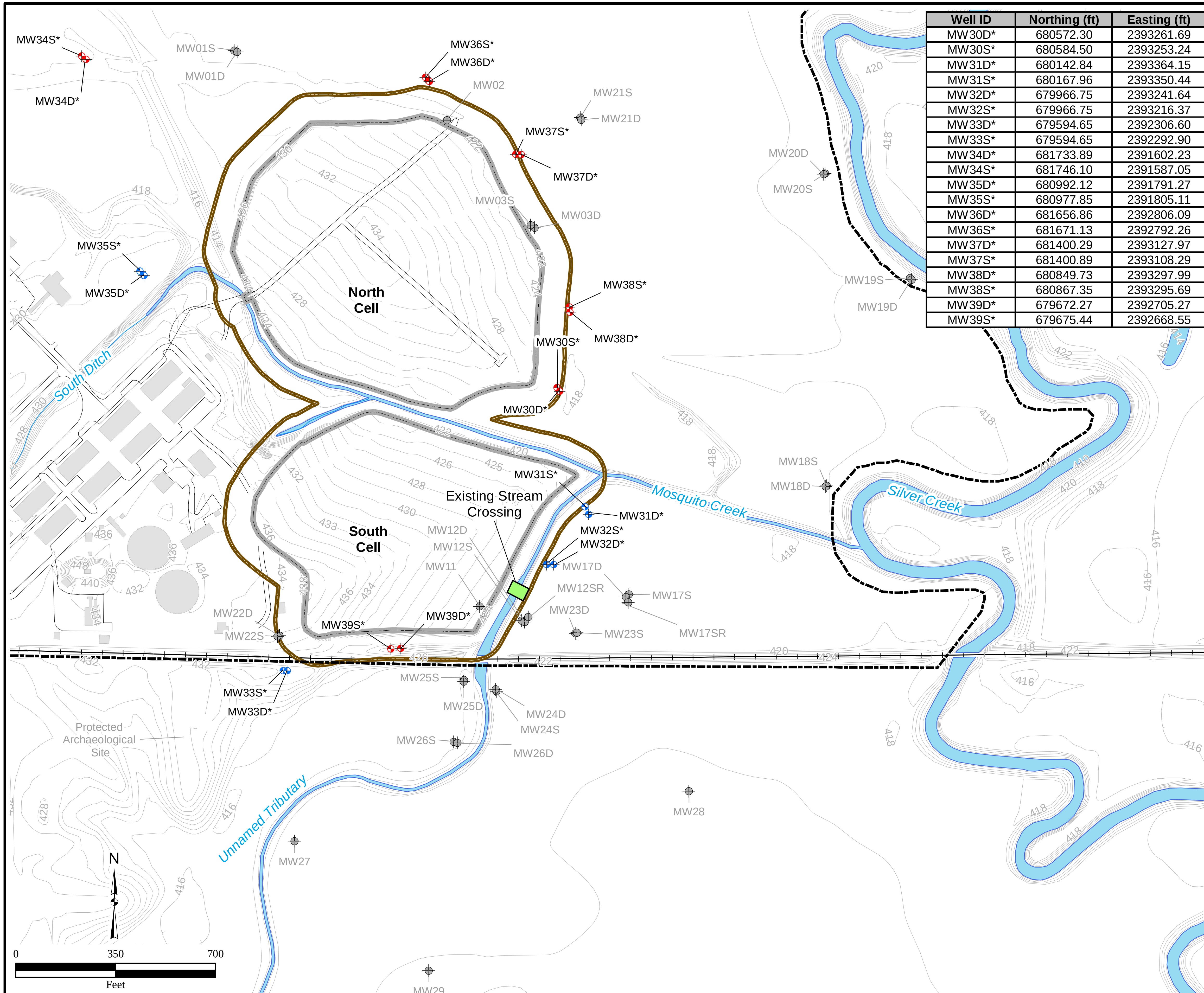
Note:
1. * = wells are proposed but not yet existing.
2. NC = No Change

This page was intentionally left blank.

FIGURE

This page was intentionally left blank.

Figure 1
LF-01 Long Term Monitoring Well Installation



Well ID	Northing (ft)	Easting (ft)
MW30D*	680572.30	2393261.69
MW30S*	680584.50	2393253.24
MW31D*	680142.84	2393364.15
MW31S*	680167.96	2393350.44
MW32D*	679966.75	2393241.64
MW32S*	679966.75	2393216.37
MW33D*	679594.65	2392306.60
MW33S*	679594.65	2392292.90
MW34D*	681733.89	2391602.23
MW34S*	681746.10	2391587.05
MW35D*	680992.12	2391791.27
MW35S*	680977.85	2391805.11
MW36D*	681656.86	2392806.09
MW36S*	681671.13	2392792.26
MW37D*	681400.29	2393127.97
MW37S*	681400.89	2393108.29
MW38D*	680849.73	2393297.99
MW38S*	680867.35	2393295.69
MW39D*	679672.27	2392705.27
MW39S*	679675.44	2392668.55

Legend

- Existing Monitoring Well
- Proposed Phase 1 Monitoring Well Pair Location
- Proposed Phase 2 Monitoring Well Pair Location
- 422 Topographic Contour (ft amsl)
- Rail Line
- Road
- Scott AFB Boundary
- Landfill Cover Boundary
- Attenuation Zone
- Existing Structure
- Surface Water

Notes:
1. The location of each well may vary based on field observations made by the field geologist.
2. Well pairs consist of a shallow well screened in Zone 2 groundwater and a deep well screened in Zone 4 groundwater.

*=proposed long term monitoring well
ft amsl=feet above mean sea level
D=deep well
IAC=Illinois Administrative Code
S=shallow well
SR=shallow replacement well

\\Gst-srv-01\HGLGIS\Scott_AFB\LF-01\Well_Installation_Techmemo\(1)LTM_Well_Installation.mxd
2/11/2013 ST, QC: 9/26/2012 PD
Source: HGL

HGL HydroGeoLogic, Inc

This page was intentionally left blank.

ATTACHMENT A

Proposed LTM Well Locations at LF-01 Technical Memorandum

This page was intentionally left blank.

ILLINOIS ENVIRONMENTAL PROTECTION AGENCY

1021 NORTH GRAND AVENUE EAST, P.O. BOX 19276, SPRINGFIELD, ILLINOIS 62794-9276 • (217) 782-2829
PAT QUINN, GOVERNOR JOHN J. KIM, DIRECTOR

217.785.7728

January 28, 2013

Ms. Richelle Neff Collingham
Environmental Restoration Program Manager
375th CES/CEAN
701 Hangar Road, Bldg. 531
Scott AFB, IL 62225-5035

Re: Proposed LTM Well Location
Tech Memo, Landfill LF-01

1630105014 – St. Clair Co.
Scott Air Force Base
Superfund/Technical Reports

Dear Ms. Collingham:

The Illinois Environmental Protection Agency (Illinois EPA) is in receipt of an e-mail dated November 20, 2012 regarding the *Proposed LTM Well Location at LF-01 Technical Memorandum.* The e-mail contains two additional well network modeling scenarios (#6 and #7) prepared by HydroGeoLogic, Inc. (HGL) on behalf of Scott Air Force Base under the FY08 Performance Based Contract with the U.S. Army Corps of Engineers.

On October 29, 2012 Illinois EPA transmitted its evaluation of the original LTM well location Tech Memo; identifying LTM monitoring well configuration Run #1, identified in Table 1 and Figure 1 of the Tech Memo as the preferred well network configuration for LF-01. Run #1 consists of 28 wells with a modeled monitoring efficiency of 99.5% (i.e., estimated to fail detecting releases from 3 out of 653 source grids). In its current counter-proposal, HGL identifies Scenario #7 as the well configuration it would like to implement on behalf of the Air Force. Scenario #7 excludes wells IEPA-MW-03 and IEPA-MW-05 and has an estimated monitoring efficiency of 98.5% (failed to detect 9 of 653 source elements).

As communicated in our October correspondence, Illinois EPA guidance calls for 99% monitoring efficiency for well networks designed for new landfills. Both Run#1 in the original Tech Memo and more recently developed Scenario #6 (excludes IEPA-MW-03, fails to detect 3 of 653 source elements and has an estimated efficiency of 99.5%) meet Illinois EPA's efficiency criterion. Based upon HGL's modeling results, Illinois EPA finds the use of either the original Run #1 or Scenario #6 acceptable for the LF-01 LTM well network.

Illinois EPA requests that the Tech Memo be revised to select either Run #1 or Scenario #6 and that it incorporate reference to Illinois EPA's 99% efficiency goal found in its guidance document LPC-PA2. The Tech Memo then should be appended to the Operation and Maintenance (O&M) Plan for the LF-01.

Should you have any question or require further assistance concerning this letter, do not hesitate to contact me at the number above or by e-mail at Paul.Lake@illinois.gov.

Sincerely,

Paul T. Lake, Remedial Project Manager
Federal Site Remediation Section
Bureau of Land

PTL:cah:rac:h:/site files/irp/Scott AFB/LF-01/LF01 LTM Well Network Scenarios 6 +7_012813.docx

cc: Bruce Little, HGL
Caleb Moore, BAS
Brian Siegfried, AFCEE

PROPOSED LONG TERM MONITORING WELL LOCATIONS AT LF-01
SCOTT AIR FORCE BASE, ILLINOIS

TO:	Paul Lake, Illinois Environmental Protection Agency
FROM:	Robert Bird, HydroGeoLogic, Inc. (HGL) Task Manager
CC:	Doug Simpleman, Project Manager, U.S. Army Corps of Engineers-Omaha District Richelle Collingham, Program Manager, Scott Air Force Base Lynn Kessler, HGL, Project Manager
DATE:	February 20, 2013
SUBJECT:	Proposed Long Term Monitoring Well Locations at LF-01, Scott Air Force Base, Illinois
CONTRACT NO:	W91238-06-D-0021-DK02
CLIN NO:	0001

As part of the Landfill 01 (LF-01) landfill construction activities, a monitoring well network will be installed to comply with the postclosure care requirements specified in 35 Illinois Administrative Code (IAC) 811. At the request of the Illinois Environmental Protection Agency (IEPA), the Monitoring Analysis Package (MAP) Version 1.1 modeling program was used to facilitate the design of the monitoring well network. MAP Version 1.1 includes three modeling tools: the Plume Generation Model (PLUME), the Monitoring Efficiency Model (MEMO) and the Contamination Probability Model (COPRO).

PLUME is the plume generation model that computes the analytical solution of a two dimensional transport problem from a contaminant source. The analytical solution provided by Domenico and Robbins in 1985 and modified by Domenico in 1987, is an approximate analytical solution for advection-dispersive transport that has been used in several contaminant transport packages including BIOSCREEN, BIOCHLOR, FOOTPRINT and REMChlor.

MEMO is the monitoring well optimization model. It requires user inputs for site geometry, hydrogeology, and initial well locations for determining the efficiency of the monitoring well network. The efficiency, as the principle output of MEMO, is defined by the ratio of the area from which a release would be detected to the total potential source area. The higher the efficiency, the more effective the monitoring network. When using MEMO, the user may choose an acceptable level of efficiency, such as 90 percent (%), and manipulate well locations to achieve such efficiency.

MEMO solution can be achieved by two methods: (1) the “Buffer Zone with Advection Time Limit” method (default choice), and (2) the “Advection Time Only” method. In the first method, contaminants originate randomly from any of the discretized source zone elements and travel towards the buffer zone boundary. If the contaminant has reached any of the monitoring wells at the time the contaminant reaches the buffer zone boundary (defined as the “critical time"), then the detection is successful; otherwise, the detection failed. The efficiency is calculated as the ratio of total successful detections to the total number of releases (the total number of the source zone elements). In the second method, the maximum detection time is specified rather than using the model computed "critical time", and the efficiency is calculated using the method described in method (1).

COPRO is the contamination probability model that maps the probability of a downgradient area being contaminated, given a single random source release from the source area. This model is not relevant to the current task and will not be discussed further.

MODEL CONCEPTUALIZATION AND PARAMETER SETTINGS

Source Zone

Based on the site geometry and the hydraulic gradient, the landfill is divided into two source zones: Source Zone 1 that represents the North Cell and Source Zone 2 which represents the South Cell. Source Zone boundaries are delineated along the landfill boundary and the two source zones share the same boundary along Mosquito Creek.

Line of Compliance and Buffer Zone Boundary

The line of compliance is an optional input which provides a line along which wells can be easily moved and is irrelevant to the model result. The line of compliance was set at 100 feet from the landfill edge. Solution method (1) "Buffer Zone with Advection Time Limit" requires defining a buffer zone boundary, which is the limit to which a plume may extend before it should be detected by a monitoring well. This boundary is important to the model solution. The further the buffer zone boundary is from the source area, the bigger the spread of the plume when the plume reaches the boundary, and the more likely the plume will be detected by the monitoring network. Therefore, choosing a buffer zone closer to the landfill boundary is a more conservative approach. For this problem, the buffer zone boundary is set to be 200 feet from the landfill edge.

Source Zone Grid Size and Buffer Zone Array Space

MEMO requires the source zone to be divided into small elements/grids using user specified grid spacing. The finer the discretization the more accurate is the solution, but this increases computational time. For this problem, a grid spacing of 50 feet was chosen and the two source zones were divided into a total of 653 grid blocks.

The array spacing determines how fine the buffer zone boundary will be discretized. Similar to the source zone, decreasing the value of array spacing improves the accuracy, but increases computational time. The same order of magnitude is recommended for both source zone grid spacing and buffer zone array spacing. Therefore, a buffer zone array spacing of 25 feet was chosen for the model.

Hydraulic Gradient Zones

For the model, two hydraulic gradient zones were specified for two source zones. The North Cell has a hydraulic gradient of 330 degree counter-clockwise from the x-axis (Northwest to Southeast). The South Cell has a hydraulic gradient of 340 degree counter-clockwise from the x-axis. The hydraulic gradient value is 0.005 feet per foot (ft/ft) (HGL, 2012), which was used to compute the linear (contaminant) velocity below.

Linear (Contaminant) Velocity

Hydraulic conductivity (K) tests performed in the proximity of the site provide K values in the order of 5e-4 centimeters per second (cm/s) (1.4 feet per day (feet/day) (MWH, 2003a). A groundwater model conducted in the same region produced a calibrated K of 20 feet/day (AMC White paper). For a similar type of soil in the same region, slug tests provided K values ranging from 1.05 to 5.81 feet/day (MWH, 2003b). For the model, as an approximation, HGL used a K of 8 feet/day. Given the hydraulic gradient of 0.005, an assumed porosity of 0.4 and a retardation factor of 1 (no retardation), the linear (contaminant) velocity was calculated to be 8*0.005/(0.4*1)=0.1 feet/day.

Dispersivities

Longitudinal dispersivity (αL) and transverse dispersivity (αT) determine the lateral and transverse spreading of the plume. These are critical parameters for computing the efficiency when using MEMO. Dispersivities are scale-dependant and cannot be achieved by lab test. Therefore, there are three different approaches for estimating longitudinal dispersivity (αL) (West, et al., 2007):

- αL = 10% of plume length
- αL = 10% of L, where L is the longitudinal distance to the reference point (i.e., the x distance at any observation point) according to Pickens and Grisak (1981)
- αL = 0.83(log10 L)2.414 according to Xu and Eckstein (1995)

Any one of these methods is acceptable. For this model, HGL used the first method. An approximate plume length of 1,500 feet was chosen and the αL was calculated to be 150 feet.

Data presented by Gelhar et al (1985) and Isherwood (1981) suggested a ratio of longitudinal to transverse dispersivity of 10. A larger ratio produces a thinner plume and is a more conservative method. Therefore, a ratio of 10 was chosen, and the transverse dispersitivity was computed to be 15 feet for the model.

Diffusion Coefficient

Molecular diffusion is usually much smaller compared to dispersion and convection except in a very impermeable aquifer. The aquifer at LF-01 is not impermeable; therefore, the molecular diffusion coefficient was assumed to be zero.

First Order Decay Coefficient

The Decay coefficient was set to zero in the model (default setting).

Source Width

MAP computes the 2D solution for a line source. The length of the line source is considered the width of the source. This width depends on the type of leak and on the amount of lateral spreading in the vadose zone prior to reaching the water table. For the model, the contaminant is leaked from a landfill; therefore, the default value of 20 feet was considered appropriate for the source width.

Dilution Contour Level (Cd/C0)

The dilution contour level (Cd/C0) is used to define the contaminant plume. Cd is the detection limit of the indicator compound. C0 is the concentration of the indicator compound at the source. The smaller the Cd/C0 value is, the larger the spreading of the plume, which results in a higher efficiency for the monitoring network. Therefore, for the same given Cd, a smaller C0 value provides a more conservative estimate. The choice of the dilution contour is subject to user judgment and is very critical to the model solution.

For the model, vinyl-chloride was selected as the representative indicator compound. A detection limit of 0.5 parts per billion (ppb) was used for Cd. Measured concentrations of 109 and 285 ppb have been detected in the source zone. The conservative choice of 109 ppb was used and the Cd/C0 was calculated to be 0.5/109 = 0.0046.

RESULTS AND CONCLUSION

Sensitivity Analysis

To better understand the model parameters and settings, a sensitivity analysis was conducted on several model parameters. The results are summarized below:

- A sensitivity analysis was conducted for dispersitivity. The model default αL of 70 feet and αT of 10 feet was used to compare with HGL's choice of 150 feet and 15 feet. The difference in efficiencies between the default dispersivities and HGL's input values was below 1%.

- A sensitivity analysis was conducted on buffer zone grid spacing. Spacings of 100, 50, 25, and 10 feet were tested. The model runs using the first three spacings produce identical solutions. The model run using a spacing of 10 failed for unknown reasons. A sensitivity analysis was also conducted on source zone grid spacing. Spacings of 50 and 25 feet were tested. Both spacings produced identical efficiencies, but the run with a spacing 25 feet took much longer to process.
- A sensitivity analysis was conducted on contaminant velocity. Velocities of 1, 0.1, and 0.01 foot/day were tested. As expected, the solutions are irrelevant to the contaminant velocity when plumes have enough time to reach the buffer boundary (as in the case of 1 and 0.1 foot/day). If the velocity is set at 0.01 foot/day, many plumes cannot reach the buffer boundary before the maximum convection time (36,500 days) expires, which are classified as non-detected plumes by MEMO. As a result, the detection efficiency is reduced, but these are not true non-detection failures of the monitoring network.

Model Solution

Multiple MEMO model runs were conducted using the well locations provided in Table 1 and illustrated in Figure 1. Results of the runs are summarized in Table 2 and described below. Figures 2 through 8 illustrate model results in finer detail. In the figures, the brown line, green line, and red line indicate the source zone boundary, the line of compliance, and the buffer zone boundary, respectively. The red diamonds and numbers together indicate the monitoring well location and well ID. The red blocks within the source boundary represent the non-detect source elements where the contaminants originate and travel to reach the buffer zone boundary, but are not detected by the monitoring well network.

- Run #1 is conducted with all the wells listed in Table 1. The monitoring efficiency is 99.5% (failed to detect 3 of 653 source elements). Figure 2 shows most of the source zone elements are detected except a few element blocks, two of which are in the North Cell along the east boundary, and one is in the South Cell along the south boundary.
- Run #2 uses all of the Run #1 parameters excluding the five IEPA-MW wells. The efficiency is reduced to 81.9%. As shown in Figure 3, a large area of the source zone becomes nondetect due to the lack of monitoring wells downgradient.
- Run #3 uses all of the Run #1 parameters, but removes IEPA-MW1, IEPA-MW3, and IEPA-MW5 from the well network. As shown in Figure 4, the efficiency increases to 97.5%.
- Run #4 uses all of the Run #1 parameters, but removes IEPA-MW1, IEPA-MW2, IEPA-MW3, IEPA-MW5 and LF01-MW30S/D, and adds a new well between the proposed locations of IEPA-MW2 and LF01-MW30S/D. Compared to Run #3, only one well is reduced from the monitoring network, but the efficiencies dropped to 84.5%. The non-detection areas and the well locations are shown in Figure 5.
- Run #5 uses all of the Run #1 parameters, but removes IEPA-MW1, IEPA-MW3, IEPA-MW4, IEPA-MW5 and LF01-MW31S/D, and adds a new well between the proposed locations of IEPA-MW4 and LF01-MW31S/D. Compared to Run #3, only

one well is reduced from the monitoring well network and the efficiencies dropped to 92.2%. The non-detect areas and the well locations are shown in Figure 6.

- Run #6 uses wells listed in Table 1 excluding IEPA-MW3. The monitoring efficiency is 99.5% (failed to detect 3 of 653 source elements). Figure 7 shows most of the source zone elements are detected except a few element blocks, two of which are in the North Cell along the east boundary, and one is in the South Cell along the south boundary, same as Run #1.
- Run #7 uses wells listed in Table 1 excluding IEPA-MW3 and IEPA-MW5. The monitoring efficiency is 98.5% (failed to detect 9 of 653 source elements). Figure 8 shows most of the source zone elements are detected except a few element blocks, two of which are in the North Cell along the east boundary, and in the South Cell along the south boundary.

SUMMARY

According to the IEPA's guidance document LPC-PA2, a monitoring efficiency of 99% is required for well networks designed for new landfills. Runs #1 (99.5% efficiency) and #6 (99.5% efficiency) meet IEPA's efficiency criterion. Based on the model results, HGL recommends the well configuration associated with Run #6. In a letter dated January 28, 2013, IEPA approved of the well configuration from Run #6 as the LF-01 LTM well network. Under this configuration, all wells listed in Table 1, excluding the proposed IEPA well IEPA-MW3, would be used.

REFERENCES

Domenico, P.A. and G.A. Robbins, 1985. “A New Method of Contaminant Plume Analysis.” *Ground Water*, 23, 4: 476–485.

Domenico, P.A, 1987. “An Analytical Model for Multidimensional Transport of a Decaying Contaminant Species.” *Journal of Hydrology*, 91: 49–58.

Gelhar, L.W., A. Mantoglou, C. Welty, and K.R. Rehfelt, 1985. A Review of Field Scale Physical Solute Transport Processes in Saturated and Unsaturated Porous Media. Electrical Power Research Institute, RP-2485-05, Palo Alto, California.

HydroGeoLogic, Inc. (HGL), 2012. Draft-Final Feasibility Study Report Addendum for Groundwater at Site LF-01, Scott Air Force Base, Illinois. March.

Isherwood, D., 1981, Geoscience Data Base Handbook for Modeling a Nuclear Waste Repository, Vol.1, NUREG/CR-0912, U.S. Nuclear Regulatory Agency, Washington, D.C.

MWH, 2003a. Final RI/FS for LF01 and FT03, Scott Air Force Base, St. Clair County, Illinois. April.

MWH, 2003b, Draft RCRA Facility Investigation / Corrective Measures Study for Site SS-14 (Consolidated Aircraft Maintenance Squadron). Scott Air Force Base, St. Clair County, IL. January.

Pickens, J. and G. Grisak, 1981. “Scale-Dependent Dispersion in a Stratified Granular Aquifer.” *Water Resources Research*, 17, 4: 1191–1211.

Xu, M. and Y. Eckstein, 1995. “Use of Weighted Least-Squares Method in Evaluation of the Relationship Between Dispersivity and Field Scale.” *Ground Water*, 33, 6: 905–908.

West, M.R., B.H. Kueper, and M.J. Ungs, 2007. “On the Use and Error of Approximation in the Domenico (1987) Solution.” *Ground Water*, 45, 2: 126–135.

AMC, *White Paper* on. *Groundwater Flow and Transport Model for Site SS-14*. (Consolidated Aircraft Maintenance Squadron) Scott Air. Force Base, Illinois. 03/24/2010. http://www.amcadminrec.com/pdfs/scott2/AR-1043.pdf

ATTACHMENTS

Table 1 Well Locations Used in MAP Model

Table 2 Summary of MEMO Results

Figure 1 Post-Closure Groundwater Monitoring Network

Figure 2 Model Run #1 Results, 99.5% Efficiency (3/653)

Figure 3 Model Run #2 Results, 81.9% Efficiency (118/653)

Figure 4 Model Run #3 Results, 97.5% Efficiency (16/653)

Figure 5 Model Run #4 Results, 84.5% Efficiency (101/653)

Figure 6 Model Run #5 Results, 92.2% Efficiency (51/653)

Figure 7 Model Run #6 Results, 99.5% Efficiency (3/653)

Figure 8 Model Run #7 Results, 98.5% Efficiency (9/653)

ATTACHMENTS

This page was intentionally left blank.

Table 1
Well Locations Used in MAP Model

Well Name	X - Easting (feet)	Y - Northing (feet)
LF01-MW01D	2392131.63	681760.03
LF01-MW01S	2392121.45	681763.87
LF01-MW03D	2393175.14	681143.75
LF01-MW03S	2393160.34	681152.73
LF01-MW12D	2393139.04	679765.02
LF01-MW12SR	2393152.84	679782.31
LF01-MW21D	2393337.76	681521.29
LF01-MW21S	2393333.90	681529.29
LF01-MW25D	2392925.74	679555.28
LF01-MW25S	2392927.24	679561.82
LF01-MW30D*	2393258.70	680577.12
LF01-MW30S*	2393260.17	680591.39
LF01-MW31D*	2393364.35	680150.67
LF01-MW31S*	2393365.03	680167.02
LF01-MW33D*	2392309.70	679592.72
LF01-MW33S*	2392292.09	679598.19
LF01-MW34S*	2392123.67	680275.42
LF01-MW34D*	2392138.86	680263.21
LF01-MW35D*	2391791.27	680992.12
LF01-MW35S*	2391805.11	680977.85
LF01-MW36S*	2392792.26	681671.13
LF01-MW36D*	2392806.09	681656.86
IEPA-MW1*	2393122.08	681384.71
IEPA-MW2*	2393295.69	680867.35
IEPA-MW3*	2393318.26	680405.54
IEPA-MW4*	2393243.61	679974.99
IEPA-MW5*	2392703.68	679659.01

Note: * wells are proposed but not yet existing.

Table 2
Summary of MEMO Results

Run #	Efficiency (Percent)	Nondetect / Total Source Elements	Total # of Monitoring Wells	Monitoring Wells	Source Width
1	99.5%	3/653	28	All Wells in Table 1	20 feet
2	81.9%	118/653	23	All Table 1 Wells, except IEPA-MW1 to -MW5	20 feet
3	97.5%	16/653	25	All Table 1 Wells, except IEPA-MW1, -MW3, and -MW5	20 feet
4	84.5%	101/653	24	All Table 1 Wells, except IEPA-MW1, -MW2, -MW3, -MW5 and LF01-MW30S/D. Added a new well between IEPA-MW2 and LF01-MW30S/D	20 feet
5	92.2%	51/653	24	All Table 1 Wells, except IEPA-MW1, -MW3, -MW4, -MW5 and LF01-MW31S/D. Added a new well between IEPA-MW4 and LF01-MW31S/D	20 feet
6	99.5%	3/653	27	All Table 1 Wells, except IEPA-MW3	20 feet
7	98.5%	9/653	26	All Table 1 Wells, except IEPA-MW3 and IEPA-MW5	20 feet

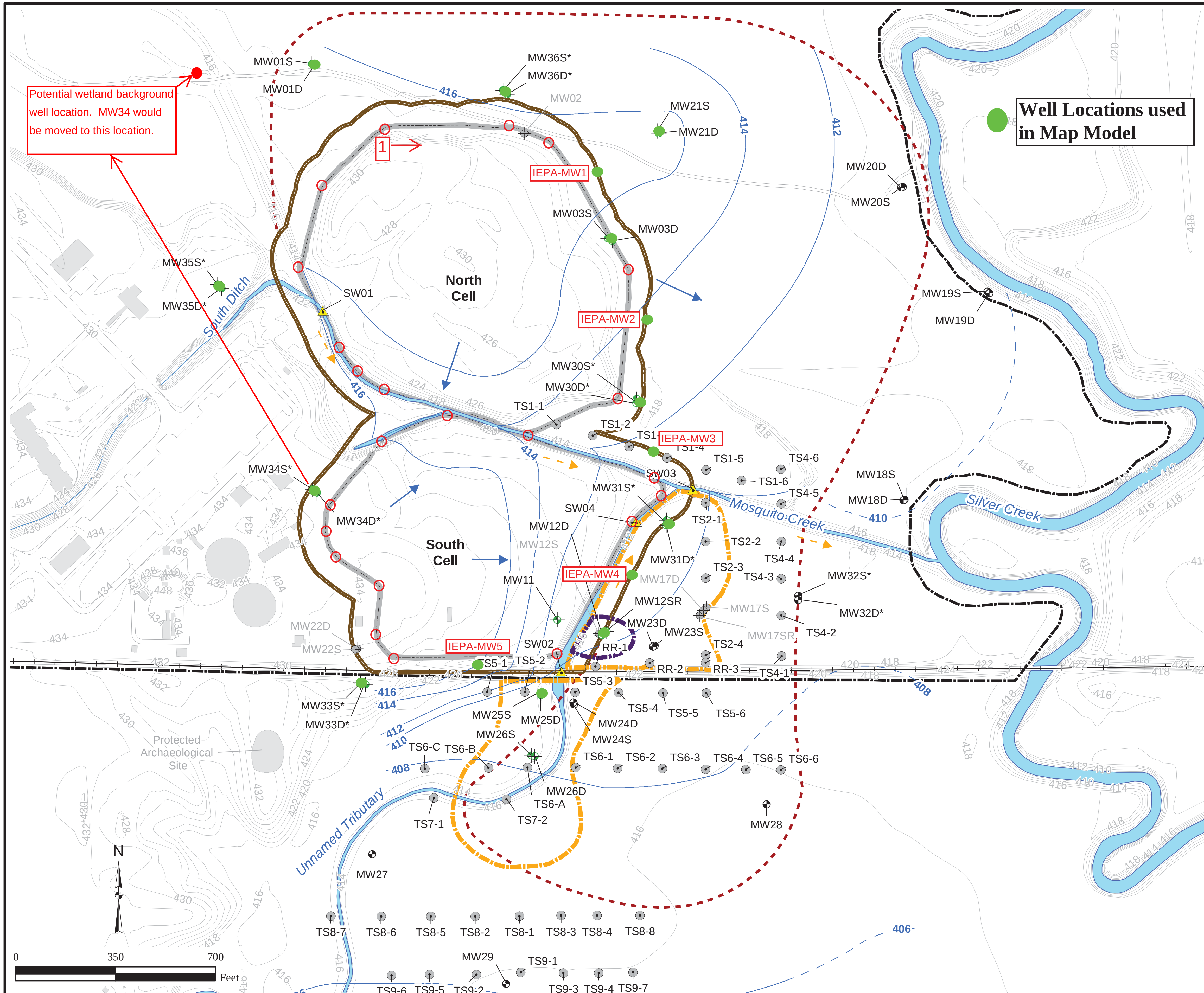


Figure 1
Post-Closure Groundwater Monitoring Network

Legend

- Well Not Included in Groundwater Monitoring Network
- Groundwater COC Monitoring Well
- Landfill Performance Monitoring Well
- Groundwater DPT Boring
- Surface Water Monitoring Point
- Groundwater Flow Direction
- Surface Water Flow Direction
- 408 — Groundwater Contour Line (feet amsl) (dashed where inferred)
- 422 — Elevation Contour Line (feet)
- Rail Line
- Scott AFB Boundary
- Landfill Cover Boundary
- Attenuation Zone
- Extent of Shallow Groundwater Requiring Remedial Action for Iron (Dashed where inferred)
- Deep Transmissive Zone Treatment Area
- Shallow Transmissive Zone Treatment Area
- Surface Water

Notes:

1. Landfill performance wells will be installed near the perimeter of the zone of attenuation to meet requirements of 35 IAC 811.318.
2. Groundwater flow directions based on elevations in the shallow transmissive zone measured on November 7th and 8th, 2011.
3. Elevation contours are based on pre-construction conditions.

* = proposed long term monitoring well
amsl = above mean sea level
D = deep well
IAC = Illinois Administrative Code
S = shallow well
SR = shallow replacement well

\\gst-srv-01\Graphics\SA13\LF01\Post_Closure\(2-1)Monitoring_Landfill_Performance_GW_COCs.mxd
8/21/2012 ST
Source: HGL

HGL HydroGeoLogic, Inc

This page was intentionally left blank.

Figure 2
Model Run #1 Results, 99.5% Efficiency (3/653)

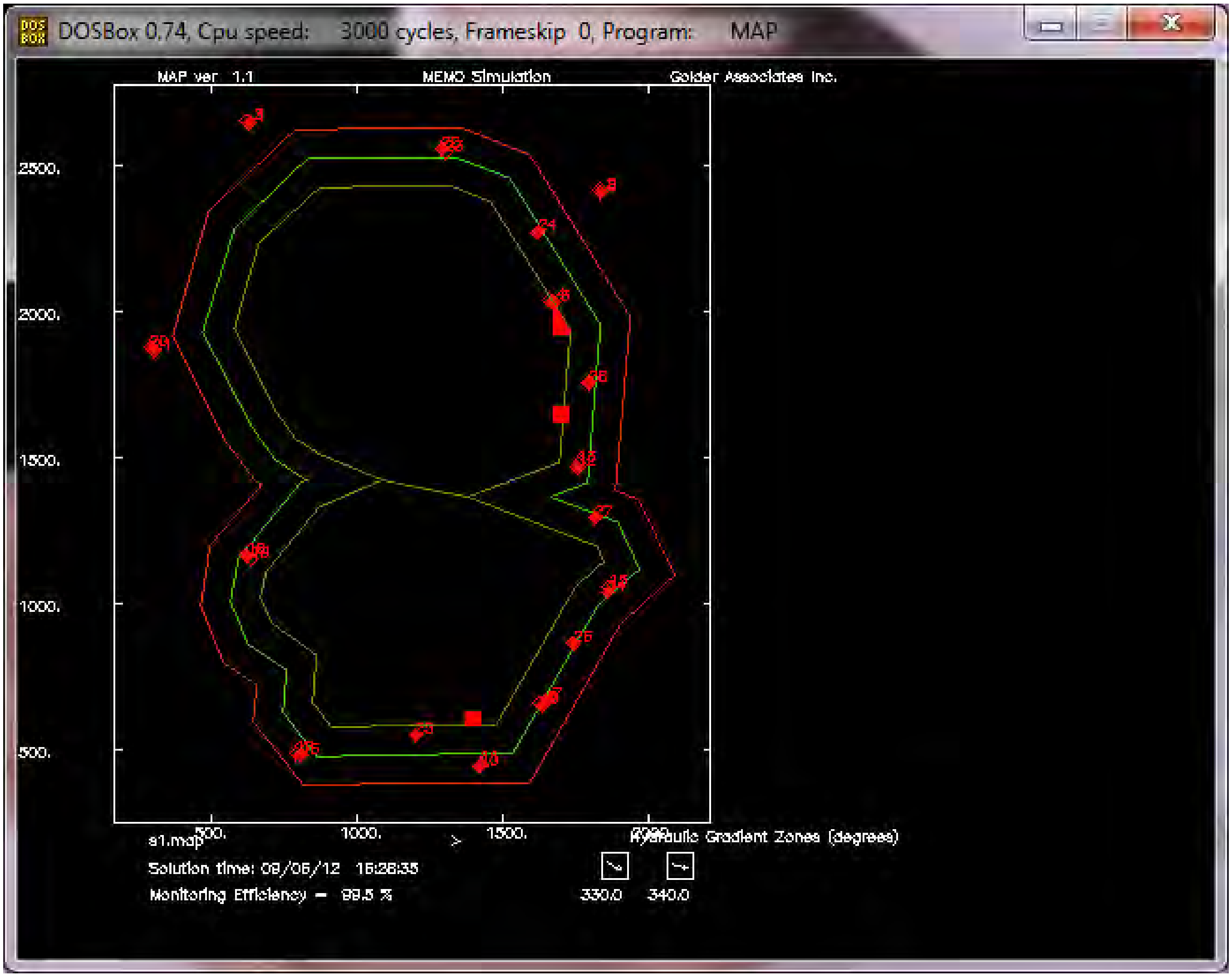


Figure 3
Model Run #2 Results, 81.9% Efficiency (118/653)

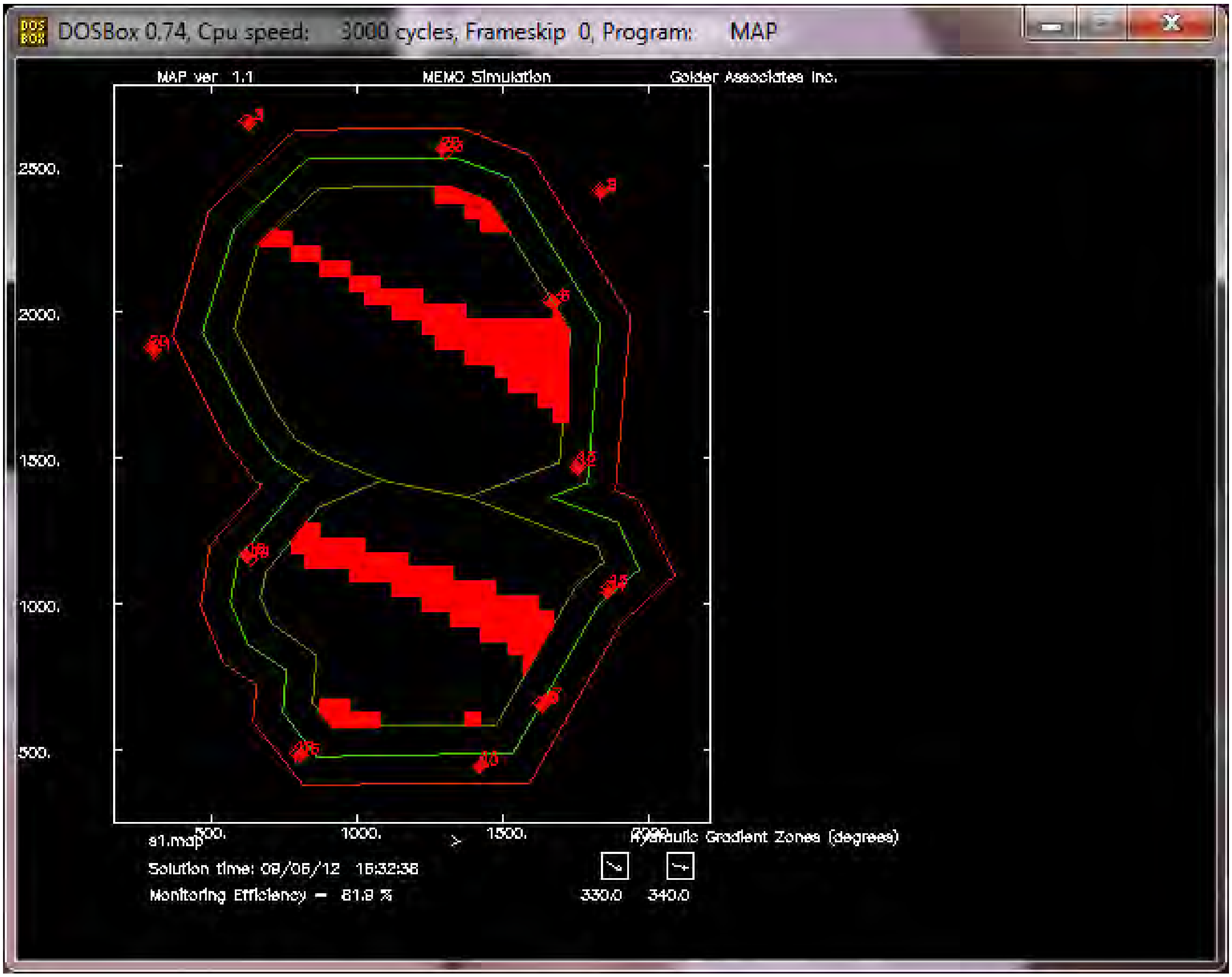


Figure 4
Model Run #3 Results, 97.5% Efficiency (16/653)

Figure 5
Model Run #4 Results, 84.5% Efficiency (101/653)

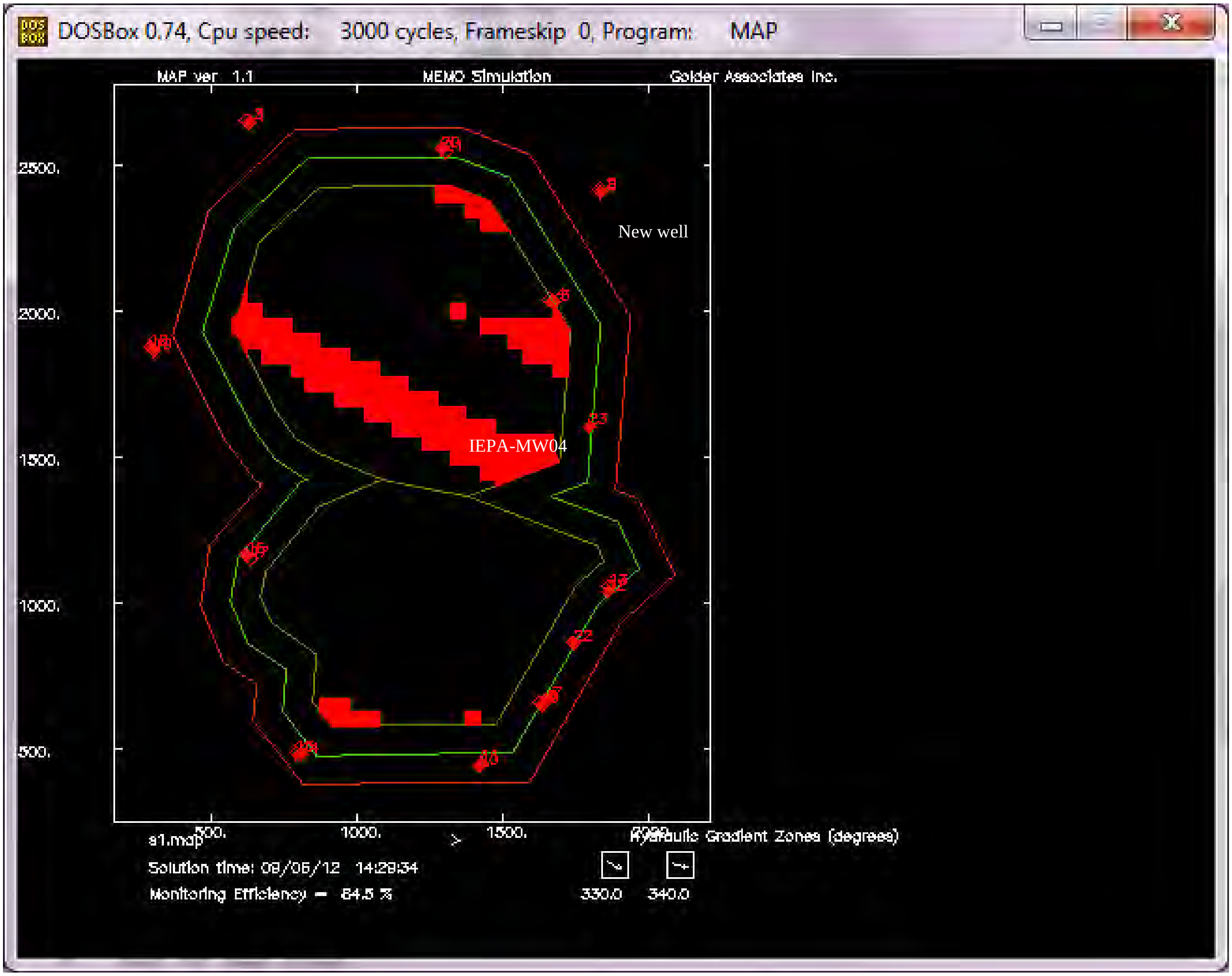


Figure 6
Model Run #5 Results, 92.2% Efficiency (51/653)

Figure 7
Model Run #6 Results, 99.5% Efficiency (3/653)

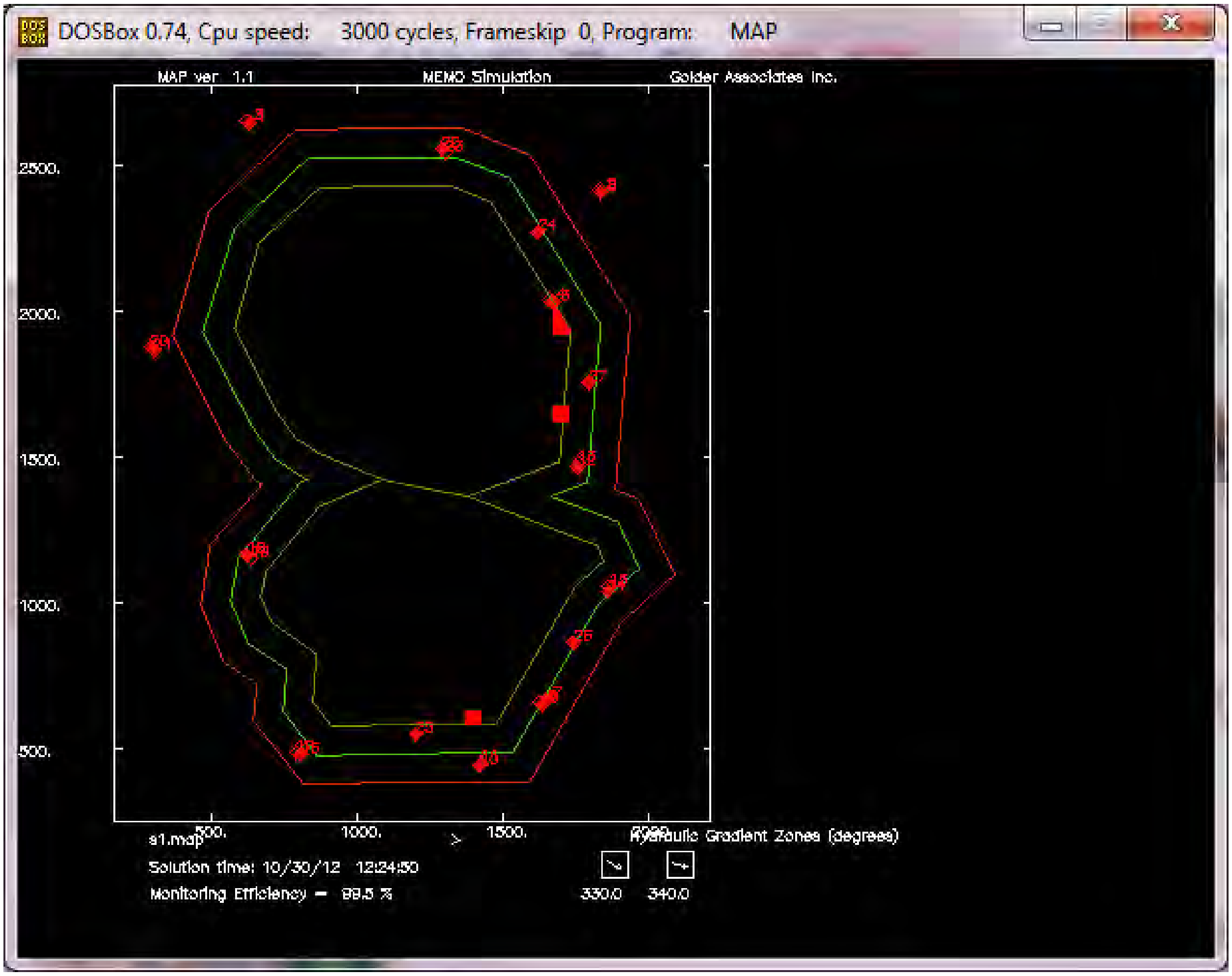
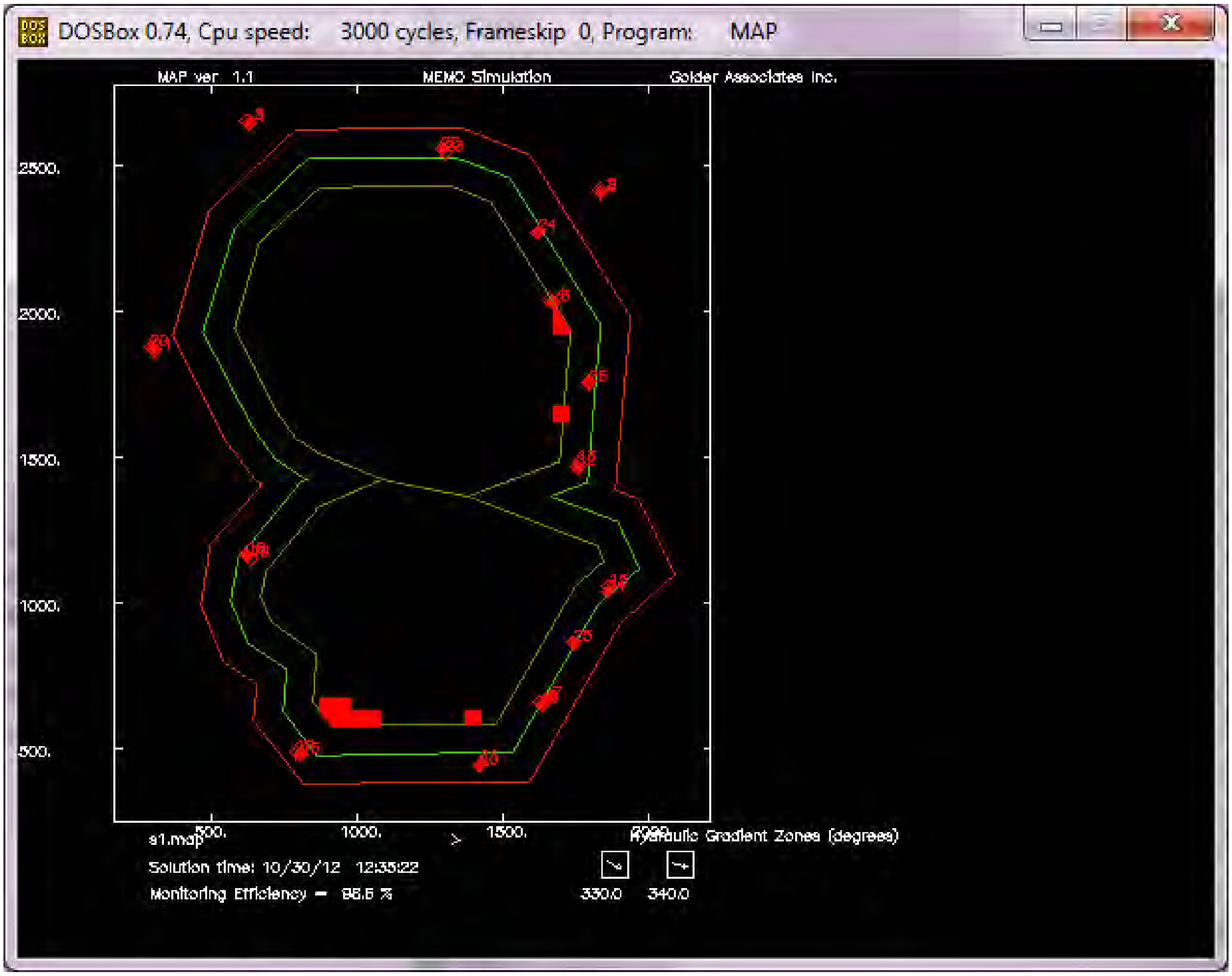


Figure 8
Model Run #7 Results, 98.5% Efficiency (9/653)



This page was intentionally left blank.

SCOTT AFB BASEWIDE GROUNDWATER DATASET SUMMARY

This page was intentionally left blank.

Statistical Summary
Scott AFB Basewide Groundwater Dataset
Scott AFB, IL

Analyte	Number of Data Points	Number of Detects	Minimum Detected Value (μg/L)	Maximum Detected Value (μg/L)	Mean Detected Value (μg/L)	Median Detected Value (μg/L)	UCL Statistic	Distribution	95% UCL (ug/L)
Silver*	35	1	0.99	0.99	0.99	0.99	--	--	--
Aluminum	27	16	28.6	546	134	94.3	95% KM (t) UCL	G	129
Arsenic*	35	4	2.5	19.1	8	5.2	--	--	--
Barium	28	28	56	1100	166	121	95% Chebyshev (mean, Sd) UCL	NP	325
Beryllium*	35	2	1.9	2.5	2.2	2.2	--	--	--
Calcium	26	26	37900	520000	105000	91900	95% Chebyshev (mean, Sd) UCL	NP	182,000
Cadmium*	35	2	0.5	0.54	0.52	0.52	--	--	--
Cobalt	28	7	1	9.8	2.96	2	95% KM (% Bootstrap) UCL	NP	2.56
Chromium*	35	5	0.69	2.1	1.25	1.1	--	--	--
Copper*	35	4	2.1	5.2	3.22	2.8	--	--	--
Iron	28	15	33.7	26000	2370	466	99% KM (Chebyshev) UCL	LN	10,600
Mercury*	38	6	0.067	0.21	0.15	0.155	--	--	--
Potassium	28	28	206	10000	1640	925	95% H-UCL	LN	2,450
Magnesium	28	28	5500	50500	33100	32700	95% Student's-t UCL	N	37,000
Manganese	32	32	1.9	6120	580	111	95% Adjusted Gamma UCL	G	1,020
Molybdenum*	3	3	6	8.4	7.43	7.9	--	--	--
Sodium	29	29	4980	99000	36400	33300	95% Approximate Gamma UCL	G	44,900
Nickel	35	16	1.1	48	8.35	4.5	95% KM (t) UCL	G	6.97
Lead*	35	6	1.7	3	2.26	2.05	--	--	--
Antimony*	35	2	3.7	4.6	4.15	4.15	--	--	--
Selenium*	35	6	0.6	6	3.25	3.15	--	--	--
Vanadium	28	19	0.59	4.4	2.26	2.3	95% KM (t) UCL	N	2.27
Zinc	35	12	2.8	114	17.8	8.6	95% KM (t) UCL	G	13.6

* Six or fewer detections; insufficient detection frequency to support UCL calculation.

UCL = Upper confidence level

-- = not applicable

µg/L = micrograms per liter

NP = nonparametric

LN = lognormal

G = gamma

N = normal

KM = Kaplan-Meier

H-UCL = UCL based upon Land’s H-statistic

This page was intentionally left blank.

FINAL PAGE

ADMINISTRATIVE RECORD

FINAL PAGE